

3. The Motion of Rigid Bodies



Figure 22: Wolfgang Pauli and Niels Bohr stare in wonder at a spinning top.

Having now mastered the technique of Lagrangians, this section will be one big application of the methods. The systems we will consider are the spinning motions of extended objects. As we shall see, these can often be counterintuitive. Certainly Pauli and Bohr found themselves amazed!

We shall consider extended objects that don't have any internal degrees of freedom. These are called “rigid bodies”, defined to be a collection of N points constrained so that the distance between the points is fixed. i.e.

$$|\mathbf{r}_i - \mathbf{r}_j| = \text{constant} \quad (3.1)$$

for all $i, j = 1, \dots, N$. A simple example is a dumbbell (two masses connected by a light rod), or the pyramid drawn in the figure. In both cases, the distances between the masses is fixed.

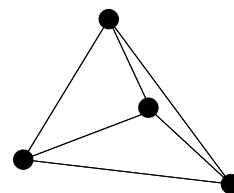


Figure 23:

Often we will work with continuous, rather than discrete, bodies simply by replacing $\sum_i m_i \rightarrow \int d\mathbf{r} \rho(\mathbf{r})$ where $\rho(\mathbf{r})$ is the density of the object. A rigid body has six degrees of freedom

3 Translation + 3 Rotation

The most general motion of a free rigid body is a translation plus a rotation about some point P . In this section we shall develop the techniques required to describe this motion.

3.1 Kinematics

Consider a body fixed at a point P . The most general allowed motion is a rotation about P . To describe this, we specify positions in a *fixed space frame* $\{\tilde{\mathbf{e}}_a\}$ by embedding a *moving body frame* $\{\mathbf{e}_a\}$ in the body so that $\{\mathbf{e}_a\}$ moves with the body.

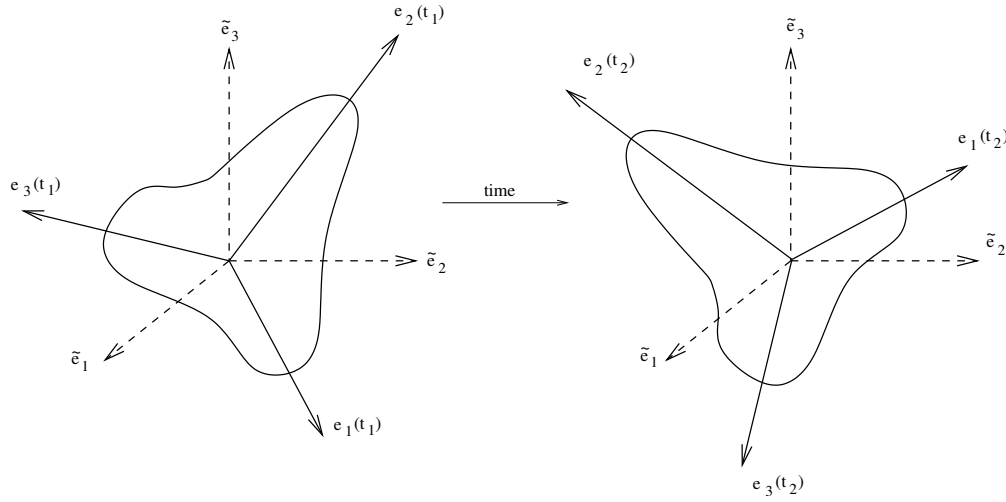


Figure 24: The fixed space frame and the moving body frame.

Both axes are orthogonal, so we have

$$\tilde{\mathbf{e}}_a \cdot \tilde{\mathbf{e}}_b = \delta_{ab} \quad , \quad \mathbf{e}_a(t) \cdot \mathbf{e}_b(t) = \delta_{ab} \quad (3.2)$$

We will soon see that there is a natural choice of the basis $\{\mathbf{e}_a\}$ in the body.

Claim: For all t , there exists a unique orthogonal matrix $R(t)$ with components $R_{ab}(t)$ such that $\mathbf{e}_a(t) = R_{ab}(t)\tilde{\mathbf{e}}_b$

Proof: $\mathbf{e}_a \cdot \mathbf{e}_b = \delta_{ab} \Rightarrow R_{ac}R_{bd}\tilde{\mathbf{e}}_c \cdot \tilde{\mathbf{e}}_d = \delta_{ab} \Rightarrow R_{ac}R_{bc} = \delta_{ab}$ or, in other words, $(R^T R)_{ab} = \delta_{ab}$ which is the statement that R is orthogonal. The uniqueness of R follows by construction: $R_{ab} = \mathbf{e}_a \cdot \tilde{\mathbf{e}}_b$. $\square$

So as the rigid body rotates it is described by a time dependent orthogonal 3×3 matrix $R(t)$. This matrix also has the property that its determinant is 1. (The other possibility is that its determinant is -1 which corresponds to a rotation and a reflection $\mathbf{e}_a \rightarrow -\mathbf{e}_a$). Conversely, every one-parameter family $R(t)$ describes a possible motion of the body. We have

$$C = \text{Configuration Space} = \text{Space of } 3 \times 3 \text{ Special Orthogonal Matrices} \equiv SO(3)$$

A 3×3 matrix has 9 components but the condition of orthogonality $R^T R = 1$ imposes 6 relations, so the configuration space C is 3 dimensional and we need 3 generalised coordinates to parameterise C . We shall describe a useful choice of coordinates, known as Euler angles, in section 3.5.

3.1.1 Angular Velocity

Any point $\mathbf{r}$ in the body can be expanded in either the space frame or the body frame:

$$\begin{aligned} \mathbf{r}(t) &= \tilde{r}_a(t) \tilde{\mathbf{e}}_a && \text{in the space frame} \\ &= r_a \mathbf{e}_a(t) && \text{in the body frame} \end{aligned} \quad (3.3)$$

where $\tilde{r}_b(t) = r_a R_{ab}(t)$. Taking the time derivative, we have

$$\begin{aligned} \frac{d\mathbf{r}}{dt} &= \frac{d\tilde{r}_a}{dt} \tilde{\mathbf{e}}_a && \text{in the space frame} \\ &= r_a \frac{d\mathbf{e}_a(t)}{dt} && \text{in the body frame} \\ &= r_a \frac{dR_{ab}}{dt} \tilde{\mathbf{e}}_b \end{aligned} \quad (3.4)$$

Alternatively, we can ask how the body frame basis itself changes with time,

$$\frac{d\mathbf{e}_a}{dt} = \frac{dR_{ab}}{dt} \tilde{\mathbf{e}}_b = \frac{dR_{ab}}{dt} R_{bc}^{-1} \mathbf{e}_c \equiv \omega_{ac} \mathbf{e}_c \quad (3.5)$$

where, in the last equality, we have defined $\omega_{ac} = \dot{R}_{ab}(R^{-1})_{bc} = \dot{R}_{ab}R_{cb}$ using the fact that $R^T R = 1$.

Claim: $\omega_{ac} = -\omega_{ca}$ i.e. ω is antisymmetric.

Proof: $R_{ab}R_{cb} = \delta_{ac} \Rightarrow \dot{R}_{ab}R_{cb} + R_{ab}\dot{R}_{cb} = 0 \Rightarrow \omega_{ac} + \omega_{ca} = 0$ □

Since ω_{ac} is antisymmetric, we can use it to define an object with a single index (which we will also call ω) using the formula

$$\omega_a = \frac{1}{2}\epsilon_{abc}\omega_{bc} \quad (3.6)$$

so that $\omega_3 = \omega_{12}$ and so on. We treat these ω_a as the components of a vector *in the body frame*, so $\boldsymbol{\omega} = \omega_a \mathbf{e}_a$. Then finally we have our result for the change of the body frame basis with time

$$\frac{d\mathbf{e}_a}{dt} = -\epsilon_{abc}\omega_b \mathbf{e}_c = \boldsymbol{\omega} \times \mathbf{e}_a \quad (3.7)$$

where, in the second equality, we have used the fact that our body frame axis has a “right-handed” orientation, meaning $\mathbf{e}_a \times \mathbf{e}_b = \epsilon_{abc}\mathbf{e}_c$. The vector $\boldsymbol{\omega}$ is called the *instantaneous angular velocity* and its components ω_a are measured with respect to the body frame.

Since the above discussion was a little formal, let’s draw a picture to uncover the physical meaning of $\boldsymbol{\omega}$. Consider a displacement of a given point $\mathbf{r}$ in the body by rotating an infinitesimal amount $d\phi$ about an axis $\hat{\mathbf{n}}$. From the figure, we see that $|d\mathbf{r}| = |\mathbf{r}| d\phi \sin \theta$. Moreover, this displacement is perpendicular to $\mathbf{r}$ since the distance to P is fixed by the definition of a rigid body. So we have

$$d\mathbf{r} = d\boldsymbol{\phi} \times \mathbf{r} \quad (3.8)$$

with $d\boldsymbol{\phi} = \hat{\mathbf{n}}d\phi$. “Dividing” this equation by dt , we have the result

$$\dot{\mathbf{r}} = \boldsymbol{\omega} \times \mathbf{r} \quad (3.9)$$

where $\boldsymbol{\omega} = d\boldsymbol{\phi}/dt$ is the instantaneous angular velocity. In general, both the axis of rotation $\hat{\mathbf{n}}$ and the rate of rotation $d\phi/dt$ will change over time.

Aside: One could define a slightly different type of angular velocity by looking at how the space frame coordinates $\tilde{r}_a(t)$ change with time, rather than the body frame axes $\mathbf{e}_a$. Since we have $\tilde{r}_b(t) = r_a R_{ab}(t)$, performing the same steps as above, we have

$$\dot{\tilde{r}}_b = r_a \dot{R}_{ab} = \tilde{r}_a (R^{-1} \dot{R})_{ab} \quad (3.10)$$

which tempts us to define a different type of angular velocity, sometimes referred to as “convective angular velocity” by $\Omega_{ab} = R_{ac}^{-1} \dot{R}_{cb}$ which has the R^{-1} and $\dot{R}$ in a different order. Throughout our discussion of rigid body motion, we will only deal with the original $\omega = \dot{R}R^{-1}$.

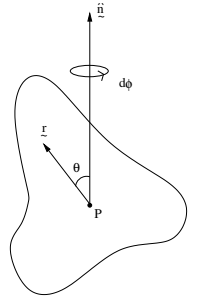


Figure 25:

3.1.2 Path Ordered Exponentials

In the remainder of this chapter, we will devote much effort to determine the angular velocity vector $\boldsymbol{\omega}(t)$ of various objects as they spin and turn. But how do we go from this to the rotation $R(t)$? As described above, we first turn the vector $\boldsymbol{\omega} = \omega_a \mathbf{e}_a$ into a 3×3 antisymmetric matrix $\omega_{ab} = \epsilon_{abc} \omega_c$. Then, from this, we get the rotation matrix R by solving the differential equation

$$\dot{\omega} = \frac{dR}{dt} R^{-1} \quad (3.11)$$

If ω and R were scalar functions of time, then we could simply integrate this equation to get the solution

$$R(t) = \exp \left(\int_0^t \omega(t') dt' \right) \quad (3.12)$$

which satisfies the initial condition $R(0) = 1$. But things are more complicated because both ω and R are matrices. Let's first describe how we take the exponential of a matrix. This is defined by the Taylor expansion. For any matrix M , we have

$$\exp(M) \equiv 1 + M + \frac{1}{2}M^2 + \dots \quad (3.13)$$

As our first guess for the solution to the matrix equation (3.11), we could try the scalar solution (3.12) and look at what goes wrong. If we take the time derivative of the various terms in the Taylor expansion of this putative solution, then problems first arise when we hit the $\frac{1}{2}M^2$ type term. The time derivative of this reads

$$\frac{1}{2} \frac{d}{dt} \left(\int_0^t \omega(t') dt' \right)^2 = \frac{1}{2} \omega(t) \left(\int_0^t \omega(t') dt' \right) + \frac{1}{2} \left(\int_0^t \omega(t') dt' \right) \omega(t) \quad (3.14)$$

We would like to show that $\dot{R} = \omega R$. The first term on the right-hand side looks good since it appears in the Taylor expansion of ωR . But the second term isn't right. The problem is that we cannot commute $\omega(t)$ past $\omega(t')$ when $t' \neq t$. For this reason, equation (3.12) is *not* the solution to (3.11) when ω and R are matrices. But it does give us a hint about how we should proceed. Since the problem is in the ordering of the matrices, the correct solution to (3.11) takes a similar form as (3.12), but with a different ordering. It is the *path ordered exponential*,

$$R(t) = P \exp \left(\int_0^t \omega(t') dt' \right) \quad (3.15)$$

where the P in front means that when we Taylor expand the exponential, all matrices are ordered so that later times appear on the left. In other words

$$R(t) = 1 + \int_0^t \omega(t') dt' + \int_0^{t''} \int_{t'}^t \omega(t'') \omega(t') dt' dt'' + \dots \quad (3.16)$$

The double integral is taken over the range $0 < t' < t'' < t$. If we now differentiate this double integral with respect to t , we get just the one term $\omega(t) \left(\int_0^t \omega(t') dt' \right)$, instead of the two that appear in (3.14). It can be checked that the higher terms in the Taylor expansion also have the correct property if they are ordered so that matrices evaluated at later times appear to the left in the integrals. This type of path ordered integral comes up frequently in theories involving non-commuting matrices, including the standard model of particle physics.

As an aside, the rotation matrix R is a member of the Lie group $SO(3)$, the space of 3×3 orthogonal matrices with unit determinant. The antisymmetric angular velocity matrix ω , corresponding to an instantaneous, infinitesimal rotation, lives in the Lie algebra $so(3)$.

3.2 The Inertia Tensor

Let's look at the kinetic energy for a rotating body. We can write

$$\begin{aligned} T &= \frac{1}{2} \sum_i m_i \dot{\mathbf{r}}_i^2 \\ &= \frac{1}{2} \sum_i m_i (\boldsymbol{\omega} \times \mathbf{r}_i) \cdot (\boldsymbol{\omega} \times \mathbf{r}_i) \\ &= \frac{1}{2} \sum_i m_i ((\boldsymbol{\omega} \cdot \boldsymbol{\omega})(\mathbf{r}_i \cdot \mathbf{r}_i) - (\mathbf{r}_i \cdot \boldsymbol{\omega})^2) \end{aligned} \quad (3.17)$$

Or, in other words, we can write the kinetic energy of a rotating body as

$$T = \frac{1}{2} \omega_a I_{ab} \omega_b \quad (3.18)$$

where I_{ab} , $a, b = 1, 2, 3$ are the components of the *inertia tensor* measured in the body frame, defined by

$$I_{ab} = \sum_i m_i ((\mathbf{r}_i \cdot \mathbf{r}_i) \delta_{ab} - (\mathbf{r}_i)_a (\mathbf{r}_i)_b) \quad (3.19)$$

Note that $I_{ab} = I_{ba}$ so the inertia tensor is symmetric. Moreover, the components are independent of time since they are measured with respect to the body frame. For

continuous bodies, we have the analogous expression

$$I = \int d^3\mathbf{r} \rho(\mathbf{r}) \begin{pmatrix} y^2 + z^2 & -xy & -xz \\ -xy & x^2 + z^2 & -yz \\ -xz & -yz & x^2 + y^2 \end{pmatrix} \quad (3.20)$$

Since I_{ab} is a symmetric real matrix, we can diagonalise it. This means that there exists an orthogonal matrix O such that $OIO^T = I'$ where I' is diagonal. Equivalently, we can rotate the body frame axis $\{\mathbf{e}_a\}$ to coincide with the eigenvectors of I (which are $\{O\mathbf{e}_a\}$) so that, in this frame, the inertia tensor is diagonal. These preferred body axes, in which I is diagonal, are called the *principal axes*. In this basis,

$$I = \begin{pmatrix} I_1 & & \\ & I_2 & \\ & & I_3 \end{pmatrix} \quad (3.21)$$

The eigenvalues I_a are called the *principal moments of inertia*. The kinematical properties of a rigid body are fully determined by its mass, principal axes, and moments of inertia. Often the principal axes are obvious by symmetry.

Claim: The I_a are real and positive.

Proof: If $\mathbf{c}$ is an arbitrary vector, then

$$I_{ab}c^ac^b = \sum_i m_i(r_i^2c^2 - (\mathbf{r}_i \cdot \mathbf{c})^2) \geq 0 \quad (3.22)$$

with equality only if all the $\mathbf{r}_i$ lie on a line. If $\mathbf{c}$ is the a^{th} eigenvector of I then this result becomes $I_{ab}c^ac^b = I_a|\mathbf{c}|^2$ which tells us $I_a \geq 0$. $\square$

Example: The Rod

Consider the inertia tensor of a uniform rod of length l and mass M about its centre. The density of the rod is $\rho = M/l$. By symmetry, we have $I = \text{diag}(I_1, I_1, 0)$ where

$$I_1 = \int_{-l/2}^{l/2} \rho x^2 dx = \frac{1}{12}Ml^2 \quad (3.23)$$

Example: The Disc

Now consider a uniform disc of radius r and mass M . We take the z axis to be perpendicular to the disc and measure I about its centre of mass. Again we know that $I = \text{diag}(I_1, I_2, I_3)$. The density of the disc is $\rho = M/\pi r^2$, so we have

$$I_1 = \int \rho y^2 d^2x \quad , \quad I_2 = \int \rho x^2 d^2x$$

so $I_1 = I_2$ by symmetry, while

$$I_3 = \int \rho(x^2 + y^2) d^2x$$

Therefore

$$I_3 = I_1 + I_2 = 2\pi\rho \int_0^r r'^3 dr' = \frac{1}{2}Mr^2 \quad (3.24)$$

So the moments of inertia are $I_1 = I_2 = \frac{1}{4}Mr^2$ and $I_3 = \frac{1}{2}Mr^2$.

3.2.1 Parallel Axis Theorem

The inertia tensor depends on what point P in the body is held fixed. In general, if we know I about a point P it is messy to compute it about some other point P' . But it is very simple if P happens to coincide with the centre of mass of the object.

Claim: If P' is displaced by $\mathbf{c}$ from the centre of mass, then

$$(I_{\mathbf{c}})_{ab} = (I_{c.of.m})_{ab} + M(c^2\delta_{ab} - \mathbf{c}_a\mathbf{c}_b) \quad (3.25)$$

Proof:

$$\begin{aligned} (I_{\mathbf{c}})_{ab} &= \sum_i m_i \{(\mathbf{r}_i - \mathbf{c})^2 \delta_{ab} - (\mathbf{r}_i - \mathbf{c})_a (\mathbf{r}_i - \mathbf{c})_b\} \\ &= \sum_i m_i \{r_i^2 \delta_{ab} - (\mathbf{r}_i)_a (\mathbf{r}_i)_b + [-2\mathbf{r}_i \cdot \mathbf{c} \delta_{ab} + (\mathbf{r}_i)_a \mathbf{c}_b + (\mathbf{r}_i)_b \mathbf{c}_a] + (c^2 \delta_{ab} - \mathbf{c}_a \mathbf{c}_b)\} \end{aligned} \quad (3.26)$$

But the terms in square brackets that are linear in $\mathbf{r}_i$ vanish if $\mathbf{r}_i$ is measured from the centre of mass since $\sum_i m_i \mathbf{r}_i = 0$. $\square$

The term $M(c^2\delta_{ab} - \mathbf{c}_a\mathbf{c}_b)$ is the inertia tensor we would find if the whole body was concentrated at the centre of mass.

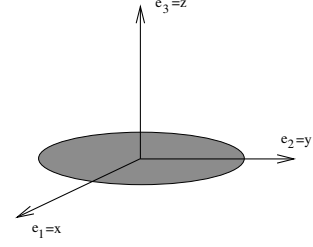


Figure 26:

Example: The Rod Again

The inertia tensor of the rod about one of its ends is $I_1 = \frac{1}{12}Ml^2 + M(l/2)^2 = \frac{1}{3}Ml^2$.

Example: The Disc Again

Consider measuring the inertia tensor of the disc about a point displaced by $\mathbf{c} = (c, 0, 0)$ from the centre. We have

$$\begin{aligned} I_{\mathbf{c}} &= M \begin{pmatrix} \frac{1}{4}r^2 & & \\ & \frac{1}{4}r^2 & \\ & & \frac{1}{2}r^2 \end{pmatrix} + M \left[\begin{pmatrix} c^2 & & \\ & c^2 & \\ & & c^2 \end{pmatrix} - \begin{pmatrix} c^2 & & \\ & 0 & \\ & & 0 \end{pmatrix} \right] \\ &= M \begin{pmatrix} \frac{1}{4}r^2 & & \\ & \frac{1}{4}r^2 + c^2 & \\ & & \frac{1}{2}r^2 + c^2 \end{pmatrix} \end{aligned}$$

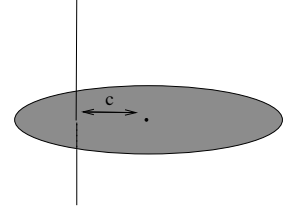


Figure 27:

3.2.2 Angular Momentum

The angular momentum $\mathbf{L}$ about a point P can also be described neatly in terms of the inertia tensor. We have

$$\begin{aligned} \mathbf{L} &= \sum_i m_i \mathbf{r}_i \times \dot{\mathbf{r}}_i \\ &= \sum_i m_i \mathbf{r}_i \times (\boldsymbol{\omega} \times \mathbf{r}_i) \\ &= \sum_i m_i (r_i^2 \boldsymbol{\omega} - (\boldsymbol{\omega} \cdot \mathbf{r}_i) \mathbf{r}_i) \\ &= I \boldsymbol{\omega} \end{aligned} \tag{3.27}$$

In the body frame, we can write $\mathbf{L} = L_a \mathbf{e}_a$ to get

$$L_a = I_{ab} \omega_b \tag{3.28}$$

where $\boldsymbol{\omega} = \omega_a \mathbf{e}_a$. Note that in general, $\boldsymbol{\omega}$ is *not* equal to $\mathbf{L}$: the spin of the body and its angular momentum point in different directions. This fact will lead to many of the peculiar properties of spinning objects.

3.3 Euler's Equations

So far we have been discussing the rotation of a body fixed at a point P . In this section we will be interested in the rotation of a free body suspended in space - for example, a satellite or the planets. Thankfully, this problem is identical to that of an object fixed at a point. Let's show why this is the case and then go on to analyse the motion.

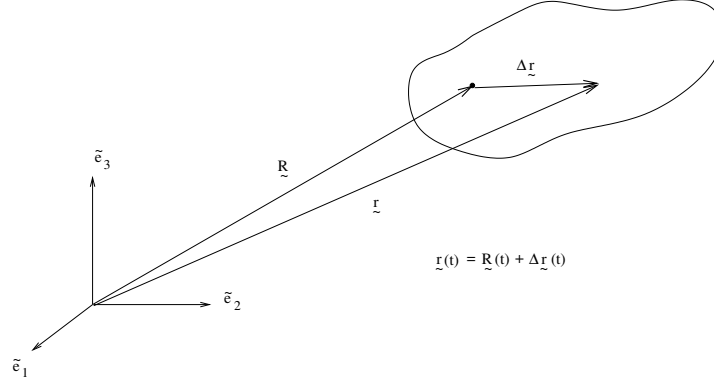


Figure 28:

The most general motion of a body is an overall translation superposed with a rotation. We could take this rotation to be about any point in the body (or, indeed, a point outside the body). But it is useful to consider the rotation to be about the center of mass. We can write the position of a particle in the body as

$$\mathbf{r}_i(t) = \mathbf{R}(t) + \Delta\mathbf{r}_i(t) \quad (3.29)$$

where $\Delta\mathbf{r}_i$ is the position measured from the centre of mass. Then examining the kinetic energy (which, for a free body, is all there is)

$$\begin{aligned} T &= \frac{1}{2} \sum_i m_i \dot{\mathbf{r}}_i^2 \\ &= \sum_i m_i \left[\frac{1}{2} \dot{\mathbf{R}}^2 + \dot{\mathbf{R}} \cdot (\boldsymbol{\omega} \times \Delta\mathbf{r}_i) + \frac{1}{2} (\boldsymbol{\omega} \times \Delta\mathbf{r}_i)^2 \right] \\ &= \frac{1}{2} M \dot{\mathbf{R}}^2 + \frac{1}{2} \boldsymbol{\omega}_a I_{ab} \boldsymbol{\omega}_b \end{aligned} \quad (3.30)$$

where we've used the fact that $\sum_i m_i \Delta\mathbf{r}_i = 0$. So we find that the dynamics separates into the motion of the centre of mass $\mathbf{R}$, together with rotation about the centre of mass. This is the reason that the analysis of the last section is valid for a free object.

3.3.1 Euler's Equations

From now on, we shall neglect the center of mass and concentrate on the rotation of the rigid body. Since the body is free, its angular momentum must be conserved. This gives us the vector equation

$$\frac{d\mathbf{L}}{dt} = 0 \quad (3.31)$$

Let's expand this in the body frame. we have

$$\begin{aligned} 0 &= \frac{d\mathbf{L}}{dt} = \frac{dL_a}{dt} \mathbf{e}_a + L_a \frac{d\mathbf{e}_a}{dt} \\ &= \frac{dL_a}{dt} \mathbf{e}_a + L_a \boldsymbol{\omega} \times \mathbf{e}_a \end{aligned} \quad (3.32)$$

This simplifies if we choose the body axes $\{\mathbf{e}_a\}$ to coincide with the the principal axes. Using $L_a = I_{ab}\omega_b$, we can then write $L_1 = I_1\omega_1$ and so on. The equations of motion (3.32) are now three non-linear coupled first order differential equations,

$$\begin{aligned} I_1\dot{\omega}_1 + \omega_2\omega_3(I_3 - I_2) &= 0 \\ I_2\dot{\omega}_2 + \omega_3\omega_1(I_1 - I_3) &= 0 \\ I_3\dot{\omega}_3 + \omega_1\omega_2(I_2 - I_1) &= 0 \end{aligned} \quad (3.33)$$

These are *Euler's Equations*.

We can extend this analysis to include a torque $\boldsymbol{\tau}$. The equation of motion becomes $\dot{\mathbf{L}} = \boldsymbol{\tau}$ and we can again expand in the body frame along the principal axes to derive Euler's equations (3.33), now with the components of the torque on the RHS.

3.4 Free Tops

“To those who study the progress of exact science, the common spinning-top is a symbol of the labours and the perplexities of men.”

James Clerk Maxwell, no less

In this section, we'll analyse the motion of free rotating bodies (known as free tops) using Euler's equation.

We start with a trivial example: the sphere. For this object, $I_1 = I_2 = I_3$ which means that the angular velocity $\boldsymbol{\omega}$ is parallel to the angular momentum $\mathbf{L}$. Indeed, Euler's equations tell us that ω_a is a constant in this case and the sphere continues to spin around the same axis you start it on. To find a more interesting case, we need to look at the next simplest object.

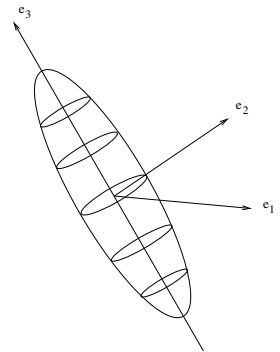


Figure 29:

3.4.1 The Symmetric Top

The symmetric top is an object with $I_1 = I_2 \neq I_3$. A typical example is drawn in figure 29. Euler's equations become

$$\begin{aligned} I_1\dot{\omega}_1 &= \omega_2\omega_3(I_1 - I_3) \\ I_2\dot{\omega}_2 &= -\omega_1\omega_3(I_1 - I_3) \\ I_3\dot{\omega}_3 &= 0 \end{aligned} \quad (3.34)$$

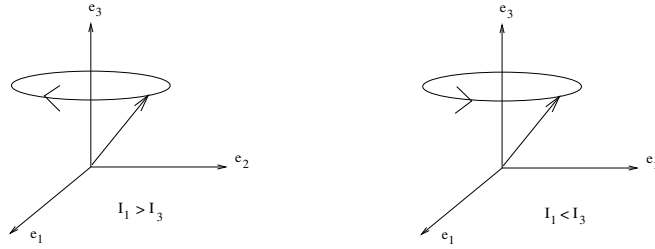


Figure 30: The precession of the spin: the direction of precession depends on whether the object is short and fat ($I_3 > I_1$) or tall and skinny ($I_3 < I_1$)

So, in this case, we see that ω_3 , which is the spin about the symmetric axis, is a constant of motion. In contrast, the spins about the other two axes are time dependent and satisfy

$$\dot{\omega}_1 = \Omega\omega_2 \quad , \quad \dot{\omega}_2 = -\Omega\omega_1 \quad (3.35)$$

where

$$\Omega = \omega_3(I_1 - I_3)/I_1 \quad (3.36)$$

is a constant. These equations are solved by

$$(\omega_1, \omega_2) = \omega_0(\sin \Omega t, \cos \Omega t) \quad (3.37)$$

for any constant ω_0 . This means that, in the body frame, the direction of the spin is not constant: it *precesses* about the $\mathbf{e}_3$ axis with frequency Ω . The direction of the spin depends on the sign on Ω or, in other words, whether $I_1 > I_3$ or $I_1 < I_3$. This is drawn in figure 30.

In an inertial frame, this precession of the spin looks like a wobble. To see this, recall that $\mathbf{L}$ has a fixed direction. Since both ω_3 and L_3 are constant in time, the $\mathbf{e}_3$ axis must stay at a fixed angle with respect to the $\mathbf{L}$ and $\boldsymbol{\omega}$. It rotates about the $\mathbf{L}$ axis as shown in figure 31. We'll examine this wobble more in the next section.

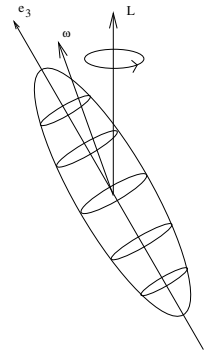


Figure 31:

3.4.2 Example: The Earth's Wobble

The spin of the earth causes it to bulge at the equator so it is no longer a sphere but can be treated as a symmetric top. It is an oblate ellipsoid, with $I_3 > I_1$, and is spherical to roughly 1 part in 300, meaning

$$\frac{I_1 - I_3}{I_1} \approx -\frac{1}{300} \quad (3.38)$$

Of course, we know the magnitude of the spin ω_3 : it is $\omega_3 = (1 \text{ day})^{-1}$. This information is enough to calculate the frequency of the earth's wobble; from (3.36), it should be

$$\Omega_{\text{earth}} = \frac{1}{300} \text{ day}^{-1} \quad (3.39)$$

This calculation was first performed by Euler in 1749 who predicted that the Earth completes a wobble every 300 days. Despite many searches, this effect wasn't detected until 1891 when Chandler re-analysed the data and saw a wobble with a period of 427 days. It is now known as the *Chandler wobble*. It is very small! The angular velocity ω intercepts the surface of the earth approximately 10 metres from the North pole and precesses around it. More recent measurements place the frequency at 435 days, with the discrepancy between the predicted 300 days and observed 435 days due to the fact that the Earth is not a rigid body, but is flexible because of tidal effects. Less well understood is why these same tidal effects haven't caused the wobble to dampen and disappear completely. There are various theories about what keeps the wobble alive, from earthquakes to fluctuating pressure at the bottom of the ocean.

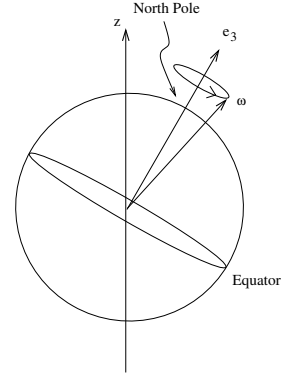


Figure 32:

3.4.3 The Asymmetric Top: Stability

The most general body has no symmetries and $I_1 \neq I_2 \neq I_3 \neq I_1$. The rotational motion is more complicated but there is a simple result that we will describe here. Consider the case where the spin is completely about one of the principal axes, say $\mathbf{e}_1$. i.e.

$$\omega_1 = \Omega \quad , \quad \omega_2 = \omega_3 = 0 \quad (3.40)$$

This solves Euler's equations (3.33). The question we want to ask is: what happens if the spin varies slightly from this direction? To answer this, consider small perturbations about the spin

$$\omega_1 = \Omega + \eta_1 \quad , \quad \omega_2 = \eta_2 \quad , \quad \omega_3 = \eta_3 \quad (3.41)$$

where η_a , $a = 1, 2, 3$ are all taken to be small. Substituting this into Euler's equations and ignoring terms of order η^2 and higher, we have

$$\begin{aligned} I_1 \dot{\eta}_1 &= 0 \\ I_2 \dot{\eta}_2 &= \Omega \eta_3 (I_3 - I_1) \end{aligned} \tag{3.42}$$

$$I_3 \dot{\eta}_3 = \Omega \eta_2 (I_1 - I_2) \tag{3.43}$$

We substitute the third equation into the second to find an equation for just one of the perturbations, say η_2 ,

$$I_2 \ddot{\eta}_2 = \frac{\Omega^2}{I_3} (I_3 - I_1)(I_1 - I_2) \eta_2 \equiv A \eta_2 \tag{3.44}$$

The fate of the small perturbation depends on the sign of the quantity A . We have two possibilities

- $A < 0$: In this case, the disturbance will oscillate around the constant motion.
- $A > 0$: In this case, the disturbance will grow exponentially.

Examining the definition of A , we find that the motion is unstable if

$$I_2 < I_1 < I_3 \quad \text{or} \quad I_3 < I_1 < I_2 \tag{3.45}$$

with all other motions stable. In other words, a body will rotate stably about the axis with the largest or the smallest moment of inertia, but not about the intermediate axis. Pick up a tennis racket and try it for yourself!

3.4.4 The Asymmetric Top: Poinot Construction

The analytic solution for the general motion of an asymmetric top is rather complicated, involving Jacobian elliptic functions. But there's a nice geometrical way of viewing the motion due to Poinot.

We start by working in the body frame. There are two constants of motion: the kinetic energy T and the magnitude of the angular momentum $\mathbf{L}^2$. In terms of the angular velocity, they are

$$2T = I_1 \omega_1^2 + I_2 \omega_2^2 + I_3 \omega_3^2 \tag{3.46}$$

$$\mathbf{L}^2 = I_1^2 \omega_1^2 + I_2^2 \omega_2^2 + I_3^2 \omega_3^2 \tag{3.47}$$

Each of these equations defines an ellipsoid in $\boldsymbol{\omega}$ space. The motion of the vector $\boldsymbol{\omega}$ is constrained to lie on the intersection of these two ellipsoids. The first of these ellipsoids, defined by

$$\frac{I_1}{2T} \omega_1^2 + \frac{I_2}{2T} \omega_2^2 + \frac{I_3}{2T} \omega_3^2 = 1 \quad (3.48)$$

is known as the *inertia ellipsoid* (or, sometimes, the inertia quadric). If we fix the kinetic energy, we can think of this abstract ellipsoid as embedded within the object, rotating with it.

The inertia ellipsoid is drawn in figure 33, where we've chosen $I_1 > I_2 > I_3$ so that the major axis is ω_3 and the minor axis is ω_1 . The lines drawn on the figure are the intersection of the inertia ellipsoid with the other ellipsoid, defined by (3.47), for various values of $\mathbf{L}^2$. Since this has the same major and minor axes as the inertia ellipsoid (because $I_1^2 > I_2^2 > I_3^2$), the intersection lines are small circles around the ω_1 and ω_3 axes, but two lines passing through the ω_2 axis. For fixed T and $\mathbf{L}^2$, the vector $\boldsymbol{\omega}$ moves along one of the intersection lines. This provides a pictorial demonstration of the fact we learnt in the previous subsection: an object will spin in a stable manner around the principal axes with the smallest and largest moments of inertia, but not around the intermediate axis. The path that $\boldsymbol{\omega}$ traces on the inertia ellipsoid is known as the *polhode* curve. We see from the figure that the polhode curves are always closed, and motion in the body frame is periodic.

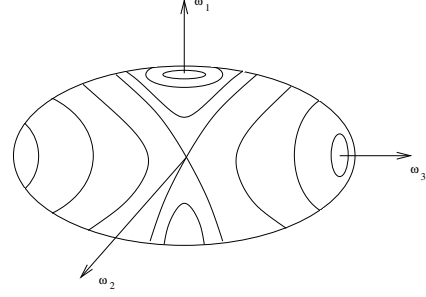


Figure 33:

So much for the body frame. What does all this look like in the space frame? The vector $\mathbf{L}$ is a constant of motion. Since the kinetic energy $2T = \mathbf{L} \cdot \boldsymbol{\omega}$ is also constant, we learn that $\boldsymbol{\omega}$ must lie in a fixed plane perpendicular to $\mathbf{L}$. This is known as the *invariable plane*. The inertia ellipsoid touches the invariable plane at the point defined by the angular velocity vector $\boldsymbol{\omega}$. Moreover, the invariable plane is always tangent to the inertial ellipsoid at the point $\boldsymbol{\omega}$. To see this, note that the angular momentum can be written as

$$\mathbf{L} = \nabla_{\boldsymbol{\omega}} T \quad (3.49)$$

where the gradient operator is in $\boldsymbol{\omega}$ space, i.e. $\nabla_{\boldsymbol{\omega}} = (\partial/\partial\omega_1, \partial/\partial\omega_2, \partial/\partial\omega_3)$. But recall that the inertia ellipsoid is defined as a level surface of T , so equation (3.49) tells

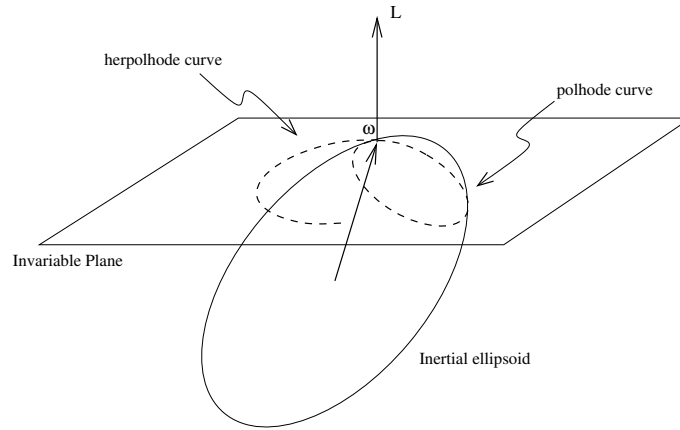


Figure 34: The inertia ellipsoid rolling around on the invariable plane, with the polhode and herpolhode curves drawn for a fixed time period.

us that the angular momentum $\mathbf{L}$ is always perpendicular to the ellipsoid. This, in turn, ensures that the invariable plane is always tangent to the ellipsoid. In summary, the angular velocity traces out two curves: one on the inertia ellipsoid, known as the *polhode curve*, and another on the invariable plane, known as the *herpolhode curve*. The body moves as if it is embedded within the inertia ellipsoid, which rolls around the invariable plane without slipping, with the center of the ellipsoid a constant distance from the plane. The motion is shown in figure 34. Unlike the polhode curve, the herpolhode curve does not necessarily close.

An Example: The Asteroid Toutatis

Astronomical objects are usually symmetric, but there's an important exception wandering around our solar system, depicted in figure² 35. This is the asteroid Toutatis. In September 2004 it passed the earth at a distance of about four times that to the moon. This is (hopefully!) the closest any asteroid will come for the next 60 years. The orbit of Toutatis is thought to be chaotic, which could potentially be bad news for Earth a few centuries from now. As you can see from the picture, its tumbling motion is complicated. It is aperiodic. The pictures show the asteroid at intervals of a day. The angular momentum vector $\mathbf{L}$ remains fixed and vertical throughout the motion. The angular velocity $\boldsymbol{\omega}$ traces out the herpolhode curve in the horizontal plane, perpendicular to $\mathbf{L}$. The angular momentum vector $\boldsymbol{\omega}$ also traces out a curve over the asteroid's

²This picture was created by Scott Hudson of Washington State University and was taken from <http://www.solarviews.com/eng/toutatis.htm> where you can find many interesting facts about the asteroid.

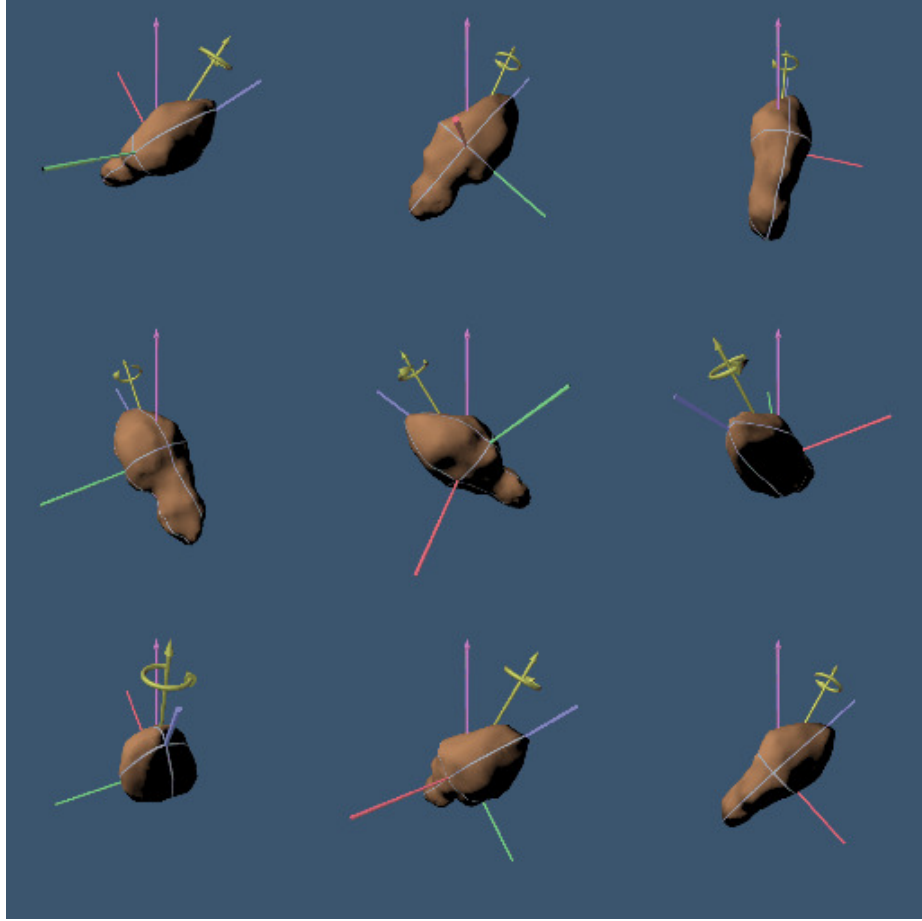


Figure 35: By Toutatis! The three principal axes are shown in red, green and blue (without arrows). The angular momentum $\mathbf{L}$ is the vertical, purple arrow. The angular velocity $\boldsymbol{\omega}$ is the circled, yellow arrow.

surface: this is the polhode curve. It has a period of 5.4 days which you can observe by noting that $\boldsymbol{\omega}$ has roughly the same orientation relative to the principal axes every five to six days.

However, there are further effects at play in a real object like Toutatis which is not spinning around a principal axis. Various stresses and strains lead to dissipation. This means that the angular velocity $\boldsymbol{\omega}$ does not quite follow the polhode curve in Figure 33. Instead it begins close to the major axis ω_3 and slowly spirals towards the minor axis ω_1 . This is why we see so few wobbling asteroids.

3.5 Euler's Angles

So far we've managed to make quite a lot of progress working just with the angular velocity ω_a and we haven't needed to introduce an explicit parametrization of the configuration space C . But to make further progress we're going to need to do this. We will use a choice due to Euler which often leads to simple solutions.

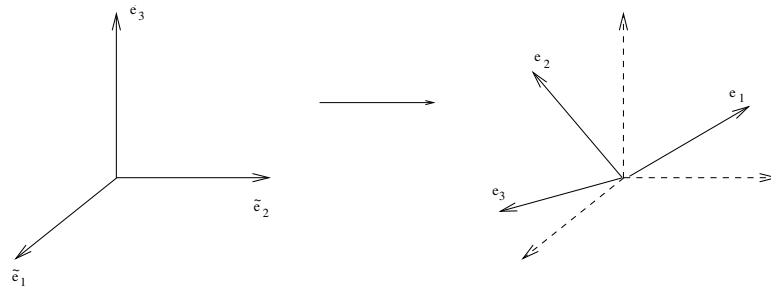


Figure 36: The rotation from space frame $\{\tilde{\mathbf{e}}_a\}$ to body frame $\{\mathbf{e}_a\}$.

A general rotation of a set of axis is shown in Figure 36. We'd like to construct a way of parameterizing such a rotation. The way to do this was first described by Euler:

Euler's Theorem:

An arbitrary rotation may be expressed as the product of 3 successive rotations about 3 (in general) different axes.

Proof: Let $\{\tilde{\mathbf{e}}_a\}$ be space frame axes. Let $\{\mathbf{e}_a\}$ be body frame axes. We want to find the rotation R so that $\mathbf{e}_a = R_{ab}\tilde{\mathbf{e}}_b$. We can accomplish this in three steps

$$\{\tilde{\mathbf{e}}_a\} \xrightarrow{R_3(\phi)} \{\mathbf{e}'_a\} \xrightarrow{R_1(\theta)} \{\mathbf{e}''_a\} \xrightarrow{R_3(\psi)} \{\mathbf{e}_a\} \quad (3.50)$$

Let's look at these step in turn.

Step 1: Rotate by ϕ about the $\tilde{\mathbf{e}}_3$ axis. So $\mathbf{e}'_a = R_3(\phi)_{ab}\tilde{\mathbf{e}}_b$ with

$$R_3(\phi) = \begin{pmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (3.51)$$

This is shown in Figure 37.

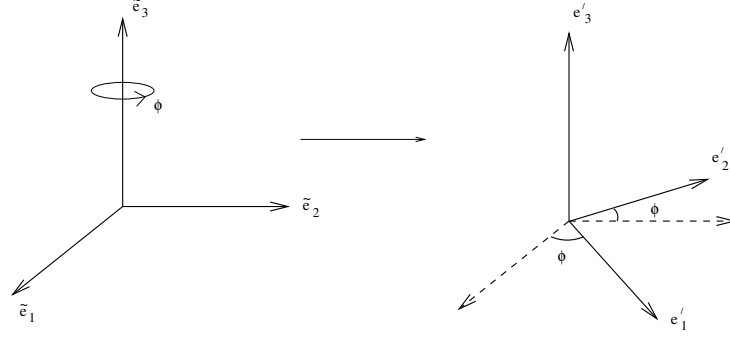


Figure 37: Step 1: Rotate around the space-frame axis $\tilde{\mathbf{e}}_3$.

Step 2: Rotate by θ about the *new* axis $\mathbf{e}'_1$. This axis $\mathbf{e}'_1$ is sometimes called the “line of nodes”. We write $\mathbf{e}''_a = R_1(\theta)_{ab}\mathbf{e}'_b$ with

$$R_1(\theta) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos \theta & \sin \theta \\ 0 & -\sin \theta & \cos \theta \end{pmatrix} \quad (3.52)$$

This is shown in Figure 38

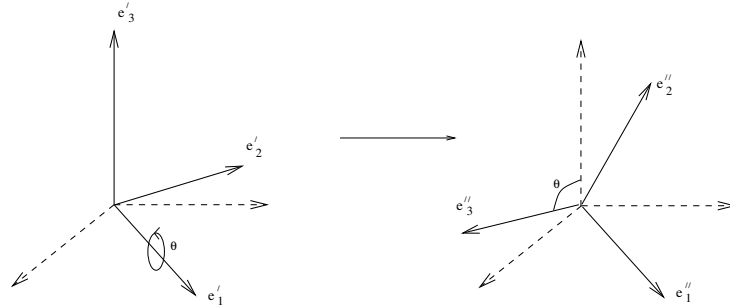


Figure 38: Step 2: Rotate around the new axis axis $\mathbf{e}'_1$.

Step 3: Rotate by ψ about the *new new* axis $\mathbf{e}''_3$ so $\mathbf{e}_a = R_3(\psi)_{ab}\mathbf{e}''_b$ with

$$R_3(\psi) = \begin{pmatrix} \cos \psi & \sin \psi & 0 \\ -\sin \psi & \cos \psi & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad (3.53)$$

This is shown in Figure 39.

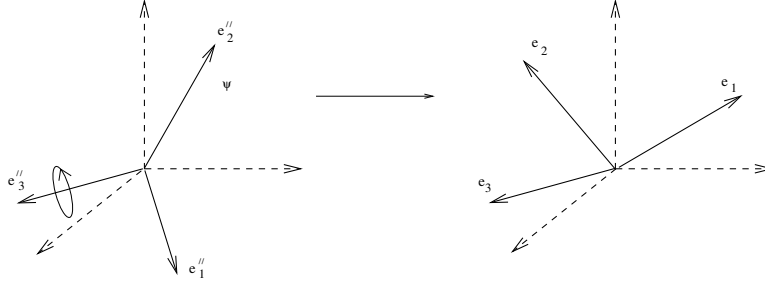


Figure 39: Step 3: Rotate around the latest axis $\mathbf{e}''_3$.

So putting it all together, we have

$$R_{ab}(\phi, \theta, \psi) = [R_3(\psi)R_1(\theta)R_3(\phi)]_{ab} \quad (3.54)$$

□

The angles ϕ, θ, ψ are the Euler angles. If we write out the matrix $R(\phi, \theta, \psi)$ longhand, it reads

$$R = \begin{pmatrix} \cos \psi \cos \phi - \cos \theta \sin \phi \sin \psi & \sin \phi \cos \psi + \cos \theta \sin \psi \cos \phi & \sin \theta \sin \psi \\ -\cos \phi \sin \psi - \cos \theta \cos \psi \sin \phi & -\sin \psi \sin \phi + \cos \theta \cos \psi \cos \phi & \sin \theta \cos \psi \\ \sin \theta \sin \phi & -\sin \theta \cos \phi & \cos \theta \end{pmatrix}$$

Note: Recall that we may expand a vector $\mathbf{r}$ either in the body frame $\mathbf{r} = r_a \mathbf{e}_a$, or in the space frame $\mathbf{r} = \tilde{r}_a \tilde{\mathbf{e}}_a$. The above rotations can be equally well expressed in terms of the coordinates r_a rather than the basis $\{\mathbf{e}_a\}$: we have $\tilde{r}_b = r_a R_{ab}$. Be aware that some books choose to describe the Euler angles in terms of the coordinates r_a which they write in vector form. In some conventions this can lead to an apparent reversal in the ordering of the three rotation matrices.

3.5.1 Leonhard Euler (1707-1783)

As is clear from the section headings, the main man for this chapter is Euler, by far the most prolific mathematician of all time. As well as developing the dynamics of rotations, he made huge contributions to the fields of number theory, geometry, topology, analysis and fluid dynamics. For example, the lovely equation $e^{i\theta} = \cos \theta + i \sin \theta$ is due to Euler. In 1744 he was the first to correctly present a limited example of the calculus of variations (which we saw in section 2.1) although he generously gives credit to a rather botched attempt by his friend Maupertuis in the same year. Euler also invented much of the modern notation of mathematics: $f(x)$ for a function; e for exponential; π for, well, π and so on.

Euler was born in Basel, Switzerland and held positions in St Petersburg, Berlin and, after falling out with Frederick the Great, St Petersburg again. He was pretty much absorbed with mathematics day and night. Upon losing the sight in his right eye in his twenties he responded with: “Now I will have less distraction”. Even when he went completely blind later in life, it didn’t slow him down much as he went on to produce over half of his total work. The St Petersburg Academy of Science continued to publish his work for a full 50 years after his death.

3.5.2 Angular Velocity

There’s a simple expression for the instantaneous angular velocity $\boldsymbol{\omega}$ in terms of Euler angles. To derive this, we could simply plug (3.54) into the definition of angular velocity (3.5). But this is tedious, and a little bit of thought about what this means physically will get us there quicker. Consider the motion of a rigid body in an infinitesimal time dt during which

$$(\psi, \theta, \phi) \rightarrow (\psi + d\psi, \theta + d\theta, \phi + d\phi) \quad (3.55)$$

From the definition of the Euler angles, the angular velocity must be of the form

$$\boldsymbol{\omega} = \dot{\phi} \tilde{\mathbf{e}}_3 + \dot{\theta} \mathbf{e}'_1 + \dot{\psi} \mathbf{e}_3 \quad (3.56)$$

But we can express the first two vectors in terms of the body frame. They are

$$\begin{aligned} \tilde{\mathbf{e}}_3 &= \sin \theta \sin \psi \mathbf{e}_1 + \sin \theta \cos \psi \mathbf{e}_2 + \cos \theta \mathbf{e}_3 \\ \mathbf{e}'_1 &= \cos \psi \mathbf{e}_1 - \sin \psi \mathbf{e}_2 \end{aligned} \quad (3.57)$$

from which we can express $\boldsymbol{\omega}$ in terms of the Euler angles in the body frame axis

$$\boldsymbol{\omega} = [\dot{\phi} \sin \theta \sin \psi + \dot{\theta} \cos \psi] \mathbf{e}_1 + [\dot{\phi} \sin \theta \cos \psi - \dot{\theta} \sin \psi] \mathbf{e}_2 + [\dot{\psi} + \dot{\phi} \cos \theta] \mathbf{e}_3 \quad (3.58)$$

By playing a similar game, we can also express $\boldsymbol{\omega}$ in the space frame axis.

3.5.3 The Free Symmetric Top Revisited

In section 3.4 we studied the free symmetric top working in the *body frame* and found a constant spin ω_3 while, as shown in equation (3.37), ω_1 and ω_2 precess with frequency

$$\Omega = \omega_3 \frac{(I_1 - I_3)}{I_1} \quad (3.59)$$

But what does this look like in the *space frame*? Now that we have parametrised motion in the space frame in terms of Euler angles, we can answer this question. This is simplest if we choose the angular momentum $\mathbf{L}$ to lie along the $\tilde{\mathbf{e}}_3$ space-axis. Then, since we have already seen that $\mathbf{e}_3$ sits at a fixed angle to $\mathbf{L}$, from the figure we see that $\dot{\theta} = 0$. Now we could either use the equations (3.58) or, alternatively, just look at figure 40, to see that we should identify $\Omega = \dot{\psi}$.

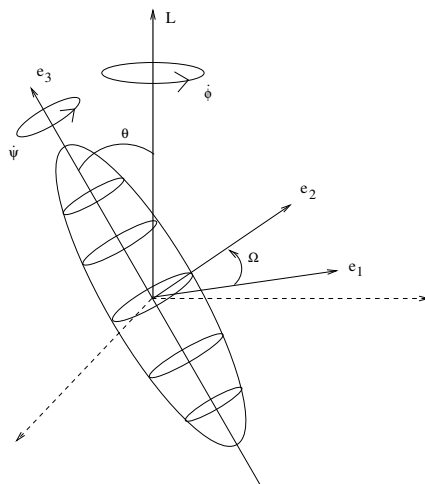


Figure 40: Euler angles for the free symmetric top when $\mathbf{L}$ coincides with $\tilde{\mathbf{e}}_3$

But we know from (3.58) that the expression for ω_3 (which, remember, is the component of $\boldsymbol{\omega}$ in the body frame) in terms of Euler angles is $\omega_3 = \dot{\psi} + \dot{\phi} \cos \theta$ so, substituting for $\Omega = \dot{\psi}$, we find the precession frequency

$$\dot{\phi} = \frac{I_3 \omega_3}{I_1 \cos \theta} \quad (3.60)$$

An Example: The Wobbling Plate

The physicist Richard Feynman tells the following story:

“I was in the cafeteria and some guy, fooling around, throws a plate in the air. As the plate went up in the air I saw it wobble, and I noticed the red medallion of Cornell on the plate going around. It was pretty obvious to me that the medallion went around faster than the wobbling.

I had nothing to do, so I start figuring out the motion of the rotating plate. I discover that when the angle is very slight, the medallion rotates twice as fast as the wobble rate – two to one. It came out of a complicated equation!

I went on to work out equations for wobbles. Then I thought about how the electron orbits start to move in relativity. Then there’s the Dirac equation in electrodynamics. And then quantum electrodynamics. And before I knew it....the whole business that I got the Nobel prize for came from that piddling around with the wobbling plate.”

Feynman was right about quantum electrodynamics. But what about the plate? We can look at this easily using what we've learnt. The spin of the plate is ω_3 , while the precession, or wobble, rate is $\dot{\phi}$ which is given in (3.60). To calculate this, we need the moments of inertia for a plate. But we figured this out for the disc in Section 3.2 where we found that $I_3 = 2I_1$. We can use this to see that $\dot{\psi} = -\omega_3$ for this example and so, for slight angles θ , have

$$\dot{\phi} \approx -2\dot{\psi} \quad (3.61)$$

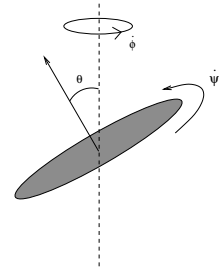


Figure 41:

Or, in other words, the wobble rate is twice as fast as the spin of the plate. It's the opposite to how Feynman remembers!

There is another elegant and simple method you can use to see that Feynman was wrong: you can pick up a plate and throw it. It's hard to see that the wobble to spin ratio is exactly two. But it's easy to see that it wobbles faster than it spins.

3.6 The Heavy Symmetric Top

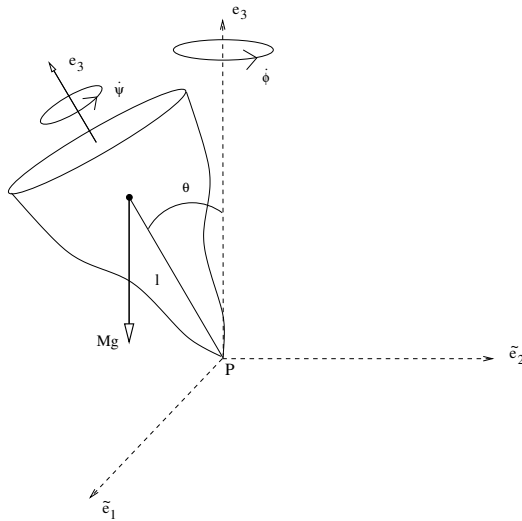


Figure 42: The heavy top with its Euler angles

The “heavy” in the title of this section means that the top is acted upon by gravity. We'll deal only with a symmetric top, pinned at a point P which is a distance l from the centre of mass. This system is drawn in the figure. The principal axes are $\mathbf{e}_1$, $\mathbf{e}_2$

and $\mathbf{e}_3$ and we have $I_1 = I_2$. From what we have learnt so far, it is easy to write down the Lagrangian:

$$\begin{aligned} L &= \frac{1}{2}I_1(\omega_1^2 + \omega_2^2) + \frac{1}{2}I_3\omega_3^2 - Mgl \cos \theta \\ &= \frac{1}{2}I_1(\dot{\theta}^2 + \sin^2 \theta \dot{\phi}^2) + \frac{1}{2}I_3(\dot{\psi} + \cos \theta \dot{\phi})^2 - Mgl \cos \theta \end{aligned} \quad (3.62)$$

A quick examination of this equation tells us that both ψ and ϕ are ignorable coordinates. This gives us the constants of motion p_ψ and p_ϕ , where

$$p_\psi = I_3(\dot{\psi} + \cos \theta \dot{\phi}) = I_3\omega_3 \quad (3.63)$$

This is the angular momentum about the symmetry axis $\mathbf{e}_3$ of the top. The angular velocity $\boldsymbol{\omega}_3$ about this axis is simply called the *spin* of the top and, as for the free symmetric top, it is a constant. The other constant of motion is

$$p_\phi = I_1 \sin^2 \theta \dot{\phi} + I_3 \cos \theta (\dot{\psi} + \dot{\phi} \cos \theta) \quad (3.64)$$

As well as these two conjugate momenta, the total energy E is also conserved

$$E = T + V = \frac{1}{2}I_1(\dot{\theta}^2 + \dot{\phi}^2 \sin^2 \theta) + \frac{1}{2}I_3\omega_3^2 + Mgl \cos \theta \quad (3.65)$$

To simplify these equations, let's define the two constants

$$a = \frac{I_3\omega_3}{I_1} \quad \text{and} \quad b = \frac{p_\phi}{I_1} \quad (3.66)$$

Then we can write

$$\dot{\phi} = \frac{b - a \cos \theta}{\sin^2 \theta} \quad (3.67)$$

and

$$\dot{\psi} = \frac{I_1 a}{I_3} - \frac{(b - a \cos \theta) \cos \theta}{\sin^2 \theta} \quad (3.68)$$

So if we can solve $\theta = \theta(t)$ somehow, then we can always integrate these two equations to get $\phi(t)$ and $\psi(t)$. But first we have to figure out what θ is doing. To do this, let's define the “reduced energy” $E' = E - \frac{1}{2}I_3\omega_3^2$. Then, since E and ω_3 are constant, so is E' . We have

$$E' = \frac{1}{2}I_1\dot{\theta}^2 + V_{\text{eff}}(\theta) \quad (3.69)$$

where the effective potential is

$$V_{\text{eff}}(\theta) = \frac{I_1(b - a \cos \theta)^2}{2 \sin^2 \theta} + Mgl \cos \theta \quad (3.70)$$

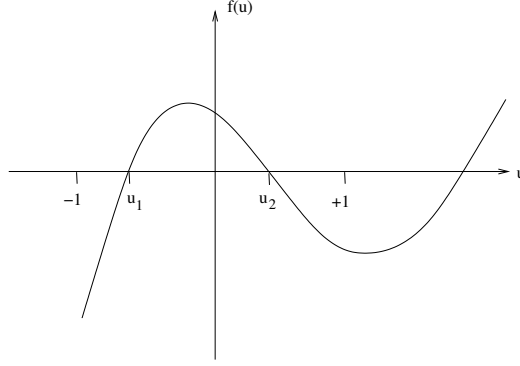


Figure 43:

So we've succeeded in getting an equation (3.69) purely in terms of θ . To simplify the analysis, let's define the new coordinate

$$u = \cos \theta \quad (3.71)$$

Clearly $-1 \leq u \leq 1$. We'll also define two further constants to help put the equations in the most concise form

$$\alpha = \frac{2E'}{I_1} \quad \text{and} \quad \beta = \frac{2Mgl}{I_1} \quad (3.72)$$

With all these redefinitions, the equations of motion (3.67), (3.68) and (3.69) can be written as

$$\dot{u}^2 = (1 - u^2)(\alpha - \beta u) - (b - au)^2 \equiv f(u) \quad (3.73)$$

$$\dot{\phi} = \frac{b - au}{1 - u^2} \quad (3.74)$$

$$\dot{\psi} = \frac{I_1 a}{I_3} - \frac{u(b - au)}{1 - u^2} \quad (3.75)$$

We could take the square root of equation (3.73) and integrate to reduce the problem to quadrature. The resulting integral is known as an “elliptic integral”. But, rather than doing this, there's a better way to understand the physics qualitatively.

Note that the function $f(u)$ defined in (3.73) is a cubic polynomial that behaves as

$$f(u) \rightarrow \begin{cases} +\infty & \text{as } u \rightarrow \infty \\ -\infty & \text{as } u \rightarrow -\infty \end{cases} \quad (3.76)$$

and $f(\pm 1) = -(b \mp a)^2 \leq 0$. So if we plot the function $f(u)$, it looks like figure 43

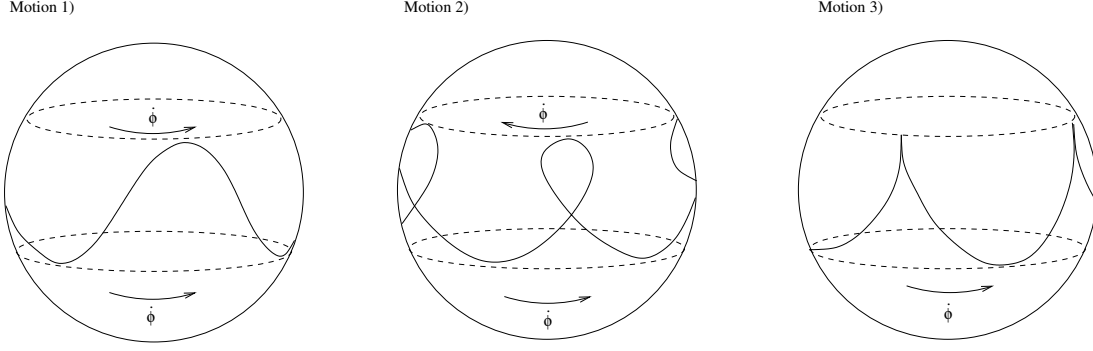


Figure 44: The three different types of motion depend on the direction of precession at the extremal points.

The physical range is $\dot{u}^2 = f(u) > 0$ and $-1 \leq u \leq 1$ so we find that, like in the spherical pendulum and central force problem, the system is confined to lie between the two roots of $f(u)$.

There are three possibilities for the motion depending on the sign of $\dot{\phi}$ at the two roots $u = u_1$ and $u = u_2$ as determined by (3.74). These are

- $\dot{\phi} > 0$ at both $u = u_1$ and $u = u_2$
- $\dot{\phi} > 0$ at $u = u_1$, but $\dot{\phi} < 0$ at $u = u_2$
- $\dot{\phi} > 0$ at $u = u_1$ and $\dot{\phi} = 0$ at $u = u_2$

The different paths of the top corresponding to these three possibilities are shown in figure 44. Motion in ϕ is called *precession* while motion in θ is known as *nutation*.

3.6.1 Letting the Top go

The last of these three motions is not as unlikely as it may first appear. Suppose we spin the top and let it go at some angle θ . What happens? We have the initial conditions

$$\begin{aligned}
 \dot{\theta}_{t=0} = 0 &\Rightarrow f(u_{t=0}) = 0 \\
 &\Rightarrow u_{t=0} = u_2 \\
 \text{and } \dot{\phi}_{t=0} = 0 &\Rightarrow b - au_{t=0} = 0 \\
 &\Rightarrow u_{t=0} = \frac{b}{a}
 \end{aligned} \tag{3.77}$$

Remember also that the quantity

$$p_\phi = I_1 \dot{\phi} \sin^2 \theta + I_3 \omega_3 \cos \theta = I_3 \omega_3 \cos \theta_{t=0} \quad (3.78)$$

is a constant of motion. We now have enough information to figure out the qualitative motion of the top. Firstly, it starts to fall under the influence of gravity, so θ increases. But as the top falls, $\dot{\phi}$ must turn and increase in order to keep p_ϕ constant. Moreover, we also see that the direction of the precession $\dot{\phi}$ must be in the same direction as the spin ω_3 itself. What we get is motion of the third kind.

3.6.2 Uniform Precession

Can we make the top precess with bobbing up and down? i.e. with $\dot{\theta} = 0$ and $\dot{\phi}$ constant. For this to happen, we would need the function $f(u)$ to have a single root u_0 lying in the physical range $-1 \leq u_0 \leq +1$. This root must satisfy,

$$\begin{aligned} f(u_0) &= (1 - u_0^2)(\alpha - \beta u_0) - (b - a u_0)^2 = 0 \\ \text{and } f'(u_0) &= -2u_0(\alpha - \beta u_0) - \beta(1 - u_0^2) + 2a(b - a u_0) = 0 \end{aligned} \quad (3.79)$$

Combining these, we find $\frac{1}{2}\beta = a\dot{\phi} - \dot{\phi}^2 u_0$. Substituting the definitions $I_1 a = I_3 \omega_3$ and $\beta = 2Mgl/I_1$ into this expression, we find

$$Mgl = \dot{\phi}(I_3 \omega_3 - I_1 \dot{\phi} \cos \theta_0) \quad (3.80)$$

The interpretation of this equation is as follows: for a fixed value of ω_3 (the spin of the top) and θ_0 (the angle at which you let it go), we need to give exactly the right push $\dot{\phi}$ to make the top spin without bobbing. In fact, since equation (3.80) is quadratic in $\dot{\phi}$, there are two frequencies with which the top can precess without bobbing.

Of course, these “slow” and “fast” precessions only exist if equation (3.80) has any solutions at all. Since it is quadratic, this is not guaranteed, but requires

$$\omega_3 > \frac{2}{I_3} \sqrt{Mgl I_1 \cos \theta_0} \quad (3.81)$$

So we see that, for a given θ_0 , the top has to be spinning fast enough in order to have uniform solutions. What happens if it's not spinning fast enough? Well, the top falls over!

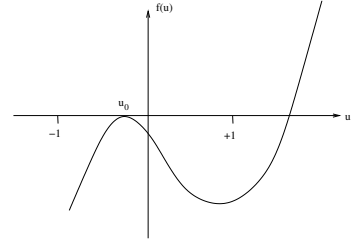


Figure 45:

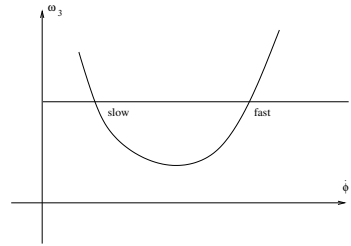


Figure 46:

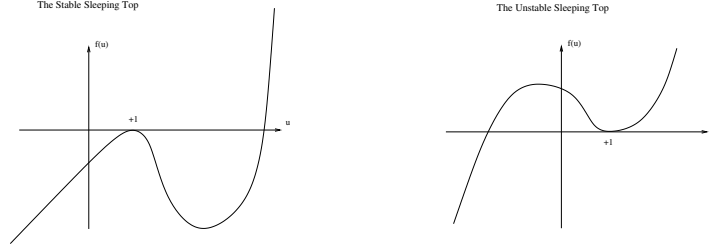


Figure 47: The function $f(u)$ for the stable and unstable sleeping top.

3.6.3 The Sleeping Top

Suppose we start the top spinning in an upright position, with

$$\theta = \dot{\theta} = 0 \quad (3.82)$$

When it spins upright, it is called a *sleeping top*. The question we want to answer is: will it stay there? Or will it fall over? From (3.73), we see that the function $f(u)$ must have a root at $\theta = 0$, or $u = +1$: $f(1) = 0$. From the definitions (3.66) and (3.72), we can check that $a = b$ and $\alpha = \beta$ in this situation and $f(u)$ actually has a double zero at $u = +1$,

$$f(u) = (1 - u)^2(\alpha(1 + u) - a^2) \quad (3.83)$$

The second root of $f(u)$ is at $u_2 = a^2/\alpha - 1$. There are two possibilities

1: $u_2 > 1$ or $\omega_3^2 > 4I_1Mgl/I_3^2$. In this case, the graph of $f(u)$ is drawn in first in figure 47. This motion is stable: if we perturb the initial conditions slightly, we will perturb the function $f(u)$ slightly, but the physical condition that we must restrict to the regime $f(u) > 0$ means that the motion will continue to be trapped near $u = 1$

2: $u_2 < 1$ or $\omega_3^2 < 4I_1Mgl/I_3^2$. In this case, the function $f(u)$ looks like the second figure of 47. Now the top is unstable; slight changes in the initial condition allow a large excursion.

In practice, the top spins upright until it is slowed by friction to $I_3\omega_3 = 2\sqrt{I_1Mgl}$, at which point it starts to fall and precess.

3.6.4 The Precession of the Equinox

The Euler angles for the earth are drawn in figure 48. The earth spins at an angle of $\theta = 23.5^\circ$ to the plane of its orbit around the sun (known as the plane of the elliptic).

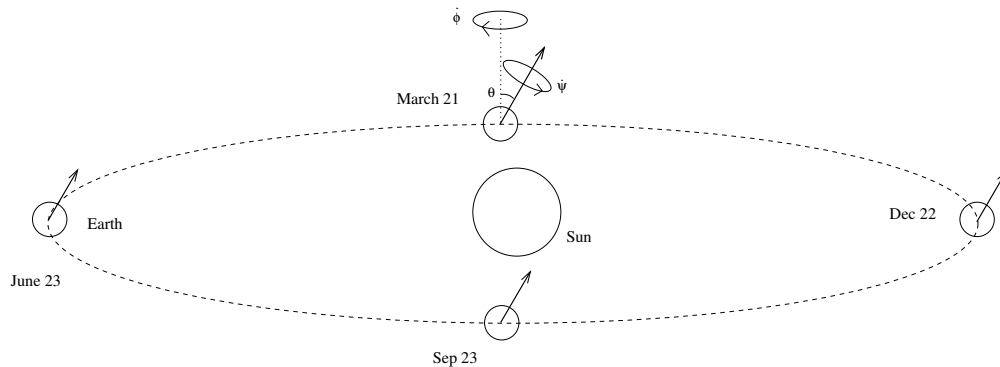


Figure 48: The precession of the earth.

The spin of the earth is $\dot{\psi} = (\text{day})^{-1}$. This causes the earth to bulge at the equator so it is no longer a sphere, but rather a symmetric top. In turn, this allows the moon and sun to exert a torque on the earth which produces a precession $\dot{\phi}$. Physically this means that the direction in which the north pole points traces a circle in the sky and what we currently call the “pole star” will no longer be in several thousand years time. It turns out that this precession is “retrograde” i.e. opposite to the direction of the spin.

One can calculate the precession $\dot{\phi}$ of the earth due to the moon and sun using the techniques described in the chapter. But the calculation is rather long and we won’t go over it in this course (see the book by Hand and Finch if you’re interested). Instead, we will use a different technique to calculate the precession of the earth: astrology!³

To compute the precession of the earth, the first fact we need to know is that Jesus was born in the age of Pisces. This doesn’t mean that Jesus looked up Pisces in his daily horoscope (while scholars are divided over the exact date of his birth, he seems to exhibit many traits of a typical Capricorn) but rather refers to the patch of the sky in which the sun appears during the first day of spring. Known in astronomical terms as the “vernal equinox”, this day of the year is defined by the property that the sun sits directly above the equator at midday. As the earth precesses, this event takes place at a slightly different point in its orbit each year, with a slightly different backdrop of stars as a result. The astrological age is defined to be the background constellation in which the sun rises during vernal equinox.

³I learnt about this fact from John Baez’ website where you can find lots of well written explanations of curiosities in mathematical physics: <http://math.ucr.edu/home/baez/README.html>.

It is easy to remember that Jesus was born in the age of Pisces since the fish was used as an early symbol for Christianity. The next fact that we need to know is that we're currently entering the age of Aquarius (which anyone who has seen the musical Hair will know). So we managed to travel backwards one house of the zodiac in 2,000 years. We've got to make it around 12 in total, giving us a precession time of $2,000 \times 12 = 24,000$ years. The actual value of the precession is 25,700 years. Our calculation is pretty close considering the method!

The earth also undergoes other motion. The value of θ varies from 22.1° to 24.5° over a period of 41,000 years, mostly due to the effects of the other planets. These also affect the eccentricity of the orbit over a period of 105,000 years.

3.7 The Motion of Deformable Bodies

Take a lively cat. (Not one that's half dead like Schrödinger's). Hold it upside down and drop it. The cat will twist its body and land sprightly on its feet. Yet it doesn't do this by pushing against anything and its angular momentum is zero throughout. If the cat were rigid, such motion would be impossible since a change in orientation for a rigid body necessarily requires non-vanishing angular momentum. But the cat isn't rigid (indeed, it can be checked that dead cats are unable to perform this feat) and bodies that can deform are able to reorient themselves without violating the conservation of angular momentum. In this section we'll describe some of the beautiful mathematics that lies behind this. I should warn you that this material is somewhat more advanced than the motion of rigid bodies. The theory described below was first developed in the late 1980s in order to understand how micro-organisms swim⁴.

3.7.1 Kinematics

We first need to describe the configuration space $\mathcal{C}$ of a deformable body. We factor out translations by insisting that all bodies have the same center of mass. Then the configuration space $\mathcal{C}$ is the space of all shapes with some orientation.

Rotations act naturally on the space $\mathcal{C}$ (they simply rotate each shape). This allows us to define the smaller *shape space* $\tilde{\mathcal{C}}$ so that any two configurations in $\mathcal{C}$ which are related by a rotation are identified in $\tilde{\mathcal{C}}$. In other words, any two objects that have the same shape, but different orientation, are described by different points in $\mathcal{C}$, but the same point in $\tilde{\mathcal{C}}$. Mathematically, we say $\tilde{\mathcal{C}} \cong \mathcal{C}/SO(3)$.

⁴See A. Shapere and F. Wilczek, "Geometry of Self-Propulsion at Low Reynolds Number", J. Fluid Mech. **198** 557 (1989) . This is the same Frank Wilczek who won the 2004 Nobel prize for his work on quark interactions.

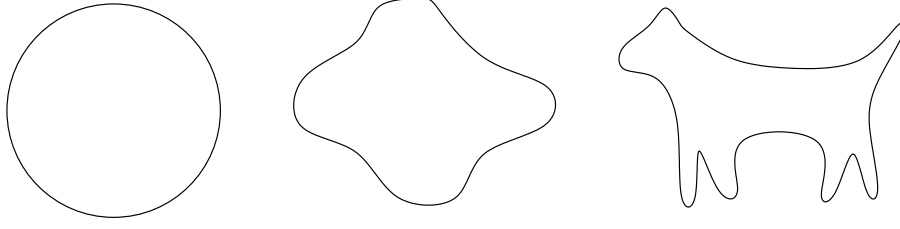


Figure 49: Three possible shapes of a deformable object.

We can describe this in more detail for a body consisting of N point masses, each with position $\mathbf{r}_i$. Unlike in section 3.1, we do not require that the distances between particles are fixed, i.e. $|\mathbf{r}_i - \mathbf{r}_j| \neq \text{constant}$. (However, there may still be some restrictions on the $\mathbf{r}_i$). The configuration space $\mathcal{C}$ is the space of all possible configurations $\mathbf{r}_i$. For each different shape in $\mathcal{C}$, we pick a representative $\tilde{\mathbf{r}}_i$ with some, fixed orientation. It doesn't matter what representative we choose — just as long as we pick one. These variables $\tilde{\mathbf{r}}_i$ are coordinates on the space shape $\tilde{\mathcal{C}}$. For each $\mathbf{r}_i \in \mathcal{C}$, we can always find a rotation matrix $R \in SO(3)$ such that

$$\mathbf{r}_i = R \tilde{\mathbf{r}}_i \quad (3.84)$$

As in section 3.1, we can always do this to continuous bodies. In this case, the configuration space $\mathcal{C}$ and the shape space $\tilde{\mathcal{C}}$ may be infinite dimensional. Examples of different shapes for a continuously deformable body are shown in figure 49.

We want to understand how an object rotates as it changes shape keeping its angular momentum fixed (for example, keeping $\mathbf{L} = 0$ throughout). The first thing to note is that we can't really talk about the rotation between objects of different shapes. (How would you say that the the third object in figure 49 is rotated with respect to the first or the second?). Instead, we should think of an object moving through a sequence of shapes before returning to its initial shape. We can then ask if there's been a net rotation. As the object moves through its sequence of shapes, the motion is described by a time dependent $\tilde{\mathbf{r}}_i(t)$, while the corresponding change through the configuration space is

$$\mathbf{r}_i(t) = R(t) \tilde{\mathbf{r}}(t) \quad (3.85)$$

where the 3×3 rotation matrix $R(t)$ describes the necessary rotation to go from our fixed orientation of the shape $\tilde{\mathbf{r}}$ to the true orientation. As in section 3.1.1, we can define the 3×3 anti-symmetric matrix that describes the instantaneous angular velocity of

the object. In fact, it will for once prove more useful to work with the “convective angular velocity” defined around equation (3.10)

$$\Omega = R^{-1} \frac{dR}{dt} \quad (3.86)$$

This angular velocity is non-zero due to the changing shape of the object, rather than the rigid rotation that we saw before. Let’s do a quick change of notation and write coordinates on the shape space $\tilde{\mathcal{C}}$ as x^A , with $A = 1, \dots, 3N$ instead of in vector notation $\tilde{\mathbf{r}}_i$, with $i = 1, \dots, N$. Then, since Ω is linear in time derivatives, we can write

$$\Omega = \Omega_A(x) \dot{x}^A \quad (3.87)$$

The component $\Omega_A(x)$ is the 3×3 angular velocity matrix induced if the shape changes from x^A to $x^A + \delta x^A$. It is independent of time: all the time dependence sits in the $\dot{x}^A$ factor which tells us how the shape is changing. The upshot is that for each shape $x \in \tilde{\mathcal{C}}$, we have a 3×3 anti-symmetric matrix Ω_A associated to each of the $A = 1, \dots, 3N$ directions in which the shape can change.

However, there is an ambiguity in defining the angular velocity Ω . This comes about because of our arbitrary choice of reference orientation when we picked a representative $\tilde{\mathbf{r}}_i \in \tilde{\mathcal{C}}$ for each shape. We could quite easily have picked a different orientation,

$$\tilde{\mathbf{r}}_i \rightarrow S(x^A) \tilde{\mathbf{r}}_i \quad (3.88)$$

where $S(x^A)$ is a rotation that, as the notation suggests, can vary for each shape x^A . If we pick this new set of representative orientations, then the rotation matrix R defined in (3.85) changes: $R(t) \rightarrow R(t) S^{-1}(x^A)$. Equation (3.86) then tells us that the angular velocity also change as

$$\Omega_A \rightarrow S \Omega_A S^{-1} + S \frac{\partial S^{-1}}{\partial x^A} \quad (3.89)$$

This ambiguity is related to the fact that we can’t define the meaning of rotation between two different shapes. Nonetheless, we will see shortly that when we come to compute the net rotation of the same shape, this ambiguity will disappear, as it must. Objects such as Ω_A which suffer an ambiguity of form (3.89) are extremely important in modern physics and geometry. They are known as non-abelian *gauge potentials* to physicists, or as *connections* to mathematicians.

3.7.2 Dynamics

So far we've learnt how to describe the angular velocity Ω of a deformable object. The next step is to see how to calculate Ω . We'll now show that, up to the ambiguity described in (3.89), the angular velocity Ω is specified by the requirement that the angular momentum $\mathbf{L}$ of the object is zero.

$$\begin{aligned}\mathbf{L} &= \sum_i m_i \mathbf{r}_i \times \dot{\mathbf{r}}_i \\ &= \sum_i m_i \left[(R\tilde{\mathbf{r}}_i) \times (R\dot{\tilde{\mathbf{r}}}_i) + (R\tilde{\mathbf{r}}_i) \times (\dot{R}\tilde{\mathbf{r}}_i) \right] = 0\end{aligned}\quad (3.90)$$

In components this reads

$$L_a = \epsilon_{abc} \sum_i m_i \left[R_{bd} R_{ce} (\tilde{\mathbf{r}}_i)_d (\dot{\tilde{\mathbf{r}}}_i)_e + R_{bd} \dot{R}_{ce} (\tilde{\mathbf{r}}_i)_d (\tilde{\mathbf{r}}_i)_e \right] = 0 \quad (3.91)$$

The vanishing $\mathbf{L} = 0$ is enough information to determine the following result:

Claim: The 3×3 angular velocity matrix $\Omega_{ab} = R_{ac}^{-1} \dot{R}_{cb}$ is given by

$$\Omega_{ab} = \epsilon_{abc} \tilde{I}_{cd}^{-1} \tilde{L}_d \quad (3.92)$$

where $\tilde{I}$ is the instantaneous inertia tensor of the shape described by $\tilde{\mathbf{r}}_i$,

$$\tilde{I}_{ab} = \sum_i m_i ((\tilde{\mathbf{r}}_i \cdot \tilde{\mathbf{r}}_i) \delta_{ab} - (\tilde{\mathbf{r}}_i)_a (\tilde{\mathbf{r}}_i)_b) \quad (3.93)$$

and $\tilde{L}_a$ is the apparent angular momentum

$$\tilde{L}_a = \epsilon_{abc} \sum_i m_i (\tilde{\mathbf{r}}_i)_b (\dot{\tilde{\mathbf{r}}}_i)_c \quad (3.94)$$

Proof: We start by multiplying L_a by ϵ_{afg} . We need to use the fact that if we multiply two ϵ -symbols, we have $\epsilon_{abc}\epsilon_{afg} = (\delta_{bf}\delta_{cg} - \delta_{bg}\delta_{cf})$. Then

$$\begin{aligned}\epsilon_{afg} L_a &= \sum_i m_i \left[R_{fd} R_{ge} (\tilde{\mathbf{r}}_i)_d (\dot{\tilde{\mathbf{r}}}_i)_e - R_{gd} R_{fe} (\tilde{\mathbf{r}}_i)_d (\dot{\tilde{\mathbf{r}}}_i)_e \right. \\ &\quad \left. - R_{gd} \dot{R}_{fe} (\tilde{\mathbf{r}}_i)_d (\tilde{\mathbf{r}}_i)_e + R_{fd} \dot{R}_{ge} (\tilde{\mathbf{r}}_i)_d (\tilde{\mathbf{r}}_i)_e \right] = 0\end{aligned}\quad (3.95)$$

Now multiply by $R_{fb} R_{gc}$. Since R is orthogonal, we know that $R_{fb} R_{fd} = \delta_{bd}$ which, after contracting a bunch of indices, gives us

$$R_{fb} R_{gc} \epsilon_{afg} L_a = \sum_i m_i \left[(\tilde{\mathbf{r}}_i)_b (\dot{\tilde{\mathbf{r}}}_i)_c - (\tilde{\mathbf{r}}_i)_c (\dot{\tilde{\mathbf{r}}}_i)_b - \Omega_{bd} (\tilde{\mathbf{r}}_i)_c (\tilde{\mathbf{r}}_i)_d + \Omega_{cd} (\tilde{\mathbf{r}}_i)_b (\tilde{\mathbf{r}}_i)_d \right] = 0$$

This is almost in the form that we want, but the indices aren't quite contracted in the right manner to reproduce (3.92). One can try to play around to get the indices working right, but at this stage it's just as easy to expand out the components explicitly. For example, we can look at

$$\begin{aligned}
\tilde{L}_1 &= \sum_i m_i [(\tilde{\mathbf{r}}_i)_2(\dot{\mathbf{r}}_i)_3 - ((\tilde{\mathbf{r}}_i)_3(\dot{\mathbf{r}}_i)_2)] \\
&= \sum_i m_i [\Omega_{21}(\tilde{\mathbf{r}}_i)_3(\tilde{\mathbf{r}}_i)_1 + \Omega_{23}(\tilde{\mathbf{r}}_i)_3(\tilde{\mathbf{r}}_i)_3 - \Omega_{31}(\tilde{\mathbf{r}}_i)_2(\tilde{\mathbf{r}}_i)_1 - \Omega_{32}(\tilde{\mathbf{r}}_i)_2(\tilde{\mathbf{r}}_i)_2] \\
&= \tilde{I}_{11}\Omega_{23} + \tilde{I}_{12}\Omega_{31} + \tilde{I}_{13}\Omega_{12} = \frac{1}{2} \epsilon_{abc} \tilde{I}_{1a} \Omega_{bc}
\end{aligned} \tag{3.96}$$

where the first equality is the definition of $\tilde{L}_1$, while the second equality uses our result above, and the third equality uses the definition of $\tilde{I}$ given in (3.93). There are two similar equations, which are summarised in the formula

$$\tilde{L}_a = \frac{1}{2} \epsilon_{bcd} \tilde{I}_{ab} \Omega_{cd} \tag{3.97}$$

Multiplying both sides by $\tilde{I}^{-1}$ gives us precisely the claimed result (3.92). This concludes the proof. $\square$.

To summarise: a system with no angular momentum that can twist and turn and change its shape has an angular velocity (3.92) where $\tilde{\mathbf{r}}_i(t)$ is the path it chooses to take through the space of shapes. This is a nice formula. But what do we do with it? We want to compute the net rotation R as the body moves through a sequence of shapes and returns to its starting point at a time T later. This is given by solving (3.86) for R . The way to do this was described in section 3.1.2. We use *path ordered exponentials*

$$R = \tilde{P} \exp \left(\int_0^T \Omega(t) dt \right) = \tilde{P} \exp \left(\oint \Omega_A dx^A \right) \tag{3.98}$$

The path ordering symbol $\tilde{P}$ puts all matrices evaluated at later times to the right. (This differs from the ordering in section 3.1.2 where we put later matrices to the left. The difference arises because we're working with the angular velocity $\Omega = R^{-1}\dot{R}$ instead of the angular velocity $\omega = \dot{R}R^{-1}$). In the second equality above, we've written the exponent as an integral around a closed path in shape space. Here time has dropped out. This tells us an important fact: it doesn't matter how quickly we perform the change of shapes — the net rotation of the object will be the same.

In particle physics language, the integral in (3.98) is called a “Wilson loop”. We can see how the rotation fares under the ambiguity (3.87). After some algebra, you can find that the net rotation R of an object with shape x^A is changed by

$$R \rightarrow S(x^A) R S(x^A)^{-1} \tag{3.99}$$

This is as it should be: the S^{-1} takes the shape back to our initial choice of standard orientation; the matrix R is the rotation due to the change in shape; finally S puts us back to the new, standard orientation. So we see that even though the definition of the angular velocity is plagued with ambiguity, when we come to ask physically meaningful questions — such as how much has a shape rotated — the ambiguity disappears. However, if we ask nonsensical questions — such as the rotation between two different shapes — then the ambiguity looms large. In this manner, the theory contains a rather astonishing new ingredient: it lets us know what are the sensible questions to ask! Quantities for which the ambiguity (3.87) vanishes are called *gauge invariant*.

In general, it's quite hard to explicitly compute the integral (3.98). One case where it is possible is for infinitesimal changes of shape. Suppose we start with a particular shape x_A^0 , and move infinitesimally in a loop in shape space:

$$x_A(t) = x_A^0 + \alpha_A(t) \quad (3.100)$$

Then we can Taylor expand our angular velocity components,

$$\Omega_A(x(t)) = \Omega_A(x^0) + \left. \frac{\partial \Omega_A}{\partial x^B} \right|_{x^0} \alpha_B \quad (3.101)$$

Expanding out the rotation matrix (3.98) and taking care with the ordering, one can show that

$$\begin{aligned} R &= 1 + \frac{1}{2} F_{AB} \oint \alpha_A \dot{\alpha}_B dt + \mathcal{O}(\alpha^3) \\ &= 1 + \frac{1}{2} \int F_{AB} dA_{AB} + \mathcal{O}(\alpha^3) \end{aligned} \quad (3.102)$$

where F_{AB} is anti-symmetric in the shape space indices A and B , and is a 3×3 matrix (the $a, b = 1, 2, 3$ indices have been suppressed) given by

$$F_{AB} = \frac{\partial \Omega_A}{\partial x^B} - \frac{\partial \Omega_B}{\partial x^A} + [\Omega_A, \Omega_B] \quad (3.103)$$

It is known as the *field strength* to physicists (or the *curvature* to mathematicians). It is evaluated on the initial shape x_A^0 . The second equality in (3.102) gives the infinitesimal rotation as the integral of the field strength over the area traversed in shape space. This field strength contains all the information one needs to know about the infinitesimal rotations of objects induced by changing their shape.

One of the nicest things about the formalism described above is that it mirrors very closely the mathematics needed to describe the fundamental laws of nature, such as the strong and weak nuclear forces and gravity. They are all described by “non-abelian gauge theories”, with an object known as the gauge potential (analogous to Ω_A) and an associated field strength.

Chapter 8

Angular momentum, Part I

(Constant $\hat{\mathbf{L}}$)

The angular momentum of a point mass, relative to a given origin, is defined by

$$\mathbf{L} = \mathbf{r} \times \mathbf{p}. \quad (8.1)$$

For a collection of particles, the total $\mathbf{L}$ is simply the sum of the $\mathbf{L}$'s of all the particles. The vector $\mathbf{r} \times \mathbf{p}$ is a useful thing to study because it has many nice properties. One of these is the conservation law presented in Theorem 7.1, which allowed us to introduce the “effective potential” in Section 7.2. And later in this chapter we will introduce the concept of *torque*, $\boldsymbol{\tau}$, which appears in the bread-and-butter statement, $\boldsymbol{\tau} = d\mathbf{L}/dt$ (analogous to Newton's $\mathbf{F} = d\mathbf{p}/dt$ law).

There are two basic types of angular momentum problems in the world. Since the solution to any rotational problem invariably comes down to using $\boldsymbol{\tau} = d\mathbf{L}/dt$, as we will see, we must determine how $\mathbf{L}$ changes in time. And since $\mathbf{L}$ is a vector, it can change because (1) its length changes, or (2) its direction changes (or through some combination of these effects). In other words, if we write $\mathbf{L} = L\hat{\mathbf{L}}$, where $\hat{\mathbf{L}}$ is the unit vector in the $\mathbf{L}$ direction, then $\mathbf{L}$ can change because L changes, or because $\hat{\mathbf{L}}$ changes, or both.

The first of these cases, that of constant $\hat{\mathbf{L}}$, is the easily understood one. Consider a spinning record, with the center chosen as the origin. The vector $\mathbf{L} = \sum \mathbf{r} \times \mathbf{p}$ is perpendicular to the record, because every term in the sum has this property. If we give the record a tangential force in the proper direction, then it will speed up (in a precise way which we will soon determine). There is nothing mysterious going on here. If we push on the record, it goes faster. $\mathbf{L}$ points in the same direction as before, but now simply with a larger magnitude. In fact, in this type of problem, we can completely forget that $\mathbf{L}$ is a vector. We can deal just with its magnitude L , and everything will be fine. This first case is the subject of the present chapter.

The second case, however, where $\mathbf{L}$ changes direction, can get rather confusing. This is the subject of the following chapter, where we will talk about twirling tops and other kinds of spinning objects that have a tendency to make one's head spin too. In these situations, the entire point is that $\mathbf{L}$ is actually a

vector. And unlike in the constant- $\hat{\mathbf{L}}$ case, we really have to visualize things in three dimensions to see what's going on.¹

The angular momentum of a point mass is given by the simple expression in Eq. (8.1). But in order to deal with setups in the real world, which invariably consist of many particles, we must learn how to calculate the angular momentum of an extended object. This is the task of the Section 8.1. In this chapter, we'll deal only with rotations around the z axis, or an axis parallel to the z axis. We'll save the general 3-D motion for Chapter 9.

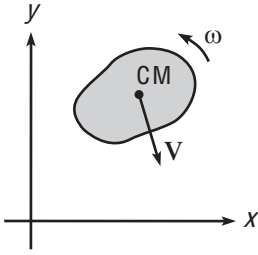


Fig. 8.1

8.1 Pancake object in x - y plane

Consider a flat rigid body undergoing arbitrary motion (both translating and spinning) in the x - y plane; see Fig. 8.1. What is the angular momentum of this body, relative to the origin of the coordinate system?² If we imagine the body to consist of particles of mass m_i , then the angular momentum of the entire body is the sum of the angular momenta of each m_i , which is $\mathbf{L}_i = \mathbf{r}_i \times \mathbf{p}_i$. So the total angular momentum is

$$\mathbf{L} = \sum_i \mathbf{r}_i \times \mathbf{p}_i. \quad (8.2)$$

For a continuous distribution of mass, we would have an integral instead of a sum. $\mathbf{L}$ depends on the locations and momenta of the masses. The momenta in turn depend on how fast the body is translating and spinning. Our goal here is to find the dependence of $\mathbf{L}$ on the distribution and motion of the constituent masses. The result will involve the geometry of the body in a specific way, as we will show.

In this section, we will deal only with pancake-like objects that move in the x - y plane. We will calculate $\mathbf{L}$ relative to the origin, and we will also derive an expression for the kinetic energy. We will deal with non-pancake objects in Section 8.2. Note that since both the $\mathbf{r}$ and $\mathbf{p}$ of all the masses in our pancake-like objects always lie in the x - y plane, the vector $\mathbf{L} = \sum \mathbf{r} \times \mathbf{p}$ always points in the $\hat{\mathbf{z}}$ direction. As mentioned above, this fact is what makes these pancake cases easy to deal with. $\mathbf{L}$ changes only because its length changes, not its direction. So when we eventually get to the $\boldsymbol{\tau} = d\mathbf{L}/dt$ equation, it will take on a simple form. Let's first look at a special case, and then we'll look at general motion in the x - y plane.

¹ The difference between these two cases is essentially the same as the difference between the two basic $\mathbf{F} = d\mathbf{p}/dt$ cases. The vector $\mathbf{p}$ can change because its magnitude changes, in which case we have $F = ma$ (assuming that m is constant). Or, $\mathbf{p}$ can change because its direction changes, in which case we have the centripetal-acceleration statement, $F = mv^2/r$. (Or there could be a combination of these effects.) The former case seems a bit more intuitive than the latter.

² Remember, $\mathbf{L}$ is defined relative to a chosen origin, because it has the vector $\mathbf{r}$ in it. So it makes no sense to ask what $\mathbf{L}$ is, without specifying what origin we've chosen.

8.1.1 Rotation about the z axis

The pancake in Fig. 8.2 is pivoted at the origin and rotates with angular speed ω around the z axis, in the counterclockwise direction (as viewed from above). Consider a little piece of the body, with mass dm and position (x, y) . This little piece travels in a circle around the origin with speed $v = \omega r$, where $r = \sqrt{x^2 + y^2}$. Therefore, the angular momentum of this piece (relative to the origin) equals $\mathbf{L} = \mathbf{r} \times \mathbf{p} = r(v dm)\hat{\mathbf{z}} = dm r^2 \omega \hat{\mathbf{z}}$. The $\hat{\mathbf{z}}$ direction arises from the cross product of the (orthogonal) vectors $\mathbf{r}$ and $\mathbf{p}$. The angular momentum of the entire body is therefore

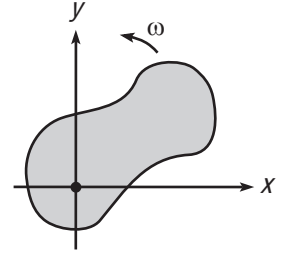


Fig. 8.2

$$\mathbf{L} = \int r^2 \omega \hat{\mathbf{z}} dm = \int (x^2 + y^2) \omega \hat{\mathbf{z}} dm, \quad (8.3)$$

where the integration runs over the area of the body. If the density of the object is constant, as is usually the case, then we have $dm = \rho dx dy$. If we define the *moment of inertia* around the z axis to be

$$I_z \equiv \int r^2 dm = \int (x^2 + y^2) dm, \quad (8.4)$$

then the z component of $\mathbf{L}$ is

$$L_z = I_z \omega, \quad (8.5)$$

and both L_x and L_y are zero. In the case where the rigid body is made up of a collection of point masses m_i in the x - y plane, the moment of inertia in Eq. (8.4) takes the discretized form,

$$I_z \equiv \sum_i m_i r_i^2. \quad (8.6)$$

Given any rigid body in the x - y plane, we can calculate I_z . And given ω , we can then multiply it by I_z to find L_z . In Section 8.3.1, we'll get some practice calculating various moments of inertia.

What is the kinetic energy of our object? We need to add up the energies of all the little pieces. A little piece has energy $dm v^2/2 = dm(r\omega)^2/2$. So the total kinetic energy is

$$T = \int \frac{r^2 \omega^2}{2} dm. \quad (8.7)$$

With our definition of I_z in Eq. (8.4), this becomes

$$T = \frac{I_z \omega^2}{2}. \quad (8.8)$$

This is easy to remember, because it looks a lot like the kinetic energy of a point mass, $mv^2/2$.

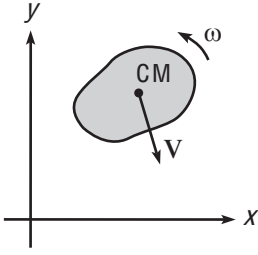


Fig. 8.3

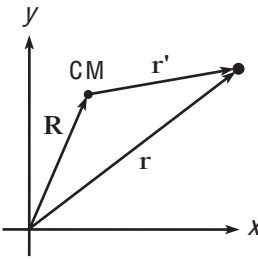


Fig. 8.4

8.1.2 General motion in x - y plane

How do we deal with general motion in the x - y plane? For the motion in Fig. 8.3, where the object is both translating and spinning, the various pieces of mass don't travel in circles around the origin, so we can't write $v = \omega r$ as we did above. It turns out to be highly advantageous to write the angular momentum, $\mathbf{L}$, and the kinetic energy, T , in terms of the center-of-mass (CM) coordinates and the coordinates relative to the CM. The expressions for $\mathbf{L}$ and T take on very nice forms when written this way, as we now show.

Let the position of the CM relative to a fixed origin be $\mathbf{R} = (X, Y)$. And let the position of a given point relative to the CM be $\mathbf{r}' = (x', y')$. Then the position of the given point relative to the fixed origin is $\mathbf{r} = \mathbf{R} + \mathbf{r}'$ (see Fig. 8.4). Let the velocity of the CM be $\mathbf{V}$, and let the velocity relative to the CM be $\mathbf{v}'$. Then $\mathbf{v} = \mathbf{V} + \mathbf{v}'$. Let the body rotate with angular speed ω' around the CM (around an instantaneous axis parallel to the z axis, so that the pancake remains in the x - y plane at all times).³ Then $v' = \omega' r'$.

Let's look at $\mathbf{L}$ first. Let M be the total mass of the pancake. The angular momentum relative to the origin is

$$\begin{aligned}
 \mathbf{L} &= \int \mathbf{r} \times \mathbf{v} \, dm \\
 &= \int (\mathbf{R} + \mathbf{r}') \times (\mathbf{V} + \mathbf{v}') \, dm \\
 &= \int \mathbf{R} \times \mathbf{V} \, dm + \int \mathbf{r}' \times \mathbf{v}' \, dm \quad (\text{cross terms vanish; see below}) \\
 &= M\mathbf{R} \times \mathbf{V} + \left(\int r'^2 \omega' \, dm \right) \hat{\mathbf{z}} \\
 &\equiv \mathbf{R} \times \mathbf{P} + \left(I_z^{\text{CM}} \omega' \right) \hat{\mathbf{z}}.
 \end{aligned} \tag{8.9}$$

In going from the second to third line above, the cross terms, $\int \mathbf{r}' \times \mathbf{V} \, dm$ and $\int \mathbf{R} \times \mathbf{v}' \, dm$, vanish by definition of the CM, which says that $\int \mathbf{r}' \, dm = 0$. (See Eq. (5.58); basically, the position of the CM in the CM frame is zero.) This implies that $\int \mathbf{v}' \, dm = d(\int \mathbf{r}' \, dm)/dt$ also equals zero. And since we can pull the constant vectors $\mathbf{V}$ and $\mathbf{R}$ out of the above integrals, we are therefore left with zero. The quantity I_z^{CM} in the final result is the moment of inertia around an axis through the CM, parallel to the z axis. Equation (8.9) is a very nice result, and it is important enough to be called a theorem. In words, it says:

Theorem 8.1 *The angular momentum (relative to the origin) of a body can be found by treating the body like a point mass located at the CM and finding the*

³ What we mean here is the following. Consider a coordinate system whose origin is the CM and whose axes are parallel to the fixed x and y axes. Then the pancake rotates with angular speed ω' with respect to this system.

angular momentum of this point mass relative to the origin, and by then adding on the angular momentum of the body relative to the CM.⁴

Note that if we have the special case where the CM travels around the origin in a circle with angular speed Ω (so that $V = \Omega R$), then Eq. (8.9) becomes $\mathbf{L} = (MR^2\Omega + I_z^{\text{CM}}\omega')\hat{\mathbf{z}}$.

Now let's look at T . The kinetic energy is

$$\begin{aligned}
 T &= \int \frac{1}{2}v^2 dm \\
 &= \int \frac{1}{2}|\mathbf{V} + \mathbf{v}'|^2 dm \\
 &= \frac{1}{2} \int V^2 dm + \frac{1}{2} \int v'^2 dm \quad (\text{cross term vanishes; see below}) \\
 &= \frac{1}{2}MV^2 + \frac{1}{2} \int r'^2 \omega'^2 dm \\
 &\equiv \frac{1}{2}MV^2 + \frac{1}{2}I_z^{\text{CM}}\omega'^2.
 \end{aligned} \tag{8.10}$$

In going from the second to third line above, the cross term $\int \mathbf{V} \cdot \mathbf{v}' dm = \mathbf{V} \cdot \int \mathbf{v}' dm$ vanishes by definition of the CM, as in the above calculation of $\mathbf{L}$. Again, Eq. (8.10) is a very nice result. In words, it says:

Theorem 8.2 *The kinetic energy of a body can be found by treating the body like a point mass located at the CM, and by then adding on the kinetic energy of the body due to the motion relative to the CM.*⁵

To calculate E , my dear class,
 Just add up two things, and you'll pass.
 Take the CM point's E ,
 And then add on with glee,
 The E 'round the center of mass.

Example (Cylinder on a ramp): A cylinder of mass m , radius r , and moment of inertia $I = (1/2)mr^2$ (this is the I for a solid cylinder around its center, as we'll see in Section 8.3.1) rolls without slipping down a plane inclined at an angle θ . What is the acceleration of the center of the cylinder?

⁴ This theorem works only if we use the CM as the location of the imagined point mass. True, in the above analysis we could have chosen a point P other than the CM, and then written things in terms of the coordinates of P and the coordinates relative to P (which could also be described by a rotation). But then the cross terms in Eq. (8.9) wouldn't vanish, and we'd end up with an unenlightening mess.

⁵ We already knew this from Section 5.6.2. It's just that now we know that the kinetic energy in the CM frame takes the form of $I_z^{\text{CM}}\omega'^2/2$.

Solution: We'll use conservation of energy to determine the speed v of the center of the cylinder after it has moved a distance d down the plane, and then we'll read off a from the standard constant-acceleration kinematic relation, $v = \sqrt{2ad}$.

The loss in potential energy of the cylinder is $mgd \sin \theta$. This shows up as kinetic energy, which equals $mv^2/2 + I\omega^2/2$ from Theorem 8.2. But the nonslipping condition is $v = \omega r$. Therefore, $\omega = v/r$, and conservation of energy gives

$$\begin{aligned} mgd \sin \theta &= \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 \\ &= \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\left(\frac{v}{r}\right)^2 \\ &= \frac{3}{4}mv^2. \end{aligned} \quad (8.11)$$

So the speed as a function of distance is $v = \sqrt{(4/3)gd \sin \theta}$. Hence, $v = \sqrt{2ad}$ gives $a = (2/3)g \sin \theta$, which is independent of r .

REMARKS: Our answer is $2/3$ of the $g \sin \theta$ result for a block sliding down a frictionless plane. It is smaller because there is kinetic energy “wasted” in the rotational motion. Alternatively, it is smaller because there is a friction force pointing up the plane (to provide the torque necessary to get the cylinder rotating, but we'll talk about torque in Section 8.4), so this decreases the net force down the plane.

If we let I take the general form $I = \beta mr^2$, where β is a numerical factor, then you can show that the acceleration becomes $a = \sqrt{g \sin \theta / (1 + \beta)}$. So if $\beta = 0$ (all the mass is at the center), then we simply have $a = g \sin \theta$, so the cylinder behaves just like a sliding block. If $\beta = 1$ (all the mass is on the rim), then $a = (1/2)g \sin \theta$. If $\beta \rightarrow \infty$,⁶ then $a \rightarrow 0$. After reading Section 8.4, you can think about these special cases alternatively in terms of the forces and torques involved.

Although this problem can also be solved by using force and torque (the task of Exercise 8.37), the conservation-of-energy method is generally quicker in more complicated problems of this type, as you can see by doing, for example, Exercises 8.28 and 8.46. ♣

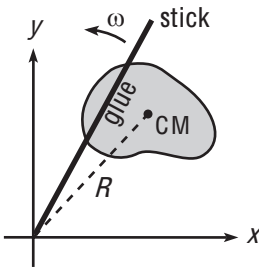


Fig. 8.5

8.1.3 The parallel-axis theorem

Consider the special case where the CM rotates around the origin at the same rate as the body rotates around the CM. This can be achieved, for example, by gluing a stick across the pancake and pivoting one end of the stick at the origin; see Fig. 8.5. In this special case, we have the simplified situation where all points in the pancake travel in circles around the origin. Let their angular speed be ω . Then the speed of the CM is $V = \omega R$, so Eq. (8.9) gives the angular momentum around the origin as

$$L_z = (MR^2 + I_z^{\text{CM}})\omega. \quad (8.12)$$

⁶ This can be obtained by adding on long spokes protruding from the cylinder and having them pass through a deep groove in the plane, or by having a spool with a very small inner radius roll down a thin plane with only its inner “axle” rolling on the plane.

In other words, the moment of inertia around the origin is

$$I_z = MR^2 + I_z^{\text{CM}}. \quad (8.13)$$

This is the *parallel-axis theorem*. It says that once you've calculated the moment of inertia of an object around the axis passing through the CM (namely I_z^{CM}), then if you want to calculate the moment of inertia around a parallel axis, you simply have to add on MR^2 , where R is the distance between the two axes, and M is the mass of the object. Note that since the parallel-axis theorem is a special case of the result in Eq. (8.9), it is valid *only* with the CM, and not with any other point. The parallel-axis theorem actually holds for arbitrary nonplanar objects too, as we'll see in Section 8.2. And we'll also derive a more general form of the theorem in Chapter 9.

We can also look at the kinetic energy in this special case where the CM rotates around the origin at the same rate as the body rotates around the CM. Using $V = \omega R$ in Eq. (8.10), we find

$$T = \frac{1}{2} (MR^2 + I_z^{\text{CM}}) \omega^2 = \frac{1}{2} I_z \omega^2. \quad (8.14)$$

Example (A stick): Let's verify the parallel-axis theorem for a stick of mass m and length ℓ , in the case where we want to compare the moment of inertia around an axis through an end (perpendicular to the stick) with the moment of inertia around an axis through the CM (perpendicular to the stick).

For convenience, let $\rho = m/\ell$ be the density. The moment of inertia around an axis through an end is

$$I^{\text{end}} = \int_0^\ell x^2 dm = \int_0^\ell x^2 \rho dx = \frac{1}{3} \rho \ell^3 = \frac{1}{3} (\rho \ell) \ell^2 = \frac{1}{3} m \ell^2. \quad (8.15)$$

The moment of inertia around an axis through the CM is

$$I^{\text{CM}} = \int_{-\ell/2}^{\ell/2} x^2 dm = \int_{-\ell/2}^{\ell/2} x^2 \rho dx = \frac{1}{12} \rho \ell^3 = \frac{1}{12} (\rho \ell) \ell^2 = \frac{1}{12} m \ell^2. \quad (8.16)$$

This is consistent with the parallel-axis theorem, Eq. (8.13), because

$$I^{\text{end}} = m \left(\frac{\ell}{2} \right)^2 + I^{\text{CM}}. \quad (8.17)$$

Remember that this works only with the CM. If we instead want to compare I^{end} with the I around a point, say, $\ell/6$ from that end, then we cannot say that they differ by $m(\ell/6)^2$. But we *can* compare each of them to I^{CM} and say that they differ by $(\ell/2)^2 - (\ell/3)^2 = 5\ell^2/36$.

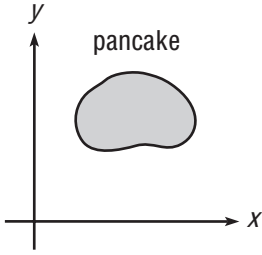


Fig. 8.6

8.1.4 The perpendicular-axis theorem

This theorem is valid *only* for pancake objects. Consider a pancake object in the x - y plane (see Fig. 8.6). Then the *perpendicular-axis theorem* says that

$$I_z = I_x + I_y, \quad (8.18)$$

where I_x and I_y are defined analogously to the I_z in Eq. (8.4). That is, to find I_x , imagine spinning the object around the x axis at angular speed ω , and then define $I_x \equiv L_x/\omega$. (Only the distance from the x axis matters in calculating the speed of a given point. So the fact that the object has extent along the x direction and is therefore not a pancake in the y - z plane is irrelevant. The following section has further discussion of this.) Likewise for I_y . In other words,

$$I_x \equiv \int (y^2 + z^2) dm, \quad I_y \equiv \int (z^2 + x^2) dm, \quad I_z \equiv \int (x^2 + y^2) dm. \quad (8.19)$$

To prove the perpendicular-axis theorem, we simply use the fact that $z = 0$ for our pancake object. Equation (8.19) then gives $I_z = I_x + I_y$. In the limited number of situations where this theorem is applicable, it can save you some trouble. A few examples are given in Section 8.3.1.

8.2 Nonplanar objects

In Section 8.1, we restricted the discussion to pancake objects in the x - y plane. However, nearly all the results we derived carry over to nonplanar objects, provided that the axis of rotation is parallel to the z axis, and provided that we are concerned only with L_z , and not L_x or L_y . So let's drop the pancake assumption and run through the results we obtained above.

First, consider an object rotating around the z axis. Let the object have extension in the z direction. If we imagine slicing the object into pancakes parallel to the x - y plane, then Eqs. (8.4) and (8.5) correctly give the L_z for each pancake. And since the L_z of the whole object is the sum of the L_z 's of all the pancakes, we see that the I_z of the whole object is the sum of the I_z 's of all the pancakes. The difference in the z values of the pancakes is irrelevant. Therefore, for *any* object rotating around the z axis, we have

$$I_z = \int (x^2 + y^2) dm, \quad \text{and} \quad L_z = I_z \omega, \quad (8.20)$$

where the integration runs over the entire volume of the body. We'll calculate the I_z for many nonplanar objects in Section 8.3.1. Note that even though Eq. (8.20) gives the L_z for an arbitrary object, the analysis in this chapter is still not completely general because (1) we are restricting the axis of rotation to be the (fixed) z axis, and (2) even with this restriction, an object outside the x - y plane might

have nonzero x and y components of $\mathbf{L}$, but we found only the z component in Eq. (8.20). This second fact is strange but true. We'll deal with it in great detail in Section 9.2.

As far as the kinetic energy goes, the T for a nonplanar object rotating around the z axis is still given by Eq. (8.8), because we can obtain the total T by adding up the T 's of all the pancake slices.

Also, Eqs. (8.9) and (8.10) continue to hold for a nonplanar object in the case where the CM is translating while the object is spinning around it (or more precisely, spinning around an axis parallel to the z axis and passing through the CM). The velocity $\mathbf{V}$ of the CM can actually point in any direction, and these two equations are still valid. But we'll assume in this chapter that all velocities are in the x - y plane.

Lastly, the parallel-axis theorem still holds for a nonplanar object (the derivation using Eq. (8.9) is the same). But as mentioned in Section 8.1.4, the perpendicular-axis theorem does *not* hold. This is the one instance where we need the planar assumption.

Finding the CM

The center of mass has come up repeatedly in this chapter. For example, when we used the parallel-axis theorem, we needed to know where the CM was. In some cases, such as with a stick or a disk, the location is obvious. But in other cases, it isn't so clear. So let's get a little practice calculating the location of the CM. Depending on whether the mass distribution is discrete or continuous, the position of the CM is defined by (see Eq. (5.58))

$$\mathbf{R}_{\text{CM}} = \frac{\sum \mathbf{r}_i m_i}{M}, \quad \text{or} \quad \mathbf{R}_{\text{CM}} = \frac{\int \mathbf{r} dm}{M}, \quad (8.21)$$

where M is the total mass. We'll do an example with a continuous mass distribution here. As is often the case with problems involving an integral, the main step in the solution is deciding how you want to slice up the object to do the integral.

Example (Hemispherical shell): Find the location of the CM of a hollow hemispherical shell, with uniform mass density and radius R .

Solution: By symmetry, the CM is located on the line above the center of the base. So our task reduces to finding the height, y_{CM} . Let the mass density be σ . We'll slice the hemisphere up into horizontal rings, described by the angle θ above the horizontal, as shown in Fig. 8.7. If the angular thickness of a ring is $d\theta$, then its mass is

$$dm = \sigma dA = \sigma (\text{length})(\text{width}) = \sigma (2\pi R \cos \theta)(R d\theta). \quad (8.22)$$

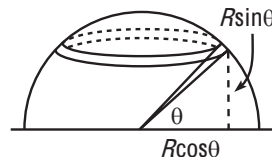


Fig. 8.7

All points on the ring have a y value of $R \sin \theta$. Therefore,

$$\begin{aligned}
 y_{\text{CM}} &= \frac{1}{M} \int y \, dm = \frac{1}{(2\pi R^2)\sigma} \int_0^{\pi/2} (R \sin \theta) (2\pi R^2 \sigma \cos \theta \, d\theta) \\
 &= R \int_0^{\pi/2} \sin \theta \cos \theta \, d\theta \\
 &= \frac{R \sin^2 \theta}{2} \Big|_0^{\pi/2} \\
 &= \frac{R}{2}.
 \end{aligned} \tag{8.23}$$

The simple factor of $1/2$ here is nice, but it's not all that obvious. It comes from the fact that each value of y is represented equally. If you solved the problem by doing a dy integral instead of a $d\theta$ one, you would find that there is the same area (and hence the same mass) in each ring of vertical height dy . In short, as y increases, the larger tilt of the surface cancels out the smaller radius of the rings, yielding the same area. You are encouraged to work this out.

The calculation of a CM is very similar to the calculation of a moment of inertia. Both involve an integration over the mass of an object, but the former has one power of a length in the integrand, whereas the latter has two powers.

8.3 Calculating moments of inertia

8.3.1 Lots of examples

Let's now calculate the moments of inertia of various objects, around specified axes. We'll use ρ to denote the mass density (per unit length, area, or volume, as appropriate), and we'll assume that this density is uniform throughout the object. For the more complicated objects in the list below, it is generally a good idea to slice them up into pieces for which I is already known. The problem then reduces to integrating over these known I 's. There is usually more than one way to do this slicing. For example, a sphere can be looked at as a series of concentric shells or a collection of disks stacked on top of each other. In the examples below, you may want to play around with slicings other than the ones given. Consider at least a few of these examples to be problems and try to work them out for yourself.

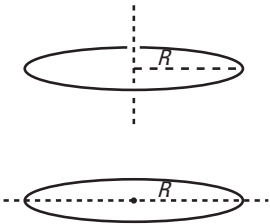


Fig. 8.8

1. A ring of mass M and radius R (axis through center, perpendicular to plane; Fig. 8.8):

$$I = \int r^2 \, dm = \int_0^{2\pi} R^2 \rho R \, d\theta = (2\pi R \rho) R^2 = \boxed{MR^2}, \tag{8.24}$$

as expected, because all of the mass is a distance R from the axis.

2. A ring of mass M and radius R (axis through center, in plane; Fig. 8.8): The distance from the axis is (the absolute value of) $R \sin \theta$. Therefore,

$$I = \int r^2 dm = \int_0^{2\pi} (R \sin \theta)^2 \rho R d\theta = \frac{1}{2} (2\pi R \rho) R^2 = \boxed{\frac{1}{2} MR^2}, \quad (8.25)$$

where we have used $\sin^2 \theta = (1 - \cos 2\theta)/2$. You can also find I by using the perpendicular-axis theorem. In the notation of Section 8.1.4, we have $I_x = I_y$, by symmetry. Therefore, $I_z = 2I_x$. Using $I_z = MR^2$ from Example 1 then gives $I_x = MR^2/2$.

3. A disk of mass M and radius R (axis through center, perpendicular to plane; Fig. 8.9):

$$I = \int r^2 dm = \int_0^{2\pi} \int_0^R r^2 \rho r dr d\theta = (R^4/4) 2\pi \rho = \frac{1}{2} (\rho \pi R^2) R^2 = \boxed{\frac{1}{2} MR^2}. \quad (8.26)$$

You can save the (trivial) step of integrating over θ by considering the disk to be made up of many concentric rings, and invoking Example 1. The mass of each ring is $\rho 2\pi r dr$. Integrating over the rings gives $I = \int_0^R (\rho 2\pi r dr) r^2 = \pi R^4 \rho / 2 = MR^2/2$, as above. Slicing up the disk is fairly inconsequential in this example, but it will save you some trouble in others.

4. A disk of mass M and radius R (axis through center, in plane; Fig. 8.9): Slice the disk up into rings, and use Example 2.

$$I = \int_0^R (1/2)(\rho 2\pi r dr) r^2 = (R^2/4) \rho \pi = \frac{1}{4} (\rho \pi R^2) R^2 = \boxed{\frac{1}{4} MR^2}. \quad (8.27)$$

Or just use Example 3 and the perpendicular-axis theorem.

5. A thin uniform rod of mass M and length L (axis through center, perpendicular to rod; Fig. 8.10): We already found this I and the next one in Section 8.1.3, but we'll include them here for completeness.

$$I = \int x^2 dm = \int_{-L/2}^{L/2} x^2 \rho dx = \frac{1}{12} (\rho L) L^2 = \boxed{\frac{1}{12} ML^2}. \quad (8.28)$$

6. A thin uniform rod of mass M and length L (axis through end, perpendicular to rod; Fig. 8.10):

$$I = \int x^2 dm = \int_0^L x^2 \rho dx = \frac{1}{3} (\rho L) L^2 = \boxed{\frac{1}{3} ML^2}. \quad (8.29)$$

7. A spherical shell of mass M and radius R (any axis through center; Fig. 8.11): Let's slice the sphere into horizontal ring-like strips. In spherical coordinates, the radius of a ring is given by $r = R \sin \theta$, where θ is the angle down from the north pole.

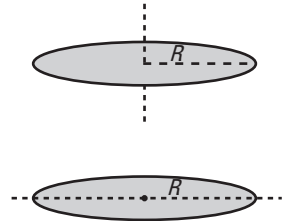


Fig. 8.9

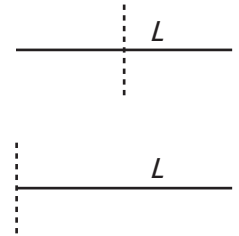


Fig. 8.10

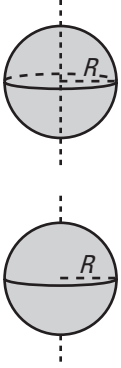


Fig. 8.11

The area of a strip is then $2\pi(R \sin \theta)R d\theta$. Using $\int \sin^3 \theta = \int \sin \theta (1 - \cos^2 \theta) = -\cos \theta + \cos^3 \theta / 3$, we have

$$I = \int r^2 dm = \int_0^\pi (R \sin \theta)^2 2\pi \rho (R \sin \theta) R d\theta = 2\pi \rho R^4 \int_0^\pi \sin^3 \theta d\theta \\ = 2\pi \rho R^4 (4/3) = \frac{2}{3} (4\pi R^2 \rho) R^2 = \boxed{\frac{2}{3} MR^2}. \quad (8.30)$$

An alternate and slick way of deriving this is to use the following result, similar in spirit to the perpendicular-axis theorem: Adding up the three moments of inertia in Eq. (8.19) gives

$$I_x + I_y + I_z = 2 \int (x^2 + y^2 + z^2) dm = 2 \int r^2 dm. \quad (8.31)$$

If we apply this to a spherical shell, where r always equals the radius R , then the right-hand side is $2MR^2$. And since the I 's are all equal by symmetry, they must all be $2MR^2/3$.

8. A solid sphere of mass M and radius R (any axis through center; Fig. 8.11): A sphere is made up of concentric spherical shells. The volume of a shell is $4\pi r^2 dr$. Using Example 7, we have

$$I = \int_0^R (2/3)(4\pi \rho r^2 dr) r^2 = (R^5/5)(8\pi \rho/3) = \frac{2}{5} (4\pi R^3 \rho/3) R^2 = \boxed{\frac{2}{5} MR^2}. \quad (8.32)$$

The task of Exercise 8.33 is to derive this result by slicing the sphere into horizontal disks.

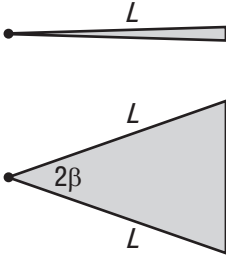


Fig. 8.12

9. An infinitesimally thin triangle of mass M and length L (axis through tip, perpendicular to plane; Fig. 8.12): Let the base have length a , where a is infinitesimally small. Then a tiny vertical slice a distance x from the tip has length $a(x/L)$. If the slice has thickness dx , then it is essentially a point mass of mass $dm = \rho ax dx/L$. Therefore,

$$I = \int x^2 dm = \int_0^L x^2 \rho ax/L dx = \frac{1}{2} (\rho aL/2) L^2 = \boxed{\frac{1}{2} ML^2}, \quad (8.33)$$

because $aL/2$ is the area of the triangle. This has the same form as the disk in Example 3, because a disk is made up of many of these triangles.

10. An isosceles triangle of mass M , vertex angle 2β , and common-side length L (axis through tip, perpendicular to plane; Fig. 8.12): Let h be the altitude of the triangle (so $h = L \cos \beta$). Slice the triangle into thin strips parallel to the base. Let x be the distance from a strip to the vertex. Then the length of a strip is $\ell = 2x \tan \beta$, and its mass is $dm = \rho(2x \tan \beta dx)$. Using Example 5 above, along with the parallel-axis theorem, we have

$$I = \int dm \left(\frac{\ell^2}{12} + x^2 \right) = \int_0^h (\rho 2x \tan \beta dx) \left(\frac{(2x \tan \beta)^2}{12} + x^2 \right) \\ = 2\rho \tan \beta \int_0^h \left(1 + \frac{\tan^2 \beta}{3} \right) x^3 dx = 2\rho \tan \beta \left(1 + \frac{\tan^2 \beta}{3} \right) \frac{h^4}{4}. \quad (8.34)$$

But the area of the entire triangle is $h^2 \tan \beta$, so we have $I = (Mh^2/2)(1 + (1/3) \tan^2 \beta)$. In terms of $L = h/\cos \beta$, this is

$$I = (ML^2/2)(\cos^2 \beta + \sin^2 \beta/3) = \boxed{\frac{1}{2}ML^2\left(1 - \frac{2}{3}\sin^2 \beta\right)}. \quad (8.35)$$

11. A regular N -gon of mass M and “radius” R (axis through center, perpendicular to plane; Fig. 8.13): The N -gon is made up of N isosceles triangles, so we can use Example 10, with $\beta = \pi/N$. The masses of the triangles simply add, so if M is the mass of the whole N -gon, we have

$$I = \boxed{\frac{1}{2}MR^2\left(1 - \frac{2}{3}\sin^2 \frac{\pi}{N}\right)}. \quad (8.36)$$

We can list the values of I for a few N . With the shorthand notation $(N, I/MR^2)$, Eq. (8.36) gives $(3, \frac{1}{4})$, $(4, \frac{1}{3})$, $(6, \frac{5}{12})$, $(\infty, \frac{1}{2})$. These values of I form a nice arithmetic progression.

12. A rectangle of mass M and sides of length a and b (axis through center, perpendicular to plane; Fig. 8.13): Let the z axis be perpendicular to the plane. We know that $I_x = Mb^2/12$ and $I_y = Ma^2/12$ (because the extent of an object along an axis doesn’t affect the moment around that axis, when written in terms of the total M). So the perpendicular-axis theorem gives

$$I_z = I_x + I_y = \boxed{\frac{1}{12}M(a^2 + b^2)}. \quad (8.37)$$

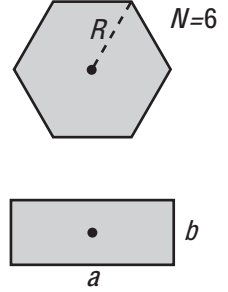


Fig. 8.13

8.3.2 A neat trick

For some objects with certain symmetries, it’s possible to calculate the moment of inertia without doing any integrals. The only things we need are a scaling argument and the parallel-axis theorem. We’ll illustrate this technique by finding the I for a stick around its center (Example 5 above). You’ll find other applications in the problems for this chapter.

In the present example, the basic trick is to compare the I for a stick of length L with the I for a stick of length $2L$ (and same density ρ). A quick scaling argument shows that the latter is eight times the former. This is true because the integral $\int x^2 dm = \int x^2 \rho dx$ has three powers of x in it (yes, the dx counts). So a change of variables, $x = 2y$, brings in a factor of $2^3 = 8$. Equivalently, if we imagine expanding the smaller stick into the larger one, then a corresponding piece in the larger stick will be twice as far from the axis, and also twice as massive. The integral $\int x^2 dm$ therefore increases by a factor of $2^2 \cdot 2 = 8$.

The technique is most easily illustrated with pictures. If we denote the moment of inertia of an object by a picture of the object, with a dot signifying the axis,

then we have:

$$\begin{aligned}
 \text{---} \overset{L}{\bullet} \text{---} \overset{L}{\bullet} \text{---} &= 8 \text{---} \overset{L}{\bullet} \text{---} \\
 \text{---} \bullet \text{---} &= 2 \bullet \text{---} \\
 \bullet \text{---} &= \text{---} \bullet + M \left(\frac{L}{2} \right)^2
 \end{aligned}$$

The first line comes from the scaling argument, the second line comes from the fact that moments of inertia simply add (the left-hand side is two copies of the right-hand side, attached at the pivot), and the third line comes from the parallel-axis theorem. Equating the right-hand sides of the first two equations gives

$$\bullet \text{---} = 4 \text{---} \bullet$$

Plugging this expression for $\bullet \text{---}$ into the third equation gives the desired result,

$$\text{---} \bullet = \frac{1}{12} ML^2$$

Note that sooner or later we must use real live numbers, which enter here through the parallel-axis theorem. Using only scaling arguments isn't sufficient, because they provide only linear equations homogeneous in the I 's, and therefore give no means of picking up the proper dimensions. (For an interesting account of Galileo's discovery of scaling laws, see Peterson (2002).)

Once you've mastered this trick and applied it to the fractal objects in Problem 8.8, you can impress your friends by saying that you know how to "use scaling arguments, along with the parallel-axis theorem, to calculate moments of inertia of objects with fractal dimension." And you never know when that might come in handy!

8.4 Torque

We will now show that under certain conditions (stated below), the rate of change of angular momentum is equal to a certain quantity, τ , which we call the *torque*. That is, $\tau = d\mathbf{L}/dt$. This is the rotational analog of our old friend $\mathbf{F} = d\mathbf{p}/dt$ involving linear momentum. The basic idea here is straightforward, but there are two subtle issues. One deals with internal forces within a collection of particles. The other deals with the possible acceleration of the origin (the point relative to which the torque and angular momentum are calculated). To keep things straight, we'll prove the general result by dealing with three increasingly complicated situations.

Our derivation of $\boldsymbol{\tau} = d\mathbf{L}/dt$ here holds for completely general motion, so we can take the result and use it in the following chapter, too. If you wish, you can construct a more specific proof of $\boldsymbol{\tau} = d\mathbf{L}/dt$ for the special case where the axis of rotation is parallel to the z axis. But since the general proof is no more difficult, we'll present it here in this chapter and do it once and for all.

8.4.1 Point mass, fixed origin

Consider a point mass at position $\mathbf{r}$ relative to a fixed origin (see Fig. 8.14). The time derivative of the angular momentum, $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, is

$$\begin{aligned} \frac{d\mathbf{L}}{dt} &= \frac{d}{dt}(\mathbf{r} \times \mathbf{p}) \\ &= \frac{d\mathbf{r}}{dt} \times \mathbf{p} + \mathbf{r} \times \frac{d\mathbf{p}}{dt} \\ &= \mathbf{v} \times (m\mathbf{v}) + \mathbf{r} \times \mathbf{F} \\ &= 0 + \mathbf{r} \times \mathbf{F}, \end{aligned} \tag{8.38}$$

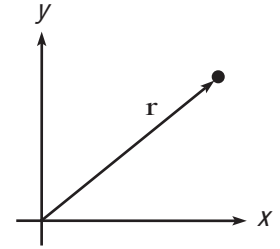


Fig. 8.14

where $\mathbf{F}$ is the force acting on the particle. This is the same calculation as in Theorem 7.1, except that here we are considering an arbitrary force instead of a central one. If we define the *torque* on the particle as

$$\boldsymbol{\tau} \equiv \mathbf{r} \times \mathbf{F}, \tag{8.39}$$

then Eq. (8.38) becomes

$$\boldsymbol{\tau} = \frac{d\mathbf{L}}{dt}. \tag{8.40}$$

It is understood that the $\mathbf{r}$ in the torque is measured with respect to the same origin as the $\mathbf{r}$ in the angular momentum.

8.4.2 Extended mass, fixed origin

In an extended object, there are internal forces acting on the various pieces of the object, in addition to whatever external forces exist. For example, the external force on a given molecule in a body might come from gravity, while the internal forces come from the adjacent molecules. How do we deal with these different types of forces?

In what follows, we will deal only with internal forces that are central forces, so that the force between two objects is directed along the line between them. This is a valid assumption for the pushing and pulling forces between molecules in a solid. (It isn't valid, for example, when dealing with magnetic forces. But we won't be interested in such things here.) We will invoke Newton's third law,

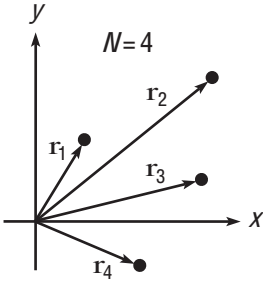


Fig. 8.15

which says that the force that particle 1 applies to particle 2 is equal and opposite to the force that particle 2 applies to particle 1.

For concreteness, let's assume that we have a collection of N discrete particles labeled by the index i (see Fig. 8.15). In the continuous case, we would need to replace the following sums with integrals. The total angular momentum of the system is

$$\mathbf{L} = \sum_{i=1}^N \mathbf{r}_i \times \mathbf{p}_i. \quad (8.41)$$

The force acting on each particle is $\mathbf{F}_i^{\text{ext}} + \mathbf{F}_i^{\text{int}} = d\mathbf{p}_i/dt$. Therefore,

$$\begin{aligned} \frac{d\mathbf{L}}{dt} &= \frac{d}{dt} \sum_i \mathbf{r}_i \times \mathbf{p}_i \\ &= \sum_i \frac{d\mathbf{r}_i}{dt} \times \mathbf{p}_i + \sum_i \mathbf{r}_i \times \frac{d\mathbf{p}_i}{dt} \\ &= \sum_i \mathbf{v}_i \times (m\mathbf{v}_i) + \sum_i \mathbf{r}_i \times (\mathbf{F}_i^{\text{ext}} + \mathbf{F}_i^{\text{int}}) \\ &= 0 + \sum_i \mathbf{r}_i \times \mathbf{F}_i^{\text{ext}} \\ &\equiv \sum_i \boldsymbol{\tau}_i^{\text{ext}}. \end{aligned} \quad (8.42)$$

The second-to-last line follows because $\mathbf{v}_i \times \mathbf{v}_i = 0$, and also because $\sum_i \mathbf{r}_i \times \mathbf{F}_i^{\text{int}} = 0$, as you can show in Problem 8.9. In other words, the internal forces provide no net torque. This is quite reasonable. It basically says that a rigid object with no external forces won't spontaneously start rotating.

Note that the right-hand side involves the *total* external torque acting on the body, which may come from forces acting at many different points. Note also that nowhere did we assume that the particles are rigidly connected to each other. Equation (8.42) still holds even if there is relative motion among the particles. But in that case, it's usually hard to get a handle on $\mathbf{L}$, because it doesn't take the nice $I\omega$ form.

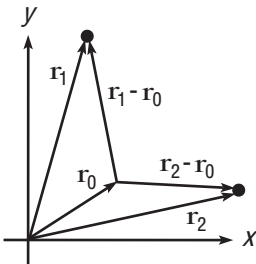


Fig. 8.16

8.4.3 Extended mass, nonfixed origin

Let the position of the origin be $\mathbf{r}_0$ (see Fig. 8.16), and let the positions of the particles be $\mathbf{r}_i$. The vectors $\mathbf{r}_0$ and $\mathbf{r}_i$ are measured with respect to a given fixed coordinate system. The total angular momentum of the system, relative to the

(possibly accelerating) origin $\mathbf{r}_0$, is⁷

$$\mathbf{L} = \sum_i (\mathbf{r}_i - \mathbf{r}_0) \times m_i (\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_0). \quad (8.43)$$

Therefore,

$$\begin{aligned} \frac{d\mathbf{L}}{dt} &= \frac{d}{dt} \left(\sum_i (\mathbf{r}_i - \mathbf{r}_0) \times m_i (\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_0) \right) \\ &= \sum_i (\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_0) \times m_i (\dot{\mathbf{r}}_i - \dot{\mathbf{r}}_0) + \sum_i (\mathbf{r}_i - \mathbf{r}_0) \times m_i (\ddot{\mathbf{r}}_i - \ddot{\mathbf{r}}_0) \\ &= 0 + \sum_i (\mathbf{r}_i - \mathbf{r}_0) \times (\mathbf{F}_i^{\text{ext}} + \mathbf{F}_i^{\text{int}} - m_i \ddot{\mathbf{r}}_0), \end{aligned} \quad (8.44)$$

because $m_i \ddot{\mathbf{r}}_i$ is the net force (namely $\mathbf{F}_i^{\text{ext}} + \mathbf{F}_i^{\text{int}}$) acting on the i th particle. But a quick corollary to Problem 8.9 is that the term involving $\mathbf{F}_i^{\text{int}}$ vanishes (as you should check). And since $\sum m_i \mathbf{r}_i = M\mathbf{R}$ (where $M = \sum m_i$ is the total mass, and $\mathbf{R}$ is the position of the center of mass), we have

$$\frac{d\mathbf{L}}{dt} = \left(\sum_i (\mathbf{r}_i - \mathbf{r}_0) \times \mathbf{F}_i^{\text{ext}} \right) - M(\mathbf{R} - \mathbf{r}_0) \times \ddot{\mathbf{r}}_0. \quad (8.45)$$

The first term here is the external torque, measured relative to the origin $\mathbf{r}_0$. The second term is something we wish would go away. And indeed, it usually does. It vanishes if any of the following three conditions is satisfied.

1. $\mathbf{R} = \mathbf{r}_0$, that is, the origin is the CM.
2. $\ddot{\mathbf{r}}_0 = \mathbf{0}$, that is, the origin is not accelerating.
3. $(\mathbf{R} - \mathbf{r}_0)$ is parallel to $\ddot{\mathbf{r}}_0$. This condition is rarely invoked.

If any of these conditions is satisfied, then we are free to write

$$\frac{d\mathbf{L}}{dt} = \sum_i (\mathbf{r}_i - \mathbf{r}_0) \times \mathbf{F}_i^{\text{ext}} \equiv \sum_i \boldsymbol{\tau}_i^{\text{ext}}. \quad (8.46)$$

In other words, we can equate the total external torque with the rate of change of the total angular momentum. An immediate corollary of this result is:

Corollary 8.3 *If the total external torque on a system is zero, then its angular momentum is conserved. In particular, the angular momentum of an isolated system (one that is subject to no external forces) is conserved.*

⁷ More precisely, we are calculating the angular momentum relative to a coordinate system whose origin is $\mathbf{r}_0$ and whose axes remain parallel to the fixed axes. If we allowed for a rotation of the axes, then we would have to deal with all the fictitious-force issues that are the subject of Chapter 10. As it is, we will still end up dealing with one fictitious force (see the remark below).

Everything up to this point is valid for arbitrary motion. The particles can be moving relative to each other, and the various $\mathbf{L}_i$'s can point in different directions, etc. But let's now restrict the motion. In the present chapter, we are dealing only with cases where $\hat{\mathbf{L}}$ is constant (taken to point in the z direction). Therefore, $d\mathbf{L}/dt = d(L\hat{\mathbf{L}})/dt = (dL/dt)\hat{\mathbf{L}}$. If in addition we consider only rigid objects (where the relative distances among the particles are fixed) that undergo pure rotation around a given point, then $L = I\omega$, which gives $dL/dt = I\dot{\omega} \equiv I\alpha$. Taking the magnitude of both sides of Eq. (8.46) then gives

$$\tau_{\text{ext}} = I\alpha. \quad (8.47)$$

Invariably, we will calculate angular momentum and torque around either the CM or a fixed point (or a point that moves with constant velocity, but this doesn't come up often). These are the "safe" origins, in the sense that Eq. (8.46) holds. As long as you always use one of these safe origins, you can simply apply Eq. (8.46) and not worry much about its derivation.

REMARK: You'll probably never end up invoking the third condition above, but it's interesting to note that there's a simple way of understanding it in terms of accelerating reference frames. This is the topic of Chapter 10, so we're getting a little ahead of ourselves here, but the reasoning is as follows. Let $\mathbf{r}_0$ be the origin of a reference frame that is accelerating with acceleration $\ddot{\mathbf{r}}_0$. Then all objects in this accelerating frame feel a mysterious fictitious force of $-m\ddot{\mathbf{r}}_0$. For example, on a train accelerating to the right with acceleration a , you feel a strange force of ma pointing to the left. If you don't counter this with another force (by grabbing a handle, for example), then you will fall over. This fictitious force acts just like a gravitational force, because it is proportional to the mass. Therefore, it effectively acts at the CM, producing a torque of $(\mathbf{R} - \mathbf{r}_0) \times (-M\ddot{\mathbf{r}}_0)$. This is the second term in Eq. (8.45). This term vanishes if the CM is directly "above" or "below" (as far as the fictitious gravitational force is concerned) the origin, in other words, if $(\mathbf{R} - \mathbf{r}_0)$ is parallel to $\ddot{\mathbf{r}}_0$. See Problem 10.8 for further discussion of this in terms of fictitious forces.

There is one common situation where the third condition can be invoked. Consider a wheel rolling without slipping on the ground. Mark a dot on the rim. At the instant this dot is in contact with the ground, it is a valid choice for the origin. This is true because $(\mathbf{R} - \mathbf{r}_0)$ points vertically. And $\ddot{\mathbf{r}}_0$ also points vertically, because a dot on a rolling wheel traces out a cycloid. Right before the dot hits the ground, it is moving straight downward. And right after it hits the ground, it is moving straight upward. But having said this, there's usually nothing to be gained by picking such an origin. So the safe thing to do is to always pick your origin to be either the CM or a fixed point, even if the third condition holds.

For conditions that number but three,
We say, "Torque is dL by dt ."
But though they're all true,
I'll stick to just two;
It's CM's and fixed points for me! ♣

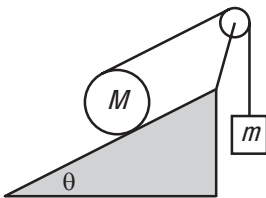


Fig. 8.17

Example: A string wraps around a uniform cylinder of mass M , which rests on a fixed plane. The string passes up over a massless pulley and is connected to a mass m , as shown in Fig. 8.17. Assume that the cylinder rolls without slipping on the plane, and that the string is parallel to the plane. What is the acceleration of the mass m ?

What is the condition on the ratio M/m for which the cylinder accelerates down the plane?

First solution: The friction, tension, and gravitational forces are shown in Fig. 8.18. Define positive a_1 , a_2 , and α as shown. These three accelerations, along with T and F , are five unknowns. We therefore need to produce five equations. They are:

1. $F = ma$ on $m \implies T - mg = ma_2$.
2. $F = ma$ on $M \implies Mg \sin \theta - T - F = Ma_1$.
3. $\tau = I\alpha$ on M (around the CM) $\implies FR - TR = (MR^2/2)\alpha$.
4. Nonslipping condition $\implies \alpha = a_1/R$.
5. Conservation of string $\implies a_2 = 2a_1$.

A few comments on these equations: The normal force and the gravitational force perpendicular to the plane cancel, so we can ignore them. We have picked positive F to point up the plane, but if it happens to point down the plane and thereby turn out to be negative, that's fine (but it won't); we don't need to worry about which way it really points. In (3), we are using the CM of the cylinder as our origin, but we can also use a fixed point; see the second solution below. In (5), we have used the fact that the top of a rolling wheel moves twice as fast as the center. This is true because the top has the same speed relative to the center as the center has relative to the ground.

We can go about solving these five equations in various ways. Three of the equations involve only two variables, so it's not so bad. (3) and (4) give $F - T = Ma_1/2$. Adding this to (2) gives $Mg \sin \theta - 2T = 3Ma_1/2$. Using (1) to eliminate T , and using (5) to write a_1 in terms of a_2 , then gives

$$Mg \sin \theta - 2(mg + ma_2) = \frac{3Ma_2}{4} \implies a_2 = \frac{(M \sin \theta - 2m)g}{(3/4)M + 2m}. \quad (8.48)$$

And $a_1 = a_2/2$. We see that a_1 is positive (that is, the cylinder rolls down the plane) if $M/m > 2/\sin \theta$. If $\theta \rightarrow 0$, then this gives $M/m \rightarrow \infty$, which makes sense. If $\theta \rightarrow \pi/2$, then $M/m \rightarrow 2$. The basic reason for this is that the friction force F , in addition to the tension T , is holding the cylinder up (the coefficient of friction must be very large in this case if no slipping is to occur).

Second solution: In using $\tau = dL/dt$, we can also pick a fixed point as our origin, instead of the CM. The most sensible point is one that is located somewhere along the plane. The $Mg \sin \theta$ force now provides a torque, but the friction does not. And the lever arm for the tension is now $2R$. The angular momentum of the cylinder with respect to a point on the plane is $L = I\omega + Mv_1R$, where the second term comes from the angular momentum due to the object being treated like a point mass at the CM. So $dL/dt = I\alpha + Ma_1R$, and $\tau = dL/dt$ gives

$$(Mg \sin \theta)R - T(2R) = (MR^2/2)\alpha + Ma_1R. \quad (8.49)$$

This turns out to be the sum of the third equation plus R times the second equation in the first solution. We therefore obtain the same result.

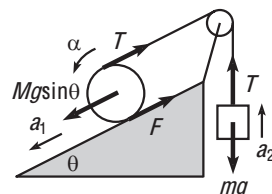


Fig. 8.18

8.5 Collisions

In Section 5.7, we looked at collisions involving point particles or otherwise nonrotating objects. The fundamental ingredients there that enabled us to solve problems were conservation of linear momentum and conservation of energy (if the collision was elastic). With conservation of angular momentum now at our disposal, we can extend our study of collisions to ones with rotating objects. The additional fact of conservation of L is compensated for by the new degree of freedom of the rotation. Therefore, provided that the problem is set up properly, we will still have the same number of equations as unknowns.

Conservation of energy can be used in a collision only if it is elastic (by definition). But conservation of angular momentum is similar to conservation of linear momentum, in that it can *always* be used (see the remark below), assuming that the system is isolated. However, conservation of L is a little different from conservation of p , because we have to pick an origin before we can proceed. In view of the three conditions that are necessary for Corollary 8.3 to hold, we must pick our origin to be either a fixed point or the CM of the system (we'll ignore the third condition, since it's rarely used). If we unwisely choose an accelerating point, then $\boldsymbol{\tau} = d\mathbf{L}/dt$ does *not* hold, so we have no right to claim that $d\mathbf{L}/dt$ equals zero just because the torque is zero (as it is for an isolated system). In collision problems, it is easy to fall into the trap of picking an accelerating point as your origin. For example, you might choose the center of a stick as your origin. But if another object collides with the stick, then the center will accelerate, making it an invalid choice for the origin.

REMARK: As far as conservation goes, the way that E differs from $\mathbf{p}$ and $\mathbf{L}$ is that energy can be hidden in the microscopic motion of molecules in the body, in the form of heat. This motion consists of little vibrations with small amplitudes but high speeds. The energy of these vibrations can be large enough to be on the same order of magnitude as the overall energy of the system. But because the vibrations are too small to see, it appears that energy is lost. However, note that even though they're too small to see, you can still feel them with your hand, as heat.

Linear momentum, however, can't be hidden. If an object (not necessarily rigid) has nonzero momentum, then it has to be moving as a whole, and there's no way around that. In short, since $P = MV_{\text{CM}}$, we see that if P is nonzero, then V_{CM} is also. So the motion must be on a macroscopic scale; there's no way for it to be hidden on a microscopic scale.

With angular momentum, things are a little trickier. If the object is rigid, then it can't have hidden angular momentum, for reasons similar to those in the linear momentum case; since $L = I\omega$, we see that if L is nonzero, then ω is also. However, if the object isn't rigid (consider, say, a gas of particles), then it turns out that it *can* theoretically have hidden angular momentum in microscopic motion, although in practice it ends up being too small to notice. This hidden angular momentum can arise from little swirling regions throughout the system. In contrast with linear momentum, it is possible to have angular momentum without any overall motion. So in this respect, this microscopic swirling motion is similar to the microscopic vibrational motion that yields hidden energy. There are, however, three main differences.

First, if we're assuming that the swirling motion takes place on a microscopic scale, then the r in $L = mrv$ is very small, and this leads to a negligible L . This argument doesn't hold for E , because the energy of the vibrations doesn't involve r . Instead, it involves only v , in the form of

$mv^2/2$, so it can end up being large. Second, there's no easy way to start up the circular motion of many little swirls by means of a collision, in contrast with the easily started random linear motion that makes up the energy of heat; you just smash two things together. And third, in the case where the object is rigid, the molecules can easily vibrate, but they can't rotate indefinitely because this would involve ripping apart the bonds between adjacent molecules. The issue is that in vibrational motion all coordinates remain small, whereas in rotational motion they don't, because θ eventually becomes large.

There is, however, one very common phenomenon, namely magnetism, that (in a sense) is an exception to all three of the above points. Although magnetism isn't an angular momentum, it does come (roughly speaking) from the "circular" motion of electrons around the nuclei in atoms. (In general, it actually comes more from the "spin" of the electrons than their orbital motion around the nuclei, but let's not worry about that here. To get everything right, we'd have to think in terms of quantum mechanics, anyway. Let's just work in a rough classical approximation.) Electrons throughout a magnetic material move in tiny little correlated loops. We can escape the above three points of reasoning because first, the magnetic field involves the electric charge e , and this is large enough (on the scale of things) to cancel out the smallness of the r factor. (The actual angular momentum of the electrons in a magnetic material is negligible because their mass is so small. There isn't a large quantity like e to save the day.) Second, it is quite easy to get the electrons moving in correlated circles by means of magnetic forces; there's no need to smash things together. And third, the electrons are free to move around in little circles (in a classical sense) in atoms without ripping things apart. ♣

It is important to remember that you are free to choose your origin from the legal possibilities of fixed points or the CM. Since it is generally the case that one choice is better than others (in that it makes the calculations easier), you should take advantage of this freedom. Let's do two examples. First an elastic collision, and then an inelastic one.

Example (Elastic collision): A mass m travels perpendicular to a stick of mass m and length ℓ , which is initially at rest. At what location should the mass collide elastically with the stick, so that the mass and the center of the stick move with equal speeds after the collision?

Solution: Let the initial speed of the mass be v_0 . We have three unknowns in the problem (see Fig. 8.19), namely the desired distance from the middle of the stick, h ; the final (equal) speeds of the stick and the mass, v ; and the final angular speed of the stick, ω . We can solve for these three unknowns by using our three available conservation laws:

- Conservation of p :

$$mv_0 = mv + mv \implies v = \frac{v_0}{2}. \quad (8.50)$$

- Conservation of E : Remembering that the energy of the stick equals the energy of the rotational motion around the center, plus the energy of the effective point mass at the center, we have

$$\frac{mv_0^2}{2} = \frac{m}{2} \left(\frac{v_0}{2} \right)^2 + \left[\frac{m}{2} \left(\frac{v_0}{2} \right)^2 + \frac{1}{2} \left(\frac{m\ell^2}{12} \right) \omega^2 \right] \implies \omega = \frac{\sqrt{6}v_0}{\ell}. \quad (8.51)$$

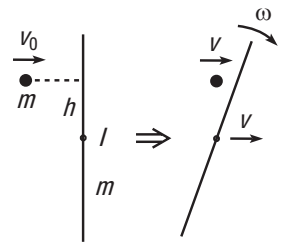


Fig. 8.19

- Conservation of L : Let's pick our origin to be the fixed point in space that coincides with the initial location of the center of the stick. Then conservation of L gives

$$mv_0h = m\left(\frac{v_0}{2}\right)h + \left[\left(\frac{m\ell^2}{12}\right)\omega + 0\right]. \quad (8.52)$$

The zero here comes from the fact that the CM of the stick moves directly away from the origin, so there is no contribution to L from the first of the two parts in Theorem 8.1. Plugging the ω from Eq. (8.51) into Eq. (8.52) gives

$$\frac{1}{2}mv_0h = \left(\frac{m\ell^2}{12}\right)\left(\frac{\sqrt{6}v_0}{\ell}\right) \implies h = \frac{\ell}{\sqrt{6}}. \quad (8.53)$$

You are encouraged to solve this problem again with a different choice of origin, for example, the fixed point that coincides with the spot where the mass hits the stick, or the CM of the entire system.

REMARK: After the stick makes half of a revolution, it will hit the backside of m . The resulting motion will have the stick sitting at rest (both translationally and rotationally) and the mass moving with its initial speed v_0 . You can show this by working through the second collision, using the quantities we found above. Or you can just use the fact that this scenario certainly satisfies conservation of p , E , and L with the initial conditions, so it must be what happens (because the quadratic conservation statements have only two solutions, and the other one corresponds to the intermediate motion). Note that the time it takes the stick to make half of a revolution is $\pi/\omega = \pi\ell/\sqrt{6}v_0$. So the stick travels a distance of $(v_0/2)(\pi\ell/\sqrt{6}v_0) = (\pi\ell/2\sqrt{6})$ before it ends up at rest. This distance is independent of v_0 (which follows from dimensional analysis). ♣

Let's now look at an inelastic collision, where one object sticks to another. We won't be able to use conservation of E now. But conservation of p and L will be sufficient since there is one fewer degree of freedom in the final motion, because the objects don't move independently.

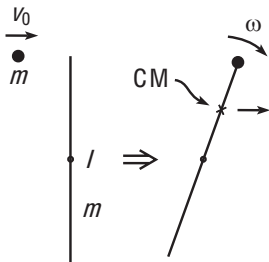


Fig. 8.20

Example (Inelastic collision): A mass m travels at speed v_0 perpendicular to a stick of mass m and length ℓ , which is initially at rest. The mass collides completely inelastically with the stick at one of its ends and sticks to it. What is the resulting angular velocity of the system?

Solution: The first thing to note is that the CM of the system is $\ell/4$ from the end, as shown in Fig. 8.20. After the collision, the system rotates about the CM as the CM moves in a straight line. Conservation of momentum quickly tells us that the speed of the CM is $v_0/2$. Also, using the parallel-axis theorem, the moment of inertia of the system around the CM is

$$I_{CM} = I_{CM}^{\text{stick}} + I_{CM}^{\text{mass}} = \left[\frac{m\ell^2}{12} + m\left(\frac{\ell}{4}\right)^2\right] + m\left(\frac{\ell}{4}\right)^2 = \frac{5}{24}m\ell^2. \quad (8.54)$$

There are now many ways to proceed, depending on what point we choose as our origin.

FIRST METHOD: Choose the origin to be the fixed point that coincides with the location of the CM right when the collision happens (that is, the point $\ell/4$ from the end of the stick). Conservation of L says that the initial L of the ball must equal the final L of the system. This gives

$$mv_0\left(\frac{\ell}{4}\right) = \left(\frac{5}{24}m\ell^2\right)\omega + 0 \implies \omega = \frac{6v_0}{5\ell}. \quad (8.55)$$

The zero here comes from the fact that the CM of the stick moves directly away from the origin, so there is no contribution to L from the first of the two parts in Theorem 8.1. Note that we didn't need to use conservation of p in this method.

SECOND METHOD: Choose the origin to be the fixed point that coincides with the initial center of the stick. Then conservation of L gives

$$mv_0\left(\frac{\ell}{2}\right) = \left(\frac{5}{24}m\ell^2\right)\omega + (2m)\left(\frac{v_0}{2}\right)\left(\frac{\ell}{4}\right) \implies \omega = \frac{6v_0}{5\ell}. \quad (8.56)$$

The right-hand side is the angular momentum of the system relative to the CM, plus the angular momentum (relative to the origin) of a point mass of mass $2m$ located at the CM.

THIRD METHOD: Choose the origin to be the CM of the system. This point moves to the right with speed $v_0/2$, along the line a distance $\ell/4$ below the top of the stick. Relative to the CM, the mass m moves to the right, and the stick moves to the left, both with speed $v_0/2$. Conservation of L gives

$$m\left(\frac{v_0}{2}\right)\left(\frac{\ell}{4}\right) + \left[0 + m\left(\frac{v_0}{2}\right)\left(\frac{\ell}{4}\right)\right] = \left(\frac{5}{24}m\ell^2\right)\omega \implies \omega = \frac{6v_0}{5\ell}. \quad (8.57)$$

The zero here comes from the fact that the stick initially has no L around its center. A fourth reasonable choice for the origin is the fixed point that coincides with the initial location of the top of the stick. You can work this one out for practice.

8.6 Angular impulse

In Section 5.5.1, we defined the *impulse*, which we'll label as $\mathcal{I}$, to be the time integral of the force applied to an object. From Newton's second law, $\mathbf{F} = d\mathbf{p}/dt$, the impulse is therefore the net change in linear momentum. That is,

$$\mathcal{I} \equiv \int_{t_1}^{t_2} \mathbf{F}(t) dt = \Delta\mathbf{p}. \quad (8.58)$$

We now define the *angular impulse*, $\mathcal{I}_\theta$, to be the time integral of the torque applied to an object. From $\boldsymbol{\tau} = d\mathbf{L}/dt$, the angular impulse is therefore the net

change in angular momentum. That is,

$$\mathcal{I}_\theta \equiv \int_{t_1}^{t_2} \boldsymbol{\tau}(t) dt = \Delta \mathbf{L}. \quad (8.59)$$

These are just definitions, devoid of any content. The place where the physics comes in is the following. Consider a situation where $\mathbf{F}(t)$ is always applied at the same position relative to the origin around which $\boldsymbol{\tau}(t)$ is calculated (this origin must be a legal one, of course). Let this position be $\mathbf{R}$. Then we have $\boldsymbol{\tau}(t) = \mathbf{R} \times \mathbf{F}(t)$. Plugging this into Eq. (8.59), and taking the constant $\mathbf{R}$ outside the integral, gives $\mathcal{I}_\theta = \mathbf{R} \times \mathcal{I}$. In other words,

$$\Delta \mathbf{L} = \mathbf{R} \times (\Delta \mathbf{p}) \quad (\text{for } \mathbf{F}(t) \text{ applied at one position}). \quad (8.60)$$

This is a very useful result. It gives the relation between the net changes in $\mathbf{L}$ and $\mathbf{p}$, as opposed to the individual values of each. Even if $\mathbf{F}(t)$ is changing in some arbitrary manner as time goes by, so that we have no idea what $\Delta \mathbf{L}$ and $\Delta \mathbf{p}$ themselves are, we still know that they are related by Eq. (8.60). In many cases, we don't have to worry about the cross product in Eq. (8.60), because the lever arm $\mathbf{R}$ is perpendicular to the change in momentum $\Delta \mathbf{p}$. In such cases, we have

$$|\Delta L| = R |\Delta p|. \quad (8.61)$$

Also, in many cases the object starts at rest, so we don't have to bother with the Δ 's. The following example is a classic application of angular impulse and Eq. (8.61).

Example (Striking a stick): A stick of mass m and length ℓ , initially at rest, is struck with a hammer. The blow is made perpendicular to the stick at one end. Let the blow occur quickly, so that the stick doesn't have time to move much while the hammer is in contact. If the CM of the stick ends up moving at speed v , what are the velocities of the ends right after the blow?

Solution: Although we have no hope of knowing exactly what $F(t)$ looks like, or the length of time it is applied, we still know from Eq. (8.61) that $\Delta L = (\ell/2)\Delta p$, where we have chosen our origin to be the CM, which gives a lever arm of $\ell/2$. Therefore, $(m\ell^2/12)\omega = (\ell/2)mv$, so the final v and ω are related by $\omega = 6v/\ell$.

The velocities of the ends right after the blow are obtained by adding (or subtracting) the rotational motion to the CM's translational motion. The rotational velocities of the ends relative to the CM are $\pm\omega(\ell/2) = \pm(6v/\ell)(\ell/2) = \pm 3v$. Therefore, the end that is hit moves with velocity $v + 3v = 4v$, and the other end moves with velocity $v - 3v = -2v$ (that is, backward).

What L was, he just couldn't tell.
 And p ? He was clueless as well.
 But despite his distress,
 He wrote down the right guess
 For their quotient: the lever arm's ℓ .

Impulse is also useful for “collisions” that occur over extended times. See, for example, Problem 8.24.

8.7 Problems

Section 8.1: Pancake object in x - y plane

8.1. Massive pulley *

Consider the Atwood's machine shown in Fig. 8.21. The masses are m and $2m$, and the pulley is a uniform disk of mass m and radius r . The string is massless and does not slip with respect to the pulley. Find the acceleration of the masses. Use conservation of energy.

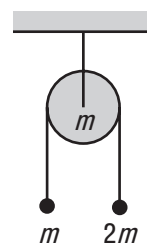


Fig. 8.21

8.2. Leaving the sphere **

A ball with moment of inertia βmr^2 rests on top of a fixed sphere. There is friction between the ball and the sphere. The ball is given an infinitesimal kick, and it rolls down without slipping. Assuming that r is much smaller than the radius of the sphere, at what point does the ball lose contact with the sphere? How does your answer change if the size of the ball is comparable to, or larger than, the size of the sphere? You may want to solve Problem 5.3 first, if you haven't already done so.

8.3. Sliding ladder ***

A ladder of length ℓ and uniform mass density stands on a frictionless floor and leans against a frictionless wall. It is initially held motionless, with its bottom end an infinitesimal distance from the wall. It is then released, whereupon the bottom end slides away from the wall, and the top end slides down the wall (see Fig. 8.22). When it loses contact with the wall, what is the horizontal component of the velocity of the center of mass?

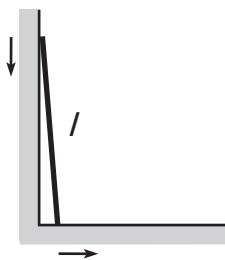


Fig. 8.22

8.4. Leaning rectangle ***

A rectangle of height $2a$ and width $2b$ rests on top of a fixed cylinder of radius R (see Fig. 8.23). The moment of inertia of the rectangle around its center is I . The rectangle is given an infinitesimal kick and then “rolls” on the cylinder without slipping. Find the equation of motion for the tilt angle of the rectangle. Under what conditions will the rectangle

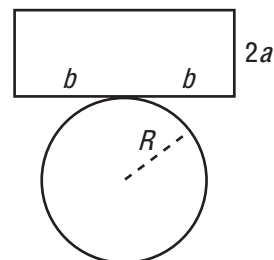


Fig. 8.23

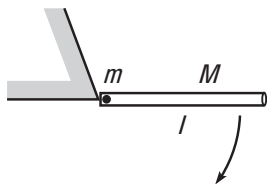


Fig. 8.24

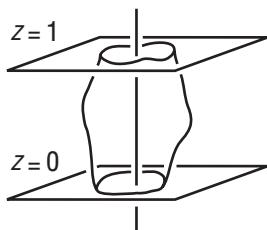


Fig. 8.25

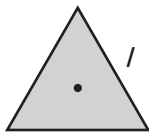
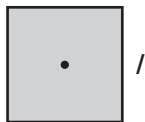


Fig. 8.26



Fig. 8.27

fall off the cylinder, and under what conditions will it oscillate back and forth? Find the frequency of these small oscillations.

8.5. Mass in a tube ***

A tube of mass M and length ℓ is free to swing around a pivot at one end. A mass m is positioned inside the (frictionless) tube at this end. The tube is held horizontal and then released (see Fig. 8.24). Let θ be the angle of the tube with respect to the horizontal, and let x be the distance the mass has traveled along the tube. Find the Euler–Lagrange equations for θ and x , and then write them in terms of θ and $\eta \equiv x/\ell$ (the fraction of the distance along the tube).

These equations can only be solved numerically, and you must pick a numerical value for the ratio $r \equiv m/M$ in order to do this. Write a program (see Section 1.4) that produces the value of η when the tube is vertical ($\theta = \pi/2$). Give this value of η for a few values of r .

Section 8.3: Calculating moments of inertia

8.6. Minimum I *

A moldable blob of matter of mass M is to be situated between the planes $z = 0$ and $z = 1$ (see Fig. 8.25) so that the moment of inertia around the z axis is as small as possible. What shape should the blob take?

8.7. Slick calculations of I **

In the spirit of Section 8.3.2, find the moments of inertia of the following objects (see Fig. 8.26).

- A uniform square of mass m and side ℓ (axis through center, perpendicular to plane).
- A uniform equilateral triangle of mass m and side ℓ (axis through center, perpendicular to plane).

8.8. Slick calculations of I for fractal objects ***

In the spirit of Section 8.3.2, find the moments of inertia of the following fractal objects. Be careful how the mass scales.

- Take a stick of length ℓ , and remove the middle third. Then remove the middle third from each of the remaining two pieces. Then remove the middle third from each of the remaining four pieces, and so on, forever. Let the final object have mass m , and let the axis be through the center, perpendicular to the stick; see Fig. 8.27.⁸

⁸ This object is the Cantor set, for those who like such things. It has no length, so the density of the remaining mass is infinite. If you suddenly develop an aversion to point masses with infinite density, simply imagine the above iteration being carried out only, say, a million times.

- (b) Take an equilateral triangle of side ℓ , and remove the “middle” triangle ($1/4$ of the area). Then remove the “middle” triangle from each of the remaining three triangles, and so on, forever. Let the final object have mass m , and let the axis be through the center, perpendicular to the plane; Fig. 8.28.
- (c) Take a square of side ℓ , and remove the “middle” square ($1/9$ of the area). Then remove the “middle” square from each of the remaining eight squares, and so on, forever. Let the final object have mass m , and let the axis be through the center, perpendicular to the plane; see Fig. 8.29.

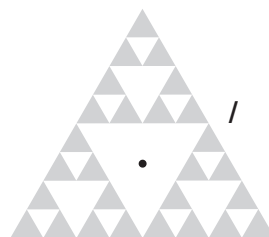


Fig. 8.28

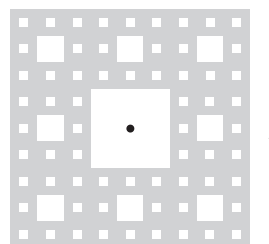


Fig. 8.29

Section 8.4: Torque

8.9. Zero torque from internal forces **

Given a collection of particles with positions $\mathbf{r}_i$, let the force on the i th particle due to all the others be $\mathbf{F}_i^{\text{int}}$. Assuming that the force between any two particles is directed along the line between them, use Newton’s third law to show that $\sum_i \mathbf{r}_i \times \mathbf{F}_i^{\text{int}} = 0$.

8.10. Removing a support *

- (a) A uniform rod of length ℓ and mass m rests on supports at its ends. The right support is quickly removed (see Fig. 8.30). What is the force from the left support immediately thereafter?
- (b) A rod of length $2r$ and moment of inertia βmr^2 rests on top of two supports, each of which is a distance d away from the center. The right support is quickly removed (see Fig. 8.30). What is the force from the left support immediately thereafter?

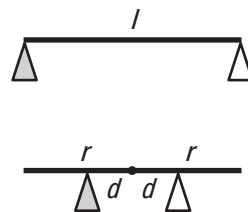


Fig. 8.30

8.11. Falling stick *

A massless stick of length b has one end pivoted on a support and the other end glued perpendicular to the middle of a stick of mass m and length ℓ .

- (a) If the two sticks are held in a horizontal plane (see Fig. 8.31) and then released, what is the initial acceleration of the CM?
- (b) If the two sticks are held in a vertical plane (see Fig. 8.31) and then released, what is the initial acceleration of the CM?

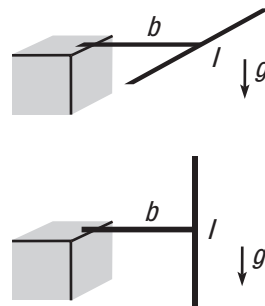


Fig. 8.31

8.12. Pulling a cylinder **

In Exercise 8.50 below, the cylinder moves directly to the right. The fact that it doesn’t have any transverse motion follows from the fact that the two segments of the string pull only to the right and therefore cannot supply a transverse force. Demonstrate this result again by explicitly

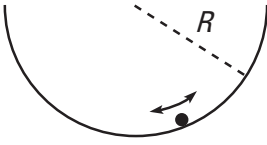


Fig. 8.32

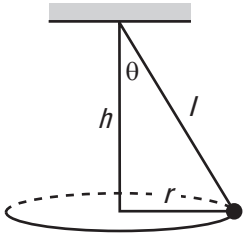


Fig. 8.33

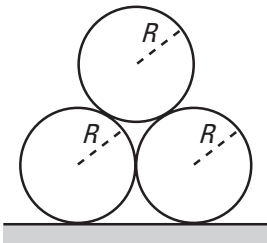


Fig. 8.34

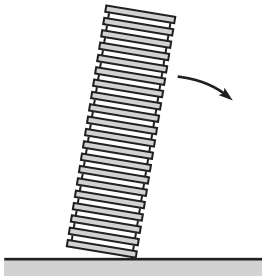


Fig. 8.35

integrating the string's force on the cylinder over the semicircle of contact. (The $N = T d\theta$ result from the "Rope wrapped around a pole" example in Section 2.1 will come in handy.)

8.13. Oscillating ball **

A small ball with radius r and uniform density rolls without slipping near the bottom of a fixed cylinder of radius R (see Fig. 8.32). What is the frequency of small oscillations? Assume $r \ll R$.

8.14. Oscillating cylinders **

A hollow cylinder of mass M_1 and radius R_1 rolls without slipping on the inside surface of another hollow cylinder of mass M_2 and radius R_2 . Assume $R_1 \ll R_2$. Both axes are horizontal, and the larger cylinder is free to rotate about its axis. What is the frequency of small oscillations?

8.15. Lengthening the string **

A mass hangs from a massless string and swings around in a horizontal circle, as shown in Fig. 8.33. The length of the string is then very slowly increased (or decreased). Let θ , ℓ , r , and h be defined as shown.

- Assuming that θ is very small, how does r depend on ℓ ?
- Assuming that θ is very close to $\pi/2$, how does h depend on ℓ ?

8.16. A triangle of cylinders ***

Three identical cylinders with moments of inertia $I = \beta m R^2$ are situated in a triangle as shown in Fig. 8.34. Find the initial downward acceleration of the top cylinder for the following two cases. Which case has a larger acceleration?

- There is friction between the bottom two cylinders and the ground (so they roll without slipping), but there is no friction between any of the cylinders.
- There is no friction between the bottom two cylinders and the ground, but there is friction between the cylinders (so they don't slip with respect to each other).

8.17. Falling chimney ****

A chimney initially stands upright. It is given a tiny kick, and it topples over. At what point along its length is it most likely to break? In doing this problem, work with the following two-dimensional simplified model of a chimney. Assume that the chimney consists of boards stacked on top of each other, and that each board is attached to the two adjacent ones with tiny rods at each end (see Fig. 8.35). The goal is to determine which rod in the chimney achieves the maximum tension. Work in the

approximation where the width of the chimney is very small compared with the height.

Section 8.5: Collisions

8.18. Ball hitting stick **

A ball of mass M collides with a stick with moment of inertia $I = \beta m \ell^2$ (relative to its center, which is its CM). The ball is initially traveling at speed V_0 perpendicular to the stick. The ball strikes the stick at a distance d from the center (see Fig. 8.36). The collision is elastic. Find the resulting translational and rotational speeds of the stick, and also the resulting speed of the ball.

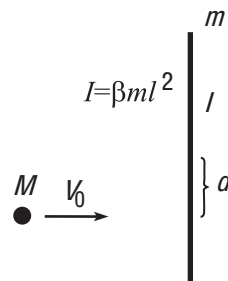


Fig. 8.36

8.19. A ball and stick theorem **

Consider the setup in Problem 8.18. Show that the relative speed of the ball and the point of contact on the stick is the same before and immediately after the collision. (This result is analogous to the “relative speed” result for a 1-D collision, Theorem 5.3 in Section 5.7.1)

Section 8.6: Angular impulse

8.20. The superball **

A ball with radius R and $I = (2/5)mR^2$ is thrown through the air. It spins around the axis perpendicular to the (vertical) plane of the motion. Call this the x - y plane. The ball bounces off a floor without slipping during the time of contact. Assume that the collision is elastic, and that the magnitude of the vertical v_y is the same before and after the bounce. Show that v'_x and ω' after the bounce are related to v_x and ω before the bounce by

$$\begin{pmatrix} v'_x \\ R\omega' \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 3 & -4 \\ -10 & -3 \end{pmatrix} \begin{pmatrix} v_x \\ R\omega \end{pmatrix}, \quad (8.62)$$

where positive v_x is to the right, and positive ω is counterclockwise.

8.21. Many bounces *

Using the result of Problem 8.20, describe what happens over the course of many superball bounces.

8.22. Rolling over a bump **

A ball with radius R and moment of inertia $I = (2/5)mR^2$ rolls with speed V_0 without slipping on the ground. It encounters a step of height h and rolls up over it. Assume that the ball sticks to the corner of

the step briefly (until the center of the ball is directly above the corner). Show that if the ball is to climb over the step, then V_0 must satisfy

$$V_0 \geq \sqrt{\frac{10gh}{7}} \left(1 - \frac{5h}{7R}\right)^{-1}. \quad (8.63)$$

8.23. Falling toast **

After buttering your toast (assumed to be a uniform rigid square of side length ℓ , for the sake of having a doable problem) one morning and holding it in a horizontal position with buttered side up, you accidentally drop it from a height H above a counter, which itself is a height h above the ground. The toast is oriented “parallel” to the counter, and as it falls, one edge barely clips the counter (elastically), causing the toast to spin. What value of H , in terms of h and ℓ , yields the unfortunate scenario where the toast makes half of a revolution, landing buttered side down on the floor? There is a special value of ℓ in terms of h . What is it, and why is it special?

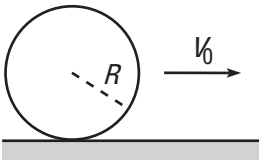


Fig. 8.37

8.24. Sliding to rolling **

A ball initially slides without rotating on a horizontal surface with friction (see Fig. 8.37). The initial speed is V_0 , and the moment of inertia around the center is $I = \beta m R^2$.

- Without knowing anything about the nature of the friction force, find the speed of the ball when it begins to roll without slipping. Also, find the kinetic energy lost while sliding.
- Now consider the special case where the coefficient of kinetic friction is μ , independent of position. At what time, and at what distance, does the ball begin to roll without slipping? Verify that the work done by friction equals the energy loss calculated in part (a). (Be careful on this.)

8.25. Lots of sticks ***

Consider a collection of rigid sticks of length $2r$, masses m_i , and moments of inertia $\beta m_i r^2$, with $m_1 \gg m_2 \gg m_3 \gg \dots$. The CM of each stick is located at its center. The sticks are placed on a horizontal frictionless surface, as shown in Fig. 8.38. The ends overlap a tiny distance in the y direction and are a tiny distance apart in the x direction. The first (heaviest) stick is given an instantaneous blow (as shown) which causes it to translate and rotate. The first stick will strike the second stick, which will then strike the third stick, and so on. Assume that all the collisions are elastic. Depending on the size of β , the speed of the n th stick will either (1) approach zero, (2) approach infinity, or

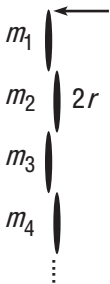


Fig. 8.38

(3) be independent of n , as $n \rightarrow \infty$. Show that the special value of β corresponding to the third of these three scenarios is $\beta = 1/3$, which happens to correspond to a uniform stick.

8.8 Exercises

Section 8.1: Pancake object in x - y plane

8.26. Swinging stick **

A uniform stick of length L is pivoted at its bottom end and is initially held vertical. It is given an infinitesimal kick, and it swings down around the pivot. After three-quarters of a turn (in the horizontal position shown in Fig. 8.39), the pivot is somehow vaporized, and the stick flies freely up in the air. What is the maximum height of the center of the stick in the resulting motion? At what angle is the stick tilted when the center reaches this maximum height?

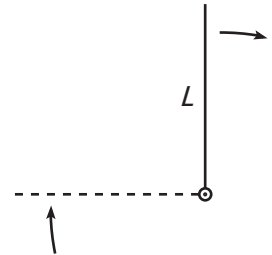


Fig. 8.39

8.27. Atwood's with a cylinder **

A massless string of negligible thickness is wrapped around a uniform cylinder of mass m and radius r . The string passes up over a massless pulley and is tied to a block of mass m at its other end, as shown in Fig. 8.40. The system is released from rest. What are the accelerations of the block and the cylinder? Assume that the string does not slip with respect to the cylinder. Use conservation of energy (after applying a quick $F = ma$ argument to show that the two objects move downward with the same acceleration).

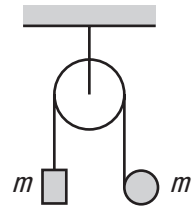


Fig. 8.40

8.28. Board and cylinders **

A board lies on top of two uniform cylinders that lie on a fixed plane inclined at an angle θ , as shown in Fig. 8.41. The board has mass m , and each of the cylinders has mass $m/2$. The system is released from rest. If there is no slipping between any of the surfaces, what is the acceleration of the board? Use conservation of energy.

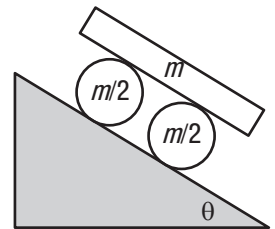


Fig. 8.41

8.29. Moving plane ***

A ball of mass m and moment of inertia $I = \beta mr^2$ is held motionless on a plane of mass M and angle of inclination θ (see Fig. 8.42). The plane rests on a frictionless horizontal surface. The ball is released. Assuming that it rolls without slipping on the plane, what is the horizontal acceleration of the plane? *Hint:* You might want to do Problem 3.8 first. But as with all the exercises in this section, use conservation of energy instead of force and torque; this problem gets extremely messy with the latter strategy.

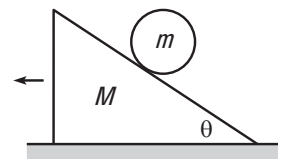


Fig. 8.42

*Section 8.2: Nonplanar objects***8.30. Semicircle CM ***

A wire is bent into a semicircle of radius R . Find the CM.

8.31. Hemisphere CM *

Find the CM of a solid hemisphere.

*Section 8.3: Calculating moments of inertia***8.32. A cone ***

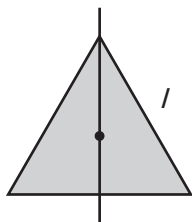
Find the moment of inertia of a solid cone (mass M , base radius R) around its symmetry axis.

8.33. A sphere *

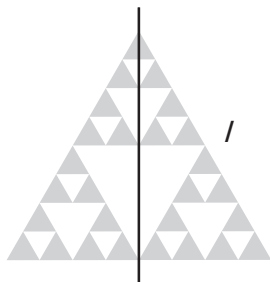
Find the moment of inertia of a solid sphere (mass M , radius R) around a diameter. Do this by slicing the sphere into disks.

8.34. A triangle, the slick way **

In the spirit of Section 8.3.2, find the moment of inertia of a uniform equilateral triangle of mass m and side ℓ , around a line joining a vertex and the opposite side; see Fig. 8.43.

**Fig. 8.43****8.35. Fractal triangle ****

Take an equilateral triangle of side ℓ , and remove the “middle” triangle ($1/4$ of the area). Then remove the “middle” triangle from each of the remaining three triangles, and so on, forever. Let the final fractal object have mass m . In the spirit of Section 8.3.2, find the moment of inertia around a line joining a vertex and the opposite side; see Fig. 8.44.

**Fig. 8.44***Section 8.4: Torque***8.36. Swinging your arms ***

You are standing on the edge of a step on some stairs, facing up the stairs. You feel yourself starting to fall backward, so you start swinging your arms around in vertical circles, like a windmill. This is what people tend to do in such a situation, but does it actually help you not to fall, or does it simply make you look silly? Explain your reasoning.

8.37. Rolling down the plane *

A ball with moment of inertia βmr^2 rolls without slipping down a plane inclined at an angle θ . What is its linear acceleration?

8.38. Coin on a plane *

A uniform coin rolls down a plane inclined at an angle θ . If the coefficient of static friction between the coin and the plane is μ , what is the largest angle θ for which the coin doesn't slip?

8.39. Accelerating plane *

A ball with $I = (2/5)MR^2$ is placed on a plane inclined at an angle θ . The plane is accelerated upwards (along its direction) with acceleration a ; see Fig. 8.45. For what value of a does the CM of the ball not move? Assume that there is sufficient friction so that the ball doesn't slip with respect to the plane.

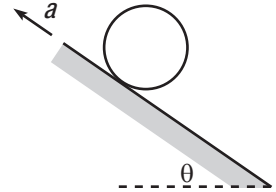


Fig. 8.45

8.40. Bowling ball on paper *

A bowling ball sits on a piece of paper on the floor. You grab the paper and pull it horizontally along the floor, with acceleration a_0 . What is the acceleration of the center of the ball? Assume that the ball does not slip with respect to the paper.

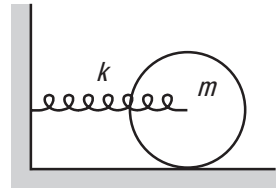


Fig. 8.46

8.41. Spring and cylinder *

The axle of a solid cylinder of mass m and radius r is connected to a spring with spring constant k , as shown in Fig. 8.46. If the cylinder rolls without slipping, what is the frequency of the oscillations?

8.42. Falling quickly *

A massless stick of length L is pivoted at one end and has a mass m attached to its other end. It is held in a horizontal position, as shown in Fig. 8.47. Where should a second mass m be attached to the stick, so that the stick falls as fast as possible when dropped?

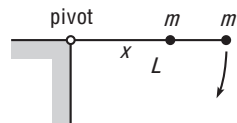


Fig. 8.47

8.43. Maximum frequency *

A pendulum is made of a uniform stick of length L . It is allowed to swing in a vertical plane. Where should the pivot be placed on the stick so that the frequency of (small) oscillations is maximum?

8.44. Massive pulley *

Solve Problem 8.1 again, but now use force and torque instead of conservation of energy.

8.45. Atwood's with a cylinder **

Solve Exercise 8.27 again, but now use force and torque instead of conservation of energy.

8.46. Board and cylinders **

Solve Exercise 8.28 again, but now use force and torque instead of conservation of energy.

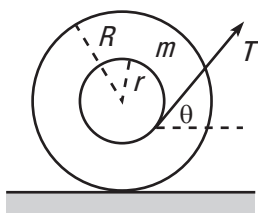


Fig. 8.48

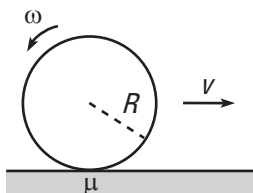


Fig. 8.49

8.47. The spool **

A spool of mass m and moment of inertia I is free to roll without slipping on a table. It has an inner radius r , and an outer radius R . If you pull on the string with tension T at an angle θ (see Fig. 8.48), what is the acceleration of the spool? Which way does it move?

8.48. Stopping the coin **

A coin stands vertically on a table. It is projected forward (in the plane of itself) with speed v and angular speed ω , as shown in Fig. 8.49. The coefficient of kinetic friction between the coin and the table is μ . What should v and ω be so that the coin comes to rest (both translationally and rotationally) a distance d from where it started?

8.49. Measuring g **

- (a) Consider an extended pendulum whose CM is a distance ℓ from the pivot, and whose moment of inertia around the pivot is I . Show that the frequency of small oscillations is $\omega = \sqrt{mg\ell/I}$, which gives $T = 2\pi/\omega = 2\pi\sqrt{I/mg\ell}$, and hence $g = 4\pi^2 I/(m\ell T^2)$. Therefore, by measuring I , m , ℓ , and T , you can determine g . However, if the pendulum has an odd shape, it may be difficult to determine I . Consider, then, the following alternative method of measuring g .
- (b) For simplicity, assume that the pendulum is planar. Pick an arbitrary point as the pivot and measure the period, T , of small oscillations. Then with the pendulum at rest, draw a vertical line through this pivot. By trial and error, find another pivot point on this line on the *same* side of the CM⁹ (you may need to extend the line with a massless extension) that yields the same period T . Let L be the sum of the lengths from these two points to the CM.¹⁰ Show that g is given by $g = 4\pi^2 L/T^2$, which is independent of m and I .

8.50. Pulling a cylinder **

A solid cylinder of mass m and radius r lies flat on a frictionless horizontal table, with a massless string running halfway around it, as shown in

⁹ The CM can be found by hanging the pendulum from a point not on the drawn line, and then drawing a vertical line through this point. The intersection of the two lines is the CM.

¹⁰ There are also two points on the other side of the CM that yield the same period (except in the special case where the two points coincide and produce the minimum period, as in Exercise 8.43, but don't worry about this). These two other points are the same distances from the CM as the original two points, as you can show. So if you happen to find all four points, you can obtain L by instead measuring the distance from the "inside" point on one side to the "outside" point on the other. This method doesn't require knowing the location of the CM.

Fig. 8.50. A mass also of mass m is attached to one end of the string, and you pull on the other end with a force T . The circumference of the cylinder is sufficiently rough so that the string does not slip with respect to it. What is the acceleration of the mass m attached to the end of the string?

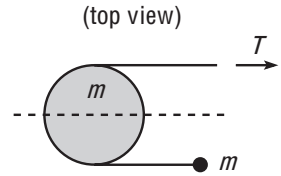


Fig. 8.50

8.51. Coin and plank **

A coin of mass M and radius R stands vertically on the right end of a horizontal plank of mass M and length L , as shown in Fig. 8.51. The system starts at rest. The plank is then pulled to the right with a constant force F . Assume that the coin does not slip with respect to the plank. What are the accelerations of the plank and coin? How far to the right does the coin move by the time the left end of the plank reaches it?

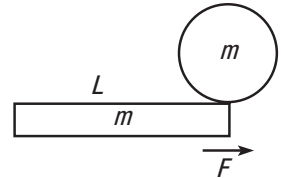


Fig. 8.51

8.52. Cylinder, board, and spring **

A board of mass m , which is free to slide on a frictionless floor, is connected by a spring (with spring constant k) to a wall. A cylinder, also of mass m (and $I = mR^2/2$), rests on top of the board, as shown in Fig. 8.52, and is free to roll without slipping on the board. If the board and the cylinder are pulled some distance to the left and then released from rest, what is the frequency of the resulting oscillatory motion?

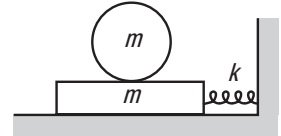


Fig. 8.52

8.53. Swirling around a cone **

A fixed hollow frictionless cone is positioned with its tip pointing down. A particle is released from rest on the inside surface. After it has slid halfway down to the tip, it bounces elastically off a platform. The platform is positioned at a 45° angle along the surface of the cone, so the particle ends up being deflected horizontally along the surface (in other words, into the page in Fig. 8.53). The particle then swirls up and around the cone before coming down. Measured from the tip of the cone, show that the ratio of the particle's maximum swirling height to the height of the platform is $(\sqrt{5} + 1)/2$, which just happens to be the golden ratio.

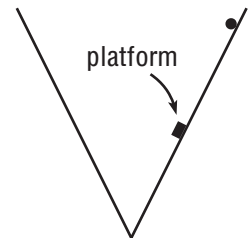


Fig. 8.53

8.54. Raising a hoop **

A bead of mass m is positioned at the top of a frictionless hoop of mass M and radius R , which stands vertically on the ground. A wall touches the hoop on its left, and a short wall of height R touches the hoop on its right, as shown in Fig. 8.54. All surfaces are frictionless. The bead is given a tiny kick, and it slides down the hoop, as shown. What is the largest value of m/M for which the hoop never rises up off the ground? *Note:* It's possible to solve this problem by using only force, but solve it here by using torque.

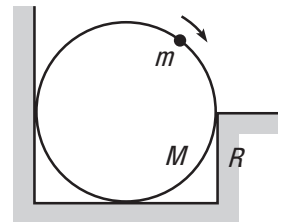


Fig. 8.54

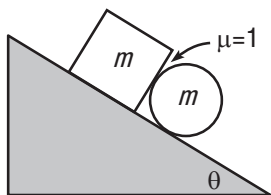


Fig. 8.55

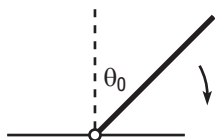


Fig. 8.56

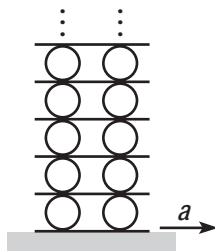


Fig. 8.57

8.55. Block and cylinder **

A block and a cylinder (with $I = \beta m R^2$), both of mass m , lie on a plane (inclined at an angle θ), touching each other as shown in Fig. 8.55. There is sufficient friction between the cylinder and the plane so that it rolls without slipping. But the bottom of the block is coated with grease, so there is no friction between it and the plane. However, there is a coefficient of kinetic friction $\mu = 1$ between the block's right face and the cylinder. What is the acceleration of the block? How does your answer compare with the result for a lone cylinder rolling down a plane? Assume that θ is small enough so that the block's bottom face remains in contact with the plane at all times; what is the condition on θ for which this is true?

8.56. Falling and sliding stick ***

One end of a uniform stick is attached to a pivot, and the pivot is free to slide along a frictionless horizontal rail. The stick is held at an initial angle θ_0 with respect to the (upward) vertical direction and then released; see Fig. 8.56. Assume that the stick can somehow swing down below the horizontal position without running into the rail (perhaps by having the pivot attached to the side of the rail, so that the stick is shifted horizontally a small distance from the rail).

- Show that when the stick is horizontal, the normal force N from the rail equals $mg/4$, independent of θ_0 .
- If $\theta_0 = 0$ (a tiny kick is allowed), show that $N = 13mg$ when the stick is at the bottom of its motion (at $\theta = \pi$).
- If $\theta_0 = 0$, show that the minimum N occurs at $\theta \approx 61.5^\circ$ and that the value is $N_{\min} \approx (0.165)mg$. You will obtain a cubic; feel free to solve it numerically.

8.57. Tower of cylinders ****

Consider the infinitely tall system of massless planks and identical massive cylinders shown in Fig. 8.57. The moment of inertia of the cylinders is $I = MR^2/2$. There are two cylinders at each level, and the number of levels is infinite. The cylinders do not slip with respect to the planks, but the bottom plank is free to slide on a table. If you pull on the bottom plank so that it accelerates horizontally with acceleration a , what is the horizontal acceleration of the bottom row of cylinders?

*Section 8.5: Collisions***8.58. Pendulum collision ***

A stick of mass m and length ℓ is pivoted at an end. It is held horizontal and then released. It swings down, and when it is vertical the free end

elastically collides with a ball, as shown in Fig. 8.58. (Assume that the ball is initially held at rest and then released a split second before the stick strikes it.) If the stick loses half of its angular velocity during the collision, what is the mass of the ball? What is the speed of the ball right after the collision?

8.59. No final rotation *

A stick of mass m and length ℓ spins around on a frictionless horizontal table, with its CM at rest (but not fixed by a pivot). A ball of mass M is placed on the table, and one end of the stick collides elastically with it, as shown in Fig. 8.59. What should M be so that after the collision the stick has translational motion, but no rotational motion?

8.60. Same final speeds *

A stick slides perpendicular to itself (without rotating) across a frictionless horizontal table and collides elastically at one of its ends with a stationary ball. Both the stick and the ball have mass m . The mass of the stick is distributed in such a way that the moment of inertia around the CM (which is at the center of the stick) is $I = A m \ell^2$, where A is some number. What should A be so that the ball moves at the same speed as the center of the stick after the collision?

8.61. Perpendicular deflection **

A mass M moves at speed V_0 perpendicular to a dumbbell at rest on a frictionless horizontal table, as shown in Fig. 8.60. The dumbbell consists of two masses m at the ends of a massless rod of length ℓ . The mass M collides elastically with one of the masses (not head-on), and afterwards it is observed that M moves perpendicular to its original direction, with speed u . What is u in terms of V_0 , m , and M ? What is the smallest value of m (in terms of M) for which this scenario is possible?

8.62. Glancing off a stick **

A frictionless stick of mass m and length ℓ lies at rest on a frictionless horizontal table. A mass km (where k is some number) moves with speed v_0 at a 45° angle to the stick and collides elastically with it very close to an end; see Fig. 8.61. What should k be so that the mass ends up moving in the y -direction, as shown? *Hint:* Remember that the stick is frictionless.

8.63. Sticking to a dumbbell *

A mass m moves at speed v perpendicular to a dumbbell at rest on a frictionless horizontal table, as shown in Fig. 8.62. The dumbbell consists of two masses also of mass m at the ends of a massless rod of length ℓ . The moving mass collides and sticks to one of the masses.

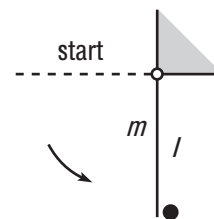


Fig. 8.58

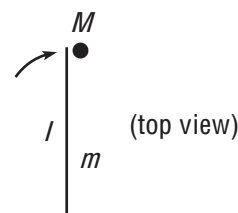


Fig. 8.59

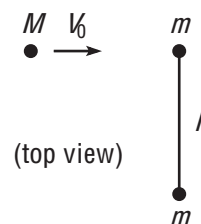


Fig. 8.60

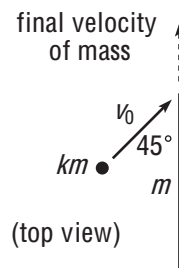


Fig. 8.61

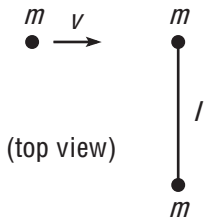


Fig. 8.62

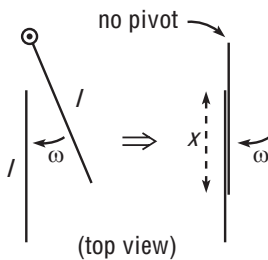


Fig. 8.63

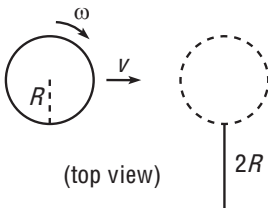


Fig. 8.64

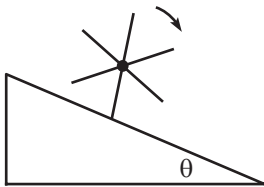


Fig. 8.65

What is the resulting ω of the system? What is the velocity of the end of the rod that has the two masses on it, after the rod has made one half of a revolution?

8.64. Colliding sticks **

On a frictionless horizontal table, a stick of mass m and length ℓ spins around a pivot at one of its ends with angular frequency ω . It collides and sticks to an identical stick, with an overlap length equal to x , as shown in Fig. 8.63. Immediately before the collision, the pivot is removed. What should x be so that after the collision the double-stick system has translational but no rotational motion?

8.65. Lollipop **

A hockey puck of mass m and radius R slides across frictionless ice, as shown in Fig. 8.64 (the view is from above). It has translational speed v to the right and rotational speed ω clockwise. It grazes the “top” end of a rod of mass m and length $2R$ which is initially at rest on the ice. It sticks to the rod, forming a rigid object that looks like a lollipop.

- In the special case of $v = R\omega$, what is the resulting angular speed of the lollipop?
- How much energy is lost during the collision? How do you explain the fact that energy is lost, given that the $v = R\omega$ condition implies that the contact point on the puck touches the rod with zero relative speed (and thus doesn’t crash into it, as is commonly the case with inelastic collisions)?
- Given ω , show that v should equal $6R\omega/5$ if you want the minimum amount of energy to be lost.

8.66. Pencil on a plane ****

This exercise deals with the terminal velocity of a “pencil” rolling down a plane. To simplify things, we’ll assume that the pencil has all its mass on its axis. And to avoid messy complications, we’ll assume that the cross section of the pencil looks like a wheel with six equally spaced massless spokes and no rim (see Fig. 8.65).¹¹ Let the length of the spokes be r , and let the plane be inclined at an angle θ . Assume that there is sufficient friction to prevent the spokes from slipping on the plane, and assume that the pencil does not bounce when a spoke hits the plane.

- Explain qualitatively why the pencil reaches a terminal (average) velocity, assuming that it remains in contact with the plane at all times.

¹¹ If the pencil instead looks like a hexagon with flat sides, then it is impossible to say how it behaves, because if the sides bow outward an infinitesimal amount then the system conserves energy, whereas if they bow inward an infinitesimal amount then it does not (for reasons you will figure out).

- (b) Assume that conditions have been set up so that the pencil eventually reaches a nonzero terminal (average) velocity, while remaining in contact with the plane at all times. Describe this terminal velocity. You may do this by stating the maximum speed of the axis in the limiting steady state.
- (c) What is the minimum value of θ for which a nonzero terminal velocity exists? An initial kick to the pencil is allowed.
- (d) What is the maximum value of θ for which the pencil remains in contact with the plane at all times? As a check on your answer, the difference between the answers in parts (c) and (d) is about 5.09° .

Section 8.6: Angular impulse

8.67. Striking a pool ball *

At what height should you horizontally strike a pool ball so that it immediately rolls without slipping?

8.68. Center of percussion *

You hold one end of a uniform stick of length L . The stick is struck with a hammer. Where should this blow occur so that the end you are holding doesn't move (immediately after the blow)? In other words, where should the blow occur so that you don't feel a "sting" in your hand? This point is called the *center of percussion*.

8.69. Another center of percussion *

You hold the top vertex of a solid equilateral triangle of side length L . The plane of the triangle is vertical. It is struck with a hammer, somewhere along the vertical axis. Where should this blow occur so that the point you are holding doesn't move (immediately after the blow)? Use the fact that the moment of inertia about any axis through the CM of an equilateral triangle is $mL^2/24$.

8.70. Not hitting the pole **

A (possibly non-uniform) stick of mass m and length ℓ lies on frictionless ice. Its midpoint (which is also its CM) touches a thin pole sticking out of the ice. One end of the stick is struck with a quick blow perpendicular to the stick, as shown in Fig. 8.66, so that the CM moves away from the pole. What is the minimum value of the stick's moment of inertia that allows the stick not to hit the pole?

8.71. Pulling the paper **

A ball sits at rest on a piece of paper on a table. You pull the paper in a straight line out from underneath the ball. You are free to pull the

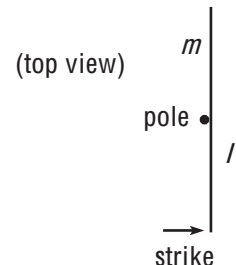


Fig. 8.66

paper in an arbitrary (straight line) manner, frontward or backward. You may even give it abrupt, jerky motions, so that the ball slips with respect to it. After the ball comes off the paper, it will eventually roll without slipping on the table. Show (and you are encouraged to experimentally verify this) that the ball in fact ends up at rest. (The generalization of this fact is given in Problem 9.29.) Is it possible to pull the paper in such a way that the ball ends up exactly where it started?

8.72. Up, down, and twisting **

A uniform stick is held horizontally and then released. At the same instant, one end is struck with a quick upwards blow. If the stick ends up horizontal when it returns to its original height, what are the possible values for the maximum height to which the center rises?

8.73. Doing work **

- A pencil of mass m and length ℓ lies at rest on a frictionless table. You push on it at its midpoint (perpendicular to it), with a constant force F for a time t . Find the final speed and the distance traveled. Verify that the work you do equals the final kinetic energy.
- Assume that you apply the same force F for the same time t as above, but that you now apply it at one of the pencil's ends (perpendicular to the pencil). Assume that t is small, so that the pencil doesn't have much time to rotate (this means that you can assume that your force is always essentially perpendicular to the pencil, as far as the torque is concerned). Find the final CM speed, the final angular speed, and the distance your hand moves. Verify that the work you do equals the final kinetic energy.

8.74. Bouncing between bricks ***

A stick of length ℓ slides on frictionless ice. It bounces elastically between two parallel fixed bricks, a distance L apart, in such a way that the same end touches both bricks, and the stick hits the bricks at an angle θ each time. See Fig. 8.67. What is θ in terms of L and ℓ (an implicit equation is fine)? Draw a reasonably accurate picture of what the situation looks like in the limit $L \ll \ell$.

What should θ be in terms of L and ℓ (again, an implicit equation is fine), if the stick makes an additional n half revolutions between the bricks? What is the minimum value of L/ℓ for which n half revolutions are possible?

8.75. Repetitive bouncing *

Using the result of Problem 8.20, what must the relation between v_x and $R\omega$ be so that a superball continually bounces back and forth between the same two points of contact on the ground?

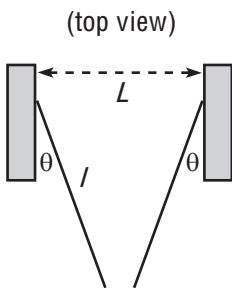


Fig. 8.67

8.76. Bouncing under a table **

You throw a superball so that it bounces off the floor, then off the underside of a table, then off the floor again. What must the initial relation between v_x and $R\omega$ be so that the ball returns to your hand, with the outward and return paths the same? Use the result of Problem 8.20, and modifications thereof.¹²

8.77. Bouncing under a table again ****

Consider the setup in the previous exercise, where we assumed that the outward and return paths were the same. Is this trajectory (where the path retraces itself) the only possible one for which the ball returns to your hand? Show¹³ that the answer is “yes” unless $t_1 = 7t_2$, where t_1 is the time the ball spends between your hand and the floor (which is the same out and back, because the magnitude of v_y isn’t changed by a bounce), and t_2 is the time the ball spends between the floor and the table (again, the same out and back). For the special case where $t_1 = 7t_2$, you will find that the ball returns to your hand for *any* initial relation between v_x and $R\omega$.¹⁴

For a ball with a general moment of inertia $I = \beta mr^2$, show that the answer is always “yes,” without exception, if $\beta \leq 1/3$ (which corresponds to a wheel with massive spokes and a massless rim). Also, show that if $\beta = 1$ (a hoop), then the $t_1 = 7t_2$ condition becomes $t_1 = t_2$. In other words, if you throw a “super-hoop” from the same height as the table, then no matter how you throw it (as long as the throw is downward and in the plane of the hoop), it will return to your hand. This is a little more believable if you look at the remark in the solution to Problem 8.20.

8.9 Solutions**8.1. Massive pulley**

The two masses have equal speeds at all times. Let v be their common speed after they have each moved a distance d . If $2m$ falls a distance d (so that m rises a distance d), then the change in potential energy is $-2mgd + 2mgd = -mgd$. The total kinetic energy is

$$K = \frac{1}{2}mv^2 + \frac{1}{2}(2m)v^2 + \frac{1}{2}I\omega^2$$

¹² You are strongly encouraged to bounce a ball in such a manner and have it magically come back to your hand. It turns out that the required value of ω is small, so a natural throw with $\omega \approx 0$ essentially gets the job done.

¹³ You will want to use Mathematica or some other aid to keep track of the matrix multiplications, especially in the second part of this exercise, which would be completely intractable otherwise.

¹⁴ I learned of this extremely bizarre fact from Howard Georgi.

$$\begin{aligned}
&= \frac{1}{2}mv^2 + \frac{1}{2}(2m)v^2 + \frac{1}{2}\left(\frac{1}{2}mr^2\right)\left(\frac{v}{r}\right)^2 \\
&= \frac{7}{4}mv^2,
\end{aligned} \tag{8.64}$$

where we have used the nonslipping condition, $v = r\omega$. Conservation of energy therefore gives

$$0 = \frac{7}{4}mv^2 - mgd \implies v = \sqrt{\frac{4}{7}gd}. \tag{8.65}$$

The usual kinematic result $v = \sqrt{2ad}$ holds here, so we obtain $a = 2g/7$.

8.2. Leaving the sphere

In this setup, as in Problem 5.3, the ball leaves the sphere when the normal force becomes zero, that is, when

$$\frac{mv^2}{R} = mg \cos \theta. \tag{8.66}$$

The only change from the solution to Problem 5.3 comes in the calculation of v . The ball now has rotational energy, so conservation of energy gives $mgR(1 - \cos \theta) = mv^2/2 + I\omega^2/2 = mv^2/2 + \beta mr^2\omega^2/2$. But the nonslipping condition is $v = r\omega$, so we have

$$\frac{1}{2}(1 + \beta)mv^2 = mgR(1 - \cos \theta) \implies v = \sqrt{\frac{2gR(1 - \cos \theta)}{1 + \beta}}. \tag{8.67}$$

Plugging this into Eq. (8.66), we see that the ball leaves the sphere when

$$\cos \theta = \frac{2}{3 + \beta}. \tag{8.68}$$

REMARKS: For $\beta = 0$, this equals $2/3$, as in Problem 5.3. For a uniform ball with $\beta = 2/5$, we have $\cos \theta = 10/17$, so $\theta \approx 54^\circ$. For $\beta \rightarrow \infty$ (for example, a spool with a very thin axle rolling down the rim of a circle), we have $\cos \theta \rightarrow 0$, so $\theta \approx 90^\circ$. This makes sense because v is always very small, since most of the energy takes the form of rotational energy. The coefficient of friction would have to be very large in this case, of course, to keep the spool from slipping near $\theta \approx 90^\circ$. ♣

If the size of the ball is comparable to, or larger than, the size of the sphere, then we must take into account the fact that the CM of the ball does not move along a circle of radius R . Instead, it moves along a circle of radius $R + r$, so Eq. (8.66) becomes

$$\frac{mv^2}{R + r} = mg \cos \theta. \tag{8.69}$$

Also, the conservation of energy equation takes the form, $mg(R + r)(1 - \cos \theta) = mv^2/2 + \beta mr^2\omega^2/2$. But v still equals $r\omega$ (because the ball can be considered to be instantaneously rotating around the contact point with angular speed ω), so the kinetic energy still equals $(1 + \beta)mv^2/2$. The conservation of energy statement is thus

$$\frac{1}{2}(1 + \beta)mv^2 = mg(R + r)(1 - \cos \theta). \tag{8.70}$$

We therefore have the same equations as above, except that R is replaced everywhere by $R + r$. But R didn't appear in the result for θ in Eq. (8.68), so the answer is unchanged.

REMARK: The method of the second solution to Problem 5.3 does *not* work in this problem, because there *is* a force available to make v_x decrease, namely the friction

force. And indeed, v_x does decrease before the rolling ball leaves the sphere. At a given value of θ , the v in the present problem is simply $1/\sqrt{1+\beta}$ times the v in Problem 5.3, so the maximum v_x is achieved here at $\cos \theta = 2/3$, just as in Problem 5.3. But the angle in Eq. (8.68) is larger than this, so v_x decreases while the ball is between these two angles. (However, see the following problem for a setup involving rotations where the max v_x is relevant.) ♣

8.3. Sliding ladder

The important point to realize in this problem is that the ladder loses contact with the wall before it hits the ground. So we need to find where this loss of contact occurs. Let $r = \ell/2$, for convenience. While the ladder is in contact with the wall, its CM moves in a circle of radius r . This follows from the fact that the median to the hypotenuse of a right triangle has half the length of the hypotenuse. Let θ be the angle between the wall and the radius from the corner to the CM (see Fig. 8.68). This is also the angle between the ladder and the wall.

We'll solve this problem by assuming that the CM always moves in a circle, and then determining the position at which the horizontal CM speed starts to decrease, that is, the point at which the normal force from the wall would have to become negative. Since the normal force of course can't be negative, this is the point where the ladder loses contact with the wall.

By conservation of energy, the kinetic energy of the ladder equals the loss in potential energy, which is $mgr(1 - \cos \theta)$. This kinetic energy can be broken up into the CM translational energy plus the rotation energy. The CM translational energy is $mr^2\dot{\theta}^2/2$, because the CM travels in a circle of radius r . The rotational energy is $I\dot{\theta}^2/2$. The same $\dot{\theta}$ applies here as in the CM translational motion, because θ is the angle between the ladder and the vertical, and is thus the angle of rotation of the ladder. Letting $I \equiv \beta mr^2$ to be general ($\beta = 1/3$ for our ladder), the conservation of energy statement is $(1+\beta)mr^2\dot{\theta}^2/2 = mgr(1 - \cos \theta)$. Therefore, the speed of the CM, which is $v = r\dot{\theta}$, equals

$$v = \sqrt{\frac{2gr(1 - \cos \theta)}{1 + \beta}}. \quad (8.71)$$

The horizontal component of this is

$$v_x = \sqrt{\frac{2gr}{1 + \beta}} \sqrt{(1 - \cos \theta) \cos \theta}. \quad (8.72)$$

Taking the derivative of $\sqrt{(1 - \cos \theta) \cos \theta}$, we see that the horizontal speed is maximum when $\cos \theta = 2/3$. Therefore the ladder loses contact with the wall when

$$\cos \theta = \frac{2}{3} \implies \theta \approx 48.2^\circ, \quad (8.73)$$

which is independent of β . This means that, for example, a dumbbell (two masses at the ends of a massless rod, with $\beta = 1$) loses contact with the wall at the same angle. Plugging the value of θ from Eq. (8.73) into Eq. (8.72), and using $\beta = 1/3$, we obtain a final horizontal speed of

$$v_x = \frac{\sqrt{2gr}}{3} \equiv \frac{\sqrt{g\ell}}{3}. \quad (8.74)$$

Note that this is $1/3$ of the $\sqrt{2gr}$ horizontal speed that the ladder would have if it were arranged (perhaps by having the top end slide down a curve) to eventually slide horizontally along the ground. You are encouraged to compare various aspects of this problem with those in Problem 8.2 and Problem 5.3.

REMARK: The normal force from the wall is zero at the start and zero at the finish, so it must reach a maximum at some intermediate value of θ . Let's find this θ . Taking the

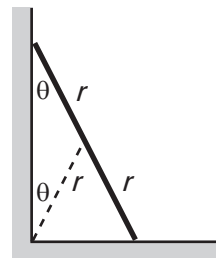


Fig. 8.68

derivative of v_x in Eq. (8.72) to find the CM's horizontal acceleration a_x , and then using $\dot{\theta} \propto \sqrt{1 - \cos \theta}$ from Eq. (8.71), we see that the force from the wall is proportional to

$$a_x \propto \frac{\sin \theta (3 \cos \theta - 2) \dot{\theta}}{\sqrt{1 - \cos \theta}} \propto \sin \theta (3 \cos \theta - 2). \quad (8.75)$$

Taking the derivative of this, we find that the force from the wall is maximum when $\cos \theta = (1 + \sqrt{19})/6 \implies \theta \approx 26.7^\circ$ ♣

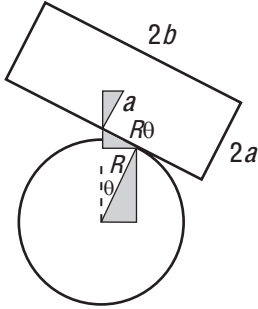


Fig. 8.69

8.4. Leaning rectangle

We must first find the position of the rectangle's CM when it has rotated through an angle θ . Using Fig. 8.69, we can obtain this position (relative to the center of the cylinder) by adding up the distances along the three shaded triangles. Because there is no slipping, the contact point has moved a distance $R\theta$ along the rectangle. We find that the position of the CM is

$$(x, y) = R(\sin \theta, \cos \theta) + R\theta(-\cos \theta, \sin \theta) + a(\sin \theta, \cos \theta). \quad (8.76)$$

We'll now use the Lagrangian method to find the equation of motion and the frequency of small oscillations. Using Eq. (8.76), you can show that the square of the speed of the CM is

$$v^2 = \dot{x}^2 + \dot{y}^2 = (a^2 + R^2\theta^2)\dot{\theta}^2. \quad (8.77)$$

The simplicity of this result suggests that there is a quicker way to obtain it. And indeed, the CM instantaneously rotates around the contact point with angular speed $\dot{\theta}$, and from Fig. 8.69, the distance to the contact point is $\sqrt{a^2 + R^2\theta^2}$. Therefore, the speed of the CM is $\omega r = \dot{\theta}\sqrt{a^2 + R^2\theta^2}$.

The Lagrangian is

$$\mathcal{L} = T - V = \frac{1}{2}m(a^2 + R^2\theta^2)\dot{\theta}^2 + \frac{1}{2}I\dot{\theta}^2 - mg((R + a)\cos \theta + R\theta \sin \theta). \quad (8.78)$$

The equation of motion is, as you can check,

$$(ma^2 + mR^2\theta^2 + I)\ddot{\theta} + mR^2\theta\dot{\theta}^2 = mga \sin \theta - mgR\theta \cos \theta. \quad (8.79)$$

Let us now consider small oscillations. Using the small-angle approximations, $\sin \theta \approx \theta$ and $\cos \theta \approx 1 - \theta^2/2$, and keeping terms up to first order in θ , we obtain

$$(ma^2 + I)\ddot{\theta} + mg(R - a)\theta = 0. \quad (8.80)$$

The coefficient of θ is positive if $a < R$. Therefore, oscillatory motion occurs if $a < R$. Note that this condition is independent of b . The frequency of small oscillations is

$$\omega = \sqrt{\frac{mg(R - a)}{ma^2 + I}}. \quad (8.81)$$

If $a \geq R$, then the rectangle falls off the cylinder.

REMARKS: Let's look at some special cases. If $I = 0$ (that is, all of the rectangle's mass is located at the CM), we have $\omega = \sqrt{g(R - a)/a^2}$. If in addition $a \ll R$, then $\omega \approx \sqrt{gR/a^2}$. You can also derive these results by considering the CM to be a point mass sliding in a parabolic potential. If the rectangle is instead a uniform horizontal stick, so that $a \ll R$, $a \ll b$, and $I \approx mb^2/3$, then we have $\omega \approx \sqrt{3gR/b^2}$. If the rectangle is a vertical stick (satisfying $a < R$), so that $b \ll a$ and $I \approx ma^2/3$, then we have $\omega \approx \sqrt{3g(R - a)/4a^2}$. If in addition $a \ll R$, then $\omega \approx \sqrt{3gR/4a^2}$.

Without doing much work, there are two other ways we can determine the condition under which there is oscillatory motion. The first is to look at the height of the CM (although this is essentially what we ended up doing in the solution above). Using small-angle approximations in Eq. (8.76), the height of the CM is $y \approx (R + a) +$

$(R - a)\theta^2/2$. Therefore, if $a < R$, the potential energy increases with θ , so the rectangle wants to decrease its θ and fall back down to the middle. But if $a > R$, the potential energy decreases with θ , so the rectangle wants to increase its θ and fall off the cylinder.

The second way is to look at the horizontal positions of the CM and the contact point. Small-angle approximations in Eq. (8.76) show that the former equals $a\theta$ and the latter equals $R\theta$. Therefore, if $a < R$ then the CM is to the left of the contact point, so the torque from gravity (relative to the contact point) makes θ decrease, and the motion is stable. But if $a > R$ then the torque from gravity makes θ increase, and the motion is unstable. ♣

8.5. Mass in a tube

The Lagrangian is

$$\mathcal{L} = \frac{1}{2} \left(\frac{1}{3} M \ell^2 \right) \dot{\theta}^2 + \left(\frac{1}{2} m x^2 \dot{\theta}^2 + \frac{1}{2} m \dot{x}^2 \right) + mgx \sin \theta + Mg \left(\frac{\ell}{2} \right) \sin \theta. \quad (8.82)$$

The Euler–Lagrange equations are then

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}} \right) &= \frac{\partial \mathcal{L}}{\partial x} \implies m\ddot{x} = m x \dot{\theta}^2 + mg \sin \theta, \\ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{\theta}} \right) &= \frac{\partial \mathcal{L}}{\partial \theta} \implies \frac{d}{dt} \left(\frac{1}{3} M \ell^2 \dot{\theta} + m x^2 \dot{\theta} \right) = \left(mgx + \frac{Mg\ell}{2} \right) \cos \theta \\ &\implies \left(\frac{1}{3} M \ell^2 + m x^2 \right) \ddot{\theta} + 2m x \dot{x} \dot{\theta} = \left(mgx + \frac{Mg\ell}{2} \right) \cos \theta. \end{aligned} \quad (8.83)$$

In terms of $\eta \equiv x/\ell$, these equations become

$$\begin{aligned} \ddot{\eta} &= \eta \dot{\theta}^2 + \tilde{g} \sin \theta, \\ (1 + 3r\eta^2) \ddot{\theta} &= \left(3r\tilde{g}\eta + \frac{3\tilde{g}}{2} \right) \cos \theta - 6r\eta \dot{\eta} \dot{\theta}, \end{aligned} \quad (8.84)$$

where $r \equiv m/M$ and $\tilde{g} \equiv g/\ell$. Below is a Maple program that numerically finds the value of η when θ equals $\pi/2$, in the case where $r = 1$. As mentioned in Problem 1.2, this value of η does not depend on g or ℓ , and hence not $\tilde{g}$. In the program, we'll denote $\tilde{g}$ by g , which we'll give the arbitrary value of 10. We'll use q for θ , and n for η . Also, we'll denote $\dot{\theta}$ by $q1$ and $\dot{\eta}$ by $n1$, etc. Even if you don't know Maple, this program should still be understandable. See Section 1.4 for more discussion of solving differential equations numerically.

```
n:=0: # initial n value
n1:=0: # initial n speed
q:=0: # initial angle
q1:=0: # initial angular speed
e:=.0001: # small time interval
g:=10: # value of g/l
r:=1: # value of m/M
while q<1.57079 do # do this process until
# the angle is pi/2
n2:=n*q1^2+g*sin(q): # the first E-L equation
q2:=( (3*r*g*n+3*g/2)*cos(q)
-6*r*n*n1*q1)/(1+3*r*n^2): # the second E-L equation
n:=n+e*n1: # how n changes
n1:=n1+e*n2: # how n1 changes
q:=q+e*q1: # how q changes
q1:=q1+e*q2: # how q1 changes
end do: # stop the process
n: # print the value of n
```

The resulting value of η is 0.378. If you actually run this program on Maple with different values of g , you will find that the result for η doesn't depend on g , as stated above. A few results for η for various values of r are, in (r, η) notation: $(0, 0.349)$, $(1, 0.378)$, $(2, 0.410)$, $(10, 0.872)$, $(20, 3.290)$. It turns out that $r \approx 11.25$ yields $\eta \approx 1$. That is, the mass m gets to the end of the tube right when the tube becomes vertical.

For η values larger than 1, we could imagine attaching a massless tubular extension on the end of the given tube. It turns out that $\eta \rightarrow \infty$ as $r \rightarrow \infty$. In this case, the mass m drops nearly straight down, causing the tube to quickly swing down to a nearly vertical position. But m ends up slightly to one side and then takes a very long time to move over to become directly below the pivot.

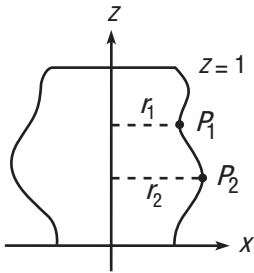


Fig. 8.70

8.6. Minimum I

The shape should be a cylinder with the z axis as its symmetry axis. A quick proof (by contradiction) is as follows. Assume that the optimal blob is not a cylinder, and consider the surface of the blob. If the blob is not a cylinder, then there exist two points on the surface, P_1 and P_2 , that are located at different distances, r_1 and r_2 , from the z axis. Assume $r_1 < r_2$ (see Fig. 8.70). If we move a small piece of the blob from P_2 to P_1 , then we decrease the moment of inertia, $\int r^2 dm$. Therefore, the proposed noncylindrical blob cannot be the one with the smallest I . In order to avoid this contradiction, all points on the surface must be equidistant from the z axis. The only blob with this property is a cylinder.

8.7. Slick calculations of I

- (a) We claim that the I for a square of side 2ℓ is 16 times the I for a square of side ℓ , assuming that the axes pass through any two corresponding points. This is true because dm is proportional to the area, which is proportional to length squared, so the corresponding dm 's differ by a factor of 4. And then there are the two powers of r in the integrand. Therefore, when changing variables from one square to the other, there are four powers of 2 in the integral $\int r^2 dm = \int r^2 \rho dx dy$.

As in Section 8.3.2, we can express the relevant relations in terms of pictures:

$$\begin{aligned}
 \boxed{\bullet}^{2\ell} &= 16 \boxed{\bullet}^{\ell} \\
 \boxed{\bullet} &= 4 \boxed{\bullet} \\
 \boxed{\bullet} &= \boxed{\bullet} + m \left(\frac{\ell}{\sqrt{2}} \right)^2
 \end{aligned}$$

The first line comes from the scaling argument, the second comes from the fact that moments of inertia simply add, and the third comes from the parallel-axis theorem. Equating the right-hand sides of the first two, and then using the third to eliminate $\boxed{\bullet}$ gives

$$\boxed{\bullet}^{\ell} = \frac{1}{6} m \ell^2$$

This agrees with the result of Example 12 in Section 8.3.1, with $a = b = \ell$.

- (b) This is again a two-dimensional object, so the I for a triangle of side 2ℓ is 16 times the I for a triangle of side ℓ , assuming that the axes pass through any two corresponding points. With pictures, we have:

$$\begin{aligned}
 \triangle_{2\ell} &= 16 \triangle_{\ell} \\
 \triangle_{2\ell} &= \triangle_{\ell} + 3(\bullet \triangle) \\
 \bullet \triangle &= \triangle_{\ell} + m\left(\frac{\ell}{\sqrt{3}}\right)^2
 \end{aligned}$$

The first line comes from the scaling argument, the second comes from the fact that moments of inertia simply add, and the third comes from the parallel-axis theorem. Equating the right-hand sides of the first two,

to eliminate $\bullet \triangle$ gives

$$\triangle_{\ell} = \frac{1}{12} m \ell^2$$

This agrees with the result of Example 11 in Section 8.3.1, with $N = 3$. The “radius” R used in that example equals $\ell/\sqrt{3}$ in the present notation.

8.8. Slick calculations of I for fractal objects

- (a) The scaling argument here is a little trickier than the one in Section 8.3.2. Our object is self-similar to an object 3 times as big, so let’s increase the length by a factor of 3 and see what happens to I . In the integral $\int x^2 dm$, the x ’s pick up a factor of 3, so this gives a factor of 9. But what happens to the dm ? Well, tripling the size of our object increases its mass by a factor of 2, because the new object is made up of two of the smaller ones, plus some empty space in the middle. So the dm picks up a factor of 2. Therefore, the I for an object of length 3ℓ is 18 times the I for an object of length ℓ , assuming that the axes pass through any two corresponding points. With pictures, we have (the following symbols denote our fractal object):

$$\begin{aligned}
 \text{---} \bullet \text{---} &= 18 \text{---} \bullet \text{---} \\
 \text{---} \bullet \text{---} &= 2 \left(\overset{\ell/2}{\bullet \text{---}} \text{---} \right) \\
 \bullet \text{---} &= \text{---} \bullet + m \ell^2
 \end{aligned}$$

The first line comes from the scaling argument, the second comes from the fact that moments of inertia simply add, and the third comes from the parallel-axis theorem. Equating the right-hand sides of the first two, and then using the

third to eliminate $\bullet - -$ gives

$$\frac{I}{\bullet - -} = \frac{1}{8} ml^2$$

This is larger than the I for a uniform stick, namely $m\ell^2/12$, because the mass here is generally farther away from the center.

REMARK: When we increase the length of our object by a factor of 3, the factor of 2 in the dm is larger than the factor of 1 relevant to a zero-dimensional object, but smaller than the factor of 3 relevant to a one-dimensional object. So in some sense our object has a dimension between 0 and 1. It is reasonable to define the fractal dimension, d , of an object as the number for which r^d is the increase in “volume” when the dimensions are increased by a factor of r . In this problem, we have $3^d = 2$, so $d = \log_3 2 \approx 0.63$. ♣

- (b) Again, the mass scales in a strange way. Let’s increase the dimensions of our object by a factor of 2 and see what happens to I . In the integral $\int x^2 dm$, the x ’s pick up a factor of 2, so this gives a factor of 4. But what happens to the dm ? Doubling the size of our object increases its mass by a factor of 3, because the new object is made up of three of the smaller ones, plus an empty triangle in the middle. So the dm picks up a factor of 3. Therefore, the I for an object of side 2ℓ is 12 times the I for an object of side ℓ , assuming that the axes pass through any two corresponding points. With pictures, we have:

$$\begin{aligned} \frac{2I}{\bullet - -} &= 12 \frac{I}{\bullet - -} \\ \frac{2I}{\bullet - -} &= 3 \left(\bullet - - \right) \\ \bullet - - &= \frac{I}{\bullet - -} + m \left(\frac{\ell}{\sqrt{3}} \right)^2 \end{aligned}$$

The first line comes from the scaling argument, the second comes from the fact that moments of inertia simply add, and the third comes from the parallel-axis theorem. Equating the right-hand sides of the first two, and then using the third

to eliminate $\bullet - -$ gives

$$\frac{I}{\bullet - -} = \frac{1}{9} m\ell^2$$

This is larger than the I for the uniform triangle in Problem 8.7, namely $m\ell^2/12$, because the mass here is generally farther away from the center. Increasing the size of our object by a factor of 2 increases the “volume” by a factor of 3, so the fractal dimension is given by $2^d = 3 \implies d = \log_2 3 \approx 1.58$.

- (c) Again, the mass scales in a strange way. Let’s increase the dimensions of our object by a factor of 3 and see what happens to I . In the integral $\int x^2 dm$, the x ’s pick up a factor of 3, so this gives a factor of 9. But what happens to the dm ?

Tripling the size of our object increases its mass by a factor of 8, because the new object is made up of eight of the smaller ones, plus an empty square in the middle. So the dm picks up a factor of 8. Therefore, the I for an object of side 3ℓ is 72 times the I for an object of side ℓ , assuming that the axes pass through any two corresponding points. With pictures, we have:

$$\begin{aligned}
 3I_{\bullet} &= 72 I_{\bullet} \\
 I_{\bullet} &= 4 \left(\bullet \square \right) + 4 \left(\bullet \square \right) \\
 \bullet \square &= \bullet + m\ell^2 \\
 \bullet \square &= \bullet + m(\sqrt{2}\ell)^2
 \end{aligned}$$

The first line comes from the scaling argument, the second comes from the fact that moments of inertia simply add, and the third and fourth come from the parallel-axis theorem. Equating the right-hand sides of the first two, and then using the third and fourth to eliminate $\bullet \square$ and $\bullet \square$ gives

$$I_{\bullet} = \frac{3}{16} m\ell^2$$

This is larger than the I for the uniform square in Problem 8.7, namely $m\ell^2/6$, because the mass here is generally farther away from the center. Increasing the size of our object by a factor of 3 increases the “volume” by a factor of 8, so the fractal dimension is given by $3^d = 8 \implies d = \log_3 8 \approx 1.89$.

8.9. Zero torque from internal forces

Let $\mathbf{F}_{ij}^{\text{int}}$ be the force that the i th particle feels due to the j th particle (see Fig. 8.71). Then

$$\mathbf{F}_i^{\text{int}} = \sum_j \mathbf{F}_{ij}^{\text{int}}. \quad (8.85)$$

The total internal torque on all the particles, relative to the chosen origin, is therefore

$$\boldsymbol{\tau}^{\text{int}} \equiv \sum_i \mathbf{r}_i \times \mathbf{F}_i^{\text{int}} = \sum_i \sum_j \mathbf{r}_i \times \mathbf{F}_{ij}^{\text{int}}. \quad (8.86)$$

But if we interchange the indices (which were labeled arbitrarily), we have

$$\boldsymbol{\tau}^{\text{int}} = \sum_j \sum_i \mathbf{r}_j \times \mathbf{F}_{ji}^{\text{int}} = - \sum_j \sum_i \mathbf{r}_j \times \mathbf{F}_{ij}^{\text{int}}, \quad (8.87)$$

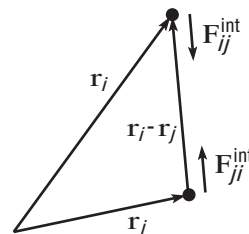


Fig. 8.71

where we have used Newton's third law, $\mathbf{F}_{ij}^{\text{int}} = -\mathbf{F}_{ji}^{\text{int}}$. Adding the two previous equations gives

$$2\boldsymbol{\tau}^{\text{int}} = \sum_i \sum_j (\mathbf{r}_i - \mathbf{r}_j) \times \mathbf{F}_{ij}^{\text{int}}. \quad (8.88)$$

But $\mathbf{F}_{ij}^{\text{int}}$ is parallel to $(\mathbf{r}_i - \mathbf{r}_j)$, by assumption. Therefore, each cross product in the sum equals zero.

The above sums might make this solution look a bit involved. But the idea is simply that the torques cancel in pairs. This is clear from Fig. 8.71, because the two forces shown are equal and opposite, and they have the same lever arm relative to the origin.

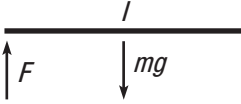


Fig. 8.72

8.10. Removing a support

- (a) **FIRST SOLUTION** Let the desired force from the left support be F , and let the downward acceleration of the stick's CM be a . Then the $F = ma$ and $\tau = I\alpha$ (relative to the fixed support; see Fig. 8.72) equations, along with the circular-motion relation between a and α , are, respectively,

$$mg - F = ma, \quad mg \frac{\ell}{2} = \left(\frac{m\ell^2}{3} \right) \alpha, \quad a = \frac{\ell}{2} \alpha. \quad (8.89)$$

The second equation gives $\alpha = 3g/2\ell$. The third equation then gives $a = 3g/4$. And the first equation then gives $F = mg/4$. Note that the right end of the stick accelerates at $2a = 3g/2$, which is larger than g .

SECOND SOLUTION Looking at torques around the CM, and also torques around the fixed support, we have, respectively,

$$F \frac{\ell}{2} = \left(\frac{m\ell^2}{12} \right) \alpha, \quad \text{and} \quad mg \frac{\ell}{2} = \left(\frac{m\ell^2}{3} \right) \alpha. \quad (8.90)$$

Dividing the first of these equations by the second gives $F = mg/4$.

- (b) **FIRST SOLUTION** As in the first solution above, we have (using the parallel-axis theorem; see Fig. 8.73)

$$mg - F = ma, \quad mgd = (\beta mr^2 + md^2)\alpha, \quad a = \alpha d. \quad (8.91)$$

Solving for F gives $F = mg/(1 + d^2/\beta r^2)$. For $d = r$ and $\beta = 1/3$, we obtain the answer in part (a).

SECOND SOLUTION As in the second solution above, looking at torques around the CM, and also torques around the fixed support, we have, respectively,

$$Fd = (\beta mr^2)\alpha, \quad \text{and} \quad mgd = (\beta mr^2 + md^2)\alpha. \quad (8.92)$$

Dividing the first of these equations by the second gives $F = mg/(1 + d^2/\beta r^2)$.

REMARKS: For the special case $d = r$, we have the following: If $\beta = 0$ (point mass in the middle) then $F = 0$; if $\beta = 1$ (dumbbell with masses at the ends) then $F = mg/2$; and if $\beta = \infty$ (masses at the ends of long massless extensions of the stick) then $F = mg$. These all make intuitive sense. In the limit $d = 0$, we have $F = mg$. And in the limit $d = \infty$ (using a massless extension), we have $F = 0$. Technically, we should be writing $d \ll \sqrt{\beta} r$ and $d \gg \sqrt{\beta} r$ here. ♣

8.11. Falling stick

- (a) Let's calculate τ and L relative to the pivot point. The torque is due to gravity, which effectively acts on the CM and has magnitude $mg b$. The moment of inertia of the stick around the horizontal axis through the pivot (and perpendicular to the massless stick) is simply mb^2 . So when the stick starts to fall, the $\tau = dL/dt$

equation is $mgb = (mb^2)\alpha$. Therefore, the initial acceleration of the CM, namely $b\alpha$, is

$$b\alpha = g, \quad (8.93)$$

which is independent of ℓ and b . This answer makes sense. The stick initially falls straight down, and the pivot provides no force because it doesn't know right away that the stick is moving.

- (b) The only change from part (a) is the moment of inertia of the stick around the horizontal axis through the pivot (and perpendicular to the massless stick). From the parallel-axis theorem, this moment is $mb^2 + m\ell^2/12$. So when the stick starts to fall, the $\tau = dL/dt$ equation is $mgb = (mb^2 + m\ell^2/12)\alpha$. Therefore, the initial acceleration of the CM is

$$b\alpha = \frac{g}{1 + (\ell^2/12b^2)}. \quad (8.94)$$

For $\ell \ll b$, this goes to g , as it should. And for $\ell \gg b$, it goes to zero, as it should. In this case, a tiny movement of the CM corresponds to a very large movement of the points far out along the stick. Therefore, by conservation of energy, the CM must be moving very slowly.

8.12. Pulling a cylinder

Consider the net force in the y direction (the transverse direction) on an infinitesimal arclength of the cylinder. Measure θ clockwise from the bottom, as shown in Fig. 8.74. The tension increases as θ increases (assuming $T_1 > T_2$), and the friction force F_f on the cylinder equals the difference dT in the tension from one end of the small arc to the other (using Newton's third law, and noting that there can be no net force on the string because it is massless). The friction force on the cylinder therefore produces a force component $dT \sin \theta$ in the y direction. From the example in Section 2.1, the normal force on the cylinder is $N = T d\theta$, which yields a force component $T d\theta \cos \theta$ in the y direction. The net y -force on the cylinder at the little arc is therefore

$$dF_y = dT \sin \theta + T d\theta \cos \theta = d(T \sin \theta). \quad (8.95)$$

The total F_y on the cylinder is thus

$$F_y = \int dF_y = \int d(T \sin \theta) = \Delta(T \sin \theta). \quad (8.96)$$

But θ runs from 0 to π , which means that $\sin \theta$ starts and ends at zero. Therefore, the total change in $T \sin \theta$ equals zero, and so $F_y = 0$, as desired. Note that nowhere in this solution did we assume anything about T . The string might be slipping, and the cylinder might be rough in some places and smooth in others, and it doesn't matter. The total F_y is still zero (as we know it must be, from the simpler reasoning that T_1 and T_2 pull only in the x direction). Physically, what happens is that N is larger in the top half of the semicircle than in the bottom half, and the resulting net downward force cancels the upward force from the friction.

The above $F_y = \Delta(T \sin \theta)$ result holds in general, and not just when the string wraps around a semicircle. Because of our convention for θ , the signs work out so that $\Delta(T \sin \theta)$ is simply the sum of the y components of the tensions T_1 and T_2 , which is the answer we expect.

8.13. Oscillating ball

Let the angle from the bottom of the cylinder to the ball be θ (see Fig. 8.75), and let F_f be the friction force. Then the tangential $F = ma$ equation is

$$F_f - mg \sin \theta = ma, \quad (8.97)$$

where we have chosen rightward to be the positive direction for a and F_f . Also, the $\tau = I\alpha$ equation (relative to the CM) is

$$-rF_f = \frac{2}{5}mr^2\alpha, \quad (8.98)$$

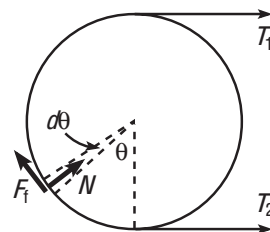


Fig. 8.74

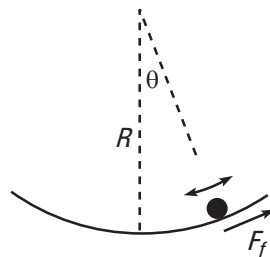


Fig. 8.75

where we have chosen clockwise to be the positive direction for α . Using the non-slipping condition $r\alpha = a$, the torque equation becomes $F_f = -(2/5)ma$. Plugging this into Eq. (8.97), and using $\sin \theta \approx \theta$, we obtain $mg\theta + (7/5)ma = 0$. Under the assumption $r \ll R$, the center of the ball moves along a circle with radius essentially equal to R , so we have $a \approx R\ddot{\theta}$. We therefore arrive at

$$\ddot{\theta} + \left(\frac{5g}{7R} \right) \theta = 0. \quad (8.99)$$

This is the equation for simple harmonic motion with frequency

$$\omega = \sqrt{\frac{5g}{7R}}. \quad (8.100)$$

In general, if the ball has a moment of inertia equal to βmr^2 , you can show that the frequency of small oscillations is $\sqrt{g/(1+\beta)R}$. Note that we needed to use two different expressions for a in this solution, namely $r\alpha$ and $R\ddot{\theta}$.

REMARKS: The answer in Eq. (8.100) is slightly smaller than the $\sqrt{g/R}$ answer for the case where the ball slides. In terms of forces, the reason for this is that the friction force causes there to be a smaller net tangential force. In terms of energy, the reason is that energy is “lost” in the rotational motion, so the ball ends up moving slower.

If we omit the $r \ll R$ assumption, then the $r\alpha = a$ relation still holds, because we can consider the ball to be instantaneously rotating around the contact point. But the $a = R\ddot{\theta}$ relation is replaced by $a = (R - r)\ddot{\theta}$, because the center of the ball moves along a circle of radius $R - r$. Therefore, the exact result for the frequency is $\omega = \sqrt{5g/7(R - r)}$. This goes to infinity as $r \rightarrow R$. ♣

8.14. Oscillating cylinders

The moments of inertia of the cylinders are $I_1 = M_1 R_1^2$ and $I_2 = M_2 R_2^2$. Let F be the force between the two cylinders, defined with rightward on the small cylinder being positive. Let θ_1 and θ_2 be the angles of rotation of the cylinders, with counterclockwise positive, relative to the position where the small cylinder is at the bottom of the big cylinder. Then the torque equations are

$$FR_1 = M_1 R_1^2 \ddot{\theta}_1, \quad \text{and} \quad FR_2 = -M_2 R_2^2 \ddot{\theta}_2. \quad (8.101)$$

We are not so much concerned with θ_1 and θ_2 as we are with the angular position that M_1 makes with the vertical. Let this angle be θ (see Fig. 8.76). In the approximation $R_1 \ll R_2$, the nonslipping condition says that $R_2\theta \approx R_2\theta_2 - R_1\theta_1$, since both sides of this equation are expressions for the arclength away from the bottom of the big cylinder. Adding Eqs. (8.101) after dividing through by the masses then gives

$$F \left(\frac{1}{M_1} + \frac{1}{M_2} \right) = -R_2 \ddot{\theta}. \quad (8.102)$$

The tangential force equation on M_1 is

$$F - M_1 g \sin \theta = M_1 (R_2 \ddot{\theta}). \quad (8.103)$$

Substituting the F from (8.102) into this equation gives, with $\sin \theta \approx \theta$,

$$\left(M_1 + \frac{1}{\frac{1}{M_1} + \frac{1}{M_2}} \right) \ddot{\theta} + \left(\frac{M_1 g}{R_2} \right) \theta = 0. \quad (8.104)$$

After simplifying, the frequency of small oscillations is

$$\omega = \sqrt{\frac{g}{R_2}} \sqrt{\frac{M_1 + M_2}{M_1 + 2M_2}}. \quad (8.105)$$

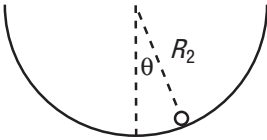


Fig. 8.76

REMARKS: In the limit $M_2 \ll M_1$, we obtain $\omega \approx \sqrt{g/R_2}$. In this case, there is essentially no friction force between the cylinders, because otherwise the “massless” M_2 would have infinite angular acceleration. So there is only a normal force, and the small cylinder essentially acts like a pendulum of length R_2 . In the limit $M_1 \ll M_2$, we obtain $\omega \approx \sqrt{g/2R_2}$. In this case, the big cylinder is essentially fixed, so we simply have the setup mentioned in the solution to Problem 8.13, with $\beta = 1$. ♣

8.15. Lengthening the string

Consider the angular momentum relative to the support point P . The forces on the mass are the tension in the string and gravity. The former provides no torque around P , and the latter provides no torque in the z direction. Therefore, L_z is constant. The motion is always approximately circular because the length of the string changes very slowly, so if we let ω_ℓ be the frequency of the circular motion when the string has length ℓ , then we can say that

$$L_z = mr^2 \omega_\ell \quad (8.106)$$

is constant. The frequency ω_ℓ can be obtained by using $F = ma$ for the circular motion. The tension in the string is essentially $mg/\cos\theta$ (to make the forces in the y direction cancel), so the horizontal radial force is $mg \tan\theta$. Therefore,

$$mg \tan\theta = mr\omega_\ell^2 = m(\ell \sin\theta)\omega_\ell^2 \implies \omega_\ell = \sqrt{\frac{g}{\ell \cos\theta}} = \sqrt{\frac{g}{h}}. \quad (8.107)$$

Plugging this into Eq. (8.106), we see that the constant value of L_z is

$$L_z = mr^2 \sqrt{\frac{g}{h}}. \quad (8.108)$$

This holds at all times, so the quantity $r^2/\sqrt{h}$ is constant. Let's look at the two cases.

- (a) For $\theta \approx 0$, we have $h \approx \ell$, so Eq. (8.108) says that $r^2/\sqrt{\ell}$ is constant. Therefore,

$$r \propto \ell^{1/4}, \quad (8.109)$$

which means that r grows very slowly as you let the string out when $\theta \approx 0$.

- (b) For $\theta \approx \pi/2$, we have $r \approx \ell$, so Eq. (8.108) says that $\ell^2/\sqrt{h}$ is constant. Therefore,

$$h \propto \ell^4, \quad (8.110)$$

which means that h grows very quickly as you let the string out when $\theta \approx \pi/2$.

Note that Eq. (8.108) says that $h \propto r^4$ for any value of θ . So no matter what θ is, if you slowly lengthen the string so that r doubles, then h increases by a factor of 16. Equivalently, if you draw the string in, the envelope of the motion of the mass is a surface of revolution generated by a curve of the form $y \propto -x^4$.

8.16. A triangle of cylinders

- (a) Let N be the normal force between the cylinders, and let F be the friction force from the ground (see Fig. 8.77). Let a_x be the initial horizontal acceleration of the right bottom cylinder (so $\alpha = a_x/R$ is its angular acceleration), and let a_y be the initial vertical acceleration of the top cylinder, with downward taken to be positive.

If we consider the torque around the center of one of the bottom cylinders, then the only relevant force is F , because N , gravity, and the normal force from the ground all point through the center. The equations for $F_x = ma_x$ on the bottom right cylinder, $F_y = ma_y$ on the top cylinder, and $\tau = I\alpha$ on the bottom right cylinder are, respectively,

$$\begin{aligned} N \cos 60^\circ - F &= ma_x, \\ mg - 2N \sin 60^\circ &= ma_y, \\ FR &= (\beta m R^2)(a_x/R). \end{aligned} \quad (8.111)$$

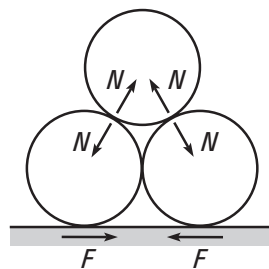


Fig. 8.77

We have four unknowns, N , F , a_x , and a_y , so we need one more equation. Fortunately, a_x and a_y are related. The contact surface between the top and bottom cylinders lies (initially) at an angle of 30° with the horizontal. Therefore, if the bottom cylinders move a distance d to the side, then the top cylinder moves a distance $d \tan 30^\circ$ downward. Hence,

$$a_x = \sqrt{3}a_y. \quad (8.112)$$

We now have four equations and four unknowns. Solving for a_y by your method of choice gives

$$a_y = \frac{g}{7 + 6\beta}. \quad (8.113)$$

- (b) Let N be the normal force between the cylinders, and let F be the friction force between the cylinders, with positive directions shown in Fig. 8.78. Let a_x be the initial horizontal acceleration of the right bottom cylinder, and let a_y be the initial vertical acceleration of the top cylinder, with downward taken to be positive. Let α be the angular acceleration of the right bottom cylinder, with counterclockwise taken to be positive. Note that α is *not* equal to a_x/R , because the bottom cylinders slip on the ground.

If we consider the torque around the center of one of the bottom cylinders, then the only relevant force is F . And from the same reasoning as in part (a), we have $a_x = \sqrt{3}a_y$. Therefore, the four equations analogous to Eqs. (8.111) and (8.112) are

$$\begin{aligned} N \cos 60^\circ - F \sin 60^\circ &= ma_x, \\ mg - 2N \sin 60^\circ - 2F \cos 60^\circ &= ma_y, \\ FR &= (\beta m R^2)\alpha, \\ a_x &= \sqrt{3}a_y. \end{aligned} \quad (8.114)$$

We have five unknowns, N , F , a_x , a_y , and α , so we need one more equation. The tricky part is relating α to a_x . One way to do this is to ignore the y motion of the top cylinder and imagine the bottom right cylinder to be rotating up and around the top cylinder, which is held fixed. In this rotational motion, the center of the bottom cylinder moves at an angle of 30° with respect to the horizontal (at the start). So if it moves an infinitesimal distance d to the right, then its center moves a distance $d / \cos 30^\circ$ up and to the right. So the angle through which the bottom cylinder rotates is $\theta = (d / \cos 30^\circ) / R = (2/\sqrt{3})(d/R)$. Bringing back in the vertical motion of the top cylinder doesn't change this result. Therefore, taking two derivatives of this relation gives

$$\alpha = \frac{2}{\sqrt{3}} \frac{a_x}{R}. \quad (8.115)$$

We now have five equations and five unknowns. Solving for a_y by your method of choice gives

$$a_y = \frac{g}{7 + 8\beta}. \quad (8.116)$$

REMARKS: If $\beta = 0$, that is, if all the mass is at the center of the cylinders, then the results in both parts (a) and (b) reduce to $g/7$. This $\beta = 0$ case is equivalent to the case of frictionless cylinders (of any mass distribution), because then nothing rotates. If $\beta \neq 0$, then the result in part (b) is smaller than that in part (a). This isn't so obvious, but the basic reason is that the bottom cylinders in part (b) take up more energy because they have to rotate slightly faster, because $\alpha = (2/\sqrt{3})(a_x/R)$ instead of $\alpha = a_x/R$. ♣

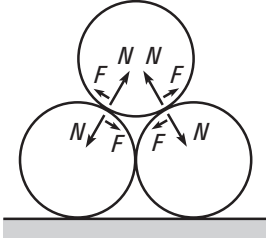


Fig. 8.78

8.17. Falling chimney

Let θ be the angle through which the chimney has fallen. Before we start dealing with the forces in the rods, let's first determine $\ddot{\theta}$ as a function of θ . Let ℓ be the height of the chimney. Then the moment of inertia around the pivot point on the ground is $m\ell^2/3$ (if we ignore the width). And the torque (around the pivot point) due to gravity is $\tau = mg(\ell/2) \sin \theta$. Therefore, $\tau = dL/dt$ gives $mg(\ell/2) \sin \theta = (1/3)m\ell^2\ddot{\theta}$, and so

$$\ddot{\theta} = \frac{3g \sin \theta}{2\ell}. \quad (8.117)$$

Let's now determine the forces in the rods. Our strategy will be to imagine that the chimney consists of a chimney of height h , with another chimney of height $\ell - h$ placed on top of it. We'll find the forces in the rods connecting these two "sub-chimneys," and then we'll maximize one of these forces (T_2 , defined below) as a function of h .

The forces on the top piece are gravity and also the forces from the two rods at each end of the bottom board. Let's break these latter forces up into transverse and longitudinal forces along the chimney. Let T_1 and T_2 be the two longitudinal components, and let F be the sum of the transverse components, as shown in Fig. 8.79. We have picked the positive directions for T_1 and T_2 so that positive T_1 corresponds to compression in the left rod, and positive T_2 corresponds to tension in the right rod (which is what the forces will turn out to be, as we'll see). It turns out that if the width (which we'll call $2r$) is much less than the height, then $T_2 \gg F$ (as we'll see below), so the tension in the right rod is essentially equal to T_2 . We will therefore be concerned with maximizing T_2 .

In writing down the force and torque equations for the top piece, we have three equations (the radial and tangential $F = ma$ equations, and $\tau = dL/dt$ around the CM), and three unknowns (F , T_1 , and T_2). If we define the fraction $f \equiv h/\ell$, then the top piece has length $(1-f)\ell$ and mass $(1-f)m$, and its CM travels in a circle of radius $(1+f)\ell/2$. Therefore, our three force and torque equations are, respectively,

$$\begin{aligned} T_2 - T_1 + (1-f)mg \cos \theta &= (1-f)m \left(\frac{(1+f)\ell}{2} \right) \dot{\theta}^2, \\ F + (1-f)mg \sin \theta &= (1-f)m \left(\frac{(1+f)\ell}{2} \right) \ddot{\theta}, \\ (T_1 + T_2)r - F \frac{(1-f)\ell}{2} &= (1-f)m \left(\frac{(1-f)^2\ell^2}{12} \right) \ddot{\theta}. \end{aligned} \quad (8.118)$$

At this point, we could plow forward and solve this system of three equations in three unknowns. But things simplify greatly in the limit $r \ll \ell$. The third equation says that $T_1 + T_2$ is of order $1/r$, and the first equation says that $T_2 - T_1$ is of order 1. These imply that $T_1 \approx T_2$, to leading order in $1/r$. Therefore, we can set $T_1 + T_2 \approx 2T_2$ in the third equation. Using this approximation, along with the value of $\ddot{\theta}$ from Eq. (8.117), the second and third equations become

$$\begin{aligned} F + (1-f)mg \sin \theta &= \frac{3}{4}(1-f^2)mg \sin \theta, \\ 2rT_2 - F \frac{(1-f)\ell}{2} &= \frac{1}{8}(1-f)^3mg \ell \sin \theta. \end{aligned} \quad (8.119)$$

This first of these equations gives

$$F = \frac{mg \sin \theta}{4}(-1 + 4f - 3f^2), \quad (8.120)$$

and then the second gives

$$T_2 \approx \frac{mg \ell \sin \theta}{8r} f(1-f)^2. \quad (8.121)$$

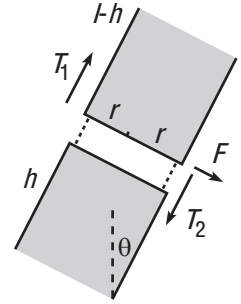


Fig. 8.79

As stated above, this is much greater than F (because $\ell/r \gg 1$), so the tension in the right rod is essentially equal to T_2 . Taking the derivative of T_2 with respect to f , we see that it is maximum at

$$f \equiv \frac{h}{\ell} = \frac{1}{3}. \quad (8.122)$$

Therefore, the chimney is most likely to break at a point one-third of the way up (assuming that the width is much less than the height). Interestingly, $f = 1/3$ makes the force F in Eq. (8.120) exactly equal to zero. For more on the falling chimney, see Madsen (1977) and Varieschi and Kamiya (2003).

8.18. Ball hitting stick

Let V , v , and ω be the speed of the ball, the speed of the stick's CM, and the angular speed of the stick, respectively, after the collision. Then conservation of momentum, angular momentum (around the fixed point that coincides with the initial center of the stick), and energy give

$$\begin{aligned} MV_0 &= MV + mv, \\ MV_0d &= MVd + \beta m\ell^2\omega, \\ MV_0^2 &= MV^2 + mv^2 + \beta m\ell^2\omega^2. \end{aligned} \quad (8.123)$$

We must solve these three equations for V , v , and ω . The first two equations quickly give $vd = \beta\ell^2\omega$. Solving for V in the first equation and plugging the result into the third, and then eliminating ω through $vd = \beta\ell^2\omega$ gives

$$v = \frac{2V_0}{1 + \frac{m}{M} + \frac{d^2}{\beta\ell^2}} \implies \omega = V_0 \frac{2\frac{d}{\beta\ell^2}V_0}{1 + \frac{m}{M} + \frac{d^2}{\beta\ell^2}}. \quad (8.124)$$

Having found v , the first equation above gives V as

$$V = V_0 \frac{1 - \frac{m}{M} + \frac{d^2}{\beta\ell^2}}{1 + \frac{m}{M} + \frac{d^2}{\beta\ell^2}}. \quad (8.125)$$

You are encouraged to check various limits of these answers. Another solution to Eq. (8.123) is of course $V = V_0$, $v = 0$, and $\omega = 0$. The initial conditions certainly satisfy conservation of p , L , and E with the initial conditions (a fine tautology, indeed). Nowhere in Eq. (8.123) does it say that the ball actually hits the stick.

8.19. A ball and stick theorem

As in the solution to Problem 8.18, we have

$$\begin{aligned} MV_0 &= MV + mv, \\ MV_0d &= MVd + I\omega, \\ MV_0^2 &= MV^2 + mv^2 + I\omega^2. \end{aligned} \quad (8.126)$$

The speed of the contact point on the stick right after the collision equals the speed of the CM plus the rotational speed relative to the CM. In other words, it equals $v + \omega d$. The desired relative speed is therefore $(v + \omega d) - V$. We can determine the value of this relative speed by solving the above three equations for V , v , and ω . Equivalently, we can just use the results of Problem 8.18. There is, however, a much more appealing method, which is the following.

The first two equations quickly give $mvd = I\omega$. The last equation may then be written as, using $I\omega^2 = (I\omega)\omega = (mvd)\omega$,

$$M(V_0 - V)(V_0 + V) = mv(v + \omega d). \quad (8.127)$$

If we now write the first equation as

$$M(V_0 - V) = mv, \quad (8.128)$$

we can divide Eq. (8.127) by Eq. (8.128) to obtain $V_0 + V = v + \omega d$, or

$$V_0 = (v + \omega d) - V, \quad (8.129)$$

as we wanted to show. In terms of velocities, the correct statement is that the final relative velocity is the negative of the initial relative velocity. In other words, $V_0 - 0 = -(V - (v + \omega d))$.

8.20. The superball

Since we are told that $|v_y|$ is unchanged by the bounce, we can ignore it when applying conservation of energy. And since the vertical impulse from the floor provides no torque around the ball's CM, we can completely ignore the y motion in this problem.

The horizontal impulse from the floor is responsible for changing both v_x and ω . With positive directions defined as in the statement of the problem, Eq. (8.61) gives

$$\begin{aligned} \Delta L &= R\Delta p \\ \implies I(\omega' - \omega) &= Rm(v'_x - v_x). \end{aligned} \quad (8.130)$$

And conservation of energy gives

$$\begin{aligned} \frac{1}{2}mv_x'^2 + \frac{1}{2}I\omega'^2 &= \frac{1}{2}mv_x^2 + \frac{1}{2}I\omega^2 \\ \implies I(\omega'^2 - \omega^2) &= m(v_x^2 - v_x'^2). \end{aligned} \quad (8.131)$$

Dividing this equation by Eq. (8.130) gives¹⁵

$$R(\omega' + \omega) = -(v'_x + v_x). \quad (8.132)$$

We can now combine this equation with Eq. (8.130) which can be rewritten as, using $I = (2/5)mR^2$,

$$\frac{2}{5}R(\omega' - \omega) = v'_x - v_x. \quad (8.133)$$

Given v_x and ω , the previous two equations are two linear equations in the two unknowns, v'_x and ω' . Solving for v'_x and ω' , and writing the result in matrix notation, gives

$$\begin{pmatrix} v'_x \\ R\omega' \end{pmatrix} = \frac{1}{7} \begin{pmatrix} 3 & -4 \\ -10 & -3 \end{pmatrix} \begin{pmatrix} v_x \\ R\omega \end{pmatrix}, \quad (8.134)$$

as desired. Note that Eq. (8.132), when written in the form of $v_x + R\omega = -(v'_x + R\omega')$, says that the relative velocity of the ball's contact point and the ground simply changes sign during the bounce.

REMARK: For a ball with a general moment of inertia $I = \beta mR^2$, you can use the above procedure to show that the matrix in Eq. (8.134) takes the general form,

$$\frac{1}{1 + \beta} \begin{pmatrix} 1 - \beta & -2\beta \\ -2 & -(1 - \beta) \end{pmatrix}. \quad (8.135)$$

¹⁵ We have divided out the trivial $\omega' = \omega$ and $v'_x = v_x$ solution, which corresponds to slipping motion on a frictionless plane. The nontrivial solution we will find shortly is the nonslipping one. Basically, to conserve energy, there must be no work done by friction. And since work is force times distance, this means that either (1) the plane is frictionless, so that the force is zero, or (2) there is no relative motion between the ball's contact point and the plane, so that the distance is zero. The latter case is the one we are concerned with here.

For $\beta = 1$ (a hoop), this becomes

$$\begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}, \quad (8.136)$$

which means that the bounce simply interchanges and negates the values of v_x and $R\omega$. In particular, if you throw a “super-hoop” sideways with no spin (that is, $R\omega = 0$), then it will bounce straight up in the air (that is, $v'_x = 0$) while spinning. ♣

8.21. Many bounces

Equation (8.62) gives the result after one bounce, so the result after two bounces is

$$\begin{aligned} \begin{pmatrix} v''_x \\ R\omega'' \end{pmatrix} &= \begin{pmatrix} 3/7 & -4/7 \\ -10/7 & -3/7 \end{pmatrix} \begin{pmatrix} v'_x \\ R\omega' \end{pmatrix} \\ &= \begin{pmatrix} 3/7 & -4/7 \\ -10/7 & -3/7 \end{pmatrix}^2 \begin{pmatrix} v_x \\ R\omega \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} v_x \\ R\omega \end{pmatrix} \\ &= \begin{pmatrix} v_x \\ R\omega \end{pmatrix}. \end{aligned} \quad (8.137)$$

The square of the matrix turns out to be the identity. Therefore, after two bounces, both v_x and ω return to their original values. The ball then repeats the motion of the previous two bounces (and so on, after every two bounces). The only difference between successive pairs of bounces is that the ball may shift horizontally. You are strongly encouraged to experimentally verify this interesting periodic behavior.

8.22. Rolling over a bump

We will use the fact that the angular momentum of the ball with respect to the corner of the step (call this point P) is unchanged by the collision. This is true because any forces exerted at point P provide zero torque around P . (The torque from gravity will be relevant during the subsequent rising-up motion. But during the instantaneous collision, L does not change.) This fact will allow us to find the energy of the ball right after the collision, which we will then require to be greater than mgh .

Breaking the initial L into the contribution relative to the CM, plus the contribution from the ball treated like a point mass located at the CM, we see that the initial angular momentum is $L = (2/5)mR^2\omega_0 + mV_0(R - h)$, where ω_0 is the initial angular speed. But the nonslipping condition tells us that $\omega_0 = V_0/R$, so we can write L as

$$L = \frac{2}{5}mRV_0 + mV_0(R - h) = mV_0\left(\frac{7R}{5} - h\right). \quad (8.138)$$

Let ω' be the angular speed of the ball around point P immediately after the collision. The parallel-axis theorem says that the moment of inertia around P is equal to $(2/5)mR^2 + mR^2 = (7/5)mR^2$. Conservation of L around P during the collision then gives

$$mV_0\left(\frac{7R}{5} - h\right) = \frac{7}{5}mR^2\omega' \implies \omega' = \frac{V_0}{R}\left(1 - \frac{5h}{7R}\right). \quad (8.139)$$

The energy of the ball right after the collision is therefore

$$E = \frac{1}{2}\left(\frac{7}{5}mR^2\right)\omega'^2 = \frac{7}{10}mV_0^2\left(1 - \frac{5h}{7R}\right)^2. \quad (8.140)$$

The ball will climb up over the step if $E \geq mgh$, which gives

$$V_0 \geq \sqrt{\frac{10gh}{7}}\left(1 - \frac{5h}{7R}\right)^{-1}. \quad (8.141)$$

REMARKS: It is possible for the ball to rise up over the step even if $h > R$, provided that the ball sticks to the corner, without slipping. (If $h > R$, the step would have to be “hollowed out” so that the ball doesn’t collide with the side of the step.) But note that $V_0 \rightarrow \infty$ as $h \rightarrow 7R/5$. For $h \geq 7R/5$, it is impossible for the ball to make it up over the step, no matter how large V_0 is. The ball will get pushed down into the ground, instead of rising up, if $h > 7R/5$.

For an object with a general moment of inertia $I = \beta mR^2$ (so $\beta = 2/5$ in our problem), you can show that the minimum initial speed is

$$V_0 \geq \sqrt{\frac{2gh}{1+\beta}} \left(1 - \frac{h}{(1+\beta)R}\right)^{-1}. \quad (8.142)$$

This decreases as β increases. It is smallest when the “ball” is a wheel with all the mass on its rim (so that $\beta = 1$), in which case it is possible for the wheel to climb up over the step even if h is close to $2R$. ♣

8.23. Falling toast

Let v_0 and v be the speeds of the CM right before and right after the collision (so we know that $v_0 = \sqrt{2gH}$). The I for a square is the same as the I for a stick, $(1/12)m\ell^2$, so $\Delta L = (\ell/2)\Delta p$ gives¹⁶

$$\left(\frac{m\ell^2}{12}\right)\omega = -\frac{\ell}{2}(mv - mv_0) \implies v_0 - v = \frac{\ell\omega}{6}. \quad (8.143)$$

Conservation of energy during the collision gives

$$\frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{m\ell^2}{12}\right)\omega^2 \implies (v_0 + v)(v_0 - v) = \frac{\ell^2\omega^2}{12}. \quad (8.144)$$

Dividing this by Eq. (8.143) gives $v_0 + v = \ell\omega/2$. We now have two linear equations for v and ω . Solving gives $v = v_0/2$ and $\omega = 3v_0/\ell$.

After the collision, the time to hit the ground is given by $vt + gt^2/2 = h$. Solving for t , setting $\omega t = \pi$ for half a rotation, and using the v and ω we just found, yields

$$\frac{3v_0}{\ell} \cdot \frac{1}{g} \left(-\frac{v_0}{2} + \sqrt{\frac{v_0^2}{4} + 2gh}\right) = \pi. \quad (8.145)$$

Factoring out a v_0^2 and then using $v_0^2 = 2gH$ gives

$$\frac{3H}{\ell} \left(\sqrt{1 + \frac{4h}{H}} - 1\right) = \pi. \quad (8.146)$$

Isolating the square root, and then squaring and solving for H gives

$$H = \frac{\pi^2 \ell^2}{6(6h - \pi\ell)}. \quad (8.147)$$

The special value of ℓ is $\ell = (6/\pi)h$ (which would be a huge piece of toast), in which case $H = \infty$. If ℓ is larger than $(6/\pi)h$, then there isn’t enough time to make half a rotation before hitting the floor. The intuitive reason for this is that since $\omega = 6v/\ell$ (from above), there is no way to increase ω without also increasing v . The toast has the best chance of making half a rotation if v is very large, because then gravity doesn’t have any time to increase v . In this limit, the toast makes half a rotation upon hitting the floor if $\pi/\omega = h/v$. Plugging in $\omega = 6v/\ell$ gives $6h = \pi\ell$, as desired.

¹⁶ The minus sign on the right-hand side comes from the fact that we’re defining the v ’s to be positive downward. Equivalently, the force from the counter increases the angular speed but decreases the linear speed.

For the reasonable values of $h = 1$ m and $\ell = 10$ cm, we obtain $H \approx 3$ mm, which is quite small. You should try this with a pencil to convince yourself that even this small distance can yield the desired (or rather, undesired) half rotation.

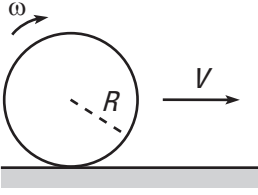


Fig. 8.80

8.24. Sliding to rolling

- (a) Define all linear quantities to be positive to the right, and all angular quantities to be positive clockwise, as shown in Fig. 8.80. Then, for example, the friction force F_f is negative. The friction force slows down the translational motion and speeds up the rotational motion, according to

$$F_f = ma, \quad \text{and} \quad -F_f R = I\alpha. \quad (8.148)$$

Eliminating F_f , and using $I = \beta m R^2$, gives $a = -\beta R \alpha$. Integrating this over time, up to the time when the ball stops slipping, gives

$$\Delta V = -\beta R \Delta \omega. \quad (8.149)$$

Note that we could have obtained this by simply using the impulse equation, Eq. (8.61). Using $\Delta V = V_f - V_0$, and $\Delta \omega = \omega_f - \omega_0 = \omega_f$, and also $\omega_f = V_f / R$ (the nonslipping condition), Eq. (8.149) gives

$$V_f = \frac{V_0}{1 + \beta}, \quad (8.150)$$

independent of the nature of F_f . F_f can depend on position, time, speed, or anything else. The relation $a = -\beta R \alpha$, and hence also Eq. (8.149), will still be true at all times.

REMARK: We can also calculate τ and L relative to a dot painted on the ground that is the contact point at a given instant. There is zero torque relative to this point. To find L , we must add the L of the CM and the L relative to the CM. Therefore, $\tau = dL/dt$ gives $0 = (d/dt)(mvR + \beta m R^2 \omega)$, and so $a = -\beta R \alpha$, as above. ♣

Using Eq. (8.150), and also the relation $\omega_f = V_f / R$, the loss in kinetic energy is

$$\begin{aligned} \Delta K &= \frac{1}{2} m V_0^2 - \left(\frac{1}{2} m V_f^2 + \frac{1}{2} I \omega_f^2 \right) \\ &= \frac{1}{2} m V_0^2 \left(1 - \frac{1}{(1 + \beta)^2} - \frac{\beta}{(1 + \beta)^2} \right) \\ &= \frac{1}{2} m V_0^2 \left(\frac{\beta}{1 + \beta} \right). \end{aligned} \quad (8.151)$$

For $\beta \rightarrow 0$, no energy is lost, which makes sense. And for $\beta \rightarrow \infty$ (a spool sliding on its axle), all the energy is lost, which also makes sense, because we essentially have a sliding block which can't rotate.

- (b) Let's first find t . The friction force is $F_f = -\mu mg$, so $F = ma$ gives $-\mu g = a$. Therefore, $\Delta V = at = -\mu gt$. But Eq. (8.150) says that $\Delta V \equiv V_f - V_0 = -V_0 \beta / (1 + \beta)$. Therefore,

$$t = \frac{\beta}{(1 + \beta)} \cdot \frac{V_0}{\mu g}. \quad (8.152)$$

For $\beta \rightarrow 0$, we have $t \rightarrow 0$, which makes sense. And for $\beta \rightarrow \infty$, we have $t \rightarrow V_0 / (\mu g)$, which equals the time a sliding block would take to stop.

Let's now find d . We have $d = V_0 t + (1/2)at^2$. Using $a = -\mu g$, and plugging in the t from Eq. (8.152), we obtain

$$d = \frac{\beta(2 + \beta)}{(1 + \beta)^2} \cdot \frac{V_0^2}{2\mu g}. \quad (8.153)$$

The two extreme cases for β check here.

To calculate the work done by friction, we might be tempted to write down the product $F_f d$, with $F_f = -\mu mg$ and d given in Eq. (8.153). But the result doesn't equal the loss in kinetic energy calculated in Eq. (8.151). What's wrong with this reasoning? The error is that the friction force does not act over a distance d . To find the distance over which F_f acts, we must find how far the surface of the ball moves relative to the ground. The speed of a dot on the ball that is instantaneously the contact point is $V_{\text{rel}}(t) = V(t) - R\omega(t) = (V_0 + at) - R\alpha t$. Using $\alpha = -a/\beta R$ and $a = -\mu g$, this becomes

$$V_{\text{rel}}(t) = V_0 - \frac{1 + \beta}{\beta} \mu g t. \quad (8.154)$$

Integrating this from $t = 0$ to the t given in Eq. (8.152) gives

$$d_{\text{rel}} = \int V_{\text{rel}}(t) dt = \frac{\beta}{1 + \beta} \cdot \frac{V_0^2}{2\mu g}. \quad (8.155)$$

The work done by friction is $F_f d_{\text{rel}} = -\mu mg d_{\text{rel}}$, which does indeed give the loss in kinetic energy given in Eq. (8.151).

8.25. Lots of sticks

Consider the collision between two sticks. Let V be the speed of the contact point on the heavy one. Since this stick is essentially infinitely heavy, we may consider it to be an infinitely heavy ball, moving at speed V . The rotational degree of freedom of the heavy stick is irrelevant, as far as the light stick is concerned. We can therefore invoke the result of Problem 8.19 to say that the relative speed of the contact points is the same before and after the collision. This implies that the contact point on the light stick picks up a speed of $2V$, because the heavy stick is essentially unaffected by the collision and keeps moving at speed V .

Let us now find the speed of the other end of the light stick. This stick receives an impulse from the heavy stick, so we can apply Eq. (8.61) to the light stick to obtain

$$\Delta L = r \Delta p \implies \beta m r^2 \omega = r(m v_{\text{CM}}) \implies r\omega = \frac{v_{\text{CM}}}{\beta}. \quad (8.156)$$

The speed of the struck (top) end is $v_{\text{top}} = r\omega + v_{\text{CM}}$, because the CM speed adds to the rotational speed. The speed of the other (bottom) end is $v_{\text{bot}} = r\omega - v_{\text{CM}}$, because the CM speed subtracts from the rotational speed.¹⁷ The ratio of these speeds is

$$\frac{v_{\text{bot}}}{v_{\text{top}}} = \frac{\frac{v_{\text{CM}}}{\beta} - v_{\text{CM}}}{\frac{v_{\text{CM}}}{\beta} + v_{\text{CM}}} = \frac{1 - \beta}{1 + \beta}. \quad (8.157)$$

This is a general result whenever you strike the end of a stick with any force. In the present problem, we have $v_{\text{top}} = 2V$. Therefore,

$$v_{\text{bot}} = V \left(\frac{2(1 - \beta)}{1 + \beta} \right). \quad (8.158)$$

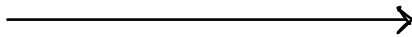
¹⁷ Since $\beta \leq 1$ for any real stick, we have $r\omega = v_{\text{CM}}/\beta \geq v_{\text{CM}}$. Therefore, $r\omega - v_{\text{CM}}$ is greater than or equal to zero.

The same analysis holds for all the other collisions. Therefore, the bottom ends of the sticks move with speeds that form a geometric progression with ratio $2(1 - \beta)/(1 + \beta)$. If this ratio is less than 1 (that is, if $\beta > 1/3$), then the speeds go to zero as $n \rightarrow \infty$. If it is greater than 1 (that is, if $\beta < 1/3$), then the speeds go to infinity as $n \rightarrow \infty$. If it equals 1 (that is, if $\beta = 1/3$), then the speeds remain equal to V and are thus independent of n , as we wanted to show. A uniform stick has $\beta = 1/3$ relative to its center (which is usually written in the form $I = m\ell^2/12$, where $\ell = 2r$).

Chapter 9

Angular momentum, Part II (General $\vec{L}$)

$\vec{L}$



chap 8 อนุกรมวิธาน chap 9 ทักษะ

In Chapter 8, we discussed situations where the direction of the vector $\vec{L}$ remained constant, and only its magnitude changed. In this chapter, we will look at more general situations where the direction of $\vec{L}$ is allowed to change. The vector nature of $\vec{L}$ will prove to be vital here, and we will arrive at all sorts of strange results for spinning tops and such things. This chapter is rather long, alas, but the general outline is that the first three sections cover general theory, then Section 9.4 introduces some actual physical setups, and then Section 9.6 begins the discussion of tops.



9.1 Preliminaries concerning rotations

9.1.1 The form of general motion

Before getting started, we should make sure we're all on the same page concerning a few important things about rotations. Because rotations generally involve three dimensions, they can often be hard to visualize. A rough drawing on a piece of paper might not do the trick. For this reason, this chapter is one of the more difficult ones in this book. But to ease into it, the next few pages consist of some definitions and helpful theorems. This first theorem describes the general form of any motion. You might consider it obvious, but it's a little tricky to prove.

Theorem 9.1 (Chasles' theorem) Consider a rigid body undergoing arbitrary motion. Pick any point P in the body. Then at any instant (see Fig. 9.1), the motion of the body can be written as the sum of the translational motion of P , plus a rotation around some axis (which may change with time) through P .¹

Proof: The motion of the body can be written as the sum of the translational motion of P , plus some other motion relative to P (this is true because relative coordinates are additive quantities). We must show that this latter motion is a rotation. This seems quite plausible, and it holds because the body is rigid; that is,

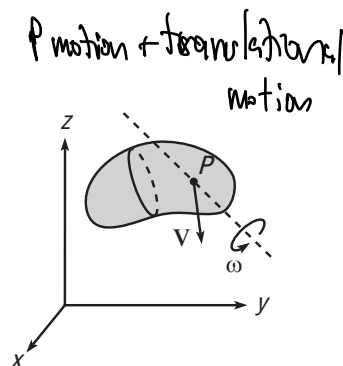


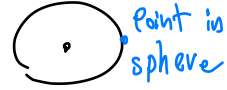
Fig. 9.1



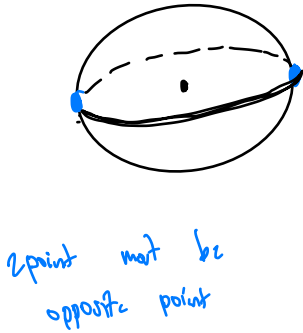
¹ In other words, a person at rest with respect to a frame whose origin is P , and whose axes are parallel to the fixed-frame axes, sees the body undergoing a rotation around some axis through P .

fixed frame axes





all points keep the same distances relative to each other. If the body weren't rigid, then this theorem wouldn't be true.



To be rigorous, consider a spherical shell fixed in the body, centered at P . The motion of the body is completely determined by the motion of the points on this sphere, so we need only examine what happens to the sphere. Because distances are preserved in the rigid body, the points on the sphere must always remain the same radial distance from P . And because we are looking at motion relative to P , we have therefore reduced the problem to the following: In what manner can a rigid sphere transform into itself? We claim that any such transformation has the property that there exist two points that end up where they started.² These two points must then be diametrically opposite points (assuming that the whole sphere doesn't end up back where it started, in which case every point ends up where it started), because distances are preserved; given one point that ends up where it started, the diametrically opposite point must also end up where it started, to maintain the distance of a diameter.

If this claim is true, then we are done, because for an infinitesimal transformation, a given point moves in only one direction, because there is no time to do any turning. So a point that ends up where it started must have remained fixed for the whole (infinitesimal) time. Therefore, all the points on the diameter joining the two fixed points must also have remained fixed the whole time, because distances are preserved. So we are left with a rotation around this axis.

This “two points ending up where they started” claim is quite believable, but nevertheless tricky to prove. Claims with these properties are always fun to think about, so I've left this one as a problem (Problem 9.2). Try to solve it on your own. ■

We will invoke this theorem repeatedly in this chapter (often without bothering to say so). Note that we are assuming that P is a point in the body, because we used the fact that P keeps the same distances from other points in the body.

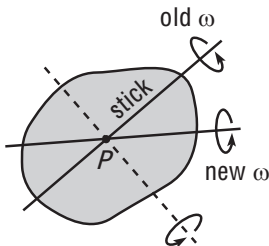


Fig. 9.2

REMARK: A situation where this theorem isn't so obvious is the following (this setup contains only rotation, with no translation of the point P). Consider an object rotating around a fixed axis, the stick shown in Fig. 9.2. But now imagine grabbing the stick and rotating it around some other axis (the dotted line shown). It isn't immediately obvious that the resulting motion is (instantaneously) a rotation around some new axis through the point P (which remains fixed). But indeed it is. We'll be quantitative about this in the “Rotating sphere” example later in this section. ♣

9.1.2 The angular velocity vector

It is extremely useful to introduce the angular velocity vector, $\boldsymbol{\omega}$, which is defined as the vector that points along the axis of rotation, and whose magnitude equals

² This claim is actually true for *any* transformation of a rigid sphere into itself, but for the present purposes we are concerned only with infinitesimal transformations, because we are looking only at what happens at a given instant in time.

the angular speed. The choice of the two possible directions along the axis is given by the right-hand rule: if you curl your right-hand fingers in the direction of the spin, then your thumb points in the direction of ω . For example, a spinning record has ω perpendicular to the record, through the center (as shown in Fig. 9.3),³ with its magnitude equal to the angular speed, ω . The points on the axis of rotation are the ones that (instantaneously) do not move. Of course, the direction of ω may change over time, so the points that were formerly on the axis may now be moving.

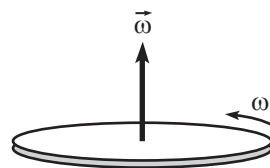


Fig. 9.3

REMARKS:

1. If you want, you can break the mold and use the left-hand rule to determine ω , as long as you use it consistently. The direction of ω will be the opposite, but that doesn't matter, because ω isn't really physical. Any physical result (for example, the velocity of a particle, given below in Theorem 9.2) will come out the same, independent of which hand you (consistently) use.

When studying vectors in school,
 You'll use your right hand as a tool.
 But look in a mirror,
 And then you'll see clearer,
 It's just like the left-handed rule.

2. The fact that we can specify a rotation by specifying a vector ω is a peculiarity to three dimensions. If we lived in one dimension, then there would be no such thing as a rotation. If we lived in two dimensions, then all rotations would take place in that plane, so we could label a rotation by simply giving its speed, ω . In three dimensions, rotations take place in $\binom{3}{2} = 3$ independent planes. And we choose to label these, for convenience, by the directions orthogonal to these planes, and by the angular speed in each plane. If we lived in four dimensions, then rotations could take place in $\binom{4}{2} = 6$ planes, so we would have to label a rotation by giving 6 planes and 6 angular speeds. Note that a vector, which has four components in four dimensions, would not do the trick. ♣

In addition to specifying the points that are instantaneously motionless, ω also easily produces the velocity of any point in the rotating object. Consider the situation where the axis of rotation passes through the origin, which we'll generally assume to be the case in this chapter, unless otherwise stated. Then we have the following theorem.

Theorem 9.2 *Given an object rotating with angular velocity ω , the velocity of a point at position $\mathbf{r}$ is given by*

$$\mathbf{v} = \omega \times \mathbf{r}. \quad (9.1)$$

Proof: Drop a perpendicular from the point in question (call it P) to the axis ω . Let Q be the foot of the perpendicular, and let $\mathbf{r}'$ be the vector from Q to P

³ It's actually meaningless to say that ω passes through the center of the record, because you can draw the vector anywhere, and it's still the same vector, as long as it has the correct magnitude and direction. Nevertheless, it's customary to draw ω along the axis of rotation and to say things like, "An object rotates around ω . . ."

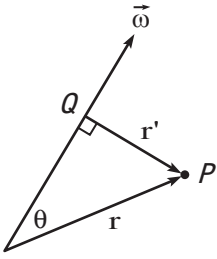


Fig. 9.4

(see Fig. 9.4). From the properties of the cross product (see Appendix B), $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ is orthogonal to $\boldsymbol{\omega}$, $\mathbf{r}$, and also $\mathbf{r}'$ because $\mathbf{r}'$ is a linear combination of $\boldsymbol{\omega}$ and $\mathbf{r}$. Therefore, the direction of $\mathbf{v}$ is correct; it is always orthogonal to $\boldsymbol{\omega}$ and $\mathbf{r}'$, so it describes circular motion around the axis $\boldsymbol{\omega}$. Also, by the right-hand rule in the cross product (or the left-hand rule, if you had chosen to be different and defined $\boldsymbol{\omega}$ that way), $\mathbf{v}$ has the proper orientation around $\boldsymbol{\omega}$, namely into the page at the instant shown. And since

$$|\mathbf{v}| = |\boldsymbol{\omega}| |\mathbf{r}| \sin \theta = \omega r', \quad (9.2)$$

we see that $\mathbf{v}$ has the correct magnitude, because $\omega r'$ is the speed of the circular motion around $\boldsymbol{\omega}$. So $\mathbf{v}$ is indeed the correct velocity vector. (If we have the special case where P lies along $\boldsymbol{\omega}$, then $\mathbf{r}$ is parallel to $\boldsymbol{\omega}$, so the cross product gives a zero result for $\mathbf{v}$, as it should.) ■

We'll make good use of Eq. (9.1) and apply it repeatedly throughout this chapter. Even if it's hard to visualize what's going on in a given rotation, all you have to do to find the speed of any point is calculate the cross product $\boldsymbol{\omega} \times \mathbf{r}$. Conversely, if the speed of every point in a body is given by $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, then the body must be undergoing a rotation with angular velocity $\boldsymbol{\omega}$, because all points on the axis $\boldsymbol{\omega}$ are motionless, and all other points move with the proper speed for this rotation.

A very nice thing about angular velocities is that they simply add. Stated more precisely:

Theorem 9.3 *Let coordinate systems S_1 , S_2 , and S_3 have a common origin. Let S_1 rotate with angular velocity $\boldsymbol{\omega}_{1,2}$ with respect to S_2 , and let S_2 rotate with angular velocity $\boldsymbol{\omega}_{2,3}$ with respect to S_3 . Then S_1 rotates (instantaneously) with angular velocity*

$$\boldsymbol{\omega}_{1,3} = \boldsymbol{\omega}_{1,2} + \boldsymbol{\omega}_{2,3} \quad (9.3)$$

with respect to S_3 .

Proof: If $\boldsymbol{\omega}_{1,2}$ and $\boldsymbol{\omega}_{2,3}$ point in the same direction, then the theorem is clear; the angular speeds just add. If, however, they don't point in the same direction, then things are a bit harder to visualize. But we can prove the theorem by making abundant use of the definition of $\boldsymbol{\omega}$.

Pick a point P_1 at rest in S_1 . Let $\mathbf{r}$ be the vector from the origin to P_1 . The velocity of P_1 (relative to a very close point P_2 at rest in S_2) due to the rotation of S_1 around $\boldsymbol{\omega}_{1,2}$ is $\mathbf{V}_{P_1 P_2} = \boldsymbol{\omega}_{1,2} \times \mathbf{r}$. The velocity of P_2 (relative to a very close point P_3 at rest in S_3) due to the rotation of S_2 around $\boldsymbol{\omega}_{2,3}$ is $\mathbf{V}_{P_2 P_3} = \boldsymbol{\omega}_{2,3} \times \mathbf{r}$, because P_2 is also located essentially at position $\mathbf{r}$. Therefore, the velocity of P_1 relative to P_3 is $\mathbf{V}_{P_1 P_2} + \mathbf{V}_{P_2 P_3} = (\boldsymbol{\omega}_{1,2} + \boldsymbol{\omega}_{2,3}) \times \mathbf{r}$. This holds for any point P_1 at rest in S_1 , so the frame S_1 rotates with angular velocity $(\boldsymbol{\omega}_{1,2} + \boldsymbol{\omega}_{2,3})$ with

respect to S_3 . We see that the proof basically comes down to the facts that (1) the linear velocities just add, as usual, and (2) the angular velocities differ from the linear velocities by a cross product with $\mathbf{r}$. ■

If $\boldsymbol{\omega}_{1,2}$ is constant in S_2 , then the vector $\boldsymbol{\omega}_{1,3} = \boldsymbol{\omega}_{1,2} + \boldsymbol{\omega}_{2,3}$ will change with respect to S_3 as time goes by, because $\boldsymbol{\omega}_{1,2}$, which is fixed in S_2 , is changing with respect to S_3 (assuming that $\boldsymbol{\omega}_{1,2}$ and $\boldsymbol{\omega}_{2,3}$ aren't parallel). But at any instant, $\boldsymbol{\omega}_{1,3}$ may be obtained by adding the present values of $\boldsymbol{\omega}_{1,2}$ and $\boldsymbol{\omega}_{2,3}$. Consider the following example.

Example (Rotating sphere): A sphere rotates with angular speed ω_3 around a stick that initially points in the $\hat{\mathbf{z}}$ direction. You grab the stick and rotate it around the $\hat{\mathbf{y}}$ axis with angular speed ω_2 . What is the angular velocity of the sphere, with respect to the lab frame, as time goes by?

Solution: In the language of Theorem 9.3, the sphere defines the S_1 frame; the stick and the $\hat{\mathbf{y}}$ axis define the S_2 frame; and the lab frame is the S_3 frame. The instant after you grab the stick, we are given that $\boldsymbol{\omega}_{1,2} = \omega_3 \hat{\mathbf{z}}$, and $\boldsymbol{\omega}_{2,3} = \omega_2 \hat{\mathbf{y}}$. Therefore, the angular velocity of the sphere with respect to the lab frame is $\boldsymbol{\omega}_{1,3} = \boldsymbol{\omega}_{1,2} + \boldsymbol{\omega}_{2,3} = \omega_3 \hat{\mathbf{z}} + \omega_2 \hat{\mathbf{y}}$, as shown in Fig. 9.5. Convince yourself that the combination of these two rotations yields zero motion for the points along the line of $\boldsymbol{\omega}_{1,3}$. As time goes by, the stick (and hence $\boldsymbol{\omega}_{1,2}$) rotates around the $\mathbf{y}$ axis, so $\boldsymbol{\omega}_{1,3} = \boldsymbol{\omega}_{1,2} + \boldsymbol{\omega}_{2,3}$ traces out a cone around the $\mathbf{y}$ axis, as shown.

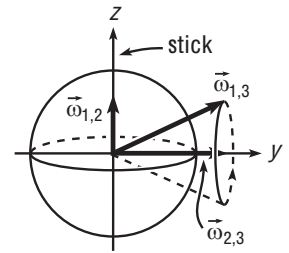


Fig. 9.5

REMARK: Note the different behavior of $\boldsymbol{\omega}_{1,3}$ for a slightly different statement of the problem: Let the sphere initially rotate with angular velocity $\omega_2 \hat{\mathbf{y}}$ around a stick, and then grab the stick and rotate it with angular velocity $\omega_3 \hat{\mathbf{z}}$. For this situation, $\boldsymbol{\omega}_{1,3}$ initially points in the same direction as in the original statement of the problem (it initially equals $\omega_2 \hat{\mathbf{y}} + \omega_3 \hat{\mathbf{z}}$). But as time goes by, it is now the horizontal component (defined by the stick) of $\boldsymbol{\omega}_{1,3}$ that changes, so $\boldsymbol{\omega}_{1,3} = \boldsymbol{\omega}_{1,2} + \boldsymbol{\omega}_{2,3}$ traces out a cone around the $\mathbf{z}$ axis, as shown in Fig. 9.6. ♣

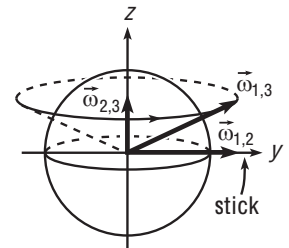


Fig. 9.6

An important point concerning rotations is that they are defined with respect to a *coordinate system*. It makes no sense to ask how fast an object is rotating with respect to a certain point, or even a certain axis. Consider, for example, an object rotating with angular velocity $\boldsymbol{\omega} = \omega_3 \hat{\mathbf{z}}$ with respect to the lab frame. Saying only, “The object has angular velocity $\boldsymbol{\omega} = \omega_3 \hat{\mathbf{z}}$,” is not sufficient, because someone standing in the frame of the object would measure $\boldsymbol{\omega} = 0$, and would therefore be very confused by your statement. Throughout this chapter, we’ll try to remember to state the coordinate system with respect to which $\boldsymbol{\omega}$ is measured. But if we forget, the default frame is the lab frame.

This section was definitely a bit abstract, so don’t worry too much about it at the moment. The best strategy is perhaps to read on, and then come back for a second pass after digesting a few more sections. At any rate, we’ll be

discussing many other aspects (probably more than you'd ever want to know) of $\boldsymbol{\omega}$ in Section 9.7.2, so you're assured of getting a lot more practice with it. For now, if you want to strain some brain cells thinking about $\boldsymbol{\omega}$ vectors, you are encouraged to solve Problem 9.3, and also to look at the three given solutions.

9.2 The inertia tensor

Given an object undergoing general motion, the *inertia tensor* is what relates the angular momentum, $\mathbf{L}$, to the angular velocity, $\boldsymbol{\omega}$. This tensor (which is just a fancy name for “matrix” in this context) depends on the geometry of the object, as we'll see. In finding the $\mathbf{L}$ due to general motion, we'll follow the strategy of Section 8.1. We'll first look at the special case of rotation around an axis through the origin, then we'll look at the most general possible motion.

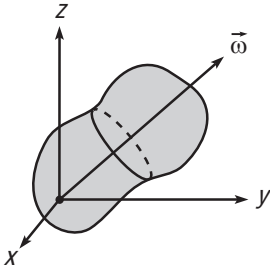


Fig. 9.7

9.2.1 Rotation around an axis through the origin

The three-dimensional object in Fig. 9.7 rotates with angular velocity $\boldsymbol{\omega}$. Consider a little piece of the body, with mass dm and position $\mathbf{r}$. The velocity of this piece is $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, so its angular momentum (relative to the origin) is $\mathbf{r} \times \mathbf{p} = (dm)\mathbf{r} \times \mathbf{v} = (dm)\mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r})$. The angular momentum of the entire body is therefore

$$\mathbf{L} = \int \mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r}) dm, \quad (9.4)$$

where the integration runs over the volume of the body. In the case where the rigid body is made up of a collection of point masses m_i , the angular momentum is

$$\mathbf{L} = \sum_i m_i \mathbf{r}_i \times (\boldsymbol{\omega} \times \mathbf{r}_i). \quad (9.5)$$

The double cross product in Eqs. (9.4) and (9.5) looks a bit intimidating, but it's actually not so bad. First, we have

$$\begin{aligned} \boldsymbol{\omega} \times \mathbf{r} &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ \omega_1 & \omega_2 & \omega_3 \\ x & y & z \end{vmatrix} \\ &= (\omega_2 z - \omega_3 y)\hat{\mathbf{x}} + (\omega_3 x - \omega_1 z)\hat{\mathbf{y}} + (\omega_1 y - \omega_2 x)\hat{\mathbf{z}}. \end{aligned} \quad (9.6)$$

We're using the notation ω_1 instead of ω_x , etc., because there are already enough x, y, z 's floating around here. The double cross product is then

$$\begin{aligned} \mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r}) &= \begin{vmatrix} \hat{\mathbf{x}} & \hat{\mathbf{y}} & \hat{\mathbf{z}} \\ x & y & z \\ (\omega_2 z - \omega_3 y) & (\omega_3 x - \omega_1 z) & (\omega_1 y - \omega_2 x) \end{vmatrix} \\ &= (\omega_1(y^2 + z^2) - \omega_2 xy - \omega_3 zx)\hat{\mathbf{x}} \end{aligned}$$

$$\begin{aligned}
& + \left(\omega_2(z^2 + x^2) - \omega_3 yz - \omega_1 xy \right) \hat{\mathbf{y}} \\
& + \left(\omega_3(x^2 + y^2) - \omega_1 zx - \omega_2 yz \right) \hat{\mathbf{z}}.
\end{aligned} \tag{9.7}$$

The angular momentum in Eq. (9.4) may therefore be written in the concise matrix form,

$$\begin{aligned}
\begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} &= \begin{pmatrix} \int (y^2 + z^2) & -\int xy & -\int zx \\ -\int xy & \int (z^2 + x^2) & -\int yz \\ -\int zx & -\int yz & \int (x^2 + y^2) \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} \\
&\equiv \begin{pmatrix} I_{xx} & I_{xy} & I_{xz} \\ I_{yx} & I_{yy} & I_{yz} \\ I_{zx} & I_{zy} & I_{zz} \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} \\
&\equiv \mathbf{I}\boldsymbol{\omega}.
\end{aligned} \tag{9.8}$$

For the sake of clarity, we have not bothered to write the dm part of each integral (and we'll continue to drop it for most of the remainder of this section). The matrix $\mathbf{I}$ is called the *inertia tensor*. If the word “tensor” scares you, just ignore it. $\mathbf{I}$ is simply a matrix. It acts on a vector (the angular velocity) and produces another vector (the angular momentum).

Example (Cube with origin at corner): Calculate the inertia tensor for a solid cube of mass M and side length L , with the coordinate axes parallel to the edges of the cube, and the origin at a corner (see Fig. 9.8).

Solution: Due to the symmetry of the cube, there are only two integrals we need to calculate in Eq. (9.8). The diagonal entries are all equal to $\int (y^2 + z^2) dm$, and the off-diagonal entries are all equal to $-\int xy dm$. With $dm = \rho dx dy dz$, and $\rho = M/L^3$, these two integrals are

$$\begin{aligned}
\int_0^L \int_0^L \int_0^L (y^2 + z^2) \rho dx dy dz &= \rho L^2 \int_0^L y^2 dy + \rho L^2 \int_0^L z^2 dz = \frac{2}{3} ML^2, \\
-\int_0^L \int_0^L \int_0^L xy \rho dx dy dz &= -\rho L \int_0^L x dx \int_0^L y dy = -\frac{ML^2}{4}.
\end{aligned} \tag{9.9}$$

Therefore,

$$\mathbf{I} = ML^2 \begin{pmatrix} 2/3 & -1/4 & -1/4 \\ -1/4 & 2/3 & -1/4 \\ -1/4 & -1/4 & 2/3 \end{pmatrix}. \tag{9.10}$$

Having found $\mathbf{I}$, we can calculate the angular momentum associated with any given angular velocity. If, for example, the cube is rotating around the z axis with angular speed ω , then we can apply the matrix $\mathbf{I}$ to the vector $(0, 0, \omega)$ to find that the angular momentum is $\mathbf{L} = ML^2 \omega (-1/4, -1/4, 2/3)$. Note the somewhat odd fact

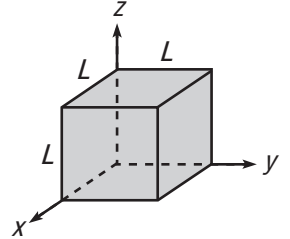


Fig. 9.8

that L_x and L_y are nonzero, even though the rotation is only around the z axis. We'll discuss this issue after the following remarks.

REMARKS:

1. The inertia tensor in Eq. (9.8) is a rather formidable-looking object. You will therefore be very pleased to hear that you rarely have to use it. It's nice to know that it's there if you need it, but the concept of *principal axes* (discussed in Section 9.3) provides a way to avoid using the inertia tensor (or more precisely, to greatly simplify it) and is therefore much more useful in solving problems.
2. $\mathbf{I}$ is a symmetric matrix, which is a fact that will be important in Section 9.3. There are therefore only six independent entries, instead of nine.
3. In the case where the rigid body is made up of a collection of point masses m_i , the entries in the matrix are just sums. For example, the upper left entry is $\sum m_i(y_i^2 + z_i^2)$.
4. $\mathbf{I}$ depends only on the geometry of the object, and not on ω .
5. To construct an $\mathbf{I}$, you not only need to specify the origin, you also need to specify the x, y, z axes of your coordinate system. And the basis vectors must be orthogonal, because the cross product calculation above is valid only for an orthonormal basis. If someone else comes along and chooses a different orthonormal basis (but the same origin), then her $\mathbf{I}$ will have different *entries*, as will her ω , as will her $\mathbf{L}$. But her ω and $\mathbf{L}$ will be exactly the same *vectors* as your ω and $\mathbf{L}$. They will appear different only because they are written in a different coordinate system. A vector is what it is, independent of how you choose to look at it. If you each point your arm in the direction of what you calculate $\mathbf{L}$ to be, then you will both be pointing in the same direction.
6. For the case of a pancake object rotating in the x - y plane, we have $z = 0$ for all points in the object. And $\omega = \omega_3 \hat{\mathbf{z}}$, so $\omega_1 = \omega_2 = 0$. The only nonzero term in the $\mathbf{L}$ in Eq. (9.8) is therefore $L_3 = \int (x^2 + y^2) dm \omega_3$, which is simply the $L_z = I_z \omega$ result we found in Eq. (8.5). ♣

This is all perfectly fine. Given any rigid body, we can calculate $\mathbf{I}$ (relative to a given origin, using a given set of axes). And given ω , we can then apply $\mathbf{I}$ to it to find $\mathbf{L}$. But what do these entries in $\mathbf{I}$ really mean? How do we interpret them? Note, for example, that ω_3 appears not only in L_3 in Eq. (9.8), but also in L_1 and L_2 . But ω_3 is relevant to rotations around the z axis, so what in the world is it doing in L_1 and L_2 ? Consider the following examples.

Example 1 (Point mass in the x - y plane): Consider a point mass m traveling in a circle of radius r (centered at the origin) in the x - y plane, with frequency ω_3 , as shown in Fig. 9.9. Using $\omega = (0, 0, \omega_3)$, $x^2 + y^2 = r^2$, and $z = 0$ in Eq. (9.8) (with a discrete sum of only one object, instead of the integrals), the angular momentum with respect to the origin is

$$\mathbf{L} = (0, 0, mr^2\omega_3). \quad (9.11)$$

The z component is $mr(r\omega_3) = mrv$, as it should be. And the x and y components are zero, as they should be. This case where $\omega_1 = \omega_2 = 0$ and $z = 0$ is simply the case we studied in Chapter 8, as mentioned in Remark 6 above.

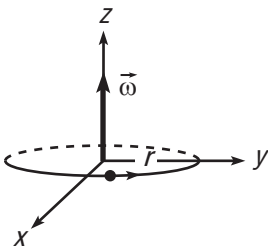


Fig. 9.9

Example 2 (Point mass in space): Consider a point mass m traveling in a circle of radius r , with frequency ω_3 . But now let the circle be centered at the point $(0, 0, z_0)$, with the plane of the circle parallel to the x - y plane, as shown in Fig. 9.10. Using $\boldsymbol{\omega} = (0, 0, \omega_3)$, $x^2 + y^2 = r^2$, and $z = z_0$ in Eq. (9.8), the angular momentum with respect to the origin is

$$\mathbf{L} = m\omega_3(-xz_0, -yz_0, r^2). \quad (9.12)$$

The z component is $mr^2\omega_3$, as it should be. But surprisingly, we have nonzero L_1 and L_2 , even though the mass is just rotating around the z axis. $\mathbf{L}$ does *not* point along $\boldsymbol{\omega}$ here. What's going on?

Consider an instant when the mass is in the y - z plane, as shown in Fig. 9.10. The velocity of the mass is then in the $-\hat{x}$ direction. Therefore, the particle most certainly has angular momentum around the y axis, as well as the z axis. Someone looking at a split-second movie of the mass at this point can't tell whether it's rotating around the y axis, the z axis, or undergoing some complicated motion. But the past and future motion is irrelevant; at any instant in time, as far as the angular momentum goes, we are concerned only with what is happening at this instant.

At this instant, the angular momentum around the y axis is $L_2 = -mz_0v$, because z_0 is the distance from the y axis, and the minus sign comes from the right-hand rule. Using $v = \omega_3 r = \omega_3 y$, we have $L_2 = -mz_0\omega_3 y$, in agreement with Eq. (9.12). Also, at this instant, L_1 is zero, because the velocity is parallel to the x axis. This agrees with Eq. (9.12), since $x = 0$. As an exercise, you can check that Eq. (9.12) is also correct when the mass is at a general point (x, y, z_0) .

We see that, for example, the $I_{yz} \equiv -\int yz$ entry in $\mathbf{I}$ tells us how much the ω_3 component of the angular velocity contributes to the L_2 component of the angular momentum. And due to the symmetry of $\mathbf{I}$, the $I_{yz} = I_{zy}$ entry in $\mathbf{I}$ also tells us how much the ω_2 component of the angular velocity contributes to the L_3 component of the angular momentum. In the former case, if we group the product of the various quantities as $-\int (\omega_3 y)z$, we see that this is simply the appropriate component of the velocity times the distance from the y axis. In the latter case with $-\int (\omega_2 z)y$, it is the opposite grouping. But in both cases there is one factor of y and one factor of z , hence the symmetry in $\mathbf{I}$.

REMARK: For a point mass, $\mathbf{L}$ is actually more easily obtained by just calculating $\mathbf{L} = \mathbf{r} \times \mathbf{p}$. The result for the instant shown in Fig. 9.10 is drawn in Fig. 9.11, where it is clear that $\mathbf{L}$ has both y and z components, and thus also clear that $\mathbf{L}$ doesn't point along $\boldsymbol{\omega}$. For a more complicated object, the tensor $\mathbf{I}$ is generally used, because it is necessary to perform the integral of the $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ contributions over the entire object, and the tensor has this integral built into it. At any rate, whatever method you use, you will find that except in special circumstances (see Section 9.3), $\mathbf{L}$ doesn't point along $\boldsymbol{\omega}$.

Consider the vector of $\mathbf{L}$,
And that of $\boldsymbol{\omega}$ as well.
The erroneous claim
That they must aim the same
Is a view that you've got to dispel! ♣

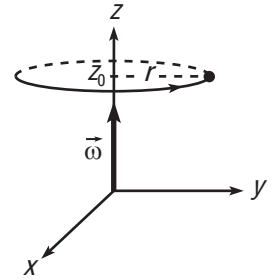


Fig. 9.10

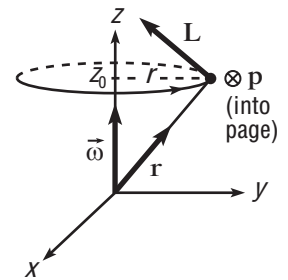


Fig. 9.11

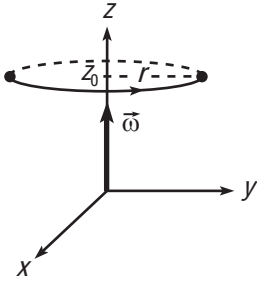


Fig. 9.12

Example 3 (Two point masses): Let's now add another point mass m to the previous example. Let it travel in the same circle, at the diametrically opposite point, as shown in Fig. 9.12. Using $\omega = (0, 0, \omega_3)$, $x^2 + y^2 = r^2$, and $z = z_0$ in Eq. (9.8), you can show that the angular momentum with respect to the origin is

$$\mathbf{L} = 2m\omega_3(0, 0, r^2). \quad (9.13)$$

Since $v = \omega_3 r$, the z component is $2mr v$, as it should be. And L_1 and L_2 are zero, unlike in the previous example, because these components of the $\mathbf{L}$'s of the two particles cancel. This occurs because of the symmetry of the masses around the z axis, which causes the L_{zx} and L_{zy} entries in the inertia tensor to vanish; they are each the sum of two terms, with opposite x values, or opposite y values. Alternatively, you can just note that adding on the mirror-image $\mathbf{L}$ vector in Fig. 9.10 produces canceling x and y components.

Let's now look at the kinetic energy of our object, which is rotating around an axis passing through the origin. To find this, we must add up the kinetic energies of all the little pieces. A little piece has energy $(dm) v^2/2 = dm |\omega \times \mathbf{r}|^2/2$. Therefore, using Eq. (9.6), the total kinetic energy is

$$T = \frac{1}{2} \int \left((\omega_2 z - \omega_3 y)^2 + (\omega_3 x - \omega_1 z)^2 + (\omega_1 y - \omega_2 x)^2 \right) dm. \quad (9.14)$$

Multiplying this out, we see (after a little work) that we can write T as

$$\begin{aligned} T &= \frac{1}{2} (\omega_1, \omega_2, \omega_3) \cdot \begin{pmatrix} \int (y^2 + z^2) & -\int xy & -\int zx \\ -\int xy & \int (z^2 + x^2) & -\int yz \\ -\int zx & -\int yz & \int (x^2 + y^2) \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} \\ &= \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{I} \boldsymbol{\omega} = \frac{1}{2} \boldsymbol{\omega} \cdot \mathbf{L}. \end{aligned} \quad (9.15)$$

If $\boldsymbol{\omega} = \omega_3 \hat{\mathbf{z}}$, then this reduces to $T = I_{zz} \omega_3^2/2$, which agrees with the result in Eq. (8.8), with a slight change in notation.

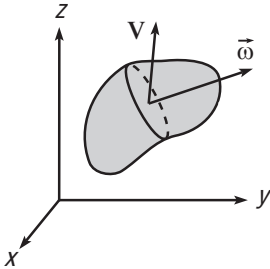


Fig. 9.13

9.2.2 General motion

How do we deal with general motion in space? That is, what if an object is both translating and rotating? For the motion in Fig. 9.13, the various pieces of mass aren't traveling in circles around the origin, so we can't write $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$, as we did prior to Eq. (9.4).

To determine $\mathbf{L}$ (relative to the origin), and also the kinetic energy T , we will use Theorem 9.1 to write the motion as the sum of a translation plus a rotation. In applying the theorem, we may choose any point in the body to be the point P in the theorem. However, only in the case where P is the object's CM can we extract anything useful, as we'll see. The theorem then says that the motion of

the body is the sum of the motion of the CM plus a rotation around the CM. So let the CM move with velocity $\mathbf{V}$, and let the body instantaneously rotate with angular velocity $\boldsymbol{\omega}'$ around the CM (that is, with respect to the frame whose origin is the CM, and whose axes are parallel to the fixed-frame axes).⁴

Let the position of the CM relative to the origin be $\mathbf{R} = (X, Y, Z)$, and let the position of a given piece of mass relative to the CM be $\mathbf{r}' = (x', y', z')$. Then $\mathbf{r} = \mathbf{R} + \mathbf{r}'$ is the position of a piece of mass relative to the origin (see Fig. 9.14). Let the velocity of a piece of mass relative to the CM be $\mathbf{v}'$ (so $\mathbf{v}' = \boldsymbol{\omega}' \times \mathbf{r}'$). Then $\mathbf{v} = \mathbf{V} + \mathbf{v}'$ is the velocity relative to the origin.

Let's look at $\mathbf{L}$ first. The angular momentum is

$$\begin{aligned} \mathbf{L} &= \int \mathbf{r} \times \mathbf{v} \, dm = \int (\mathbf{R} + \mathbf{r}') \times (\mathbf{V} + (\boldsymbol{\omega}' \times \mathbf{r}')) \, dm \\ &= \int (\mathbf{R} \times \mathbf{V}) \, dm + \int \mathbf{r}' \times (\boldsymbol{\omega}' \times \mathbf{r}') \, dm \\ &= M(\mathbf{R} \times \mathbf{V}) + \mathbf{L}_{\text{CM}}, \end{aligned} \quad (9.16)$$

where the cross terms vanish because the integrands are linear in $\mathbf{r}'$. More precisely, the integrals involve $\int \mathbf{r}' \, dm$, which is zero by definition of the CM (because $\int \mathbf{r}' \, dm / M$ is the position of the CM relative to the CM, which is zero). $\mathbf{L}_{\text{CM}}$ is the angular momentum relative to the CM.⁵

We see that as in the pancake case in Section 8.1.2, the angular momentum (relative to the origin) of a body can be found by treating the body like a point mass located at the CM and finding the angular momentum of this point mass relative to the origin, and by then adding on the angular momentum of the body relative to the CM. Note that these two parts of the angular momentum need not point in the same direction, as they did in the case of the pancake moving in the x - y plane.

Now let's look at T . The kinetic energy is

$$\begin{aligned} T &= \int \frac{1}{2} v^2 \, dm = \int \frac{1}{2} |\mathbf{V} + \mathbf{v}'|^2 \, dm \\ &= \int \frac{1}{2} V^2 \, dm + \int \frac{1}{2} v'^2 \, dm \\ &= \frac{1}{2} M V^2 + \int \frac{1}{2} |\boldsymbol{\omega}' \times \mathbf{r}'|^2 \, dm \\ &\equiv \frac{1}{2} M V^2 + \frac{1}{2} \boldsymbol{\omega}' \cdot \mathbf{L}_{\text{CM}}, \end{aligned} \quad (9.17)$$

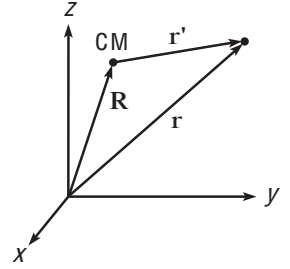


Fig. 9.14

⁴ It's not necessary to put the prime on the $\boldsymbol{\omega}$ here, because the angular velocity vector in the CM frame is the same as in the lab frame. But we'll use the prime just because we'll have primes on the other CM quantities below.

⁵ By this, we mean the angular momentum as measured in the coordinate system whose origin is the CM, and whose axes are parallel to the fixed-frame axes.

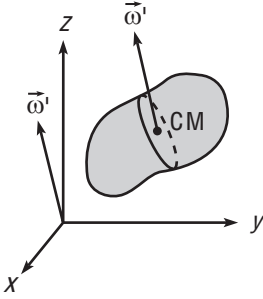


Fig. 9.15

where the last line follows from the steps leading to Eq. (9.15). The cross term $\int \mathbf{V} \cdot \mathbf{v}' dm = \int \mathbf{V} \cdot (\boldsymbol{\omega}' \times \mathbf{r}') dm$ vanishes because the integrand is linear in $\mathbf{r}'$ and thus yields a zero integral, by definition of the CM. As in the pancake case in Section 8.1.2, the kinetic energy of a body can be found by treating the body like a point mass located at the CM, and by then adding on the kinetic energy of the body due to the rotation around the CM.

9.2.3 The parallel-axis theorem

Consider the special case where the CM rotates around the origin with the same angular velocity at which the body rotates around the CM (see Fig. 9.15), that is, $\mathbf{V} = \boldsymbol{\omega}' \times \mathbf{R}$. This can be achieved, for example, by piercing the body with the base of a rigid “T” and then rotating the T and the body around the (fixed) line of the “upper” part of the T (the origin must pass through this line). We then have the nice situation where all points in the body travel in fixed circles around the axis of rotation. Mathematically, this follows from $\mathbf{v} = \mathbf{V} + \mathbf{v}' = \boldsymbol{\omega}' \times \mathbf{R} + \boldsymbol{\omega}' \times \mathbf{r}' = \boldsymbol{\omega}' \times \mathbf{r}$. Dropping the prime on $\boldsymbol{\omega}$, Eq. (9.16) becomes

$$\mathbf{L} = M\mathbf{R} \times (\boldsymbol{\omega} \times \mathbf{R}) + \int \mathbf{r}' \times (\boldsymbol{\omega} \times \mathbf{r}') dm \quad (9.18)$$

Expanding the double cross products as in the steps leading to Eq. (9.8), we can write this as

$$\begin{aligned} \begin{pmatrix} L_1 \\ L_2 \\ L_3 \end{pmatrix} &= M \begin{pmatrix} Y^2 + Z^2 & -XY & -ZX \\ -XY & Z^2 + X^2 & -YZ \\ -ZX & -YZ & X^2 + Y^2 \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} \\ &+ \begin{pmatrix} \int (y'^2 + z'^2) & -\int x'y' & -\int z'x' \\ -\int x'y' & \int (z'^2 + x'^2) & -\int y'z' \\ -\int z'x' & -\int y'z' & \int (x'^2 + y'^2) \end{pmatrix} \begin{pmatrix} \omega_1 \\ \omega_2 \\ \omega_3 \end{pmatrix} \\ &\equiv (\mathbf{I}_R + \mathbf{I}_{CM})\boldsymbol{\omega}. \end{aligned} \quad (9.19)$$

This is the generalized parallel-axis theorem. It says that once you’ve calculated $\mathbf{I}_{CM}$ relative to the CM, then if you want to calculate $\mathbf{I}$ relative to another point, you simply have to add on the $\mathbf{I}_R$ matrix, obtained by treating the object like a point mass at the CM. So you have to compute six extra numbers (there are six, instead of nine, because the $\mathbf{I}_R$ matrix is symmetric) instead of just the one MR^2 in the parallel-axis theorem in Chapter 8, given in Eq. (8.13). Problem 9.4 gives another derivation of the parallel-axis theorem, without mentioning the angular velocity.

REMARK: The name “parallel-axis” theorem is actually a misnomer here. The inertia tensor isn’t associated with one particular axis, as the moment of inertia in Chapter 8 was. The moment of inertia is just one of the diagonal entries (associated with a given axis) in the inertia tensor. The inertia tensor depends on the entire coordinate system. So in that sense we should call this

the “parallel-axes” theorem, because the coordinate axes in the CM frame are assumed to be parallel to the ones in the fixed frame. At any rate, the point is that the parallel-axis theorem in Chapter 8 dealt with shifting the axis, whereas the present theorem deals with shifting the origin (and hence all three axes in general). ♣

As far as the kinetic energy goes, if $\boldsymbol{\omega}$ and $\boldsymbol{\omega}'$ are equal, so that $\mathbf{V} = \boldsymbol{\omega}' \times \mathbf{R}$, then Eq. (9.17) gives (dropping the prime on $\boldsymbol{\omega}$)

$$T = \frac{1}{2}M|\boldsymbol{\omega} \times \mathbf{R}|^2 + \int \frac{1}{2}|\boldsymbol{\omega} \times \mathbf{r}'|^2 dm. \quad (9.20)$$

Performing the steps leading to Eq. (9.15), this becomes

$$T = \frac{1}{2}\boldsymbol{\omega} \cdot (\mathbf{I}_R + \mathbf{I}_{CM})\boldsymbol{\omega} = \frac{1}{2}\boldsymbol{\omega} \cdot \mathbf{L}. \quad (9.21)$$

9.3 Principal axes

The cumbersome expressions in the previous section may seem a bit unsettling, but it turns out that usually we can get by without them. The strategy for avoiding all of the above mess is to use the *principal axes* of a body, which we will define below.

In general, the inertia tensor $\mathbf{I}$ in Eq. (9.8) has nine nonzero entries, of which six are independent due to the symmetry of $\mathbf{I}$. In addition to depending on the origin chosen, the inertia tensor depends on the set of orthonormal basis vectors chosen for the coordinate system; the x, y, z variables in the integrals in $\mathbf{I}$ depend, of course, on the coordinate system they’re measured with respect to. Given a blob of material, and given an arbitrary origin,⁶ any orthonormal set of basis vectors is usable, but there is one special set that makes all our calculations very nice. These special basis vectors are called the *principal axes*. They can be defined in various equivalent ways:

- The principal axes are the orthonormal basis vectors for which $\mathbf{I}$ is diagonal, that is, for which⁷

$$\mathbf{I} = \begin{pmatrix} I_1 & 0 & 0 \\ 0 & I_2 & 0 \\ 0 & 0 & I_3 \end{pmatrix}. \quad (9.22)$$

I_1, I_2 , and I_3 are called the *principal moments*. For many objects, it is quite obvious what the principal axes are. For example, consider a uniform rectangle in the x - y plane. Pick the origin to be the CM, and let the x and y axes be parallel to the sides. Then the

⁶ The CM is often chosen to be the origin, but it need not be. There are principal axes associated with any origin.

⁷ Technically, we should be writing I_{11} or I_{xx} instead of I_1 , etc., in this matrix, because the one-index object I_1 looks like the component of a vector, not a matrix. But the two-index notation gets cumbersome, so we’ll be sloppy and just use I_1 , etc.

principal axes are clearly the x , y , and z axes, because all the off-diagonal elements in the inertia tensor in Eq. (9.8) vanish, by symmetry. For example, $I_{xy} \equiv -\int xy \, dm$ equals zero, because for every point (x, y) in the rectangle, there is a corresponding point $(-x, y)$, so the contributions to $\int xy \, dm$ cancel in pairs. Also, the integrals involving z are identically zero, because $z = 0$.

- A principal axis is an axis $\hat{\omega}$ for which $\mathbf{I}\hat{\omega} = I\hat{\omega}$. That is, a principal axis is a special direction with the property that if ω points along it, then so does $\mathbf{L}$. The principal axes of an object are then the orthonormal set of three vectors $\hat{\omega}_1, \hat{\omega}_2, \hat{\omega}_3$ with the property that

$$\mathbf{I}\hat{\omega}_1 = I_1\hat{\omega}_1, \quad \mathbf{I}\hat{\omega}_2 = I_2\hat{\omega}_2, \quad \mathbf{I}\hat{\omega}_3 = I_3\hat{\omega}_3. \quad (9.23)$$

The three statements in Eq. (9.23) are equivalent to Eq. (9.22), because the vectors $\hat{\omega}_1, \hat{\omega}_2$, and $\hat{\omega}_3$ are simply $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ in the frame in which they are the basis vectors.

- Consider an object rotating around a fixed axis with constant angular speed. Then this axis is a principal axis if there is no need for any torque. So in some sense, the object is “happy” to spin around a principal axis. A set of three orthonormal axes, each of which has this property, is by definition what we call a set of principal axes.

This definition of a principal axis is equivalent to the previous definition for the following reason. Assume that the object rotates around a fixed axis $\hat{\omega}_1$ for which $\mathbf{L} = \mathbf{I}\hat{\omega}_1 = I_1\hat{\omega}_1$, as in Eq. (9.23). Then since $\hat{\omega}_1$ is assumed to be fixed, we see that $\mathbf{L}$ is also fixed. Therefore, $\tau = d\mathbf{L}/dt = \mathbf{0}$.

Conversely, if the object is rotating around a fixed axis $\hat{\omega}_1$, and if $\tau = d\mathbf{L}/dt = \mathbf{0}$, then we claim that $\mathbf{L}$ points along $\hat{\omega}_1$ (that is, $\mathbf{L} = I_1\hat{\omega}_1$). This is true because if $\mathbf{L}$ does *not* point along $\hat{\omega}_1$, then imagine painting a dot on the object somewhere along the line of $\mathbf{L}$. A little while later, the dot will have rotated around the fixed vector $\hat{\omega}_1$. But the line of $\mathbf{L}$ must always pass through the dot, because we could have rotated our axes around $\hat{\omega}_1$ and started the process at a slightly later time (this argument relies on $\hat{\omega}_1$ being fixed). Therefore, we see that $\mathbf{L}$ has changed, in contradiction to the assumption that $d\mathbf{L}/dt = \mathbf{0}$. Hence, $\mathbf{L}$ must in fact point along $\hat{\omega}_1$.

For a rotation around a principal axis $\hat{\omega}$, the lack of need for any torque means that if the object is pivoted at the origin, and if the origin is the only place where any force is applied (which implies that there is zero torque around it), then the object can undergo rotation with constant angular velocity ω . If you try to set up this scenario with a nonprincipal axis, it won't work.

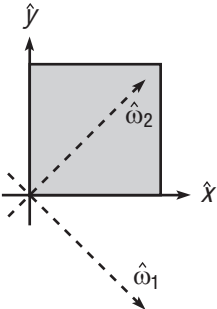


Fig. 9.16

Example (Square with origin at corner): Consider the uniform square in Fig. 9.16. In Appendix E, we show that the principal axes are the dotted lines drawn (and also the z axis perpendicular to the page). But there is no need to use the techniques in the appendix to see this, because in this new basis it is clear by symmetry that the integral $\int x_1 x_2$ is zero; for every x_1 in the integral, there is a $-x_1$. And $x_3 \equiv z$

is identically zero, which makes all the other off-diagonal terms in $\mathbf{I}$ also equal to zero. Therefore, since $\mathbf{I}$ is diagonal in this new basis, these basis vectors are the principal axes.

Furthermore, it is intuitively clear that the square will be happy to rotate around any one of these axes indefinitely. During such a rotation, the pivot will certainly be applying a *force* (if the axis is $\hat{\omega}_1$ or $\hat{z}$, but not if it is $\hat{\omega}_2$), to produce the centripetal acceleration of the CM in its circular motion. But it won't be applying a *torque* relative to the origin (because the $\mathbf{r}$ in $\mathbf{r} \times \mathbf{F}$ is $\mathbf{0}$). This is good, because for a rotation around one of these principal axes, $d\mathbf{L}/dt = \mathbf{0}$, so there is no need for any torque.

In contrast with the off-diagonal zeros in the new basis, the integral $\int xy$ in the old basis is *not* zero, because every point gives a positive contribution. So the inertia tensor is not diagonal in the old basis, which means that $\hat{x}$ and $\hat{y}$ are not principal axes. Consistent with this, it is reasonably clear that it is impossible to make the square rotate around, say, the x axis, assuming that its only contact with the outside world is through a pivot (for example, a ball and socket) at the origin. The square simply doesn't want to remain in this circular motion. Mathematically, $\mathbf{L}$ (relative to the origin) doesn't point along the x axis, so it therefore precesses around the x axis along with the square, tracing out the surface of a cone. This means that $\mathbf{L}$ is changing. But there is no torque available (relative to the origin) to provide for this change in $\mathbf{L}$. Hence, such a rotation cannot exist.

At the moment, it is not at all obvious that an orthonormal set of principal axes exists for an arbitrary object. But this is indeed the case, as stated in Theorem 9.4 below. Assuming for now that principal axes do exist, then in this basis the $\mathbf{L}$ and T in Eqs. (9.8) and (9.15) take on the particularly nice forms,

$$\begin{aligned}\mathbf{L} &= (I_1\omega_1, I_2\omega_2, I_3\omega_3), \\ T &= \frac{1}{2} (I_1\omega_1^2 + I_2\omega_2^2 + I_3\omega_3^2).\end{aligned}\tag{9.24}$$

The quantities ω_1 , ω_2 , and ω_3 here are the components of a general vector $\boldsymbol{\omega}$ written in the principal-axis basis; that is, $\boldsymbol{\omega} = \omega_1\hat{\omega}_1 + \omega_2\hat{\omega}_2 + \omega_3\hat{\omega}_3$. Equation (9.24) is a vast simplification over the general formulas in Eqs. (9.8) and (9.15). We will therefore invariably work with principal axes in the remainder of this chapter.

Note that the directions of the principal axes (relative to the body) depend only on the geometry of the body. They may therefore be considered to be painted on (or in) it. Hence, they will generally move around in space as the body rotates. For example, if the object is rotating around a principal axis, then that axis stays fixed while the other two principal axes rotate around it. In relations like $\boldsymbol{\omega} = (\omega_1, \omega_2, \omega_3)$ and $\mathbf{L} = (I_1\omega_1, I_2\omega_2, I_3\omega_3)$, the components ω_i and $I_i\omega_i$ are measured along the *instantaneous* principal axes $\hat{\omega}_i$. Since these axes change with time, it is quite possible that the components ω_i and $I_i\omega_i$ change with time, as we'll see in Section 9.5 (and onward).

Let's now state the theorem that implies that a set of principal axes does indeed exist for any body and any origin. The proof of this theorem involves a useful but rather slick technique, but it's slightly off the main line of thought, so we'll relegate it to Appendix D. Take a look at the proof if you wish, but if you want to just accept the fact that principal axes exist, that's fine.

Theorem 9.4 *Given a real symmetric 3×3 matrix, $\mathbf{I}$, there exist three orthonormal real vectors, $\hat{\boldsymbol{\omega}}_k$, and three real numbers, I_k , with the property that*

$$\mathbf{I}\hat{\boldsymbol{\omega}}_k = I_k\hat{\boldsymbol{\omega}}_k. \quad (9.25)$$

Proof: See Appendix D. ■

Since the inertia tensor in Eq. (9.8) is indeed symmetric for any body and any origin, this theorem says that we can always find three orthogonal basis vectors that satisfy Eq. (9.23). Or equivalently, we can always find three orthogonal basis vectors for which $\mathbf{I}$ is a diagonal matrix, as in Eq. (9.22). In other words, principal axes always exist. Problem 9.7 gives another way to demonstrate the existence of principal axes in the special case of a pancake object.

Invariably, it is best to work in a coordinate system that has principal axes as its basis, due to the simplicity of Eq. (9.24). And as mentioned in Footnote 6, the origin is generally chosen to be the CM, because from Section 8.4.3 the CM is one of the origins for which $\boldsymbol{\tau} = d\mathbf{L}/dt$ is a valid statement. But this choice is not necessary; there are principal axes associated with any origin.

For an object with a fair amount of symmetry, the principal axes are usually the obvious choices and can be written down by simply looking at the object (examples are given below). If, however, you are given an unsymmetrical body, then the only way to determine the principal axes is to pick an arbitrary basis, then find $\mathbf{I}$ in this basis, and then go through a diagonalization procedure. This diagonalization procedure basically consists of the steps at the beginning of the proof of Theorem 9.4 (given in Appendix D), with the addition of one more step to get the actual vectors, so we'll relegate it to Appendix E. There's no need to worry much about this method. Virtually every system you encounter will involve an object with sufficient symmetry to enable you to just write down the principal axes.

Let's now prove two very useful (and very similar) theorems.

Theorem 9.5 *If two principal moments are equal ($I_1 = I_2 \equiv I$), then any axis (through the chosen origin) in the plane of the corresponding principal axes is a principal axis, and its moment is also I . Similarly, if all three principal moments are equal ($I_1 = I_2 = I_3 \equiv I$), then any axis (through the chosen origin) in space is a principal axis, and its moment is also I .*

Proof: The first part was already proved at the end of the proof in Appendix D, but we'll do it again here. Since $I_1 = I_2 \equiv I$, we have $\mathbf{I}\mathbf{u}_1 = I\mathbf{u}_1$, and $\mathbf{I}\mathbf{u}_2 = I\mathbf{u}_2$,

where the $\mathbf{u}$'s are the principal axes. Hence, $\mathbf{I}(a\mathbf{u}_1 + b\mathbf{u}_2) = I(a\mathbf{u}_1 + b\mathbf{u}_2)$, for any a and b . Therefore, any linear combination of $\mathbf{u}_1$ and $\mathbf{u}_2$ (that is, any vector in the plane spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$) is a solution to $\mathbf{I}\mathbf{u} = I\mathbf{u}$ and is thus a principal axis, by definition.

The proof of the second part proceeds in a similar manner. Since $I_1 = I_2 = I_3 \equiv I$, we have $\mathbf{I}\mathbf{u}_1 = I\mathbf{u}_1$, $\mathbf{I}\mathbf{u}_2 = I\mathbf{u}_2$, and $\mathbf{I}\mathbf{u}_3 = I\mathbf{u}_3$. Hence, $\mathbf{I}(a\mathbf{u}_1 + b\mathbf{u}_2 + c\mathbf{u}_3) = I(a\mathbf{u}_1 + b\mathbf{u}_2 + c\mathbf{u}_3)$. Therefore, any linear combination of $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$ (that is, any vector in space) is a solution to $\mathbf{I}\mathbf{u} = I\mathbf{u}$ and is thus a principal axis, by definition.

In short, if $I_1 = I_2 \equiv I$, then $\mathbf{I}$ is the identity matrix (up to a multiple) in the space spanned by $\mathbf{u}_1$ and $\mathbf{u}_2$. And if $I_1 = I_2 = I_3 \equiv I$, then $\mathbf{I}$ is the identity matrix (up to a multiple) in the entire space. Note that it isn't required that the various $\mathbf{u}_i$ vectors be orthogonal. All we need is that they span the relevant space. ■

If two or three moments are equal, so that there is freedom in choosing the principal axes, then it is possible to pick a nonorthogonal group of them. We will, however, always choose ones that are orthogonal. So when we say "a set of principal axes," we mean an orthonormal set.

Theorem 9.6 *If a pancake object is symmetric under a rotation through an angle $\theta \neq 180^\circ$ in the x - y plane (such as a hexagon), then every axis in the x - y plane (with the origin chosen to be the center of the symmetry rotation) is a principal axis with the same moment.*

Proof: Let $\hat{\omega}_0$ be a principal axis in the plane, and let $\hat{\omega}_\theta$ be the axis obtained by rotating $\hat{\omega}_0$ through the angle θ . Then $\hat{\omega}_\theta$ is also a principal axis with the same principal moment, due to the symmetry of the object. Therefore, $\mathbf{I}\hat{\omega}_0 = I\hat{\omega}_0$, and $\mathbf{I}\hat{\omega}_\theta = I\hat{\omega}_\theta$.

Now, any vector ω in the x - y plane can be written as a linear combination of $\hat{\omega}_0$ and $\hat{\omega}_\theta$, provided that $\theta \neq 180^\circ$ (or zero, of course). That is, $\hat{\omega}_0$ and $\hat{\omega}_\theta$ span the x - y plane. Therefore, any vector ω can be written as $\omega = a\hat{\omega}_0 + b\hat{\omega}_\theta$, and so

$$\mathbf{I}\omega = \mathbf{I}(a\hat{\omega}_0 + b\hat{\omega}_\theta) = aI\hat{\omega}_0 + bI\hat{\omega}_\theta = I\omega. \quad (9.26)$$

Hence, ω is also a principal axis. Problem 9.8 gives another proof of this theorem. ■

The theorem actually holds even without the "pancake" restriction. That is, it holds for any object with a rotational symmetry around the z axis (excluding $\theta \neq 180^\circ$). This can be seen as follows. The z axis is a principal axis, because if ω points along $\hat{z}$, then $\mathbf{L}$ must also point along $\hat{z}$, by symmetry. There are therefore (at least) two principal axes in the x - y plane. Label one of these as $\hat{\omega}_0$ and proceed as above.

Let's now do some quick examples. We'll state the principal axes for the objects listed below (relative to the origin). Your task is to show that they

are correct. Usually, a quick symmetry argument shows that

$$\mathbf{I} \equiv \begin{pmatrix} \int (y^2 + z^2) & -\int xy & -\int zx \\ -\int xy & \int (z^2 + x^2) & -\int yz \\ -\int zx & -\int yz & \int (x^2 + y^2) \end{pmatrix} \quad (9.27)$$

is diagonal. In all of these examples (see Fig. 9.17), the origin for the principal axes is understood to be the origin of the given coordinate system (which is not necessarily the CM). In describing the axes, they therefore all pass through the origin, in addition to having the other properties stated.

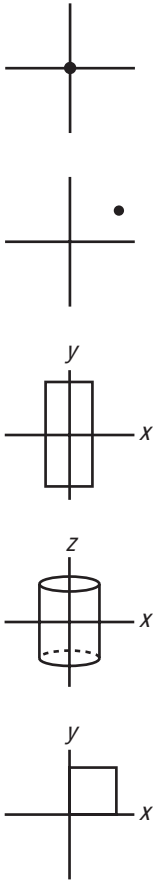


Fig. 9.17

Example 1: Point mass at the origin.

principal axes: any axes.

Example 2: Point mass at the point (x_0, y_0, z_0) .

principal axes: axis through point, any axes perpendicular to this.

Example 3: Rectangle centered at the origin, as shown.

principal axes: the x , y , and z axes.

Example 4: Cylinder with axis as z axis.

principal axes: z axis, any axes in x - y plane.

Example 5: Square with one corner at origin, as shown.

principal axes: z axis, axis through CM, axis perpendicular to this.

9.4 Two basic types of problems

The previous three sections introduced a variety of abstract concepts. We will now finally look at some actual physical systems. The concept of principal axes gives us the ability to solve many kinds of problems. Two kinds, however, come up again and again. There are variations on these, of course, but they may be generally stated as follows.

- Strike a rigid object with an impulsive (that is, quick) blow. What is the motion of the object immediately after the blow?
- An object rotates around a fixed axis. A given torque is applied. What is the frequency of the rotation? Or conversely, given the frequency, what is the required torque?

We'll work through an example for each of these problems. In both cases, the solution involves a few standard steps, so we'll write them out explicitly.

9.4.1 Motion after an impulsive blow

Problem: Consider the rigid object in Fig. 9.18. Three masses are connected by three massless rods, in the shape of an isosceles right triangle with hypotenuse length $4a$. The mass at the right angle is $2m$, and the other two masses are m . Label them A , B , C , as shown. Assume that the object is floating freely in outer space. Mass B is struck with a quick blow, directed into the page. Let the imparted impulse have magnitude $\int F dt = P$. (See Section 8.6 for a discussion of impulse and angular impulse.) What are the velocities of the three masses immediately after the blow?

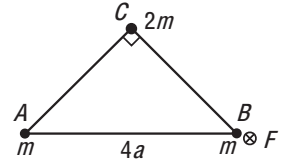


Fig. 9.18

Solution: Our strategy will be to find the angular momentum of the system (relative to the CM) using the angular impulse, and then calculate the principal moments and find the angular velocity vector (which will give the velocities relative to the CM), and then finally add on the CM motion.

The altitude from the right angle to the hypotenuse has length $2a$, and the CM is easily seen to be located at its midpoint (see Fig. 9.19). Picking the CM as our origin, and letting the plane of the paper be the x - y plane, the positions of the three masses are $\mathbf{r}_A = (-2a, -a, 0)$, $\mathbf{r}_B = (2a, -a, 0)$, and $\mathbf{r}_C = (0, a, 0)$. There are now five standard steps that we must perform.

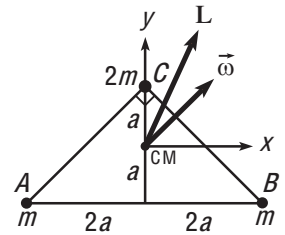


Fig. 9.19

- **Find $\mathbf{L}$:** The positive z axis is directed out of the page, so the impulse vector is $\mathbf{P} \equiv \int \mathbf{F} dt = (0, 0, -P)$. Therefore, the angular momentum of the system (relative to the CM) is

$$\begin{aligned} \mathbf{L} &= \int \boldsymbol{\tau} dt = \int (\mathbf{r}_B \times \mathbf{F}) dt = \mathbf{r}_B \times \int \mathbf{F} dt \\ &= (2a, -a, 0) \times (0, 0, -P) = aP(1, 2, 0), \end{aligned} \quad (9.28)$$

as shown in Fig. 9.19. We have used the fact that $\mathbf{r}_B$ is essentially constant during the blow (because the blow is assumed to happen very quickly) in taking $\mathbf{r}_B$ outside the integral.

- **Calculate the principal moments:** The principal axes are the x , y , and z axes, because the symmetry of the triangle makes $\mathbf{I}$ diagonal in this basis, as you can quickly check. The moments (relative to the CM) are

$$\begin{aligned} I_x &= ma^2 + ma^2 + (2m)a^2 = 4ma^2, \\ I_y &= m(2a)^2 + m(2a)^2 + (2m)0^2 = 8ma^2, \\ I_z &= I_x + I_y = 12ma^2. \end{aligned} \quad (9.29)$$

We have used the perpendicular-axis theorem to obtain I_z , although it won't be needed to solve the problem.

- **Find $\boldsymbol{\omega}$:** We now have two expressions for the angular momentum of the system. One expression is in terms of the given impulse, Eq. (9.28). The other is in terms of the

moments and the angular velocity components, Eq. (9.24). Equating these gives

$$\begin{aligned}
 (I_x\omega_x, I_y\omega_y, I_z\omega_z) &= aP(1, 2, 0) \\
 \implies (4ma^2\omega_x, 8ma^2\omega_y, 12ma^2\omega_z) &= aP(1, 2, 0) \\
 \implies (\omega_x, \omega_y, \omega_z) &= \frac{P}{4ma}(1, 1, 0), \quad (9.30)
 \end{aligned}$$

as shown in Fig. 9.19.

- **Calculate the velocities relative to the CM:** Right after the blow, the object rotates around the CM with the angular velocity found in Eq. (9.30). The velocities relative to the CM are then $\mathbf{u}_i = \boldsymbol{\omega} \times \mathbf{r}_i$. Thus,

$$\begin{aligned}
 \mathbf{u}_A &= \boldsymbol{\omega} \times \mathbf{r}_A = \frac{P}{4ma}(1, 1, 0) \times (-2a, -a, 0) = (0, 0, P/4m), \\
 \mathbf{u}_B &= \boldsymbol{\omega} \times \mathbf{r}_B = \frac{P}{4ma}(1, 1, 0) \times (2a, -a, 0) = (0, 0, -3P/4m), \quad (9.31) \\
 \mathbf{u}_C &= \boldsymbol{\omega} \times \mathbf{r}_C = \frac{P}{4ma}(1, 1, 0) \times (0, a, 0) = (0, 0, P/4m).
 \end{aligned}$$

As a check, it makes sense that $\mathbf{u}_B$ is three times as large as $\mathbf{u}_A$ and $\mathbf{u}_C$, because B is three times as far from the axis of rotation as A and C are, as you can verify by doing a little geometry in Fig. 9.19.

- **Add on the velocity of the CM:** The impulse (that is, the change in linear momentum) supplied to the whole system is $\mathbf{P} = (0, 0, -P)$. The total mass of the system is $M = 4m$. Therefore, the velocity of the CM is

$$V_{\text{CM}} = \frac{\mathbf{P}}{M} = (0, 0, -P/4m). \quad (9.32)$$

The total velocities of the masses are therefore

$$\begin{aligned}
 \mathbf{v}_A &= \mathbf{u}_A + V_{\text{CM}} = (0, 0, 0), \\
 \mathbf{v}_B &= \mathbf{u}_B + V_{\text{CM}} = (0, 0, -P/m), \quad (9.33) \\
 \mathbf{v}_C &= \mathbf{u}_C + V_{\text{CM}} = (0, 0, 0).
 \end{aligned}$$

REMARKS:

1. We see that masses A and C are instantaneously at rest immediately after the blow, and mass B acquires all of the imparted impulse. In retrospect, this is clear. Basically, it is possible for both A and C to remain at rest while B moves a tiny bit, so this is what happens. If B moves into the page by a small distance ϵ , then A and C won't know that B has moved, because their distances to B will change (assuming hypothetically that they don't move) by a distance of order only ϵ^2 . If we changed the problem and added a mass D at, say, the midpoint of the hypotenuse, then it would *not* be possible for A , C , and D to remain at rest while B moved a tiny bit. So there would have to be some other motion in addition to B 's. This setup is the topic of Exercise 9.38.
2. As time goes on, the system undergoes a rather complicated motion. What happens is that the CM moves with constant velocity while the masses rotate around it in

a messy manner. Since there are no torques acting on the system (after the initial blow), we know that $\mathbf{L}$ forever remains constant. It turns out that $\boldsymbol{\omega}$ moves around $\mathbf{L}$ while the masses rotate around this changing $\boldsymbol{\omega}$. These matters are the subject of Section 9.6, although in that discussion we restrict ourselves to symmetric tops, that is, ones with two equal moments. But these issues aside, it's good to know that we can, without too much difficulty, determine what's going on immediately after the blow.

3. The object in this problem was assumed to be floating freely in space. If we instead have an object that is pivoted at a given fixed point, then we should use this pivot as our origin. There is then no need to perform the last step of adding on the velocity of the origin (which was the CM, above), because this velocity is now zero. Equivalently, just consider the pivot to be an infinite mass, which is therefore the location of the (motionless) CM. ♣

9.4.2 Frequency of motion due to a torque

Problem: Consider a stick of length ℓ , mass m , and uniform mass density. The stick is pivoted at its top end and swings around the vertical axis. Assume that conditions have been set up so that the stick always makes an angle θ with the vertical, as shown in Fig. 9.20. What is the frequency, ω , of this motion?

Solution: Our strategy will be to find the principal moments and then the angular momentum of the system (in terms of ω), and then find the rate of change of $\mathbf{L}$, and then calculate the torque and equate it with $d\mathbf{L}/dt$. We will choose the pivot to be the origin.⁸ Again, there are five standard steps that we must perform.

- **Calculate the principal moments:** The principal axes are the axis along the stick, along with any two orthogonal axes perpendicular to the stick. So let the x and y axes be as shown in Fig. 9.21. The positive z axis then points out of the page. The moments (relative to the pivot) are $I_x = m\ell^2/3$, $I_y = 0$, and $I_z = m\ell^2/3$ (which won't be needed).
- **Find $\mathbf{L}$:** The angular velocity vector points vertically (however, see the third remark following this solution), so in the basis of the principal axes, the angular velocity vector is $\boldsymbol{\omega} = (\omega \sin \theta, \omega \cos \theta, 0)$, where ω is yet to be determined. The angular momentum of the system (relative to the pivot) is therefore

$$\mathbf{L} = (I_x \omega_x, I_y \omega_y, I_z \omega_z) = ((1/3)m\ell^2 \omega \sin \theta, 0, 0). \quad (9.34)$$

- **Find $d\mathbf{L}/dt$:** The vector $\mathbf{L}$ in Eq. (9.34) points up to the right, along the x axis (at the instant shown in Fig. 9.21), with magnitude $L = (1/3)m\ell^2 \omega \sin \theta$. As the stick rotates around the vertical axis, $\mathbf{L}$ traces out the surface of a cone. That is, the tip of $\mathbf{L}$ traces out a horizontal circle. The radius of this circle is the horizontal component of $\mathbf{L}$, which is $L \cos \theta$. The speed of the tip (which is the magnitude of $d\mathbf{L}/dt$) is therefore

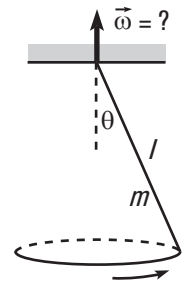


Fig. 9.20

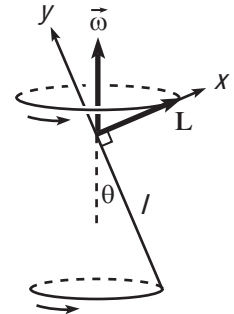


Fig. 9.21

⁸ This is a better choice than the CM because this way we won't have to worry about any messy forces acting at the pivot when computing the torque. The task of Exercise 9.41 is to work through the more complicated solution which has the CM as the origin.

$(L \cos \theta)\omega$, because $\mathbf{L}$ rotates around the vertical axis with the same frequency as the stick. So $d\mathbf{L}/dt$ has magnitude

$$\left| \frac{d\mathbf{L}}{dt} \right| = (L \cos \theta)\omega = \frac{1}{3}m\ell^2\omega^2 \sin \theta \cos \theta, \quad (9.35)$$

and it points into the page.

REMARK: With more complicated objects where $I_y \neq 0$, $\mathbf{L}$ won't point nicely along a principal axis, so the length of its horizontal component (the radius of the circle that $\mathbf{L}$ traces out) won't immediately be obvious. In this case, you can either explicitly calculate the horizontal component (see the spinning-top example in Section 9.7.5), or you can just do things the formal way by finding the rate of change of $\mathbf{L}$ via the expression $d\mathbf{L}/dt = \boldsymbol{\omega} \times \mathbf{L}$, which holds for all the same reasons that $\mathbf{v} \equiv d\mathbf{r}/dt = \boldsymbol{\omega} \times \mathbf{r}$ holds. In the present problem, we obtain

$$\begin{aligned} d\mathbf{L}/dt &= (\omega \sin \theta, \omega \cos \theta, 0) \times ((1/3)m\ell^2\omega \sin \theta, 0, 0) \\ &= (0, 0, -(1/3)m\ell^2\omega^2 \sin \theta \cos \theta), \end{aligned} \quad (9.36)$$

in agreement with Eq. (9.35). And the direction is correct, because the negative z axis points into the page. Note that we calculated this cross product in the principal-axis basis. Although these axes are changing in time, they present a perfectly good set of basis vectors at any instant. ♣

- **Calculate the torque:** The torque (relative to the pivot) is due to gravity, which effectively acts on the CM of the stick. So $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ has magnitude

$$\tau = rF \sin \theta = (\ell/2)(mg) \sin \theta, \quad (9.37)$$

and it points into the page.

- **Equate $\boldsymbol{\tau}$ with $d\mathbf{L}/dt$:** The vectors $d\mathbf{L}/dt$ and $\boldsymbol{\tau}$ both point into the page, which is good, because they had better point in the same direction. Equating their magnitudes gives

$$\frac{m\ell^2\omega^2 \sin \theta \cos \theta}{3} = \frac{mg\ell \sin \theta}{2} \implies \omega = \sqrt{\frac{3g}{2\ell \cos \theta}}. \quad (9.38)$$

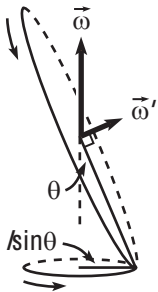


Fig. 9.22

REMARKS:

1. This frequency is slightly larger than the frequency that would arise if we instead had a mass on the end of a massless stick of length ℓ . From Problem 9.12, the frequency in that case is $\sqrt{g/\ell \cos \theta}$. So, in some sense, a uniform stick of length ℓ behaves like a mass on the end of a massless stick of length $2\ell/3$, as far as these rotations are concerned.
2. As $\theta \rightarrow \pi/2$, the frequency goes to ∞ , which makes sense. And as $\theta \rightarrow 0$, it approaches $\sqrt{3g/2\ell}$, which isn't so obvious.
3. As explained in Problem 9.1, the instantaneous $\boldsymbol{\omega}$ is not uniquely defined in some situations. At the instant shown in Fig. 9.20, the stick is moving directly into the page. What if someone else wants to think of the stick as (instantaneously) rotating around the $\boldsymbol{\omega}'$ axis perpendicular to the stick (the x axis, in the above notation), instead of the vertical axis, as shown in Fig. 9.22. What is the angular speed ω' ?

Well, if ω is the angular speed of the stick around the vertical axis, then we may view the tip of the stick as instantaneously moving in a circle of radius $\ell \sin \theta$ around the

vertical axis ω . So $\omega(\ell \sin \theta)$ is the speed of the tip of the stick. But we may also view the tip of the stick as instantaneously moving in a circle of radius ℓ around ω' , as shown. The speed of the tip is still $\omega(\ell \sin \theta)$, so the angular speed around this axis is given by $\omega' \ell = \omega(\ell \sin \theta)$. Hence $\omega' = \omega \sin \theta$, which is simply the x component of ω that we found above, right before Eq. (9.34). The moment of inertia around ω' is $m\ell^2/3$, so the angular momentum has magnitude $(m\ell^2/3)(\omega \sin \theta)$, in agreement with Eq. (9.34). And the direction is along the x axis, as it should be.

Note that although ω is not uniquely defined at any instant, $\mathbf{L} \equiv \int (\mathbf{r} \times \mathbf{p}) dm$ certainly is.⁹ Choosing ω to point vertically, as we did in the above solution, is in some sense the natural choice, because this ω doesn't change with time. ♣

9.5 Euler's equations

Consider a rigid body instantaneously rotating around an axis ω . This ω may change as time goes on, but all we care about for now is what it is at a given instant. The angular momentum is given by Eq. (9.8) as $\mathbf{L} = \mathbf{I}\omega$, where $\mathbf{I}$ is the inertia tensor, calculated with respect to a given origin and a given set of axes (and ω is written in the same basis, of course).

As usual, things are much nicer if we use the principal axes (relative to the chosen origin) as the basis vectors of our coordinate system. Since these axes are fixed with respect to the rotating object, they will rotate with respect to the fixed reference frame. In this basis, $\mathbf{L}$ takes the nice form,

$$\mathbf{L} = (I_1\omega_1, I_2\omega_2, I_3\omega_3), \quad (9.39)$$

where ω_1 , ω_2 , and ω_3 are the components of ω along the principal axes. In other words, if you take the vector $\mathbf{L}$ in space and project it onto the instantaneous principal axes, then you get the components in Eq. (9.39).

On one hand, writing $\mathbf{L}$ in terms of the rotating principal axes allows us to write it in the nice form of Eq. (9.39). But on the other hand, writing $\mathbf{L}$ in this way makes it nontrivial to determine how it changes in time, because the principal axes themselves are changing. However, it turns out that the benefits outweigh the detriments, so we will invariably use the principal axes as our basis vectors.

The goal of this section is to find an expression for $d\mathbf{L}/dt$, and to then equate this with the torque. The result will be Euler's equations in Eq. (9.45).

Derivation of Euler's equations

If we write $\mathbf{L}$ in terms of the body frame, which we'll choose to be described by the principal axes painted on the body, then $\mathbf{L}$ can change (relative to the lab frame) due to two effects. It can change because its coordinates in the body frame change, and it can also change because of the rotation of the body frame. To be precise, let $\mathbf{L}_0$ be the vector $\mathbf{L}$ at a given instant. At this instant, imagine painting the vector $\mathbf{L}_0$ onto the body frame, so that $\mathbf{L}_0$ then rotates with the body. The rate

⁹ The nonuniqueness of ω arises from the fact that $I_y = 0$ here. If all the moments are nonzero, then $(L_x, L_y, L_z) = (I_x\omega_x, I_y\omega_y, I_z\omega_z)$ uniquely determines ω , given $\mathbf{L}$.

of change of $\mathbf{L}$ with respect to the lab frame may be written in the (identically true) way,

$$\frac{d\mathbf{L}}{dt} = \frac{d(\mathbf{L} - \mathbf{L}_0)}{dt} + \frac{d\mathbf{L}_0}{dt}. \quad (9.40)$$

The second term here is simply the rate of change of a body-fixed vector, which we know is $\boldsymbol{\omega} \times \mathbf{L}_0$, which equals $\boldsymbol{\omega} \times \mathbf{L}$ at this instant. The first term is the rate of change of $\mathbf{L}$ with respect to the body frame, which we'll denote by $\delta\mathbf{L}/\delta t$. This is what someone standing fixed on the body measures. So we end up with

$$\frac{d\mathbf{L}}{dt} = \frac{\delta\mathbf{L}}{\delta t} + \boldsymbol{\omega} \times \mathbf{L}. \quad (9.41)$$

This is actually a general statement, true for any vector in any rotating frame (we'll derive it in another more mathematical way in Chapter 10). There was nothing particular about $\mathbf{L}$ that we used in the above derivation. Also, there was no need to restrict ourselves to principal axes. In words, what we've shown is that the total change equals the change relative to the rotating frame, plus the change of the rotating frame relative to the fixed frame. This is just the usual way of adding velocities when one frame moves with respect to another.

Let us now make use of our choice of the principal axes as the body axes. This will put Eq. (9.41) in a usable form. Using Eq. (9.39), we can rewrite Eq. (9.41) as

$$\frac{d\mathbf{L}}{dt} = \frac{d}{dt}(I_1\omega_1, I_2\omega_2, I_3\omega_3) + (\omega_1, \omega_2, \omega_3) \times (I_1\omega_1, I_2\omega_2, I_3\omega_3). \quad (9.42)$$

The $\delta\mathbf{L}/\delta t$ term does indeed equal $(d/dt)(I_1\omega_1, I_2\omega_2, I_3\omega_3)$, because someone in the body frame measures the components of $\mathbf{L}$ with respect to the principal axes to be $(I_1\omega_1, I_2\omega_2, I_3\omega_3)$. And $\delta\mathbf{L}/\delta t$ is by definition the rate at which these components change.

Equation (9.42) equates two vectors. As is true for any vector, these (equal) vectors have an existence that is independent of the coordinate system we choose to describe them with (Eq. (9.41) makes no reference to a coordinate system). But since we've chosen an explicit frame on the right-hand side of Eq. (9.42), we should choose the same frame for the left-hand side. We can then equate the components on the left with the components on the right. Projecting $d\mathbf{L}/dt$ onto the instantaneous principal axes, we have

$$\left(\left(\frac{d\mathbf{L}}{dt} \right)_1, \left(\frac{d\mathbf{L}}{dt} \right)_2, \left(\frac{d\mathbf{L}}{dt} \right)_3 \right) = \frac{d}{dt}(I_1\omega_1, I_2\omega_2, I_3\omega_3) + (\omega_1, \omega_2, \omega_3) \times (I_1\omega_1, I_2\omega_2, I_3\omega_3). \quad (9.43)$$

REMARK: The left-hand side looks nastier than it really is. The reason we've written it in this cumbersome way is the following (this is a remark that has to be read very slowly). We

could have written the left-hand side as $(d/dt)(L_1, L_2, L_3)$, but this might cause confusion as to whether the L_i refer to the components with respect to the rotating axes, or the components with respect to the fixed set of axes that coincide with the rotating principal axes at this instant. That is, do we project $\mathbf{L}$ onto the principal axes to obtain components, and then take the derivative of these components? Or do we take the derivative of $\mathbf{L}$ and then project onto the principal axes to obtain components? The latter is what we mean in Eq. (9.43).¹⁰ The way we've written the left-hand side of Eq. (9.43), it's clear that we're taking the derivative first. We are, after all, simply projecting Eq. (9.41) onto the principal axes. ♣

The time derivatives on the right-hand side of Eq. (9.43) are $d(I_1\omega_1)/dt = I_1\dot{\omega}_1$, etc., because the I 's are constant. Performing the cross product and equating the corresponding components on each side yields the three equations,

$$\begin{aligned}\left(\frac{d\mathbf{L}}{dt}\right)_1 &= I_1\dot{\omega}_1 + (I_3 - I_2)\omega_3\omega_2, \\ \left(\frac{d\mathbf{L}}{dt}\right)_2 &= I_2\dot{\omega}_2 + (I_1 - I_3)\omega_1\omega_3, \\ \left(\frac{d\mathbf{L}}{dt}\right)_3 &= I_3\dot{\omega}_3 + (I_2 - I_1)\omega_2\omega_1.\end{aligned}\tag{9.44}$$

We will now invoke the results of Section 8.4.3 to say that if we have chosen the origin of our rotating frame to be either a fixed point or the CM (as we always do), then we can equate $d\mathbf{L}/dt$ with the torque, $\boldsymbol{\tau}$. We therefore have

$$\begin{aligned}\tau_1 &= I_1\dot{\omega}_1 + (I_3 - I_2)\omega_3\omega_2, \\ \tau_2 &= I_2\dot{\omega}_2 + (I_1 - I_3)\omega_1\omega_3, \\ \tau_3 &= I_3\dot{\omega}_3 + (I_2 - I_1)\omega_2\omega_1.\end{aligned}\tag{9.45}$$

These are *Euler's equations*. You need to remember only one of them, because the other two can be obtained by cyclic permutation of the indices.

REMARKS:

1. We repeat that the left- and right-hand sides of Eqs. (9.45) are components that are measured with respect to the instantaneous principal axes. Let's say we do a problem, for example, where τ_3 has a constant nonzero value, and τ_1 and τ_2 are always zero (as in the example in Section 9.4.2). This doesn't mean that $\boldsymbol{\tau}$ is a constant vector. On the contrary, $\boldsymbol{\tau}$ always points along the $\hat{\mathbf{x}}_3$ vector in the rotating frame, but this vector is changing in the fixed frame (unless $\hat{\mathbf{x}}_3$ points along $\boldsymbol{\omega}$).
2. The two types of terms on the right-hand sides of Eqs. (9.44) are the two types of changes that $\mathbf{L}$ can undergo. $\mathbf{L}$ can change because its components with respect to the rotating frame change, and $\mathbf{L}$ can also change because the body is rotating around $\boldsymbol{\omega}$.

¹⁰ The former is $\delta\mathbf{L}/\delta t$, by definition. The two interpretations certainly give different results. For example, if instead of $\mathbf{L}$ we consider a vector fixed in the body (such as the $\mathbf{L}_0$ above), then the first interpretation gives a zero result, whereas the second interpretation gives a nonzero result. Considering what we mean by, say, the vector $(\omega_1, \omega_2, \omega_3)$, I think that the more logical interpretation of $(d/dt)(L_1, L_2, L_3)$ is the first one, so it should definitely be avoided.

3. Section 9.6.1 on the free symmetric top (viewed from the body frame) provides a good example of the use of Euler's equations. Another interesting application is the famed "Tennis racket theorem" (Problem 9.14).
4. It should be noted that you never *have* to use Euler's equations. You can simply start from scratch and use Eq. (9.41) each time you solve a problem. The point is that we've done the calculation of $d\mathbf{L}/dt$ once and for all, so you can just invoke the result in Eqs. (9.45). ♣

9.6 Free symmetric top

The free symmetric top is the classic example of an application of Euler's equations. Consider an object that has two of its principal moments equal, with the CM as the origin. Assume that the object is in outer space, far from any external forces. We will choose the object to have cylindrical symmetry around some axis (see Fig. 9.23), although this is not necessary; a square cross section, for example, would yield two equal moments. The principal axes are then the symmetry axis and any two orthogonal axes in the cross-section plane through the CM. Let the symmetry axis be chosen as the $\hat{\mathbf{x}}_3$ axis. Then the moments are $I_1 = I_2 \equiv I$, and I_3 .

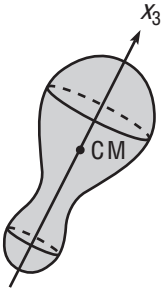


Fig. 9.23

We'll first look at things from the point of view of someone standing at rest on the body, and then we'll look at things from the point of view of someone standing at rest in an inertial frame. The mathematics involved here isn't so bad, but as with most types of top problems, it's hard to get an intuitive feel for what all the various vectors are doing. And the intuition is even more difficult in the following body-frame analysis because of the noninertial frame of reference. But let's see what we find.

9.6.1 View from the body frame

Plugging $I_1 = I_2 \equiv I$ into Euler's equations in Eq. (9.45), and using the fact that all the τ_i are zero (since there are no torques, because the top is "free"), we have

$$\begin{aligned} 0 &= I\dot{\omega}_1 + (I_3 - I)\omega_3\omega_2, \\ 0 &= I\dot{\omega}_2 + (I - I_3)\omega_1\omega_3, \\ 0 &= I_3\dot{\omega}_3. \end{aligned} \tag{9.46}$$

The last equation says that ω_3 is constant. If we then define

$$\Omega \equiv \left(\frac{I_3 - I}{I} \right) \omega_3, \tag{9.47}$$

the first two equations become

$$\dot{\omega}_1 + \Omega\omega_2 = 0, \quad \text{and} \quad \dot{\omega}_2 - \Omega\omega_1 = 0. \tag{9.48}$$

Taking the derivative of the first of these, and then using the second to eliminate $\dot{\omega}_2$, gives

$$\ddot{\omega}_1 + \Omega^2\omega_1 = 0, \tag{9.49}$$

and likewise for ω_2 . This is a good old simple-harmonic-oscillator equation. We can therefore write $\omega_1(t)$ as, say, a cosine. And then Eq. (9.48) yields a sine for $\omega_2(t)$. So we have

$$\omega_1(t) = A \cos(\Omega t + \phi), \quad \omega_2(t) = A \sin(\Omega t + \phi). \quad (9.50)$$

We see that $\omega_1(t)$ and $\omega_2(t)$ are the components of a circle in the body frame. Therefore, the ω vector traces out a cone around $\hat{x}_3$ (see Fig. 9.24), with frequency Ω , as viewed by someone standing on the body. This frequency Ω in Eq. (9.47) depends on the value of ω_3 and on the geometry of the object (through I_3 and I). But the radius, A , of the ω cone is determined by the initial values of ω_1 and ω_2 .

The angular momentum is

$$\mathbf{L} = (I_1\omega_1, I_2\omega_2, I_3\omega_3) = (IA \cos(\Omega t + \phi), IA \sin(\Omega t + \phi), I_3\omega_3), \quad (9.51)$$

so $\mathbf{L}$ also traces out a cone around $\hat{x}_3$ with frequency Ω , as viewed by someone standing on the body. This is shown in Fig. 9.24 for the case $\Omega > 0$ (that is, $I_3 > I$). In this case, $I_3 > I$ implies $L_3/L_2 > \omega_3/\omega_2$, so the $\mathbf{L}$ vector lies above the ω vector (that is, between ω and $\hat{x}_3$). An object with $I_3 > I$, such as a coin, is called an *oblate* top.

Figure 9.25 shows the case where $\Omega < 0$ (that is, $I_3 < I$). In this case, $I_3 < I$ implies $L_3/L_2 < \omega_3/\omega_2$, so the $\mathbf{L}$ vector lies below the ω vector, as shown. And since Ω is negative, the ω and $\mathbf{L}$ vectors precess around $\hat{x}_3$ in the opposite direction (clockwise, as viewed from above). An object with $I_3 < I$, such as a carrot, is called a *prolate* top.

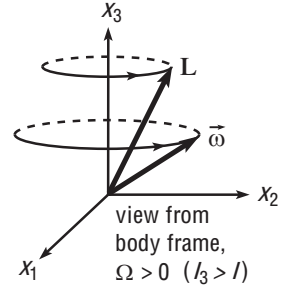


Fig. 9.24

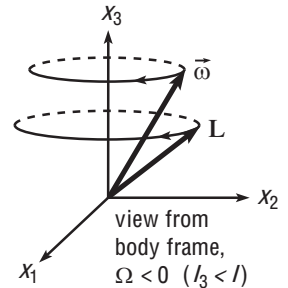


Fig. 9.25

Example (The earth): Let's consider the earth to be our top. Then $\omega_3 \approx 2\pi/(1 \text{ day})$.¹¹ The bulge at the equator (caused by the spinning of the earth) makes I_3 slightly larger than I , and it turns out that $(I_3 - I)/I \approx 1/320$. Therefore, Eq. (9.47) gives $\Omega \approx (1/320) 2\pi/(1 \text{ day})$. So the ω vector should precess around in its cone once every 320 days, as viewed by someone on the earth. The true value is more like 430 days. The difference has to do with various things, including the nonrigidity of the earth, but at least we got an answer in the right ballpark. This precession of ω is known as the "Chandler wobble."

In practice, how can we determine the direction of ω ? Simply take an extended-time photograph exposure at night. The stars will form arcs of circles. At the center of all these circles is a point that doesn't move. This is the direction of ω . Fortunately, Ω is much smaller than ω , so the ω vector doesn't change much during an exposure time of, say, an hour. So the center of the circles is essentially well defined.

How big is the ω cone, for the earth? Equivalently, what is the value of A in Eq. (9.50)? Observation has shown that the ω vector pierces the earth at a point on

¹¹ This isn't quite correct, since the earth rotates 366 times for every 365 days, due to the motion around the sun, but it's close enough for the purposes here.

the order of 10 m from the north pole, although this distance fluctuates over time.¹² Hence, $A/\omega_3 \approx (10 \text{ m})/R_E$. The half-angle of the $\boldsymbol{\omega}$ cone is therefore on the order of only 10^{-4} degrees. So if you use an extended-time photograph exposure one night to see which point in the sky stands still, and then if you do the same thing 200 nights later, you probably won't be able to tell that they're really two different points.

9.6.2 View from a fixed frame

Let's now see what our symmetric top looks like from a fixed frame. Euler's equations won't help much here, because they deal with the components of $\boldsymbol{\omega}$ in the body frame. But fortunately we can solve for the motion from scratch. In terms of the (changing) principal axes, $\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \hat{\mathbf{x}}_3$, we have

$$\boldsymbol{\omega} = (\omega_1 \hat{\mathbf{x}}_1 + \omega_2 \hat{\mathbf{x}}_2) + \omega_3 \hat{\mathbf{x}}_3, \quad (9.52)$$

$$\mathbf{L} = I(\omega_1 \hat{\mathbf{x}}_1 + \omega_2 \hat{\mathbf{x}}_2) + I_3 \omega_3 \hat{\mathbf{x}}_3.$$

Eliminating the $(\omega_1 \hat{\mathbf{x}}_1 + \omega_2 \hat{\mathbf{x}}_2)$ term from these equations gives (in terms of the Ω defined in Eq. (9.47))

$$\mathbf{L} = I(\boldsymbol{\omega} + \Omega \hat{\mathbf{x}}_3) \implies \boldsymbol{\omega} = \frac{L}{I} \hat{\mathbf{L}} - \Omega \hat{\mathbf{x}}_3, \quad (9.53)$$

where $L = |\mathbf{L}|$, and $\hat{\mathbf{L}}$ is the unit vector in the $\mathbf{L}$ direction. The linear relationship among $\mathbf{L}$, $\boldsymbol{\omega}$, and $\hat{\mathbf{x}}_3$ implies that these three vectors lie in a plane. But $\mathbf{L}$ remains fixed, because there are no torques on the system. Therefore, $\boldsymbol{\omega}$ and $\hat{\mathbf{x}}_3$ precess (as we'll see below) around $\mathbf{L}$, with the three vectors always coplanar. See Fig. 9.26 for the case $\Omega > 0$, that is, $I_3 > I$ (an oblate top), and Fig. 9.27 for the case $\Omega < 0$, that is, $I_3 < I$ (a prolate top).

What is the frequency of this precession, as viewed from the fixed frame? The rate of change of $\hat{\mathbf{x}}_3$ is $\boldsymbol{\omega} \times \hat{\mathbf{x}}_3$, because $\hat{\mathbf{x}}_3$ is fixed in the body frame, so its change comes only from rotation around $\boldsymbol{\omega}$. Therefore, Eq. (9.53) gives

$$\frac{d\hat{\mathbf{x}}_3}{dt} = \left(\frac{L}{I} \hat{\mathbf{L}} - \Omega \hat{\mathbf{x}}_3 \right) \times \hat{\mathbf{x}}_3 = \left(\frac{L}{I} \hat{\mathbf{L}} \right) \times \hat{\mathbf{x}}_3. \quad (9.54)$$

But this is simply the expression for the rate of change of a vector rotating around the fixed vector $\tilde{\boldsymbol{\omega}} \equiv (L/I) \hat{\mathbf{L}}$. The frequency of this rotation is $|\tilde{\boldsymbol{\omega}}| = L/I$. Therefore, $\hat{\mathbf{x}}_3$ precesses around the fixed vector $\mathbf{L}$ with frequency

$$\tilde{\omega} = \frac{L}{I}, \quad (9.55)$$

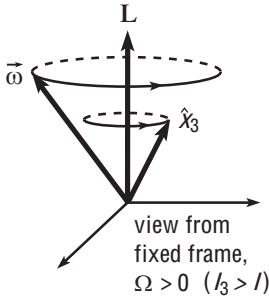


Fig. 9.26

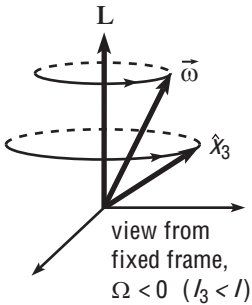


Fig. 9.27

¹² This distance could theoretically be much larger or much smaller than 10 m. It happens to be of this order due to the nature of the driving force. The present consensus for this force is pressure changes at the bottom of the ocean and in the atmosphere; see Gross (2000). Without a driving force, the amplitude would head to zero, due to the nonrigidity of the earth.

in the fixed frame. And therefore $\boldsymbol{\omega}$ does also, because it is coplanar with $\hat{\mathbf{x}}_3$ and $\mathbf{L}$.

REMARKS:

1. We just found that $\boldsymbol{\omega}$ precesses around $\mathbf{L}$ with frequency L/I . What, then, is wrong with the following reasoning: “Just as the rate of change of $\hat{\mathbf{x}}_3$ equals $\boldsymbol{\omega} \times \hat{\mathbf{x}}_3$, the rate of change of $\boldsymbol{\omega}$ should equal $\boldsymbol{\omega} \times \boldsymbol{\omega}$, which is zero. Therefore, $\boldsymbol{\omega}$ should remain constant.” The error is that the vector $\boldsymbol{\omega}$ is not fixed in the body frame. A vector $\mathbf{A}$ must be fixed in the body frame in order for its rate of change to be given by $\boldsymbol{\omega} \times \mathbf{A}$.
2. We found in Eqs. (9.51) and (9.47) that a person standing on the rotating body sees $\mathbf{L}$ (and $\boldsymbol{\omega}$) precess with frequency $\Omega \equiv \omega_3(I_3 - I)/I$ around $\hat{\mathbf{x}}_3$. But we found in Eq. (9.55) that a person standing in the fixed frame sees $\hat{\mathbf{x}}_3$ (and $\boldsymbol{\omega}$) precess with frequency L/I around $\mathbf{L}$. Are these two facts compatible? Should we have obtained the same frequency from either point of view? (Answers: yes, no).

These two frequencies are indeed consistent, as can be seen by the following reasoning. Consider the plane (call it S) containing the three vectors $\mathbf{L}$, $\boldsymbol{\omega}$, and $\hat{\mathbf{x}}_3$. We know from Eq. (9.51) that S rotates with frequency $\Omega\hat{\mathbf{x}}_3$ with respect to the body. Therefore, the body rotates with frequency $-\Omega\hat{\mathbf{x}}_3$ with respect to S . And from Eq. (9.55), S rotates with frequency $(L/I)\hat{\mathbf{L}}$ with respect to the fixed frame. Therefore, the total angular velocity of the body with respect to the fixed frame is (using the frame S as an intermediate step)

$$\boldsymbol{\omega}_{\text{total}} = \frac{L}{I}\hat{\mathbf{L}} - \Omega\hat{\mathbf{x}}_3. \quad (9.56)$$

But from Eq. (9.53), this is simply $\boldsymbol{\omega}$, as it should be. So the two frequencies in Eqs. (9.47) and (9.55) are indeed consistent.

For the earth, I_3 and I are nearly the same, so $\Omega \equiv \omega_3(I_3 - I)/I$ and L/I are quite different. L/I is roughly equal to L/I_3 , which is essentially equal to ω_3 . On the other hand, Ω is roughly equal to $(1/300)\omega_3$. Basically, an external observer sees $\boldsymbol{\omega}$ precess around its cone at roughly the rate at which the earth spins. But it's not exactly the same rate, and this difference is what causes the earth-based observer to see $\boldsymbol{\omega}$ precess with a nonzero Ω .

3. The fixed-frame precession of $\hat{\mathbf{x}}_3$ around $\mathbf{L}$ should not be confused with the “precession of the equinoxes” effect. See Problem 10.15 for a discussion of the latter. ♣

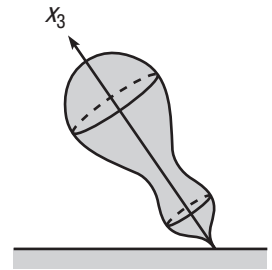


Fig. 9.28

9.7 Heavy symmetric top

Consider now a heavy symmetrical top, that is, one that spins on a table, under the influence of gravity (see Fig. 9.28). Assume that the tip of the top is fixed on the table by a pivot. We'll solve for the motion of the top in two different ways below in Sections 9.7.3 and 9.7.4. The first uses $\boldsymbol{\tau} = d\mathbf{L}/dt$, and the second uses the Lagrangian method.

9.7.1 Euler angles

For both of these methods, it is convenient to use the *Euler angles*, θ , ϕ , ψ , which are shown in Fig. 9.29 and defined as follows.

- θ : Let $\hat{\mathbf{x}}_3$ be the symmetry axis of the top. Define θ to be the angle that $\hat{\mathbf{x}}_3$ makes with the vertical $\hat{\mathbf{z}}$ axis of the fixed frame.

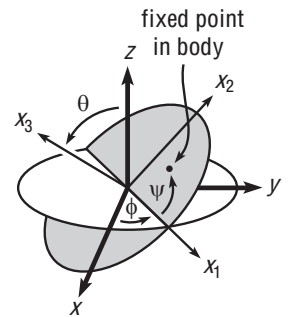


Fig. 9.29

- ϕ : Draw the plane orthogonal to $\hat{\mathbf{x}}_3$. Let $\hat{\mathbf{x}}_1$ be the intersection of this plane with the horizontal x - y plane. Define ϕ to be the angle that $\hat{\mathbf{x}}_1$ makes with the $\hat{\mathbf{x}}$ axis in the fixed frame. Note that $\hat{\mathbf{x}}_1$ is not necessarily fixed in the object.
- ψ : Let $\hat{\mathbf{x}}_2$ be orthogonal to $\hat{\mathbf{x}}_3$ and $\hat{\mathbf{x}}_1$, as shown. As with $\hat{\mathbf{x}}_1$, $\hat{\mathbf{x}}_2$ is not necessarily fixed in the object. Let frame S be the frame whose axes are $\hat{\mathbf{x}}_1$, $\hat{\mathbf{x}}_2$, and $\hat{\mathbf{x}}_3$. Define ψ to be the angle of rotation of the body around the $\hat{\mathbf{x}}_3$ axis in frame S . So $\dot{\psi}\hat{\mathbf{x}}_3$ is the angular velocity of the body with respect to S . And from the figure, we also see that the angular velocity of frame S with respect to the fixed frame is $\dot{\phi}\hat{\mathbf{z}} + \dot{\theta}\hat{\mathbf{x}}_1$.

The angular velocity of the body with respect to the fixed frame is equal to the angular velocity of the body with respect to frame S , plus the angular velocity of frame S with respect to the fixed frame. From above, we therefore have

$$\boldsymbol{\omega} = \dot{\psi}\hat{\mathbf{x}}_3 + (\dot{\phi}\hat{\mathbf{z}} + \dot{\theta}\hat{\mathbf{x}}_1). \quad (9.57)$$

It is often more convenient to rewrite $\boldsymbol{\omega}$ entirely in terms of the orthogonal $\hat{\mathbf{x}}_1$, $\hat{\mathbf{x}}_2$, $\hat{\mathbf{x}}_3$ basis vectors. Since $\hat{\mathbf{z}} = \cos\theta\hat{\mathbf{x}}_3 + \sin\theta\hat{\mathbf{x}}_2$, Eq. (9.57) gives

$$\boldsymbol{\omega} = (\dot{\psi} + \dot{\phi}\cos\theta)\hat{\mathbf{x}}_3 + \dot{\phi}\sin\theta\hat{\mathbf{x}}_2 + \dot{\theta}\hat{\mathbf{x}}_1. \quad (9.58)$$

This form of $\boldsymbol{\omega}$ is generally more useful, because $\hat{\mathbf{x}}_1$, $\hat{\mathbf{x}}_2$, $\hat{\mathbf{x}}_3$ are principal axes of the body. (We are assuming that we are working with a symmetrical top, with $I_1 = I_2 \equiv I$. This means that any axes in the $\hat{\mathbf{x}}_1$ - $\hat{\mathbf{x}}_2$ plane are principal axes.) Although $\hat{\mathbf{x}}_1$ and $\hat{\mathbf{x}}_2$ are not fixed in the object, they are still good principal axes at any instant.

9.7.2 Digression on the components of $\boldsymbol{\omega}$

The above expressions for $\boldsymbol{\omega}$ might look a little scary, but there is a very helpful diagram we can draw (see Fig. 9.30) that makes it easier to see what's going on. Let's talk a bit about this before tackling the original problem of the spinning top. The diagram is rather dense (you might even say it looks scarier than the above $\boldsymbol{\omega}$), so we'll go through it slowly. In the following discussion, we'll simplify things by setting $\dot{\theta} = 0$. All the interesting features of $\boldsymbol{\omega}$ remain. The $\dot{\theta}\hat{\mathbf{x}}_1$ component of $\boldsymbol{\omega}$ in Eqs. (9.57) and (9.58) arises simply from the easily visualizable rising and falling of the top. We will therefore concentrate on the more complicated issues, namely the components of $\boldsymbol{\omega}$ in the plane of $\hat{\mathbf{x}}_3$, $\hat{\mathbf{z}}$, and $\hat{\mathbf{x}}_2$.

With $\dot{\theta} = 0$, Fig. 9.30 shows the vector $\boldsymbol{\omega}$ in the $\hat{\mathbf{x}}_3$ - $\hat{\mathbf{z}}$ - $\hat{\mathbf{x}}_2$ plane (the way we've drawn it, $\hat{\mathbf{x}}_1$ points into the page, in contrast with Fig. 9.29). We'll refer to this figure many times in the problems for this chapter. There are numerous comments to be made about it, so we'll just list them out. The following discussion deals with the *kinematics* of $\boldsymbol{\omega}$, that is, the meaning of the various components and how they relate to each other. The discussion of the *dynamics* of $\boldsymbol{\omega}$, that is, why the components take on the values they do, given a certain physical system, is the subject of Section 9.7.3 onward.

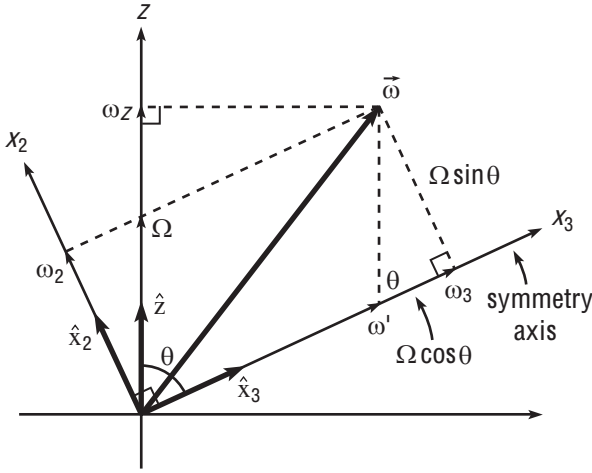


Fig. 9.30

1. If someone asks you to “decompose” ω into pieces along $\hat{z}$ and $\hat{x}_3$, what would you do? Would you draw the lines perpendicular to these axes to obtain the lengths shown (which we’ll label as ω_z and ω_3), or would you draw the lines parallel to these axes to obtain the lengths shown (which we’ll label as Ω and ω')? There is no “correct” answer to this question. The four quantities, ω_z , ω_3 , Ω , ω' simply represent different things. We will interpret each of these below, along with ω_2 (the projection of ω along $\hat{x}_2$). It turns out that Ω and ω' are the frequencies that your eye can see the easiest, while ω_2 and ω_3 are what you want to use when doing calculations involving the angular momentum. But as far as I can see, ω_z is not of much use.
2. Note that it is true that

$$\omega = \omega' \hat{x}_3 + \Omega \hat{z}, \quad (9.59)$$

but it is *not* true that $\omega = \omega_3 \hat{x}_3 + \omega_z \hat{z}$. Another true statement is

$$\omega = \omega_3 \hat{x}_3 + \omega_2 \hat{x}_2. \quad (9.60)$$

3. In terms of the Euler angles, we see by comparing Eqs. (9.59) and (9.57), with $\dot{\theta} = 0$, that

$$\omega' = \dot{\psi}, \quad \text{and} \quad \Omega = \dot{\phi}. \quad (9.61)$$

And we also have, by comparing Eqs. (9.60) and (9.58), with $\dot{\theta} = 0$,

$$\begin{aligned} \omega_3 &= \dot{\psi} + \dot{\phi} \cos \theta = \omega' + \Omega \cos \theta, \\ \omega_2 &= \dot{\phi} \sin \theta = \Omega \sin \theta. \end{aligned} \quad (9.62)$$

These are also clear from Fig. 9.30. There is therefore technically no need to introduce the new ω_2 , ω_3 , Ω , ω' definitions in Fig. 9.30, because the Euler angles are quite

sufficient. But we will be referring back to this figure many times, and it is a little easier to work with these omegas than the various combinations of the Euler angles.

4. Ω is the easiest of the frequencies to visualize. It is the frequency of precession of the top around the vertical $\hat{\mathbf{z}}$ axis.¹³ In other words, the symmetry axis $\hat{\mathbf{x}}_3$ traces out a cone (assuming $\dot{\theta} = 0$) around the $\hat{\mathbf{z}}$ axis with frequency Ω . The reason for this is the following. The vector $\boldsymbol{\omega}$ is the vector that gives the speed of any point (at position $\mathbf{r}$) fixed in the top as $\boldsymbol{\omega} \times \mathbf{r}$. Therefore, since the vector $\hat{\mathbf{x}}_3$ is fixed in the top, we can write

$$\frac{d\hat{\mathbf{x}}_3}{dt} = \boldsymbol{\omega} \times \hat{\mathbf{x}}_3 = (\omega' \hat{\mathbf{x}}_3 + \Omega \hat{\mathbf{z}}) \times \hat{\mathbf{x}}_3 = (\Omega \hat{\mathbf{z}}) \times \hat{\mathbf{x}}_3. \quad (9.63)$$

But this is precisely the expression for the rate of change of a vector rotating around the $\hat{\mathbf{z}}$ axis with frequency Ω . (This is exactly the same type of proof as the one leading to Eq. (9.54).) Note that the precession frequency around the $\hat{\mathbf{z}}$ axis is *not* ω_z . It certainly can't be ω_z , because we can imagine grabbing the symmetry axis and holding it in place, so that $\boldsymbol{\omega}$ points along $\hat{\mathbf{x}}_3$. This scenario has a nonzero ω_z , but no precession.

REMARK: In the derivation of Eq. (9.63), we basically just stripped off the part of $\boldsymbol{\omega}$ that points along the $\hat{\mathbf{x}}_3$ axis, because a rotation around $\hat{\mathbf{x}}_3$ contributes nothing to the motion of $\hat{\mathbf{x}}_3$. Note, however, that there is in fact an infinite number of ways to strip off a piece along $\hat{\mathbf{x}}_3$. For example, we can also break $\boldsymbol{\omega}$ up as, say, $\boldsymbol{\omega} = \omega_3 \hat{\mathbf{x}}_3 + \omega_2 \hat{\mathbf{x}}_2$. We then obtain $d\hat{\mathbf{x}}_3/dt = (\omega_2 \hat{\mathbf{x}}_2) \times \hat{\mathbf{x}}_3$, which means that $\hat{\mathbf{x}}_3$ is instantaneously rotating around $\hat{\mathbf{x}}_2$ with frequency ω_2 . Although this is true, it isn't as useful as the result in Eq. (9.63), because the $\hat{\mathbf{x}}_2$ axis changes with time (it precesses around $\hat{\mathbf{z}}$). The point here is that the instantaneous angular velocity vector around which the symmetry axis rotates is not well defined (Problem 9.1 discusses this issue).¹⁴ But the $\hat{\mathbf{z}}$ axis is the only one of these angular velocity vectors that is fixed. When we look at the top (or more precisely, the symmetry axis), we therefore see it precessing around the $\hat{\mathbf{z}}$ axis. ♣

5. ω' is also easy to visualize. Imagine that you are at rest in the frame that rotates around the $\hat{\mathbf{z}}$ axis with frequency Ω . Then you see the symmetry axis of the top remain perfectly still, and the only motion you see is the top spinning around this axis with frequency ω' . (This is true because $\boldsymbol{\omega} = \omega' \hat{\mathbf{x}}_3 + \Omega \hat{\mathbf{z}}$, and the rotation of your frame causes you not to see the $\Omega \hat{\mathbf{z}}$ part.) If you paint a dot somewhere on the top, then the dot traces out a fixed tilted circle, and the dot returns to, say, its maximum height at frequency ω' . A person in the lab frame sees this dot undergo a rather complicated motion but must observe the same frequency at which the dot returns to its highest point. So ω' is something quite physical in the lab frame also.
6. ω_3 is what you use to obtain the component of $\mathbf{L}$ along $\hat{\mathbf{x}}_3$, because $L_3 = I_3 \omega_3$. ω_3 is a little harder to visualize than Ω and ω' , but it is the frequency with which the top

¹³ Although we're using the same letter, this Ω doesn't have anything to do with the Ω defined in Eq. (9.47), except for the fact that they both represent the frequency of something precessing around an axis.

¹⁴ The instantaneous angular velocity of the *whole body* is well defined, of course. There is a definite line of points in the body that are instantaneously at rest. But if you look at the symmetry axis by itself, then there is an ambiguity (see Footnote 9). In short, because only one point on the axis (the bottom end), instead of a whole line, is instantaneously at rest, the instantaneous angular velocity vector can point in any direction.

instantaneously rotates, as seen by someone at rest in the frame that rotates around the instantaneous $\hat{\mathbf{x}}_2$ axis with frequency ω_2 . (This is true because $\boldsymbol{\omega} = \omega_2 \hat{\mathbf{x}}_2 + \omega_3 \hat{\mathbf{x}}_3$, and the rotation of the frame causes the person not to see the $\omega_2 \hat{\mathbf{x}}_2$ part.) This rotation is harder to visualize in the lab frame, because the $\hat{\mathbf{x}}_2$ axis changes with time.

There is one physical scenario in which ω_3 is the easily observed frequency. Imagine that the top is precessing around the $\hat{\mathbf{z}}$ axis at constant θ (we'll find in Section 9.7.5 that this is in fact a possible motion for the top), and imagine that the top has a frictionless rod protruding along its symmetry axis. If you grab the rod and stop the precession motion, so that the top is now spinning around its stationary symmetry axis, then this spinning has frequency ω_3 . This is true because when you grab the rod, your torque has no component along the $\hat{\mathbf{x}}_3$ axis (because the rod lies along this axis, and because it is frictionless). Therefore, L_3 doesn't change, and so neither does ω_3 .

7. ω_2 is similar to ω_3 , of course. ω_2 is what you use to obtain the component of $\mathbf{L}$ along $\hat{\mathbf{x}}_2$, because $L_2 = I_2 \omega_2$. It is the frequency with which the top instantaneously rotates, as seen by someone at rest in the frame that rotates around the instantaneous $\hat{\mathbf{x}}_3$ axis with frequency ω_3 . (This is true because $\boldsymbol{\omega} = \omega_2 \hat{\mathbf{x}}_2 + \omega_3 \hat{\mathbf{x}}_3$, and the rotation of the frame causes the person not to see the $\omega_3 \hat{\mathbf{x}}_3$ part.) Again, this rotation is harder to visualize in the lab frame, because the $\hat{\mathbf{x}}_3$ axis changes with time. Note that by "instantaneous $\hat{\mathbf{x}}_3$ axis," we mean the fixed axis in space that coincides with the symmetry axis at a given instant. The symmetry axis will therefore move away from this fixed axis, consistent with the fact that the person in the above-mentioned rotating frame sees the top rotate around the $\hat{\mathbf{x}}_2$ axis.

The physical scenario that produces ω_2 , analogous to the scenario that produced ω_3 above, is the following. Imagine a frictionless rod glued to the top at its tip, perpendicular to the symmetry axis, so that they form a "T." As this rod is spinning around (ignore the fact that it has to keep passing through the table), grab it at the instant it points along $\hat{\mathbf{x}}_2$. The top will then rotate with frequency ω_2 around the fixed rod. This is true for reasons analogous to the ones in the ω_3 case above.

8. ω_z is not very useful, as far as I can see. The most important thing to note about ω_z is that it is *not* the frequency of precession around the $\hat{\mathbf{z}}$ axis, even though it is the projection of $\boldsymbol{\omega}$ onto $\hat{\mathbf{z}}$. The frequency of the precession is Ω , as we found above in Eq. (9.63). A true, but somewhat useless, fact about ω_z is that if someone is at rest in the frame that rotates around the $\hat{\mathbf{z}}$ axis with frequency ω_z , then she sees all points in the top instantaneously rotating around the horizontal $\hat{\mathbf{x}}$ axis with frequency ω_x , where ω_x is the projection of $\boldsymbol{\omega}$ onto the $\hat{\mathbf{x}}$ axis. (This is true because $\boldsymbol{\omega} = \omega_x \hat{\mathbf{x}} + \omega_z \hat{\mathbf{z}}$, and the rotation of the frame causes her not to see the $\omega_z \hat{\mathbf{z}}$ part.)

9.7.3 Torque method

Let's now finally solve for the motion of a heavy top. This first method involving torque is straightforward, although a bit tedious. We include it here to (1) show that the problem can be done without resorting to a Lagrangian, and to (2) get some practice using $\boldsymbol{\tau} = d\mathbf{L}/dt$. We'll make use of the form of $\boldsymbol{\omega}$ given in

Eq. (9.58), because there it is broken up into the principal-axis components. For convenience, define $\dot{\beta} = \dot{\psi} + \dot{\phi} \cos \theta$, so that

$$\boldsymbol{\omega} = \dot{\beta} \hat{\mathbf{x}}_3 + \dot{\phi} \sin \theta \hat{\mathbf{x}}_2 + \dot{\theta} \hat{\mathbf{x}}_1. \quad (9.64)$$

Note that we've returned to the most general motion, where $\dot{\theta}$ is not necessarily zero. For our origin, we'll choose the tip of the top, which is assumed to be fixed on the table.¹⁵ Let the principal moments relative to this origin be $I_1 = I_2 \equiv I$, and I_3 . The angular momentum of the top is then

$$\mathbf{L} = I_3 \dot{\beta} \hat{\mathbf{x}}_3 + I \dot{\phi} \sin \theta \hat{\mathbf{x}}_2 + I \dot{\theta} \hat{\mathbf{x}}_1. \quad (9.65)$$

We must now calculate $d\mathbf{L}/dt$. What makes this nontrivial is the fact that the $\hat{\mathbf{x}}_1$, $\hat{\mathbf{x}}_2$, and $\hat{\mathbf{x}}_3$ unit vectors change with time (they change with θ and ϕ). But let's forge ahead and take the derivative of Eq. (9.65). Using the product rule (which works fine with the product of a scalar and a vector), we have

$$\begin{aligned} \frac{d\mathbf{L}}{dt} &= I_3 \frac{d\dot{\beta}}{dt} \hat{\mathbf{x}}_3 + I \frac{d(\dot{\phi} \sin \theta)}{dt} \hat{\mathbf{x}}_2 + I \frac{d\dot{\theta}}{dt} \hat{\mathbf{x}}_1 \\ &\quad + I_3 \dot{\beta} \frac{d\hat{\mathbf{x}}_3}{dt} + I \dot{\phi} \sin \theta \frac{d\hat{\mathbf{x}}_2}{dt} + I \dot{\theta} \frac{d\hat{\mathbf{x}}_1}{dt}. \end{aligned} \quad (9.66)$$

Using a little geometry, you can show that

$$\begin{aligned} \frac{d\hat{\mathbf{x}}_3}{dt} &= -\dot{\theta} \hat{\mathbf{x}}_2 + \dot{\phi} \sin \theta \hat{\mathbf{x}}_1, \\ \frac{d\hat{\mathbf{x}}_2}{dt} &= \dot{\theta} \hat{\mathbf{x}}_3 - \dot{\phi} \cos \theta \hat{\mathbf{x}}_1, \\ \frac{d\hat{\mathbf{x}}_1}{dt} &= -\dot{\phi} \sin \theta \hat{\mathbf{x}}_3 + \dot{\phi} \cos \theta \hat{\mathbf{x}}_2. \end{aligned} \quad (9.67)$$

As an exercise, you should verify these by making use of Fig. 9.29. In the first equation, for example, show that a change in θ causes $\hat{\mathbf{x}}_3$ to move a certain distance in the $\hat{\mathbf{x}}_2$ direction; and show that a change in ϕ causes $\hat{\mathbf{x}}_3$ to move a certain distance in the $\hat{\mathbf{x}}_1$ direction. Plugging the derivative expressions from Eq. (9.67) into Eq. (9.66) gives, after some algebra,

$$\begin{aligned} \frac{d\mathbf{L}}{dt} &= I_3 \ddot{\beta} \hat{\mathbf{x}}_3 + \left(I \ddot{\phi} \sin \theta + 2I \dot{\theta} \dot{\phi} \cos \theta - I_3 \dot{\beta} \dot{\theta} \right) \hat{\mathbf{x}}_2 \\ &\quad + \left(I \ddot{\theta} - I \dot{\phi}^2 \sin \theta \cos \theta + I_3 \dot{\beta} \dot{\phi} \sin \theta \right) \hat{\mathbf{x}}_1. \end{aligned} \quad (9.68)$$

Let's now look at the torque on the top. This arises from gravity pulling down on the CM. So from Fig. 9.29, $\boldsymbol{\tau}$ points in the $\hat{\mathbf{x}}_1$ direction and has magnitude

¹⁵ We could use the CM as our origin, but then we would have to include the complicated forces acting at the pivot point, which is difficult. But see Problem 9.19 for the case where the tip is free to slide on a frictionless table.

$Mg\ell \sin \theta$, where ℓ is the distance from the pivot to the CM. Using Eq. (9.68), the third component of $\boldsymbol{\tau} = d\mathbf{L}/dt$ quickly gives

$$\ddot{\beta} = 0. \quad (9.69)$$

Therefore, β is a constant, which we'll call ω_3 (an obvious label, in view of Eq. (9.64)). The other two components of $\boldsymbol{\tau} = d\mathbf{L}/dt$ then give

$$\begin{aligned} I\ddot{\phi} \sin \theta + \dot{\theta}(2I\dot{\phi} \cos \theta - I_3\omega_3) &= 0, \\ (Mg\ell + I\dot{\phi}^2 \cos \theta - I_3\omega_3\dot{\phi}) \sin \theta &= I\ddot{\theta}. \end{aligned} \quad (9.70)$$

We'll wait to fiddle with these equations until we have derived them again using the Lagrangian method.

9.7.4 Lagrangian method

Equation (9.15) gives the kinetic energy of the top as $T = \boldsymbol{\omega} \cdot \mathbf{L}/2$. Using Eqs. (9.64) and (9.65), we have (writing $\dot{\psi} + \dot{\phi} \cos \theta$ instead of the shorthand $\dot{\beta}$)¹⁶

$$T = \frac{1}{2}\boldsymbol{\omega} \cdot \mathbf{L} = \frac{1}{2}I_3(\dot{\psi} + \dot{\phi} \cos \theta)^2 + \frac{1}{2}I(\dot{\phi}^2 \sin^2 \theta + \dot{\theta}^2). \quad (9.71)$$

The potential energy is

$$V = Mg\ell \cos \theta, \quad (9.72)$$

where ℓ is the distance from the pivot to the CM. The Lagrangian is $\mathcal{L} = T - V$ (we'll use “ $\mathcal{L}$ ” here to avoid confusion with the angular momentum “ L ”), and so the equation of motion obtained from varying ψ is

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = \frac{\partial \mathcal{L}}{\partial \psi} \implies \frac{d}{dt}(\dot{\psi} + \dot{\phi} \cos \theta) = 0. \quad (9.73)$$

Therefore, $\dot{\psi} + \dot{\phi} \cos \theta$ is a constant. Call it ω_3 . The equations of motion obtained from varying ϕ and θ are then (making use of $\dot{\psi} + \dot{\phi} \cos \theta = \omega_3$)

$$\begin{aligned} \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\phi}} &= \frac{\partial \mathcal{L}}{\partial \phi} \implies \frac{d}{dt}(I_3\omega_3 \cos \theta + I\dot{\phi} \sin^2 \theta) = 0, \\ \frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{\theta}} &= \frac{\partial \mathcal{L}}{\partial \theta} \implies I\ddot{\theta} = (Mg\ell + I\dot{\phi}^2 \cos \theta - I_3\omega_3\dot{\phi}) \sin \theta. \end{aligned} \quad (9.74)$$

Taking the derivative in the first equation, we see that these equations are identical to those in Eq. (9.70).

¹⁶ It was ok to use β in the previous subsection. We introduced it only because it was quicker to write. But we can't use it here, because it depends on the other coordinates, and the Lagrangian method requires the use of independent coordinates. The variational proof back in Chapter 6 assumed this independence.

Note that there are two conserved quantities, arising from the facts that $\partial\mathcal{L}/\partial\psi$ and $\partial\mathcal{L}/\partial\phi$ equal zero. The conserved quantities are the angular momenta in the $\hat{\mathbf{x}}_3$ and $\hat{\mathbf{z}}$ directions, respectively. This is true because from Eq. (9.65), the former is $L_3 = I_3\omega_3$ and the latter is $L_z = L_3 \cos\theta + L_2 \sin\theta = (I_3\omega_3) \cos\theta + (I\dot{\phi} \sin\theta) \sin\theta$. These angular momenta are conserved because the torque points in the $\hat{\mathbf{x}}_1$ direction, so there is no torque in the plane spanned by $\hat{\mathbf{x}}_3$ and $\hat{\mathbf{z}}$.

9.7.5 Spinning top with $\dot{\theta} = 0$

A special case of Eqs. (9.70) occurs when $\dot{\theta} = 0$. In this case, the first of Eqs. (9.70) says that $\dot{\phi}$ is constant. The CM of the top therefore undergoes uniform circular motion in a horizontal plane. Let $\Omega \equiv \dot{\phi}$ be the frequency of this motion (this is the same notation as in Eq. (9.61)). Then the second of Eqs. (9.70) becomes

$$I\Omega^2 \cos\theta - I_3\omega_3\Omega + Mg\ell = 0. \quad (9.75)$$

This quadratic equation can be solved to yield two possible precessional frequencies, Ω , for the top. And yes, there are indeed two of them, provided that ω_3 is greater than a certain minimum value.

The previous pages in this “Heavy symmetric top” section have been a bit abstract, so let’s now take a breather and rederive Eq. (9.75) from scratch. That is, we’ll assume $\dot{\theta} = 0$ from the start of the solution, and then solve things by finding $\mathbf{L}$ and using $\boldsymbol{\tau} = d\mathbf{L}/dt$, in the spirit of Section 9.4.2. In practice, this strategy of starting from scratch is invariably the best route to take, as you’ll see in the problems and exercises for this chapter. The procedures in Sections 9.7.3 and 9.7.4 are good to know, but the technique in the following example provides a much more intuitive way of looking at things. This example is the classic “top” problem. We’ll warm up by solving it in an approximate way. Then we’ll do it for real.

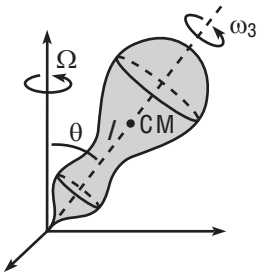


Fig. 9.31

Example (The top): A symmetric top of mass M has its CM a distance ℓ from its pivot. The moments of inertia relative to the pivot are $I_1 = I_2 \equiv I$, and I_3 . The top spins around its symmetry axis with frequency ω_3 (in the language of Section 9.7.2), and initial conditions have been set up so that the CM precesses in a circle around the vertical axis. The symmetry axis makes a constant angle θ with the vertical (see Fig. 9.31).

- Assuming that the angular momentum due to ω_3 is much larger than any other angular momentum in the problem, find an approximate expression for the frequency, Ω , of precession.
- Now solve the problem exactly. That is, find Ω by considering all of the angular momentum.

Solution:

- (a) The angular momentum (relative to the pivot) due to the spinning of the top has magnitude $L_3 = I_3\omega_3$, and it is directed along $\hat{\mathbf{x}}_3$. Let's label this angular momentum vector as $\mathbf{L}_3 \equiv L_3\hat{\mathbf{x}}_3$. As the top precesses, $\mathbf{L}_3$ traces out a cone around the vertical axis. So the tip of $\mathbf{L}_3$ moves in a circle of radius $L_3 \sin \theta$. The frequency of this circular motion is the frequency of precession, Ω . So $d\mathbf{L}_3/dt$, which is the velocity of the tip, has magnitude

$$\Omega(L_3 \sin \theta) = \Omega I_3 \omega_3 \sin \theta, \quad (9.76)$$

and it is directed into the page.

The torque (relative to the pivot) is due to gravity acting on the CM, so it has magnitude $Mg\ell \sin \theta$, and it is directed into the page. Therefore, $\boldsymbol{\tau} = d\mathbf{L}/dt$ gives

$$\Omega = \frac{Mg\ell}{I_3\omega_3}. \quad (9.77)$$

This is independent of θ , and it is inversely proportional to ω_3 .

- (b) The error in the above analysis is that we omitted the angular momentum arising from the $\hat{\mathbf{x}}_2$ (defined in Section 9.7.1) component of the angular velocity due to the precession of the top around the $\hat{\mathbf{z}}$ axis. This component has magnitude $\Omega \sin \theta$.¹⁷ Therefore, the angular momentum due to the angular velocity component in the $\hat{\mathbf{x}}_2$ direction has magnitude

$$L_2 = I\Omega \sin \theta. \quad (9.78)$$

Let's label this part of the angular momentum as $\mathbf{L}_2 \equiv L_2\hat{\mathbf{x}}_2$. The total $\mathbf{L} = \mathbf{L}_2 + \mathbf{L}_3$ is shown in Fig. 9.32. $\mathbf{L}$ precesses around in a cone, so only its horizontal component (call it $\mathbf{L}_\perp$) changes. From the figure, the length $L_\perp$ is the difference in lengths of the horizontal components of $\mathbf{L}_3$ and $\mathbf{L}_2$. Therefore,

$$L_\perp = L_3 \sin \theta - L_2 \cos \theta = I_3\omega_3 \sin \theta - I\Omega \sin \theta \cos \theta. \quad (9.79)$$

The magnitude of the rate of change of $\mathbf{L}$ is¹⁸

$$\left| \frac{d\mathbf{L}}{dt} \right| = \Omega L_\perp = \Omega (I_3\omega_3 \sin \theta - I\Omega \sin \theta \cos \theta). \quad (9.80)$$

Both $\boldsymbol{\tau}$ (which has magnitude $Mg\ell \sin \theta$) and $d\mathbf{L}/dt$ point into the page, so equating their magnitudes gives

$$I\Omega^2 \cos \theta - I_3\omega_3\Omega + Mg\ell = 0, \quad (9.81)$$

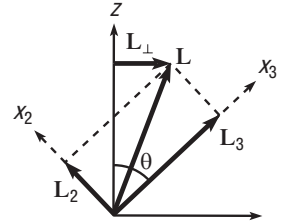


Fig. 9.32

¹⁷ The angular velocity due to the precession is $\Omega\hat{\mathbf{z}}$. We can break this up into components along the orthogonal directions $\hat{\mathbf{x}}_2$ and $\hat{\mathbf{x}}_3$. The $\Omega \cos \theta$ component along $\hat{\mathbf{x}}_3$ was already absorbed into the definition of ω_3 (see Fig. 9.30).

¹⁸ This result can also be obtained in a more formal way. Since $\mathbf{L}$ precesses with angular velocity $\Omega\hat{\mathbf{z}}$, the rate of change of $\mathbf{L}$ is $d\mathbf{L}/dt = \Omega\hat{\mathbf{z}} \times \mathbf{L}$. If you compute this cross product in the x_1, x_2, x_3 basis, you will obtain the result in Eq. (9.80).

in agreement with Eq. (9.75), as we wanted to show. The quadratic formula quickly gives the two solutions for Ω , which may be written as

$$\Omega_{\pm} = \frac{I_3 \omega_3}{2I \cos \theta} \left(1 \pm \sqrt{1 - \frac{4Mg\ell \cos \theta}{I_3^2 \omega_3^2}} \right). \quad (9.82)$$

Note that if $\theta = \pi/2$, then Eq. (9.81) is actually a linear equation, so there is only one solution for Ω , which is the one in Eq. (9.77). The reason for this is that $\mathbf{L}_2$ points vertically, so it doesn't change. Only $\mathbf{L}_3$ contributes to $d\mathbf{L}/dt$, so the approximate solution in part (a) is in fact an exact solution. Because of this simplification, a top is much easier to deal with when its symmetry axis is horizontal.

The two solutions in Eq. (9.82) are known as the *fast-precession* and *slow-precession* frequencies. For large ω_3 , you can show that the slow-precession frequency is

$$\Omega_- \approx \frac{Mg\ell}{I_3 \omega_3}, \quad (9.83)$$

in agreement with the solution found in Eq. (9.77).¹⁹ This task, along with many other interesting features of this problem (including the interpretation of the fast-precession frequency, Ω_+), is the subject of Problem 9.17, which you are encouraged to do.

9.7.6 An “explanation” of precession

The fact that a top can precess slowly around in a circle without simply falling down (as a simple pendulum would do) is rather bizarre. We showed above that this precession can be deduced perfectly well from $\boldsymbol{\tau} = d\mathbf{L}/dt$, but it would be nice if there was a more intuitive way to explain it, based at least somewhat on $\mathbf{F} = m\mathbf{a}$. Although a completely satisfactory intuitive explanation eludes me, I think the following discussion will clear some things up. This discussion will be qualitative (so no equations or numbers), but it should still suffice in explaining for the most part what's going on with precession.

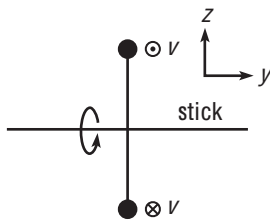


Fig. 9.33

Impulse applied to a dumbbell

Let's first look at a simple system consisting of a dumbbell with a massless stick glued perpendicular to it at its center. The dumbbell rotates around the stick which is held fixed; see Fig. 9.33. Let the y and z axes be defined as shown, with the x axis pointing out of the page. The total angular momentum points to the right, in the positive y direction. We'll ignore gravity for now. Equivalently, we can have the dumbbell pivoted at its center.

¹⁹ This is fairly clear. If ω_3 is large enough compared with Ω , then we can ignore the first term in Eq. (9.81). That is, we can ignore the effects of $\mathbf{L}_2$, which is exactly what we did in the approximate solution in part (a).

At the instant the masses are in the plane of the paper, as shown (with the top one coming out of the page and the bottom one going in), we'll apply equal and opposite small impulses to the stick, upward on the right end and downward on the left; see Fig. 9.34.²⁰ Assume that the forces are applied for infinitesimal periods of time, but that the forces are large enough so that the impulse is nonzero. What happens to the masses? In particular, what happens to the plane of their rotation?

Because the structure is rigid, the impulsive forces on the stick cause the two masses to pick up small velocity components in the $\pm y$ directions, as shown in Fig. 9.34. If the forces are applied for an infinitesimal time t , then these velocity components are of the form $v = at$ (with t very small and a very large). Additionally, the masses move a distance $d = at^2/2$ to the side. But because of the two powers of the infinitesimal time t here, this distance is negligible. In other words, the masses pick up a nonzero v , but essentially no d . This is a general result when an object is struck with a hammer: right after the blow, it has a nonzero speed, but essentially zero displacement.

A top view (with the z axis pointing out of the page) of the velocities of the two masses right after the blow is shown in Fig. 9.35. The dotted lines represent the velocity of the bottom mass which is behind (that is, below) the top mass. Now, if someone gives you these two velocities and doesn't tell you what was going on beforehand, then you will simply say that the dumbbell is rotating around in a circle that lies in the vertical plane defined by the line of the new velocity vectors in Fig 9.35. In other words, the plane of the circular motion has been rotated around the vertical z axis. A top view of the situation is shown in Fig. 9.36. From this time onward, the masses rotate around in the new vertical plane. This means that the angular momentum of the dumbbell has picked up a component in the positive x direction (out of the page in Fig. 9.34 and downward in Fig. 9.36), consistent with the fact that the torque from the two applied forces points in the positive x direction.

This example illustrates the bizarre fact that if you smack the rotation axis in one direction (vertical here), it will head off in another direction (horizontal).

Impulse applied to a symmetric top

Let's now consider the more complicated situation where instead of the above dumbbell we have a symmetric top, for example, a flat disk with a stick poking through its center. (We'll still ignore gravity for the moment.) Things are more difficult now, because if we apply our impulsive forces to the stick, the plane of the disk can't instantaneously rotate as it did above, because this would involve the "side" points on the disk moving a finite distance in zero time, as they

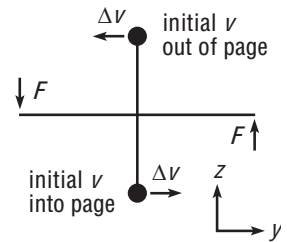


Fig. 9.34

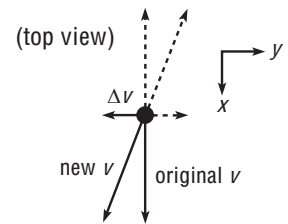


Fig. 9.35

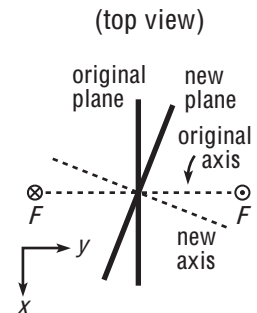


Fig. 9.36

²⁰ We're applying equal and opposite forces just so that the CM doesn't move. And we're doing this for no reason other than simplicity. The motion of the CM is irrelevant for the point we want to make in this dumbbell setup.

move from the old plane of rotation to the new plane. So what does the motion look like?

The rough answer is that our symmetric top is vaguely the same type of object as the above dumbbell, so the motion should look roughly the same. In other words, the axis of the top should end up pointing (in one way or another) a little bit in the x direction, as it did with the dumbbell. But the precise answer is that since we have a free top here, we know from the free-top discussion in Section 9.6.2 exactly what happens: the symmetry axis of the disk precesses (along with the angular velocity vector) in a thin cone around the new angular momentum vector, which points slightly out of the page due to the torque in the x direction. So although the symmetry axis of the disk doesn't point in the definite direction along $\mathbf{L}$ as it did with the dumbbell, it points *on average* along $\mathbf{L}$, which is slightly in the x direction.

Let's now bring gravity into the setup. It turns out that we can consider a heavy top's precession to be the result of a succession of many little impulsive blows to a free top that is otherwise in freefall (where gravity provides no torque). The reasoning is as follows.

Imagine that we hold our spinning top (not pivoted anywhere) and move it sideways (perpendicular to the stick) at a given constant speed; this speed will have to be chosen to be a particular value (see Footnote 23 below). And then we let go. Gravity provides a force on the CM, but zero torque around the CM, so the top is simply in freefall with the constant horizontal speed that we gave it. But let's assume that immediately after we let go, we keep the top at a constant height by applying a very quick and very small upward strike to the end of the stick, and then waiting for a very short time while the CM rises and falls in its freefall motion, and then repeating the process indefinitely (let's say we make 100 tiny strikes each second). If we arrange for the time-averaged upward force to equal mg , then the CM stays at (essentially) a constant height.

After each strike, the axis of the top undergoes its free-top precession in a very thin cone, so if the CM of the top weren't moving, the end of the stick would end up pointing slightly to the side, along the direction of the new $\mathbf{L}$.²¹ But the sideways motion of the top (due to the properly chosen original speed we gave it) exactly brings the end of the stick back to its original location; see the top view in Fig. 9.37. The dot in the figure represents a fixed point in space. We can imagine this process taking place in two steps. In step 1, the axis changes its direction due to the angular impulse relative to the CM. And in step 2, the top moves sideways (due to the original horizontal speed we gave it) during its rising and falling projectile motion. As the process repeats, the CM moves around in a circle,

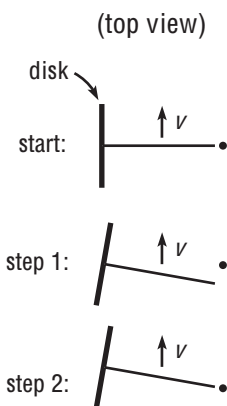


Fig. 9.37

²¹ Assume that the stick's tiny precession cone around $\mathbf{L}$ somehow damps out, so that the stick eventually points in the definite direction along $\mathbf{L}$. Since we're eventually going to consider the end of the stick to be located at a pivot, this is a reasonable assumption.

with the end of the stick remaining (essentially) fixed.²² In other words, we have recreated our precessing top.²³ And since we can consider the continuous upward force that a pivot applies to a heavy top to be a succession of quick infinitesimal strikes, we see that a heavy top precessing around a pivot is essentially the same as the above top precessing around the fixed dot in space in Fig. 9.37.

The above reasoning is only qualitative, but it should make the precession of a heavy top a little more believable. Of course, since we've argued that a heavy top pivoted at a point can be considered to be a free top undergoing a succession of impulsive and free-top motions, we've just shifted the burden of proof to an intuitive understanding of why a free top precesses the way it does. But that makes my head hurt, so I'll stop here. But at least we know that a free top should behave more or less like the dumbbell setup above, for which we showed (by actually looking at the forces) why the axis of rotation shifts to the side when vertical forces are applied.

9.7.7 Nutation

We will now solve Eq. (9.70) in a somewhat more general case, where θ is allowed to vary slightly. That is, we will consider a slight perturbation to the circular motion associated with Eq. (9.75). We will assume that ω_3 is large here, and we will assume that the original circular motion corresponds to the slow precession, so that $\dot{\phi}$ is small. Under these assumptions, we will find that the top will bounce around slightly as it travels (roughly) in a circle. This bouncing is known as *nutation*.

Since $\dot{\phi}$ is small compared with ω_3 , we can (to a good approximation) ignore the middle terms on the left-hand sides of Eqs. (9.70) to obtain

$$\begin{aligned} I\ddot{\phi} \sin \theta - \dot{\theta} I_3 \omega_3 &= 0, \\ (Mg\ell - I_3 \omega_3 \dot{\phi}) \sin \theta &= I\ddot{\theta}. \end{aligned} \quad (9.84)$$

We must somehow solve these equations for $\theta(t)$ and $\phi(t)$. Taking the derivative of the first equation and dropping the quadratic term (which is negligible for sufficiently small perturbations) gives $\ddot{\theta} = (I \sin \theta / I_3 \omega_3) d^2 \dot{\phi} / dt^2$. Substituting this expression for $\ddot{\theta}$ into the second equation gives

$$\frac{d^2 \dot{\phi}}{dt^2} + \omega_n^2 (\dot{\phi} - \Omega_s) = 0, \quad (9.85)$$

²² We'll need to apply a force directed along the axis, to provide the centripetal acceleration of the precessing CM. But this force produces no torque around the CM, so it doesn't affect any of the other aspects of the problem.

²³ If the initial conditions aren't set up properly, then the top will bounce up and down as it precesses around (this is known as *nutation*, described below in Section 9.7.7). In particular, if you hold the axis at rest and then let go, the top will initially fall straight down. So your intuition works perfectly fine here. But as time goes by, more complicated things start to happen, as explained in Remark 6 in the nutation section below.

where

$$\omega_n \equiv \frac{I_3 \omega_3}{I} \quad \text{and} \quad \Omega_s = \frac{Mg\ell}{I_3 \omega_3} \quad (9.86)$$

are, respectively, the frequency of nutation (as we shall soon see) and the slow-precession frequency given in Eq. (9.77). Shifting variables to $y \equiv \dot{\phi} - \Omega_s$ in Eq. (9.85) gives us the nice harmonic-oscillator equation, $\ddot{y} + \omega_n^2 y = 0$. Solving this and then shifting back to $\dot{\phi}$ yields

$$\dot{\phi}(t) = \Omega_s + A \cos(\omega_n t + \gamma), \quad (9.87)$$

where A and γ are determined by the initial conditions. Integrating this gives

$$\phi(t) = \Omega_s t + \left(\frac{A}{\omega_n} \right) \sin(\omega_n t + \gamma), \quad (9.88)$$

plus an irrelevant constant.

Now let's solve for $\theta(t)$. Plugging our $\phi(t)$ into the first of Eqs. (9.84) gives

$$\dot{\theta}(t) = - \left(\frac{I \sin \theta}{I_3 \omega_3} \right) A \omega_n \sin(\omega_n t + \gamma) = -A \sin \theta \sin(\omega_n t + \gamma), \quad (9.89)$$

where we have used the definition of ω_n in Eq. (9.86). Since $\theta(t)$ doesn't change much, we can set $\sin \theta \approx \sin \theta_0$, where θ_0 is, say, the initial value of $\theta(t)$. Any errors here will be second-order effects in small quantities. Integration then gives

$$\theta(t) = B + \left(\frac{A}{\omega_n} \sin \theta_0 \right) \cos(\omega_n t + \gamma), \quad (9.90)$$

where B is a constant of integration. Equations (9.88) and (9.90) show that both ϕ (neglecting the uniform $\Omega_s t$ part) and θ oscillate with frequency ω_n , and with amplitudes inversely proportional to ω_n . Note that Eq. (9.86) says that ω_n grows with ω_3 .

Example (Sideways kick): Assume that uniform circular precession is initially taking place with $\theta = \theta_0$ and $\dot{\phi} = \Omega_s$. You then give the top a quick kick along the direction of motion, so that $\dot{\phi}$ suddenly becomes $\Omega_s + \Delta\Omega$ ($\Delta\Omega$ may be positive or negative). Find $\phi(t)$ and $\theta(t)$.

Solution: This is an exercise in initial conditions. We are given the initial values of $\dot{\phi}$, $\dot{\theta}$, and θ (namely $\Omega_s + \Delta\Omega$, 0, and θ_0 , respectively), and our goal is to solve for the unknowns A , B , and γ in Eqs. (9.87), (9.89), and (9.90). $\dot{\theta}$ is initially zero, so Eq. (9.89) gives $\gamma = 0$ (or π , but this leads to the same answer). And $\dot{\phi}$ is initially $\Omega_s + \Delta\Omega$, so Eq. (9.87) gives $A = \Delta\Omega$. Finally, θ is initially θ_0 , so Eq. (9.90) gives $B = \theta_0 - (\Delta\Omega/\omega_n) \sin \theta_0$. Putting this all together, we have

$$\begin{aligned} \phi(t) &= \Omega_s t + \left(\frac{\Delta\Omega}{\omega_n} \right) \sin \omega_n t, \\ \theta(t) &= \theta_0 - \left(\frac{\Delta\Omega}{\omega_n} \sin \theta_0 \right) (1 - \cos \omega_n t). \end{aligned} \quad (9.91)$$

And for future reference (for the problems in this chapter), we'll also list the derivatives,

$$\begin{aligned}\dot{\phi}(t) &= \Omega_s + \Delta\Omega \cos \omega_n t, \\ \dot{\theta}(t) &= -\Delta\Omega \sin \theta_0 \sin \omega_n t.\end{aligned}\tag{9.92}$$

REMARKS:

1. Remember that all of this analysis holds only if $\dot{\theta}$ and $\dot{\phi}$ are small compared with ω_3 , and if θ is always near θ_0 .
2. For the initial setup we have chosen (that is, for $\dot{\theta} = 0$), Eq. (9.91) shows that θ always stays on one side of θ_0 . If $\Delta\Omega > 0$, then $\theta(t) \leq \theta_0$ for all t (that is, the top is always at a higher position, since θ is measured from the vertical). If $\Delta\Omega < 0$, then $\theta(t) \geq \theta_0$ for all t (that is, the top is always at a lower position).
3. Consider the point that is a distance ℓ from the origin and that travels in a horizontal circle with angular coordinates given by $(\phi, \theta)_{\text{avg}} = (\Omega_s t, \theta_0 - (\Delta\Omega/\omega_n) \sin \theta_0)$. This is the “average” position of the CM, in the sense that Eq. (9.91) gives the angular coordinates of the CM relative to this point as

$$(\phi, \theta)_{\text{rel}} = \left(\frac{\Delta\Omega}{\omega_n} \right) (\sin \omega_n t, \sin \theta_0 \cos \omega_n t).\tag{9.93}$$

The $\sin \theta_0$ in the second coordinate here implies that the amplitude of the θ oscillation is $\sin \theta_0$ times the amplitude of the ϕ oscillation. This is precisely the factor needed to make the CM travel in a circle, as viewed by someone riding along with coordinates $(\phi, \theta)_{\text{avg}}$, because a change in θ causes a displacement of $\ell d\theta$, whereas a change in ϕ causes a displacement of $\ell \sin \theta_0 d\phi$.

4. Figure 9.38 shows plots of $\theta(t)$ vs. $\sin \theta_0 \phi(t)$ for various values of $\Delta\Omega$. The dots at the start of each of the nine plots all represent the same starting point at $\theta = \theta_0$. The plots are stacked on top of each other for comparison only; the vertical spacing between them is meaningless. We have chosen the horizontal axis to be $\sin \theta_0 \phi(t)$ instead of $\phi(t)$, and we have chosen the vertical axis to have θ increasing downward, so that these plots are exactly the paths you would see the CM trace out in space. We have picked arbitrary values of Ω_s and ω_n to generate the plots (hence no numbers on the axes, since they wouldn't mean much), but no matter what values are picked, the shapes of the plots are still the same (for example, the $\Delta\Omega = \pm\Omega_s$ paths always have cusps). The oscillations have different frequencies and amplitudes, but both axes get scaled by the same amount (as you can check).
5. From the figure, we see that the motions associated with $\Delta\Omega$ and $-\Delta\Omega$ look the same except for being shifted vertically (relative to the starting dots, which represent the same point), and also horizontally by half a cycle. This can be seen from Eq. (9.91); changing $\Delta\Omega$ to $-\Delta\Omega$ has the effect of shifting the constant term in $\theta(t)$, and also shifting the time by half a cycle, because $\Delta\Omega \sin \omega_n t = (-\Delta\Omega) \sin(\omega_n t + \pi)$, and likewise for cosine.
6. In the $\Delta\Omega = -\Omega_s$ case, the CM starts at rest. This corresponds to holding the axis of the top at rest and then dropping it. From Fig. 9.38 we see that initially the CM simply falls straight down, as your intuition suggests. (It might not be clear from the figure, but Problem 9.25 shows that the curve is indeed vertical at the high points in the motion.) But then the strangeness of angular momentum takes over, and the top ends up bouncing and precessing instead of continuing to fall as a point mass would. Qualitatively, we can understand this bouncing and precessing as follows. Let the axis and the angular momentum initially point to the right as you hold the axis, as

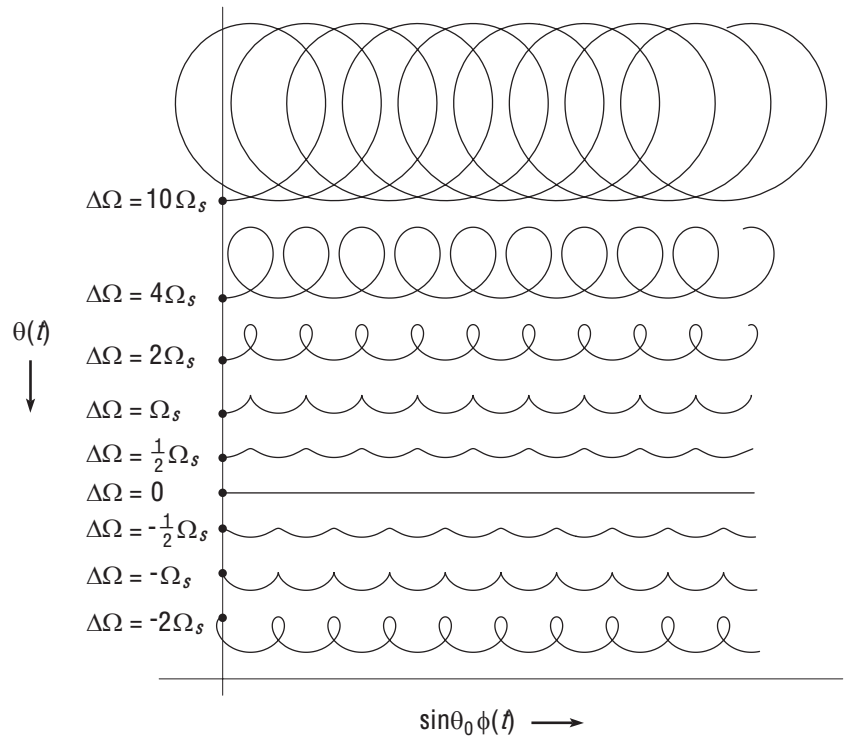


Fig. 9.38

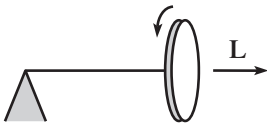


Fig. 9.39

shown in Fig. 9.39. After you let go, a number of things happen: (1) Because of the downward gravitational force, the top starts to fall. The axis therefore now points slightly downward, which means that the angular momentum picks up a downward component. Relative to the CM, the force from the pivot must therefore provide a downward torque. By the right-hand rule, this force must point into the page. (2) From $F = ma$, this inward force causes the top to accelerate into the page. This makes the axis point slightly into the page, so the angular momentum picks up a component in that direction. By the right-hand rule, there must then be an upward force from the pivot to provide the necessary torque for this. (3) This upward force slows down the downward motion and in fact eventually becomes greater than mg and causes the CM to rise back up. This means that the angular momentum increases its vertical component, so by the right-hand rule there must be a force from the pivot that points out of the page to provide the necessary torque for this. (4) This outward force then slows down the motion into the page and eventually stops it, right when the top returns to its original height. The top is momentarily at rest, and then the process repeats itself. Of course, this qualitative reasoning doesn't show that all the details work out correctly, but at least it makes the motion a little more believable.

7. If you start by holding the axis of the top at rest, as in the previous remark, the $\Delta\Omega = 0$ case corresponds to giving the axis the proper initial push into the page (instead of just dropping it, as above) so that the resulting change in angular momentum into the page requires an upward force of mg from the pivot. The CM then stays at the same height and simply moves in a horizontal circle without bouncing. ♣

9.8 Problems

Section 9.1: Preliminaries concerning rotations

9.1. Many different ω 's *

Consider a particle at the point $(a, 0, 0)$, with velocity $(0, v, 0)$. At this instant, the particle may be considered to be rotating around many different ω vectors passing through the origin. There isn't just one "correct" ω . Find all the possible ω 's (give their directions and magnitudes).

9.2. Fixed points on a sphere **

Consider a transformation of a rigid sphere into itself. Show that two points on the sphere end up where they started.

9.3. Rolling cone **

A cone rolls without slipping on a table. The half-angle at the vertex is α , and the axis has length h (see Fig. 9.40). Let the speed of the center of the base, point P in the figure, be v . What is the angular velocity of the cone with respect to the lab frame at the instant shown? There are many ways to do this problem, so you are encouraged to take a look at the three given solutions, even after solving it.

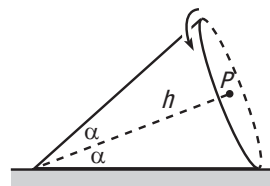


Fig. 9.40

Section 9.2: The inertia tensor

9.4. Parallel-axis theorem

Let (X, Y, Z) be the position of an object's CM, and let (x', y', z') be the position relative to the CM. Prove the parallel-axis theorem, Eq. (9.19), by setting $x = X + x'$, $y = Y + y'$, and $z = Z + z'$ in Eq. (9.8).

Section 9.3: Principal axes

9.5. A nice cylinder *

What must the ratio of height to radius of a cylinder be so that every axis is a principal axis (with the CM as the origin)?

9.6. Rotating square *

Here's an exercise in geometry. Theorem 9.5 says that if the moments of inertia around two principal axes are equal, then any axis in the plane of these axes is a principal axis. This means that the object will happily rotate around any axis in this plane, that is, no torque is needed. Demonstrate this explicitly for four equal masses in the shape of a square, with the center as the origin, which obviously has two moments equal. Assume that the masses are connected by strings to the axis, as shown in Fig. 9.41, and that they all rotate with the same ω around the axis, so that they remain in the shape of a square. Your task is to show that the tensions in the strings

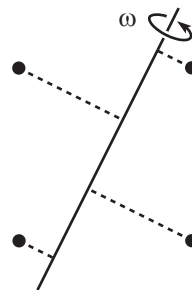


Fig. 9.41

are such that there is no net torque acting on the axis, relative to the center of the square.

9.7. Existence of principal axes for a pancake *

Given a pancake object in the x - y plane, show that there exist principal axes by considering what happens to the integral $\int xy$ when the coordinate axes are rotated about the origin by an angle of $\pi/2$.

9.8. Symmetries and principal axes for a pancake **

A rotation of the axes in the x - y plane through an angle θ transforms the coordinates according to (you can accept this)

$$\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}. \quad (9.94)$$

Use this to show that if a pancake object in the x - y plane has a symmetry under a rotation through $\theta \neq \pi$, then $\int xy = 0$ for any choice of axes, which implies that all axes (through the origin) in the plane are principal axes.

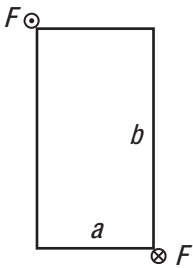


Fig. 9.42

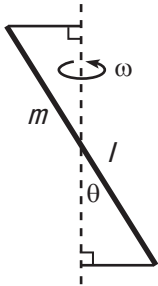


Fig. 9.43

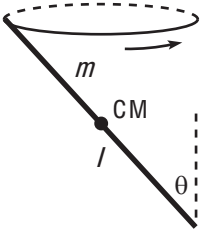


Fig. 9.44

Section 9.4: Two basic types of problems

9.9. Striking a rectangle *

A flat uniform rectangle with sides of length a and b sits in space, not rotating. You strike the corners at the ends of one diagonal, with equal and opposite forces (see Fig. 9.42). Show that the resulting initial ω points along the other diagonal.

9.10. Rotating stick **

A stick of mass m and length ℓ spins with frequency ω around an axis, as shown in Fig. 9.43. The stick makes an angle θ with the axis and is kept in its motion by two strings that are perpendicular to the axis. What is the tension in the strings? (Ignore gravity.)

9.11. Stick under a ring **

A stick of mass m and length ℓ is arranged to have its CM motionless while its top end slides in a circle on a frictionless ring, as shown in Fig. 9.44. The stick makes an angle θ with the vertical. What is the frequency of this motion?

9.12. Circular pendulum **

Consider a pendulum made of a massless rod of length ℓ with a point mass m on the end. Assume conditions have been set up so that the mass moves in a horizontal circle. Let θ be the constant angle the rod makes

with the vertical. Find the frequency, Ω , of this circular motion in three different ways.

- (a) Use $\mathbf{F} = m\mathbf{a}$. This method works only if you have a point mass. With an extended object, you have to use one of the following methods involving torque.
- (b) Use $\boldsymbol{\tau} = d\mathbf{L}/dt$ with the pendulum pivot as the origin.
- (c) Use $\boldsymbol{\tau} = d\mathbf{L}/dt$ with the mass as the origin.

9.13. Rolling in a cone **

A fixed cone stands on its tip, with its axis in the vertical direction. The half-angle at the vertex is θ . A small ring of radius r rolls without slipping on the inside surface. Assume that conditions have been set up so that (1) the point of contact between the ring and the cone moves in a circle at height h above the tip, and (2) the plane of the ring is at all times perpendicular to the line joining the point of contact and the tip of the cone (see Fig. 9.45). What is the frequency, Ω , of this circular motion? Work in the approximation where r is much smaller than the radius of the circular motion, $h \tan \theta$.

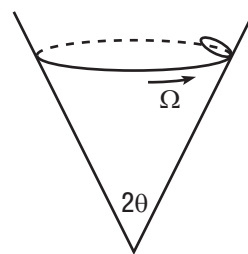


Fig. 9.45

9.14. Tennis racket theorem ***

If you try to spin a tennis racket (or a book, etc.) around any of its three principal axes, you will find that different things happen with the different axes. Assuming that the principal moments (relative to the CM) are labeled according to $I_1 > I_2 > I_3$ (see Fig. 9.46), you will find that the racket will spin nicely around the $\hat{x}_1$ and $\hat{x}_3$ axes, but it will wobble in a rather messy manner around the $\hat{x}_2$ axis. Verify this claim experimentally with a book (preferably lightweight, and wrapped with a rubber band), or a tennis racket, if you happen to study with one on hand.

Now verify this claim mathematically. The main point here is that you can't start the motion off with $\boldsymbol{\omega}$ pointing *exactly* along a principal axis. Therefore, what you want to show is that the motion around the $\hat{x}_1$ and $\hat{x}_3$ axes is *stable* (that is, small errors in the initial conditions remain small), whereas the motion around the $\hat{x}_2$ axis is *unstable* (that is, small errors in the initial conditions get larger and larger, until the motion eventually doesn't resemble a rotation around the $\hat{x}_2$ axis).²⁴ Your task is to use Euler's equations to prove these statements about stability. (Exercise 9.33 gives another derivation of this result.)

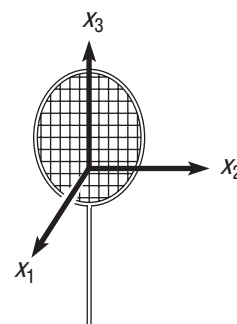


Fig. 9.46

²⁴ If you try for a long enough time, you'll probably be able to get the initial $\boldsymbol{\omega}$ pointing close enough to $\hat{x}_2$ so that the book will remain rotating (almost) around $\hat{x}_2$ for the entire time of its flight. There is, however, undoubtedly a better use for your time, as well as for the book...

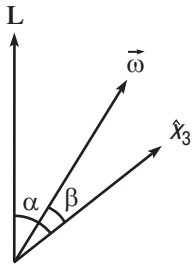


Fig. 9.47

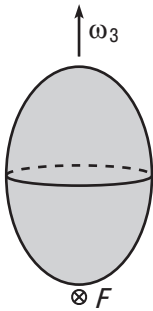


Fig. 9.48

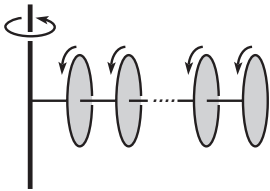


Fig. 9.49

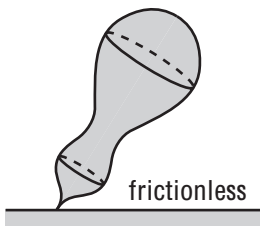


Fig. 9.50

Section 9.6: Free symmetric top

9.15. Free-top angles *

In Section 9.6.2, we showed that for a free symmetric top, the angular momentum $\mathbf{L}$, the angular velocity $\boldsymbol{\omega}$, and the symmetry axis $\hat{\mathbf{x}}_3$ all lie in a plane. Let α be the angle between $\hat{\mathbf{x}}_3$ and $\mathbf{L}$, and let β be the angle between $\hat{\mathbf{x}}_3$ and $\boldsymbol{\omega}$ (see Fig. 9.47). Find the relationship between α and β in terms of the principal moments, I and I_3 .

9.16. Staying above **

A top with $I = nI_3$, where n is a numerical factor, is initially spinning around its x_3 axis with angular speed ω_3 . You apply a strike at the bottom point, directed into the page as shown in Fig. 9.48 (imagine that you hit a little peg protruding from the bottom). What is the largest value of n for which the total $\boldsymbol{\omega}$ vector never dips below the horizontal axis in the subsequent motion, no matter how hard your strike is?

Section 9.7: Heavy symmetric top

9.17. The top **

This problem deals with the spinning top example in Section 9.7.5. It uses the result for Ω in Eq. (9.82).

- What is the minimum value of ω_3 for which circular precession is possible?
- Find approximate expressions for $\Omega_{\pm}$ when ω_3 is very large. The phrase “very large,” however, is rather meaningless. What mathematical statement should replace it?

9.18. Many tops **

N identical disks and massless sticks are arranged as shown in Fig. 9.49. Each disk is glued to the stick on its left and attached by a pivot to the stick on its right. The leftmost stick is attached by a pivot to a pole. You wish to set up circular precession with the sticks always forming a straight horizontal line. What should the relative angular speeds of the disks be so that this is possible?

9.19. Heavy top on a slippery table **

Solve the problem of a heavy symmetric top spinning on a frictionless table (see Fig. 9.50). You may do this by simply stating what modifications are needed in the derivation in Section 9.7.3 (or Section 9.7.4).

9.20. **Fixed highest point ****

Consider a top made of a uniform disk of radius R , connected to the origin by a massless stick (which is glued perpendicular to the disk) of length ℓ . Paint a dot on the top at its highest point, and label this as point P (see Fig. 9.51). You wish to set up uniform circular precession, with the stick making a constant angle θ with the vertical (θ can be chosen to be any angle between zero and π), and with P always being the highest point on the top. What is the frequency of precession, Ω ? What relation between R and ℓ must be satisfied for this motion to be possible?

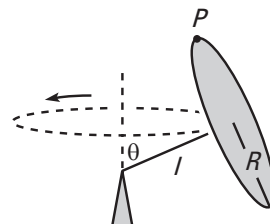


Fig. 9.51

9.21. **Basketball on a rim *****

A basketball rolls without slipping around a basketball rim in such a way that the contact points trace out a great circle on the ball, and the CM moves around in a horizontal circle with frequency Ω . The radii of the ball and rim are r and R , respectively, and the ball's radius to the contact point makes an angle θ with the horizontal (see Fig. 9.52). Assume that the ball's moment of inertia around its center is $I = (2/3)mr^2$. Find Ω .



Fig. 9.52

9.22. **Rolling lollipop *****

Consider a lollipop made of a solid sphere of mass m and radius r that is radially pierced by a massless stick. The free end of the stick is pivoted on the ground (see Fig. 9.53). The sphere rolls on the ground without slipping, with its center moving in a circle of radius R with frequency Ω . What is the normal force between the ground and the sphere?

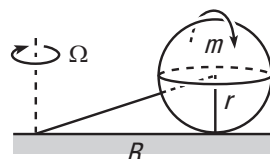


Fig. 9.53

9.23. **Rolling coin *****

Initial conditions have been set up so that a coin of radius r rolls around in a circle, as shown in Fig. 9.54. The contact point on the ground traces out a circle of radius R , and the coin makes a constant angle θ with the horizontal. The coin rolls without slipping (assume that the friction with the ground is as large as needed). What is the frequency, Ω , of the circular motion of the contact point on the ground? Show that such motion exists only if $R > (5/6)r \cos \theta$.

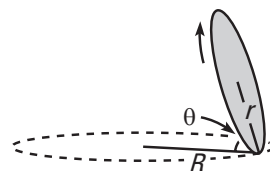


Fig. 9.54

9.24. **Wobbling coin *****

If you spin a coin around a vertical diameter on a table, it will slowly lose energy and begin a wobbling motion. The angle between the coin and the table will gradually decrease, and eventually it will come to rest. Assume that this process is slow, and consider the motion when the coin makes an angle θ with the table (see Fig. 9.55). You may assume that the CM is essentially motionless. Let R be the radius of the coin, and let Ω be the frequency at which the contact point on the table traces out its circle. Assume that the coin rolls without slipping.

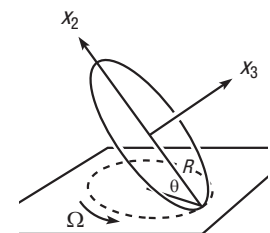


Fig. 9.55

- (a) Show that the angular velocity of the coin is $\boldsymbol{\omega} = \Omega \sin \theta \hat{\mathbf{x}}_2$, where $\hat{\mathbf{x}}_2$ points upward along the coin, directly away from the contact point.
- (b) Show that

$$\Omega = 2\sqrt{\frac{g}{R \sin \theta}}. \quad (9.95)$$

- (c) Show that Abe (or Tom, Franklin, George, John, Dwight, Susan, or Sacagawea) appears to rotate, when viewed from above, with frequency

$$2(1 - \cos \theta)\sqrt{\frac{g}{R \sin \theta}}. \quad (9.96)$$

9.25. Nutation cusps **

- (a) Using the notation and initial conditions in the example in Section 9.7.7, prove that kinks occur in nutation if and only if $\Delta\Omega = \pm\Omega_s$. A kink is where the plot of $\theta(t)$ vs. $\phi(t)$ has a discontinuity in its slope.
- (b) Prove that these kinks are in fact cusps. A cusp is a kink where the plot reverses direction in the ϕ - θ plane.

9.26. Nutation circles **

- (a) Using the notation and initial conditions in the example in Section 9.7.7, and assuming that $\omega_3 \gg \Delta\Omega \gg \Omega_s$, find (approximately) the direction of the angular momentum right after the sideways kick takes place.
- (b) Use Eq. (9.91) to show that the CM then travels (approximately) in a circle around $\mathbf{L}$. And show that this “circular” motion is just what you would expect from the free-top reasoning in Section 9.6.2, in particular, Eq. (9.55).

Additional problems

9.27. Rolling without slipping *

The standard way that a ball rolls without slipping on a flat surface is for the contact points on the ball to trace out a vertical great circle on the ball. Are there any other ways that a ball can roll without slipping?

9.28. Rolling straight? **

In some situations, such as the rolling-coin setup in Problem 9.23, the velocity of the CM of a rolling object changes direction as time goes by. Consider a uniform sphere that rolls on the ground without slipping (possibly in the nonstandard way described in the solution to Problem 9.27).

Is it possible for the CM's velocity to change direction? Justify your answer rigorously.

9.29. Ball on paper ***

A uniform ball rolls without slipping on a table (possibly in the non-standard way described in the solution to Problem 9.27). It rolls onto a piece of paper, which you then slide around in an arbitrary (horizontal) manner. You may even give the paper abrupt, jerky motions, so that the ball slips with respect to it. After you allow the ball to come off the paper, it will eventually resume rolling without slipping on the table. Show that the final velocity equals the initial velocity.

9.30. Ball on a turntable ****

A uniform ball rolls without slipping on a turntable (possibly in the nonstandard way described in the solution to Problem 9.27). As viewed from the inertial lab frame, show that the ball moves in a circle (not necessarily centered at the center of the turntable) with a frequency equal to $2/7$ times the frequency of the turntable.

9.9 Exercises

Section 9.1: Preliminaries concerning rotations

9.31. Rolling wheel **

A wheel with spokes rolls without slipping on the ground. A stationary camera takes a picture of it as it rolls by, from the side. Due to the nonzero exposure time of the camera, the spokes generally appear blurred. At what locations in the picture do the spokes *not* appear blurred? *Hint:* A common incorrect answer is that there is only one point.

Section 9.2: The inertia tensor

9.32. Inertia tensor *

Calculate the $\mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r})$ double cross product in Eq. (9.7) by using the vector identity,

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}). \quad (9.97)$$

Section 9.3: Principal axes

9.33. Tennis racket theorem **

Problem 9.14 gives the statement of the “tennis racket theorem,” and the solution there involves Euler's equations. Demonstrate the theorem here by writing down the conservation of L^2 and conservation of E statements and then using them in the following way. Produce an equation that says

that if ω_2 and ω_3 (or ω_1 and ω_2) start small, then they must remain small. And produce the analogous equation that says that if ω_1 and ω_3 start small, then they need *not* remain small. (It's another matter to show that they actually *won't* remain small. But let's not worry about that here. Anything that *can* happen generally *does* happen in physics.)

9.34. Moments for a cube **

In the spirit of Appendix E, calculate the principal moments for a solid cube of mass m and side length ℓ , with the coordinate axes parallel to the edges of the cube, and the origin at a corner.

9.35. Tilted moments **

- Consider a planar object in the x - y plane. If the x and y axes are principal axes, use the rotation matrix in Eq. (9.94) to show that the moment of inertia around the x' axis, which makes an angle θ with the x axis, is $I_{x'} = I_x \cos^2 \theta + I_y \sin^2 \theta$.
- Consider a general three-dimensional object whose principal axes are the x , y , and z axes. Consider another axis that points along the unit vector (α, β, γ) . Show that the moment of inertia around this axis is $\alpha^2 I_x + \beta^2 I_y + \gamma^2 I_z$. *Hint:* The cross product, discussed in Appendix B, provides a nice method of calculating the distance from a point to a line.

9.36. Quadrupole **

Consider an arbitrarily shaped body of mass m whose CM is at the origin. Using the law of cosines, the gravitational potential of a mass M at position $\mathbf{R}$ is

$$V(\mathbf{R}) = - \int \frac{GM dm}{\sqrt{R^2 + r^2 - 2Rr \cos \beta}}, \quad (9.98)$$

where the integration runs over the volume of the body, and β is the angle that the position vector $\mathbf{r}$ of an arbitrary point in the body makes with the vector $\mathbf{R}$.

- Assuming that all points in the body satisfy $r \ll R$, show that an approximate expression for the potential is

$$V(\mathbf{R}) \approx -\frac{GMm}{R} - \frac{GM}{2R^3} \int r^2 (3 \cos^2 \beta - 1) dm, \quad (9.99)$$

and then show that this can be written as

$$V(\mathbf{R}) \approx -\frac{GMm}{R} - \frac{GM}{2R^3} (I_1 + I_2 + I_3 - 3I_R), \quad (9.100)$$

where I_1 , I_2 , and I_3 are the moments around any three orthogonal axes (which we'll take to be principal axes in part (b)), and I_R is the moment around the axis along the $\mathbf{R}$ vector.

- (b) Consider now a planet with rotational symmetry around $\hat{\mathbf{x}}_3$, such as the earth which bulges at the equator due to the spinning. Using the result of Exercise 9.35, show that the potential in Eq. (9.100) can be written as

$$V(\mathbf{R}) \approx -\frac{GMm}{R} - \frac{GM}{2R^3}(I_3 - I)(1 - 3\cos^2\theta), \quad (9.101)$$

where $I \equiv I_1 = I_2$, and θ is the angle that $\mathbf{R}$ makes with $\hat{\mathbf{x}}_3$.

REMARK: The second term here is known as the *quadrupole* term. In electrostatics, a *dipole* consists of equal and opposite charges separated by some distance d . At a given point far away, the forces from these two charges partially cancel. But they don't exactly cancel, because the electrostatic force (which behaves like the gravitational force, with a $1/r^2$ law) depends on the distance and direction to the charges, and the two charges are located at different points. If two dipoles are oriented in opposite directions and then placed side by side, a distance d apart (so that there are charges of q and $-q$ alternating around the corners of a square) then the forces from the dipoles nearly cancel. But again, the cancellation isn't exact, because the dipoles are located in different places. This distribution of charge is called a *quadrupole*, and it is similar to the situation with a spinning (and bulging) planet, because such a planet consists of a spherical ball (which gives rise to the first term in Eq. (9.101)), plus a region of "negative" mass superimposed on the ball at the poles and a region of positive mass superimposed on the ball at the equator. Looking at the above square of charges from far out along the diagonal containing the two negative charges is similar to looking at the earth from far out along its rotation axis. ♣

Section 9.4: Two basic types of problems

9.37. Sphere and points *

A uniform sphere of mass m and radius R rotates around the vertical axis with angular speed ω . Two particles of mass $m/2$ are brought close to the sphere at diametrically opposite points, at an angle θ from the vertical, as shown in Fig. 9.56. The masses, which are initially essentially at rest, abruptly stick to the sphere. What angle does the resulting ω make with the vertical? (If you want, you can check your answer by showing that the θ that makes this angle maximum is $\sin^{-1} \sqrt{7/9} \approx 61.9^\circ$.)

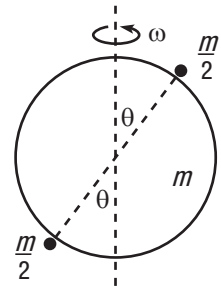


Fig. 9.56

9.38. Striking a triangle **

Consider the rigid object in Fig. 9.57. Four masses lie at the points shown on a rigid isosceles right triangle with hypotenuse length $4a$. The mass at the right angle is $3m$, and the other three masses are m . Label them A, B, C, D , as shown. Assume that the object is floating freely in outer space. Mass C is struck with a quick blow, directed into the page. Let the impulse have magnitude $\int F dt = P$. What are the velocities of all the masses immediately after the blow?

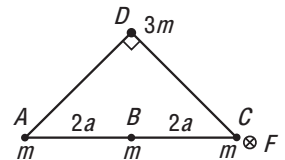


Fig. 9.57

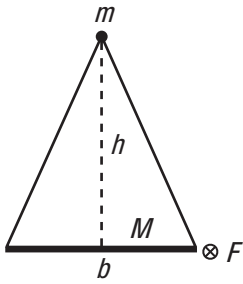


Fig. 9.58

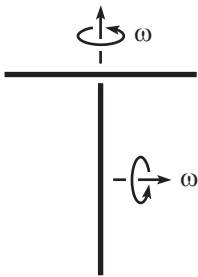


Fig. 9.59

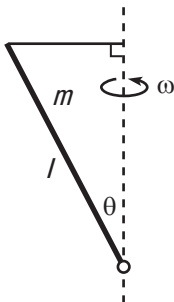


Fig. 9.60

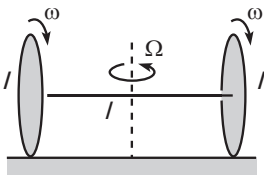


Fig. 9.61

9.39. Striking another triangle **

Consider the rigid object in Fig. 9.58. A uniform stick of mass M lies along the base of an isosceles triangle, and a mass m lies at the opposite vertex. The base has length b , and the height is h . Assume that the object is floating freely in outer space. The right end of the stick is struck with a quick blow, directed into the page. Let the impulse have magnitude $\int F dt = P$. What is the velocity of mass m immediately after the blow?

9.40. Sticking sticks **

Two identical uniform sticks spin around their stationary centers with equal angular speeds, as shown in Fig. 9.59. The bottom stick is slowly raised until its top end collides with the center of the top stick. The sticks stick together to form a rigid “T.” Assume that the collision takes place when the top stick lies in the plane of the paper. Immediately after the collision, one point (in addition to the CM) on the T will instantaneously be at rest. Where is this point?

9.41. Circling stick again **

Solve the problem in Section 9.4.2 again, but now use the CM as the origin.

9.42. Pivot and string **

A stick of mass m and length ℓ spins with frequency ω around an axis, as shown in Fig. 9.60. The stick makes an angle θ with the axis. One end is pivoted on the axis, and the other end is connected to the axis by a string that is perpendicular to the axis. What is the tension in the string, and what is the force that the pivot applies to the stick? (Ignore gravity.)

9.43. Rotating sheet **

A uniform flat rectangular sheet of mass m and side lengths a and b rotates with angular speed ω around a diagonal. What torque is required? Given a fixed area A , what should the rectangle look like if you want the required torque to be as large as possible? What is the upper bound on the torque?

9.44. Rotating axle **

Two wheels of mass m and moment of inertia I are connected by a massless axle of length ℓ , as shown in Fig. 9.61. The system rests on a frictionless surface, and the wheels rotate with frequency ω around the axle. Additionally, the whole system rotates with frequency Ω around the vertical axis through the center of the axle. What is the largest value of Ω for which both wheels stay on the ground?

9.45. Stick on a ring **

- (a) A stick of mass m and length $2r$ is arranged to make a constant angle θ with the horizontal, with its bottom end sliding in a circle on a frictionless ring of radius r , as shown in Fig. 9.62. What is the frequency of this motion? It turns out that there is a minimum θ for which this motion is possible; what is it?
- (b) If the radius of the ring is now R , what is the largest value of r/R for which this motion is possible for $\theta \rightarrow 0$?²⁵

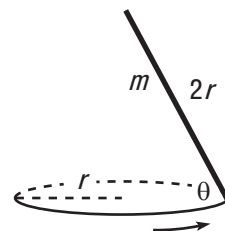


Fig. 9.62

*Section 9.6: Free symmetric top***9.46. Slightly wobbling ***

A coin of mass m and radius R is initially spinning around the axis perpendicular to its plane, with angular speed ω_3 . It is supported by a pivot at its center. You apply an infinitesimal downward strike at a point on the rim, as shown in Fig. 9.63, giving the coin an infinitesimal angular velocity component $\omega_\perp$ in the plane of the coin. When the plane of the coin returns to its original plane for the first time, what (approximately) is the orientation of the coin?

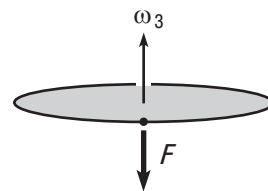


Fig. 9.63

9.47. Original orientation **

A coin of mass m and radius R is initially spinning around the axis perpendicular to its plane, with angular speed ω_3 . It is supported by a pivot at its center. You apply a (nonzero) downward strike at a point on the rim, as shown in Fig. 9.63, giving the coin an angular velocity component $\omega_\perp$ in the plane of the coin. Consider the n th time the plane of the coin returns to its original plane. What is the minimum value of n for which it is possible for the coin to have *exactly* the same orientation as when it started? What should $\omega_\perp$ be in terms of ω_3 to achieve this?

9.48. Seeing tails **

A coin (floating in outer space) of mass m and radius R is initially spinning around the axis perpendicular to its plane, with angular speed ω_3 . You view the coin directly from above, and you apply a downward strike at a point on the rim, as shown in Fig. 9.63. What is the minimum impulse, $\int F dt$, you must apply in order to be able to barely see the underside of the coin at some later time in its wobbling motion? Assuming that you apply this minimum impulse, how far will the center of the coin have moved by the time you are able to see the underside?

²⁵ You can also play around with both parts of this problem for the setup where the stick swings around below the ring, with its top end running along the ring.

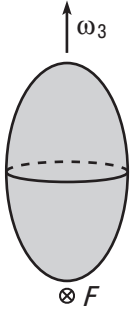


Fig. 9.64

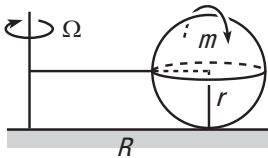


Fig. 9.65

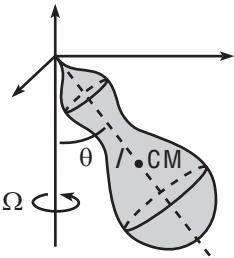


Fig. 9.66

9.49. Flipping a coin **

Imagine flipping an initially horizontal heads-up coin. If the initial angular velocity is horizontal, then the coin will rotate around this horizontal diameter the entire time in the air. The fraction of the flight time that the coin spends heads-up is therefore $1/2$. In practice, however, it is impossible to make the initial $\boldsymbol{\omega}$ be *exactly* horizontal, so assume that the initial components are $\omega_\perp$ and ω_3 , with $\omega_3 \ll \omega_\perp$. Show that in this limit, the fraction of the flight time that the coin spends heads-up is $1/2 + (4\omega_3^2)/(\pi\omega_\perp^2)$.²⁶

9.50. Dipping low **

A top with $I = 3I_3$ floats in outer space and initially spins around its x_3 axis with angular speed ω_3 . You apply a strike at the bottom point, directed into the page, as shown in Fig. 9.64, producing an angular velocity component, $\omega_\perp$, directed to the right. What should $\omega_\perp$ be in terms of ω_3 in order to have the total $\boldsymbol{\omega}$ vector dip as far below the horizontal as possible in the subsequent motion?

*Section 9.7: Heavy symmetric top***9.51. Rolling lollipop ***

Consider a lollipop made of a solid sphere of mass m and radius r that is radially pierced by a massless horizontal stick. The free end of the stick is pivoted on a pole (see Fig. 9.65), and the sphere rolls on the ground without slipping, with its center moving in a circle of radius R with frequency Ω . What is the normal force between the ground and the sphere?

9.52. Horizontal $\boldsymbol{\omega}$ **

A top (with mass m , moments I and I_3 , and distance ℓ from the pivot to CM) undergoes uniform precession, with its axis making a constant angle θ with the negative z axis, as shown in Fig. 9.66. If conditions are set up so that the top's $\boldsymbol{\omega}$ is always horizontal, what is the frequency of precession? Can such motion exist if the top is up above the horizontal, making an angle θ with the positive z axis?

²⁶ If the coin starts truly horizontal (or is at least not biased to tilt in any particular direction on average, which is still a condition that seems difficult to guarantee), and if you catch the coin in your hand to reduce random table-bouncing effects, then the result of this exercise implies that the coin is biased to come up heads. This effect is analyzed in detail in a paper by Persi Diaconis, Susan Holmes, and Richard Montgomery (to be published), who estimate that the probability of obtaining heads for a normally flipped coin is about 0.51.

9.53. Sliding lollipop ***

Consider a lollipop made of a solid sphere of mass m and radius r that is radially pierced by a massless stick. The free end of the stick is pivoted on the ground, which is frictionless (see Fig. 9.67). The sphere *slides* along the ground, with the same point on the sphere always touching the ground. The center moves in a circle of radius R with frequency Ω . Show that the normal force between the ground and the sphere is $N = mg + mr\Omega^2$, which is independent of R . Solve this by:

- Using an $\mathbf{F} = m\mathbf{a}$ argument.²⁷
- Using the more complicated $\boldsymbol{\tau} = d\mathbf{L}/dt$ argument.

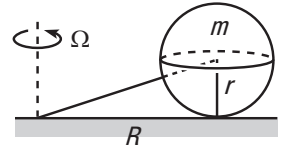


Fig. 9.67

9.54. Rolling wheel and axle ***

A massless axle has one end attached to a wheel (a uniform disk of mass m and radius r), with the other end pivoted on the ground (see Fig. 9.68). The wheel rolls on the ground without slipping, with the axle inclined at an angle θ . The point of contact on the ground traces out a circle with frequency Ω .

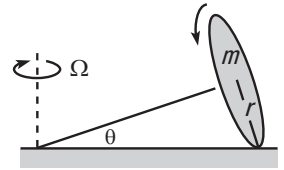


Fig. 9.68

- Show that $\boldsymbol{\omega}$ points horizontally to the right (at the instant shown), with magnitude $\omega = \Omega / \tan \theta$.
- Show that the normal force between the ground and the wheel is

$$N = mg \cos^2 \theta + mr\Omega^2 \left(\frac{1}{4} \cos \theta \sin^2 \theta + \frac{3}{2} \cos^3 \theta \right). \quad (9.102)$$

9.55. Ball under a cone ***

A hollow ball (with $I = (2/3)mR^2$) rolls without slipping on the inside surface of a fixed cone, whose tip points upward, as shown in Fig. 9.69. The angle at the vertex of the cone is 60° . Initial conditions have been set up so that the contact point on the cone traces out a horizontal circle of radius ℓ at frequency Ω , while the contact point on the ball traces out a circle of radius $R/2$. Assume that the coefficient of friction between the ball and the cone is sufficiently large to prevent slipping. What is the frequency of precession, Ω ? What does it reduce to in the limits $\ell \gg R$ and $\ell \rightarrow (\sqrt{3}/2)R$ (the ball has to fit inside the cone, of course). What relation between ℓ and R must be satisfied if the setup is to work with a solid ball with $I = (2/5)mR^2$?

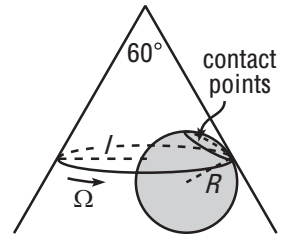


Fig. 9.69

9.56. Ball in a cone ****

A ball (with $I = (2/5)MR^2$) rolls without slipping on the inside surface of a fixed cone, whose tip points downward. The half-angle at the vertex

²⁷ This method happens to work here, due to the unusually nice nature of the sphere's motion. For more general motion (for example, in Problem 9.22, where the sphere is spinning), you must use $\boldsymbol{\tau} = d\mathbf{L}/dt$.

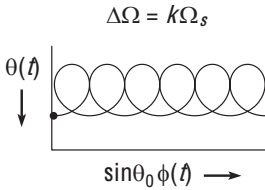


Fig. 9.70

9.57. Nutation loops **

In Fig. 9.38, the loops don't quite intersect each other in the $\Delta\Omega = 4\Omega_s$ case, but they very much do in the $\Delta\Omega = 10\Omega_s$ case (a given loop there actually intersects the two on either side). Show that the value of k for which adjacent loops in the $\Delta\Omega = k\Omega_s$ case barely touch each other (as shown in Fig. 9.70) is $k \approx 4.6033$. You will have to solve something numerically. *Hint:* The curve is vertical at the relevant points.

9.10 Solutions

9.1. Many different ω 's

We want to find all the vectors ω with the property that $\omega \times a\hat{\mathbf{x}} = v\hat{\mathbf{y}}$. Since ω is orthogonal to this cross product, ω must lie in the x - z plane. We claim that if ω makes an angle θ with the x axis and has magnitude $v/(a \sin \theta)$, then it satisfies $\omega \times a\hat{\mathbf{x}} = v\hat{\mathbf{y}}$. Indeed,

$$\omega \times a\hat{\mathbf{x}} = |\omega||a\hat{\mathbf{x}}| \sin \theta \hat{\mathbf{y}} = v\hat{\mathbf{y}}. \quad (9.103)$$

Alternatively, note that such an ω can be written as

$$\omega = \frac{v}{a \sin \theta} (\cos \theta, 0, \sin \theta) = \left(\frac{v}{a \tan \theta}, 0, \frac{v}{a} \right). \quad (9.104)$$

Only the z component here is relevant in the cross product with $a\hat{\mathbf{x}}$, so we have $\omega \times a\hat{\mathbf{x}} = (v/a)\hat{\mathbf{z}} \times a\hat{\mathbf{x}} = v\hat{\mathbf{y}}$. The answer to the problem is therefore that ω must take the form given in Eq. (9.104).

It makes sense that the magnitude of ω is $v/(a \sin \theta)$, because if we drop a perpendicular from the particle to the line of ω , we see that the particle may be considered to be instantaneously traveling in a circle of radius $r = a \sin \theta$ around ω , at speed v . And so we have $v = \omega r$, as we should. Whether the particle actually does travel in this circle is irrelevant. The past and future motion doesn't matter in finding the instantaneous ω . All we need to know is the velocity at the given instant.

A few possible ω 's are drawn in Fig. 9.71. Technically, it is possible to have $\pi < \theta < 2\pi$, but then the $v/(a \sin \theta)$ coefficient in Eq. (9.104) is negative, which means that ω really points upward in the x - z plane (physically, ω must point upward if the particle's velocity is to be in the positive y direction). So we'll assume $0 < \theta < \pi$. And since the v/a value of ω_z in Eq. (9.104) is independent of θ , all of the possible ω 's look like those in Fig. 9.72.

For $\theta = \pi/2$, we have $\omega = v/a$, which makes sense. If θ is very small, then ω is very large, because $\omega \propto 1/\sin \theta$. This also makes sense, because the particle is (instantaneously) traveling around in a very small circle at the given speed v .

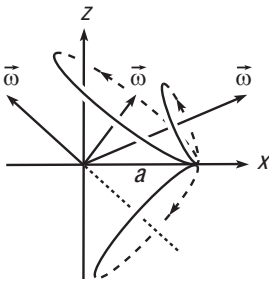


Fig. 9.71

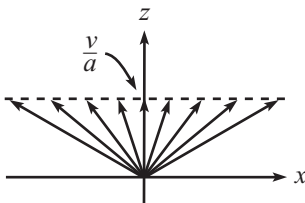


Fig. 9.72

REMARK: The point of this problem is that the particle may be in the process of having its position vector trace out a cone around one of many possible axes, or perhaps it may be undergoing a more complicated motion. If we are handed only the given information on position and velocity, then it is impossible to determine which of these motions is happening. And it is likewise impossible to uniquely determine ω . This is true for a collection of points that lie on at most one line through the origin. If the points, along with the origin, span more than a 1-D line, then ω is in fact uniquely determined (see Footnote 9). ♣

9.2. Fixed points on a sphere

FIRST SOLUTION: For the purposes of Theorem 9.1, we need only show that two points end up where they started for an *infinitesimal* transformation. But since it's possible to prove this for a general transformation, we'll consider the general case here.

Consider the point A that ends up farthest away from where it started.²⁸ Label the ending point B . Draw the great circle, C_{AB} , through A and B . Draw the great circle, C_A , that is perpendicular to C_{AB} at A ; and draw the great circle, C_B , that is perpendicular to C_{AB} at B . We claim that the transformation must take C_A to C_B . This is true for the following reason. The image of C_A is certainly a great circle through B . And this great circle must be perpendicular to C_{AB} , because otherwise there would exist another point that ended up farther away from its starting point than A did (see Fig. 9.73). Since there is only one great circle through B that is perpendicular to C_{AB} , the image of C_A must in fact be C_B .

Now consider the two points, P_1 and P_2 , where C_A and C_B intersect (any two great circles intersect in two points). Look at P_1 . The distances P_1A and P_1B are equal, because C_{AB} makes equal angles (namely 90°) with C_A and C_B . Therefore, the point P_1 is not moved by the transformation. This is true because P_1 ends up on C_B (because C_B is the image of C_A , on which P_1 started), and if it ended up at a point other than P_1 , then its final distance from B would be different from its initial distance from A . This would contradict the fact that distances are preserved on the rigid sphere. Likewise for P_2 . We have therefore found our two desired points.

Note that for a noninfinitesimal transformation, every point on the sphere may move at some time during the transformation. But what we've just shown is that two points end up back where they started, even if they've moved in the interim.

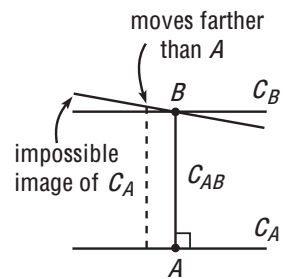


Fig. 9.73

SECOND SOLUTION: In the spirit of the above solution, we can give a simpler solution, but one that is valid only in the case of infinitesimal transformations.

Pick any point A that moves during the transformation. Draw the great circle that passes through A and is perpendicular to the direction of A 's motion. (This direction is well defined, because we are considering an infinitesimal transformation, so A doesn't have time to change direction.) All points on this great circle must move (if they move at all) perpendicular to the great circle, because otherwise their distances from A would change. But they cannot all move in the same direction, because then the center of the great circle, and hence the sphere, would move (but it is assumed to be fixed). Therefore, at least one point on the great circle moves in the direction opposite to the direction in which A moves. Therefore, by continuity, some point (and hence also its diametrically opposite point) on the great circle must remain fixed.

9.3. Rolling cone

At the risk of overdoing it, we'll present three solutions. The second and third solutions are the type that might make your head hurt, so you may want to reread them after looking at the discussion of the angular velocity vector in Section 9.7.2.

²⁸ If there is more than one such point, pick any one of them. Considering that the result of this problem is that the net motion of the sphere is a rotation around an axis, there will in fact be a whole great circle of points that move farthest.

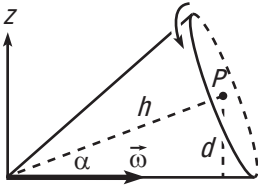


Fig. 9.74

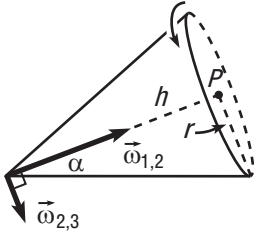


Fig. 9.75

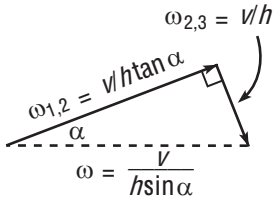


Fig. 9.76

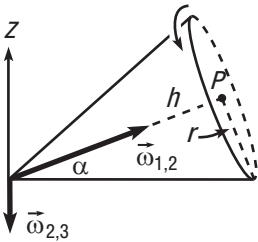


Fig. 9.77

FIRST SOLUTION: Without doing any calculations, we know that ω points along the line of contact of the cone with the table, because these are the points on the cone that are instantaneously at rest. And we know that as time goes by, ω rotates around in the horizontal plane with angular speed $v/(h \cos \alpha)$, because point P travels at speed v in a circle of radius $h \cos \alpha$ around the z axis.

The magnitude of ω can be found as follows. At a given instant, P may be considered to be rotating in a circle of radius $d = h \sin \alpha$ around ω (see Fig. 9.74). Since P moves with speed v , the angular speed of this rotation is $\omega = v/d$. Therefore,

$$\omega = \frac{v}{h \sin \alpha}. \quad (9.105)$$

SECOND SOLUTION: We can use Theorem 9.3 with the following frames. S_1 is fixed in the cone; S_3 is the lab frame; and S_2 is the frame that (instantaneously) rotates around the tilted $\omega_{2,3}$ axis shown in Fig. 9.75, at the speed such that the axis of the cone remains fixed in S_2 . The tip of $\omega_{2,3}$ traces out a circle as it precesses around the z axis, so after the cone moves a little, we will need to use a new S_2 frame. But at any moment, S_2 instantaneously rotates around the axis perpendicular to the axis of the cone.

In the language of Theorem 9.3, $\omega_{1,2}$ and $\omega_{2,3}$ point in the directions shown. We must find their magnitudes and then add the vectors to determine the angular velocity of S_1 with respect to S_3 . First, we have

$$|\omega_{2,3}| = \frac{v}{h}, \quad (9.106)$$

because point P moves (instantaneously) with speed v in a circle of radius h around $\omega_{2,3}$. We now claim that

$$|\omega_{1,2}| = \frac{v}{r} = \frac{v}{h \tan \alpha}, \quad (9.107)$$

where r is the radius of the base of the cone. This is true because someone fixed in S_2 sees the endpoint of the radius (the one drawn in Fig. 9.75) move “backward” at speed v , because it is stationary with respect to the table. Hence, the cone must be spinning with frequency v/r in S_2 .

The addition of $\omega_{1,2}$ and $\omega_{2,3}$ is shown in Fig. 9.76. The result has magnitude $v/(h \sin \alpha)$, and it points horizontally because $|\omega_{2,3}|/|\omega_{1,2}| = \tan \alpha$.

THIRD SOLUTION: We can use Theorem 9.3 with the following frames. S_1 is fixed in the cone; and S_3 is the lab frame (as in the second solution). But now let S_2 be the frame that rotates around the (negative) z axis, at the speed such that the axis of the cone remains fixed in it. Note that we can keep using this same S_2 frame as time goes by, unlike the S_2 frame in the second solution.

$\omega_{1,2}$ and $\omega_{2,3}$ point in the directions shown in Fig. 9.77. As above, we must find their magnitudes and then add the vectors to determine the angular velocity of S_1 with respect to S_3 . First, we have

$$|\omega_{2,3}| = \frac{v}{h \cos \alpha}, \quad (9.108)$$

because point P moves with speed v in a circle of radius $h \cos \alpha$ around $\omega_{2,3}$.

It's a little trickier to find $|\omega_{1,2}|$. From the point of view of someone spinning around with S_2 , the table rotates backward with frequency $|\omega_{2,3}| = v/(h \cos \alpha)$, from Eq. (9.108). Consider the circle of contact points on the table where the base of the cone touches it. This circle has a radius $h/\cos \alpha$, so someone spinning around with S_2 sees the circle move backward with speed $|\omega_{2,3}|(h/\cos \alpha) = v/\cos^2 \alpha$ around the vertical. Since there is no slipping, the contact point on the cone must also move with this speed in S_2 around the axis of the cone (which is fixed in S_2). And since the radius of the base is r , this means that the cone rotates with angular speed $v/(r \cos^2 \alpha)$ with

respect to S_2 . Therefore, using $r = h \tan \alpha$, we have

$$|\omega_{1,2}| = \frac{v}{r \cos^2 \alpha} = \frac{v}{h \sin \alpha \cos \alpha}. \quad (9.109)$$

The addition of $\omega_{1,2}$ and $\omega_{2,3}$ is shown in Fig. 9.78. The result has magnitude $v/(h \sin \alpha)$, and it points horizontally because $|\omega_{2,3}|/|\omega_{1,2}| = \sin \alpha$.

REMARK: The difference between the $\omega_{1,2}$ vectors in the second and third solutions comes down to the following fact. Let Q be the contact point on the base of the cone. Consider the ratio of the distance from Q to $\omega_{2,3}$ to the distance from P to $\omega_{2,3}$. This ratio is 1 in the second solution, but $1/\cos^2 \alpha$ in the third solution. This implies that the “backward” speed of Q relative to P , as measured in frame S_2 , is a factor $1/\cos^2 \alpha$ larger in the third solution. And since Q is the same distance r away from $\omega_{1,2}$ in both cases, we see that $\omega_{1,2}$ is a factor $1/\cos^2 \alpha$ larger in the third solution. ♣

9.4. Parallel-axis theorem

Consider one of the diagonal entries in $\mathbf{I}$, say $I_{xx} \equiv \int (y^2 + z^2)$. In terms of the new variables, this equals

$$\begin{aligned} I_{xx} &= \int \left((Y + y')^2 + (Z + z')^2 \right) = \int (Y^2 + Z^2) + \int (y'^2 + z'^2) \\ &= M(Y^2 + Z^2) + \int (y'^2 + z'^2), \end{aligned} \quad (9.110)$$

as desired. We have used the fact that the cross terms vanish because, for example, $\int Y y' = Y \int y' = 0$, by definition of the CM. Similarly, consider an off-diagonal entry in $\mathbf{I}$, say $I_{xy} \equiv -\int xy$. We have

$$I_{xy} = -\int (X + x')(Y + y') = -\int XY - \int x'y' = -M(XY) - \int x'y', \quad (9.111)$$

where the cross terms have likewise vanished. We therefore see that all of the terms in $\mathbf{I}$ take the form of those in Eq. (9.19), as desired.

9.5. A nice cylinder

By symmetry, the principal axes are the symmetry axis of the cylinder, along with any diameters in the cross-section circle through the CM. Let the equal moments around the diameters be I . Then by Theorem 9.5, if the moment around the symmetry axis also equals I , then every axis is a principal axis.

Let the mass of the cylinder be M . Let its radius be R and its height be h . Then the moment around the symmetry axis is $MR^2/2$. Let D be a diameter through the CM. The moment around D can be calculated as follows. Slice the cylinder into horizontal disks of thickness dy . Let ρ be the mass per unit height (so $\rho = M/h$). The mass of each disk is then ρdy , so the moment around a diameter through the disk is $(\rho dy)R^2/4$ (the usual result for a disk). Therefore, by the parallel-axis theorem, the moment of a disk at height y (where $-h/2 \leq y \leq h/2$) around D is $(\rho dy)R^2/4 + (\rho dy)y^2$. Hence, the moment of the entire cylinder around D is

$$I = \int_{-h/2}^{h/2} \left(\frac{\rho R^2}{4} + \rho y^2 \right) dy = \frac{\rho R^2 h}{4} + \frac{\rho h^3}{12} = \frac{MR^2}{4} + \frac{Mh^2}{12}. \quad (9.112)$$

We want this to equal $MR^2/2$. Therefore, $h = \sqrt{3}R$.

As an exercise, you can show that if the origin were instead taken to be the center of one of the circular faces, then the answer would be $h = \sqrt{3}R/2$. Note that the I in Eq. (9.112) looks just like the moment of a disk around a diameter plus the moment of a stick around its center. This is no coincidence. The integral that yielded the $Mh^2/12$ term here is the same integral that we would perform for a stick.

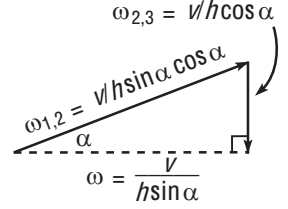


Fig. 9.78

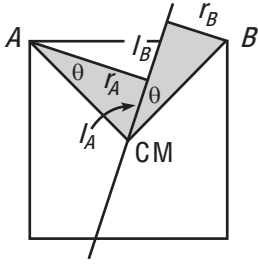


Fig. 9.79

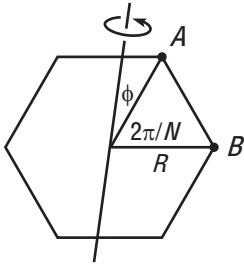


Fig. 9.80

9.6. Rotating square

Label two of the masses A and B , as shown in Fig. 9.79. Let ℓ_A be the distance along the axis from the CM to A 's string, and let r_A be the length of A 's string. Likewise for B . The force, F_A , in A 's string must account for the centripetal acceleration of A . Hence, $F_A = mr_A\omega^2$. The torque around the CM due to F_A is therefore

$$\tau_A = mr_A\ell_A\omega^2. \quad (9.113)$$

Likewise, the torque around the CM due to B 's string is $\tau_B = mr_B\ell_B\omega^2$, in the opposite direction. But the two shaded triangles in Fig. 9.79 are congruent, because they have the same hypotenuse and the same angle θ . Therefore, $\ell_A = r_B$ and $\ell_B = r_A$. Hence, $\tau_A = \tau_B$, and the torques cancel. The torques from the other two masses likewise cancel. Note that a uniform square is made up of many sets of these squares of point masses, so we've also shown that no torque is needed for a uniform square.

REMARK: For a general N -gon of point masses, Problem 9.8 below shows that any axis in the plane is a principal axis. Let's prove this here by using the above torque argument. We'll use a math trick here that involves writing a trig function (a sine) as the imaginary part of a complex exponential. Using Eq. (9.113), we see that the torque from mass A in Fig. 9.80 is $\tau_A = m\omega^2 R^2 \sin\phi \cos\phi$. Likewise, the torque from mass B is $\tau_B = m\omega^2 R^2 \sin(\phi + 2\pi/N) \cos(\phi + 2\pi/N)$, and so on. The total torque around the CM is therefore

$$\begin{aligned} \tau &= mR^2\omega^2 \sum_{k=0}^{N-1} \sin\left(\phi + \frac{2\pi k}{N}\right) \cos\left(\phi + \frac{2\pi k}{N}\right) \\ &= \frac{mR^2\omega^2}{2} \sum_{k=0}^{N-1} \sin\left(2\phi + \frac{4\pi k}{N}\right) \\ &= \frac{mR^2\omega^2}{2} \sum_{k=0}^{N-1} \text{Im}\left(e^{i(2\phi + 4\pi k/N)}\right) \\ &= \frac{mR^2\omega^2}{2} \text{Im}\left(e^{2i\phi} \left(1 + e^{4\pi i/N} + e^{8\pi i/N} + \dots + e^{4(N-1)\pi i/N}\right)\right) \\ &= \frac{mR^2\omega^2}{2} \text{Im}\left(e^{2i\phi} \left(\frac{e^{4N\pi i/N} - 1}{e^{4\pi i/N} - 1}\right)\right) \\ &= 0. \end{aligned} \quad (9.114)$$

To obtain the fifth line, we summed the geometric series. And to obtain the last line, we used the fact that $e^{4\pi i} = 1$. The one exception to this result is when $N = 2$, because the denominator in the fifth line is zero (but it's hard to have a 2-gon, anyway); this corresponds to the $\theta \neq 180^\circ$ restriction in Theorem 9.6.

Note that what we've done here is show that $\sum r_i \ell_i = 0$, which implies that the total torque is zero (which is one of the definitions of a principal axis). In terms of the chosen axis, this is equivalent to showing that $\sum xy = 0$, that is, showing that the off-diagonal terms in the inertia tensor vanish (which is simply another definition of the principal axes). ♣

9.7. Existence of principal axes for a pancake

For a pancake object, the inertia tensor $\mathbf{I}$ takes the form in Eq. (9.8), with $z = 0$. Therefore, if we can find a set of axes for which $\int xy = 0$, then $\mathbf{I}$ will be diagonal, and we will have found our principal axes. We can prove, using a continuity argument, that such a set of axes exists.

Pick a set of axes, and write down the integral $\int xy \equiv I_0$. If $I_0 = 0$, then we are done. If $I_0 \neq 0$, then rotate these axes by an angle of $\pi/2$, so that the new $\hat{\mathbf{x}}$ is the old $\hat{\mathbf{y}}$,

and the new $\hat{\mathbf{y}}$ is the old $-\hat{\mathbf{x}}$ (see Fig. 9.81). Write down the new integral $\int xy \equiv I_{xy}/2$. Since the new and old coordinates are related by $x_{\text{new}} = y_{\text{old}}$ and $y_{\text{new}} = -x_{\text{old}}$, we have $I_{xy}/2 = -I_0$. Therefore, since $\int xy$ switches sign during the (continuous) rotation of the axes, there must exist some intermediate angle for which the integral $\int xy$ is zero.

9.8. Symmetries and principal axes for a pancake

In view of the form of the inertia tensor in Eq. (9.8), we want to show that if a pancake object has a symmetry under a rotation through $\theta \neq \pi$, then $\int xy = 0$ for any set of axes (through the origin). Take an arbitrary set of axes and rotate them through an angle $\theta \neq \pi$. The new coordinates are $x' = (x \cos \theta + y \sin \theta)$ and $y' = (-x \sin \theta + y \cos \theta)$, so the new matrix entries, in terms of the old ones, are

$$\begin{aligned} I'_{xx} &\equiv \int y'^2 = I_{yy} \sin^2 \theta + 2I_{xy} \sin \theta \cos \theta + I_{xx} \cos^2 \theta, \\ I'_{yy} &\equiv \int x'^2 = I_{yy} \cos^2 \theta - 2I_{xy} \sin \theta \cos \theta + I_{xx} \sin^2 \theta, \\ I'_{xy} &\equiv - \int x'y' = I_{xy} \sin \theta \cos \theta + I_{xy} (\cos^2 \theta - \sin^2 \theta) - I_{xx} \sin \theta \cos \theta. \end{aligned} \quad (9.115)$$

If the object looks exactly like it did before the rotation, then $I'_{xx} = I_{xx}$, $I'_{yy} = I_{yy}$, and $I'_{xy} = I_{xy}$. The first two of these statements are actually equivalent (as you can show), so we'll just use the first and third. Using Eq. (9.116), along with $1 - \cos^2 \theta = \sin^2 \theta$, these two statements give

$$\begin{aligned} 0 &= -I_{xx} \sin^2 \theta + 2I_{xy} \sin \theta \cos \theta + I_{yy} \sin^2 \theta, \\ 0 &= -I_{xx} \sin \theta \cos \theta - 2I_{xy} \sin^2 \theta + I_{yy} \sin \theta \cos \theta. \end{aligned} \quad (9.116)$$

Multiplying the first of these by $\cos \theta$ and the second by $\sin \theta$, and subtracting, gives

$$2I_{xy} \sin \theta = 0. \quad (9.117)$$

Under the assumption $\theta \neq \pi$ (and $\theta \neq 0$, of course), we must therefore have $I_{xy} = 0$. Our initial axes were arbitrary; hence, any set of axes (through the origin) in the plane is a set of principal axes.

REMARK: If an object is invariant under a rotation through an angle θ , then θ must be of the form $\theta = 2\pi/N$, for some integer N (convince yourself of this). So consider a regular N -gon with “radius” R and with point masses m located at the vertices. Any object that is invariant under a rotation through $\theta = 2\pi/N$ may be considered to be built out of regular point-mass N -gons of various sizes. Theorem 9.5 implies that all axes in the plane of a regular N -gon have the same moment (the two axes in this theorem aren't required to be orthogonal). Let's demonstrate this explicitly for a regular N -gon. This method we'll use here is similar to the one in the remark in Problem 9.6, except that now we'll write a trig function (a cosine) as the real part of a complex exponential. In Fig. 9.82 the distance from mass A to the axis is $r_A = R \sin \phi$. And the distance from B to the axis is $r_B = R \sin(\phi + 2\pi/N)$, and so on for the other masses. The moment of inertia around the axis is therefore

$$\begin{aligned} I_\phi &= mR^2 \sum_{k=0}^{N-1} \sin^2 \left(\phi + \frac{2\pi k}{N} \right) \\ &= \frac{mR^2}{2} \sum_{k=0}^{N-1} \left(1 - \cos \left(2\phi + \frac{4\pi k}{N} \right) \right) \end{aligned}$$

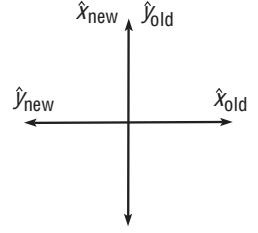


Fig. 9.81

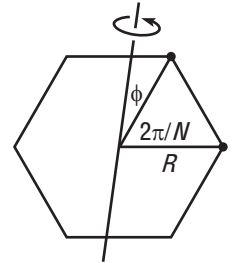


Fig. 9.82

$$\begin{aligned}
&= \frac{NmR^2}{2} - \frac{mR^2}{2} \sum_{k=0}^{N-1} \operatorname{Re} \left(e^{i(2\phi + 4\pi k/N)} \right) \\
&= \frac{NmR^2}{2} - \frac{mR^2}{2} \operatorname{Re} \left(e^{2i\phi} \left(1 + e^{4\pi i/N} + e^{8\pi i/N} + \dots + e^{4(N-1)\pi i/N} \right) \right) \\
&= \frac{NmR^2}{2} - \frac{mR^2}{2} \operatorname{Re} \left(e^{2i\phi} \left(\frac{e^{4N\pi i/N} - 1}{e^{4\pi i/N} - 1} \right) \right), \tag{9.118}
\end{aligned}$$

where we have summed the geometric series to obtain the last line. The numerator in the parentheses equals $e^{4\pi i} - 1 = 0$. And if $N \neq 2$, the denominator is not zero. Therefore, if $N \neq 2$ (which is equivalent to the $\theta \neq \pi$ restriction), we have

$$I_\phi = \frac{NmR^2}{2}, \tag{9.119}$$

which is independent of ϕ . This result of $NmR^2/2$ for the common value of all the moments makes sense, because the perpendicular-axis theorem says that this common value must be half of the moment around the axis perpendicular to the plane, which is NmR^2 . ♣

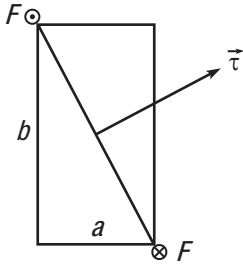


Fig. 9.83

9.9. Striking a rectangle

If the force is out of the page at the upper left corner and into the page at the lower right corner, then the torque $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ points upward to the right, as shown in Fig. 9.83, with $\boldsymbol{\tau} \propto (b, a)$. The angular momentum equals $\int \boldsymbol{\tau} dt$, so immediately after the strike, $\mathbf{L}$ is proportional to (b, a) . But the angular momentum may also be written as $\mathbf{L} = (I_x \omega_x, I_y \omega_y)$, where $I_x = mb^2/12$ and $I_y = ma^2/12$ are the principal moments. Therefore, we have

$$(I_x \omega_x, I_y \omega_y) \propto (b, a) \implies (\omega_x, \omega_y) \propto \left(\frac{b}{I_x}, \frac{a}{I_y} \right) \propto \left(\frac{b}{b^2}, \frac{a}{a^2} \right) \propto (a, b), \tag{9.120}$$

which is the direction of the other diagonal. This result checks in the special case $a = b$, and also in the limit where one of a and b is much larger than the other. Basically, if b is much larger than a , then this largeness matters more in the moment of inertia around the x axis (which is quadratic in length) than it does in the torque in the x direction (which is linear in length), so there is very little rotation around the x axis.

9.10. Rotating stick

The angular momentum around the CM can be found as follows. Break $\boldsymbol{\omega}$ up into its components along the principal axes of the stick (which are parallel and perpendicular to the stick). The moment of inertia around the stick is zero. Therefore, only the component of $\boldsymbol{\omega}$ perpendicular to the stick is relevant in calculating $\mathbf{L}$. This component is $\omega \sin \theta$, and the associated moment of inertia is $m\ell^2/12$. Hence, the angular momentum at any time has magnitude

$$L = \frac{1}{12} m\ell^2 \omega \sin \theta, \tag{9.121}$$

and it points as shown in Fig. 9.84. The tip of the vector $\mathbf{L}$ traces out a circle in a horizontal plane with frequency ω . The radius of this circle is the horizontal component of $\mathbf{L}$, which is $L_\perp \equiv L \cos \theta$. The rate of change of $\mathbf{L}$ therefore has magnitude

$$\left| \frac{d\mathbf{L}}{dt} \right| = \omega L_\perp = \omega L \cos \theta = \omega \left(\frac{1}{12} m\ell^2 \omega \sin \theta \right) \cos \theta, \tag{9.122}$$

and it is directed into the page at the instant shown.

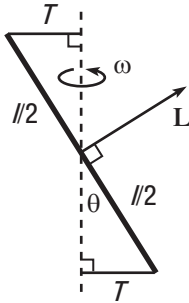


Fig. 9.84

Let the tension in the strings be T . Then the torque due to the strings is $\tau = 2T(\ell/2) \cos \theta$, directed into the page at the instant shown. Therefore, $\tau = d\mathbf{L}/dt$ gives

$$T\ell \cos \theta = \omega \left(\frac{1}{12} m\ell^2 \omega \sin \theta \right) \cos \theta \implies T = \frac{1}{12} m\ell \omega^2 \sin \theta. \quad (9.123)$$

REMARKS: For $\theta \rightarrow 0$, this goes to zero, which makes sense. For $\theta \rightarrow \pi/2$, it goes to the finite value $m\ell\omega^2/12$, which isn't so obvious. The lever arm and $|d\mathbf{L}/dt|$ are both small in this limit, and they happen to be proportional to each other as they both head to zero.

Note that if we instead had a massless stick with equal masses $m/2$ on the ends, so that the relevant moment of inertia is now $m\ell^2/4$, then our answer would be $T = (1/4)m\ell\omega^2 \sin \theta$. This makes sense if we write it as $T = (m/2) \cdot (\ell/2) \sin \theta \cdot \omega^2$, because each tension is responsible for simply keeping a mass $m/2$ moving in a circle of radius $(\ell/2) \sin \theta$ at frequency ω . This simple force argument doesn't work for the original stick because of internal forces in the stick. ♣

9.11. Stick under a ring

As in Problem 9.10, the angular momentum around the CM can be found by breaking ω up into its components along the principal axes of the stick. The reasoning is exactly the same, so the angular momentum at any time has magnitude

$$L = \frac{1}{12} m\ell^2 \omega \sin \theta, \quad (9.124)$$

and it points as shown in Fig. 9.85. The change in $\mathbf{L}$ comes from its horizontal component. This has length $L \cos \theta$ and travels in a circle at frequency ω . Hence, $|d\mathbf{L}/dt| = \omega L \cos \theta$, and it is directed into the page at the instant shown.

The torque around the CM has magnitude $mg(\ell/2) \sin \theta$, and it points into the page at the instant shown. (This torque arises from the vertical force from the ring. There is no horizontal force from the ring, because the CM does not move.) Therefore, $\tau = d\mathbf{L}/dt$ gives

$$\frac{mg\ell \sin \theta}{2} = \omega \left(\frac{m\ell^2 \omega \sin \theta}{12} \right) \cos \theta \implies \omega = \sqrt{\frac{6g}{\ell \cos \theta}}. \quad (9.125)$$

REMARKS:

1. For $\theta \rightarrow \pi/2$, this goes to infinity, which makes sense. For $\theta \rightarrow 0$, it goes to $\sqrt{6g/\ell}$, which isn't so obvious.
2. The motion in this problem isn't possible if the bottom end of the stick, instead of the top end, slides along the ring. The magnitudes of all quantities are the same as in the original problem, but the torque points in the wrong direction, as you can check. But see Exercise 9.45 for a related setup.
3. Since the middle of the stick is motionless, you might be tempted to treat the bottom half as the stick in the example in Section 9.4.2. That is, you might just want to plug in $\ell/2$ as the length in Eq. (9.38). This, however, does not yield the answer in Eq. (9.125). The error is that there are internal forces in the stick that provide torques. If a pivot were placed at the CM of the stick (connecting the two halves) in this problem, then the stick would not remain straight.
4. If we replace the stick by a massless string with equal masses $m/2$ on the ends, then the relevant moment of inertia is $m\ell^2/4$, and we obtain $\omega = \sqrt{2g/(\ell \cos \theta)}$. This is in fact simply the $\omega = \sqrt{g/[(\ell/2) \cos \theta]}$ answer for a point-mass circular pendulum of length $\ell/2$ (see Problem 9.12), because the middle of the string is motionless, and because the flexible string can't provide the internal torques mentioned in the previous paragraph. ♣

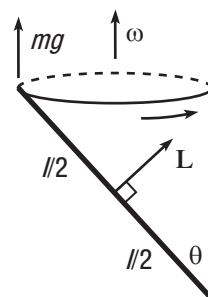


Fig. 9.85

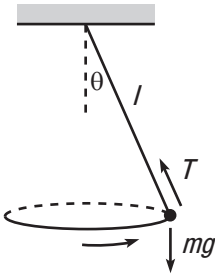


Fig. 9.86

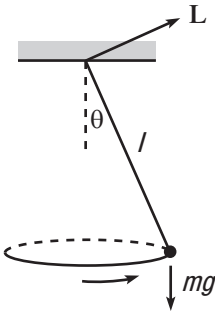


Fig. 9.87

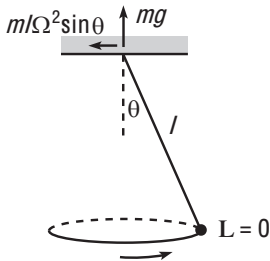


Fig. 9.88

9.12. Circular pendulum

- (a) The forces on the mass are gravity and the force from the rod (the tension), which points along the rod (see Fig. 9.86).²⁹ Since there is no vertical acceleration, we have $T \cos \theta = mg$. The unbalanced horizontal force from the tension is therefore $T \sin \theta = mg \tan \theta$. This force accounts for the centripetal acceleration, $m(\ell \sin \theta) \Omega^2$. Hence,

$$\Omega = \sqrt{\frac{g}{\ell \cos \theta}}. \quad (9.126)$$

For $\theta \approx 0$, this is the same as the $\sqrt{g/\ell}$ frequency for a simple pendulum. For $\theta \approx \pi/2$, it goes to infinity, which makes sense. Note that θ must be less than $\pi/2$ for circular motion to be possible. (This restriction doesn't hold for a spinning top with extended mass.)

- (b) The only force that applies a torque relative to the pivot is the gravitational force, so the torque is $\tau = mg \ell \sin \theta$, directed into the page at the instant shown in Fig. 9.87.

At this moment, the mass has a speed $(\ell \sin \theta) \Omega$, directed into the page. Therefore, $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ has magnitude $m \ell^2 \Omega \sin \theta$ and is directed upward to the right, as shown. The tip of $\mathbf{L}$ traces out a circle of radius $L \cos \theta$, at frequency Ω . Therefore, $d\mathbf{L}/dt$ has magnitude $\Omega L \cos \theta$ and is directed into the page. Hence, $\tau = d\mathbf{L}/dt$ gives $mg \ell \sin \theta = \Omega(m \ell^2 \Omega \sin \theta) \cos \theta$, which yields Eq. (9.126).

- (c) The only force that applies a torque relative to the mass is the force from the pivot, which has two components (see Fig. 9.88). The vertical piece is mg . Relative to the mass, this provides a torque of $mg(\ell \sin \theta)$, directed into the page. There is also the horizontal piece, which accounts for the centripetal acceleration of the mass, so it equals $m(\ell \sin \theta) \Omega^2$. Relative to the mass, this provides a torque of $m \ell \sin \theta \Omega^2 (\ell \cos \theta)$, directed out of the page.

Relative to the mass, there is no angular momentum. Therefore, $d\mathbf{L}/dt = 0$, and so there must be no torque. This means that the above two torques must cancel, which gives $mg(\ell \sin \theta) = m \ell \sin \theta \Omega^2 (\ell \cos \theta)$. This yields Eq. (9.126).

In problems that are more complicated than this one, it is often easier to work with the fixed pivot (if there is one) as the origin, instead of the CM, because then you don't have to worry about the messy pivot forces contributing to the torque.

9.13. Rolling in a cone

The forces on the ring are gravity (mg), the normal force (N) from the cone, and the friction force (F) pointing up along the cone (or perhaps down along the cone if F turns out to be negative, but we'll find that it won't). Since there is no net force in the vertical direction, we have

$$N \sin \theta + F \cos \theta = mg. \quad (9.127)$$

The inward horizontal force accounts for the centripetal acceleration, which gives

$$N \cos \theta - F \sin \theta = m(h \tan \theta) \Omega^2. \quad (9.128)$$

Solving the previous two equations for F gives

$$F = mg \cos \theta - m \Omega^2 (h \tan \theta) \sin \theta, \quad (9.129)$$

²⁹ The force from the rod must point along the rod because it is massless. If there were a tangential force on the mass, then Newton's third law would say that there would also be a tangential force on the rod. This would then produce a nonzero torque on the rod (relative to the pivot) and hence an infinite angular acceleration, because there is no gravitational torque on the (massless) rod to counter it.

with upward along the cone taken to be positive (this is just a rearrangement of the $F = ma$ equation along the cone). The torque on the ring (relative to its CM) is due only to this F , because gravity provides no torque, and N points through the center of the ring (by the second assumption in the problem). Therefore, the torque points out of the page with magnitude

$$\tau = rF = r(mg \cos \theta - m\Omega^2 h \tan \theta \sin \theta). \quad (9.130)$$

We must now find $d\mathbf{L}/dt$. Since we are assuming $r \ll h \tan \theta$, the frequency of the spinning of the ring (call it ω) is much greater than the frequency of precession, Ω . We will therefore neglect the latter in finding $\mathbf{L}$. In this approximation, $\mathbf{L}$ (relative to the CM) has magnitude $mr^2\omega$, and it points downward along the cone (for the direction of motion shown in Fig. 9.45). The horizontal component of $\mathbf{L}$ has magnitude $L_{\perp} \equiv L \sin \theta$, and it traces out a circle at frequency Ω . Therefore, $d\mathbf{L}/dt$ points out of the page with magnitude

$$\left| \frac{d\mathbf{L}}{dt} \right| = \Omega L_{\perp} = \Omega L \sin \theta = \Omega (mr^2\omega) \sin \theta. \quad (9.131)$$

The nonslipping condition is $r\omega = (h \tan \theta)\Omega$,³⁰ which gives $\omega = (h \tan \theta)\Omega/r$. Using this in Eq. (9.131) yields

$$\left| \frac{d\mathbf{L}}{dt} \right| = \Omega^2 m r h \tan \theta \sin \theta. \quad (9.132)$$

Equating this with the torque in Eq. (9.130) gives

$$\Omega = \frac{1}{\tan \theta} \sqrt{\frac{g}{2h}}. \quad (9.133)$$

REMARK: If you consider an object with moment of inertia βmr^2 (our ring has $\beta = 1$), then you can show by the above reasoning that the “2” in Eq. (9.133) is replaced by $(1 + \beta)$. This means that if we instead have a particle sliding around a frictionless cone (which is equivalent to a ring with $\beta = 0$), then the frequency is $\sqrt{g/h}/\tan \theta$, as you can check from scratch. ♣

9.14. Tennis racket theorem

ROTATION AROUND $\hat{\mathbf{x}}_1$: If the racket is rotated (nearly) around the $\hat{\mathbf{x}}_1$ axis, then the initial ω_2 and ω_3 are much smaller than ω_1 . To emphasize this, let’s relabel them as $\omega_2 \rightarrow \epsilon_2$ and $\omega_3 \rightarrow \epsilon_3$. Then Eq. (9.45) becomes (with the torque equal to zero, because gravity provides no torque around the CM)

$$\begin{aligned} 0 &= \dot{\omega}_1 - A\epsilon_2\epsilon_3, \\ 0 &= \dot{\epsilon}_2 + B\omega_1\epsilon_3, \\ 0 &= \dot{\epsilon}_3 - C\omega_1\epsilon_2, \end{aligned} \quad (9.134)$$

where we have defined (for convenience)

$$A \equiv \frac{I_2 - I_3}{I_1}, \quad B \equiv \frac{I_1 - I_3}{I_2}, \quad C \equiv \frac{I_1 - I_2}{I_3}. \quad (9.135)$$

Note that A , B , and C are all positive. This fact will be very important.

Our goal here is to show that if the ϵ ’s start out small, then they remain small. Assuming that they are small (which is true initially), the first equation says that

³⁰ This is technically not quite correct, for the same reason that the earth spins around 366 times instead of 365 times in a year. But it’s valid enough in the limit of small r .

$\dot{\omega}_1 \approx 0$, to first order in the ϵ 's. Therefore, we may assume that ω_1 is essentially constant (when the ϵ 's are small). Taking the derivative of the second equation then gives $0 = \ddot{\epsilon}_2 + B\omega_1\dot{\epsilon}_3$. Plugging the value of $\dot{\epsilon}_3$ from the third equation into this yields

$$\ddot{\epsilon}_2 = -\left(BC\omega_1^2\right)\epsilon_2. \quad (9.136)$$

Because of the negative coefficient on the right-hand side, this equation describes simple harmonic motion. Therefore, ϵ_2 oscillates sinusoidally around zero. So if it starts small, it remains small. By the same reasoning, ϵ_3 remains small.

We therefore see that $\boldsymbol{\omega} \approx (\omega_1, 0, 0)$ at all times, which implies that $\mathbf{L} \approx (I_1\omega_1, 0, 0)$ at all times. That is, $\mathbf{L}$ always points (nearly) along the $\hat{\mathbf{x}}_1$ direction (which is fixed in the racket frame). But the direction of $\mathbf{L}$ is fixed in the lab frame, because there is no torque. Therefore, the direction of $\hat{\mathbf{x}}_1$ must also be (nearly) fixed in the lab frame. In other words, the racket doesn't wobble.

ROTATION AROUND $\hat{\mathbf{x}}_3$: This calculation goes through exactly as above, except with “1” and “3” interchanged. We find that if ϵ_1 and ϵ_2 start small, they remain small.

ROTATION AROUND $\hat{\mathbf{x}}_2$: If the racket is rotated (nearly) around the $\hat{\mathbf{x}}_2$ axis, then the initial ω_1 and ω_3 are much smaller than ω_2 . As above, let's emphasize this by relabeling them as $\omega_1 \rightarrow \epsilon_1$ and $\omega_3 \rightarrow \epsilon_3$. Then as above, Eq. (9.45) becomes

$$\begin{aligned} 0 &= \dot{\epsilon}_1 - A\omega_2\epsilon_3, \\ 0 &= \dot{\omega}_2 + B\epsilon_1\epsilon_3, \\ 0 &= \dot{\epsilon}_3 - C\omega_2\epsilon_1. \end{aligned} \quad (9.137)$$

Our goal here is to show that if the ϵ 's start out small, then they do *not* remain small. Assuming that they are small (which is true initially), the second equation says that $\dot{\omega}_2 \approx 0$, to first order in the ϵ 's. So we may assume that ω_2 is essentially constant (when the ϵ 's are small). Taking the derivative of the first equation then gives $0 = \ddot{\epsilon}_1 - A\omega_2\dot{\epsilon}_3$. Plugging the value of $\dot{\epsilon}_3$ from the third equation into this yields

$$\ddot{\epsilon}_1 = \left(AC\omega_2^2\right)\epsilon_1. \quad (9.138)$$

Because of the positive coefficient on the right-hand side, this equation describes an exponentially growing motion, instead of an oscillatory one. Therefore, ϵ_1 grows quickly from its initial small value. So even if it starts small, it becomes large. By the same reasoning, ϵ_3 becomes large. Of course, once the ϵ 's become large, then our assumption of $\dot{\omega}_2 \approx 0$ isn't valid anymore. But once the ϵ 's become large, we've shown what we wanted to.

We see that $\boldsymbol{\omega}$ does *not* remain (nearly) equal to $(0, \omega_2, 0)$, which implies that $\mathbf{L}$ does *not* remain (nearly) equal to $(0, I_2\omega_2, 0)$. That is, $\mathbf{L}$ does not always point (nearly) along the $\hat{\mathbf{x}}_2$ direction (which is fixed in the racket frame). But the direction of $\mathbf{L}$ is fixed in the lab frame, because there is no torque. Therefore, the direction of $\hat{\mathbf{x}}_2$ must change in the lab frame. In other words, the racket wobbles.

9.15. Free-top angles

In terms of the principal axes, $\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \hat{\mathbf{x}}_3$, we have

$$\begin{aligned} \boldsymbol{\omega} &= (\omega_1\hat{\mathbf{x}}_1 + \omega_2\hat{\mathbf{x}}_2) + \omega_3\hat{\mathbf{x}}_3, \quad \text{and} \\ \mathbf{L} &= I(\omega_1\hat{\mathbf{x}}_1 + \omega_2\hat{\mathbf{x}}_2) + I_3\omega_3\hat{\mathbf{x}}_3. \end{aligned} \quad (9.139)$$

Let $\omega_\perp \hat{\boldsymbol{\omega}}_\perp \equiv (\omega_1\hat{\mathbf{x}}_1 + \omega_2\hat{\mathbf{x}}_2)$ be the component of $\boldsymbol{\omega}$ orthogonal to ω_3 . Then we have

$$\tan \beta = \frac{\omega_\perp}{\omega_3}, \quad \text{and} \quad \tan \alpha = \frac{I\omega_\perp}{I_3\omega_3}. \quad (9.140)$$

Therefore,

$$\frac{\tan \alpha}{\tan \beta} = \frac{I}{I_3}. \quad (9.141)$$

If $I > I_3$, then $\alpha > \beta$, and we have the situation shown in Fig. 9.89. A top with this property is called a “prolate top.” An example is an American football or a pencil.

If $I < I_3$, then $\alpha < \beta$, and we have the situation shown in Fig. 9.90. A top with this property is called an “oblate top.” An example is a coin or a Frisbee.

9.16. Staying above

If $I < I_3$ (oblate top), then the initial $\hat{x}_3$, ω , and L vectors look like those in Fig. 9.91, with L between $\hat{x}_3$ and ω . The ω vector traces out a cone around L as shown, so it always stays above the horizontal (because it starts above the horizontal).

However, if $I > I_3$ (prolate top), then the initial $\hat{x}_3$, ω , and L vectors look like those in Fig. 9.92, with ω between $\hat{x}_3$ and L . Depending on the values of I/I_3 and $\omega_{\perp}/\omega_3$ (where $\omega_{\perp}$ is the component of ω orthogonal to $\hat{x}_3$), the ω cone might stay above the horizontal axis, or it might dip below it. We are concerned with the cutoff case where it just touches the axis, as shown.

From Fig. 9.92, the half-angle of the cone is $\alpha - \beta$, so if the cone reaches down to the horizontal axis, then we have $\alpha + (\alpha - \beta) = 90^\circ$. But α and β are given by

$$\tan \beta = \frac{\omega_{\perp}}{\omega_3}, \quad \text{and} \quad \tan \alpha = \frac{L_{\perp}}{L_3} = \frac{I\omega_{\perp}}{I_3\omega_3}. \quad (9.142)$$

Therefore, the $2\alpha - \beta = 90^\circ$ condition becomes

$$2 \tan^{-1} \left(\frac{I\omega_{\perp}}{I_3\omega_3} \right) - \tan^{-1} \left(\frac{\omega_{\perp}}{\omega_3} \right) = 90^\circ. \quad (9.143)$$

With $n \equiv I/I_3$ and $x \equiv \omega_{\perp}/\omega_3$, this yields

$$\begin{aligned} 2 \tan^{-1}(nx) - \tan^{-1}(x) &= 90^\circ \\ \Rightarrow \tan(2 \tan^{-1}(nx)) &= \tan(90^\circ + \tan^{-1}(x)) \\ \Rightarrow \frac{2nx}{1 - n^2x^2} &= -\frac{1}{x} \\ \Rightarrow n(n-2)x^2 &= 1. \end{aligned} \quad (9.144)$$

We see that if $n \leq 2$, there is no solution for x . But if $n > 2$, there is a solution. Therefore, the answer to the problem is $n = 2$.

REMARKS: If n is only slightly larger than 2, then the solution for x in Eq. (9.144) (and hence $\omega_{\perp}$) is large. This means that your strike needs to be large, in order to produce a large $\omega_{\perp}$.

There are three interesting limits of what can happen in the large- n limit, so let's list them out. For these three cases, picture a thin stick spinning around its vertical x_3 axis, and then imagine striking the bottom end with different impulses.

- If $\omega_{\perp} \ll \omega_3$ and $L_{\perp} \ll L_3$, then both ω and L point nearly straight upwards, so ω traces out a very thin cone around L and is always nearly vertical. $\hat{x}_3$ is also always nearly vertical.
- If $\omega_{\perp} \ll \omega_3$ but $L_{\perp} \gg L_3$ (which is possible if n is sufficiently large), then ω points nearly straight upwards initially, whereas L points nearly horizontally. So ω traces out a very wide cone and swings down almost to the negative vertical direction. $\hat{x}_3$ also traces out a very wide cone.
- If $\omega_{\perp} \gg \omega_3$, which implies $L_{\perp} \gg L_3$, then both ω and L point nearly horizontally, so ω traces out a very thin cone around L and is always nearly horizontal.

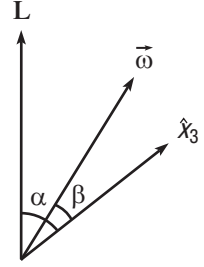


Fig. 9.89

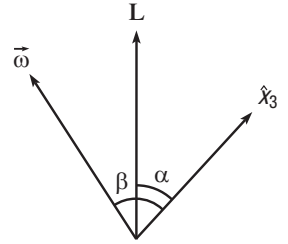


Fig. 9.90

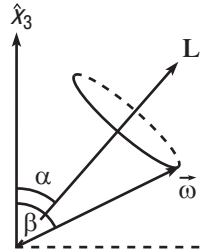


Fig. 9.91

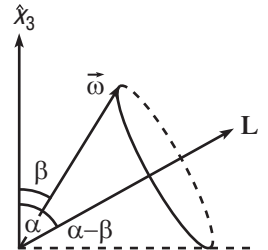


Fig. 9.92

$\hat{\mathbf{x}}_3$ still traces out a very wide cone, because it starts vertical (by assumption), independent of where $\boldsymbol{\omega}$ and $\mathbf{L}$ are. Note that your eye can't distinguish between the second and third cases here, because the $\hat{\mathbf{x}}_3$ axis is doing the same thing in both cases. ♣

9.17. The top

- (a) In order for there to exist real solutions for Ω in Eq. (9.82), the discriminant must be non-negative. If $\theta > \pi/2$, then $\cos \theta < 0$, so the discriminant is automatically positive, and any value of ω_3 is allowed. But if $\theta < \pi/2$, then the lower limit on ω_3 is

$$\omega_3 \geq \frac{\sqrt{4Mg\ell \cos \theta}}{I_3} \equiv \tilde{\omega}_3. \quad (9.145)$$

The special case of $\theta = \pi/2$ requires a limit to be taken; it turns out that there is only one (noninfinite) solution, as we'll see in part (b). Note that at the critical value of ω_3 in Eq. (9.145), Eq. (9.82) gives

$$\Omega_+ = \Omega_- = \frac{I_3 \tilde{\omega}_3}{2I \cos \theta} = \sqrt{\frac{Mg\ell}{I \cos \theta}} \equiv \Omega_0. \quad (9.146)$$

- (b) Since ω_3 has units, “large ω_3 ” is a meaningless description. What we really mean is that the fraction in the square root in Eq. (9.82) is very small compared with 1. That is, $\epsilon \equiv (4Mg\ell \cos \theta)/(I_3^2 \omega_3^2) \ll 1$. In this case, we can use $\sqrt{1 - \epsilon} \approx 1 - \epsilon/2 + \dots$ to write

$$\Omega_{\pm} \approx \frac{I_3 \omega_3}{2I \cos \theta} \left(1 \pm \left(1 - \frac{2Mg\ell \cos \theta}{I_3^2 \omega_3^2} \right) \right). \quad (9.147)$$

The two solutions for Ω are then, to leading order in ω_3 (or rather, to leading order in ϵ),

$$\Omega_+ \approx \frac{I_3 \omega_3}{I \cos \theta}, \quad \text{and} \quad \Omega_- \approx \frac{Mg\ell}{I_3 \omega_3}. \quad (9.148)$$

These are known as the “fast” and “slow” frequencies of precession, respectively. Ω_- is the approximate answer we found in Eq. (9.77). It was obtained here under the assumption $\epsilon \ll 1$, which is equivalent to

$$\omega_3 \gg \frac{\sqrt{4Mg\ell \cos \theta}}{I_3} \quad (\text{that is, } \omega_3 \gg \tilde{\omega}_3). \quad (9.149)$$

This, therefore, is the condition for the result in Eq. (9.77) to be a good approximation. If I is of the same order as I_3 , so that they are both of order $M\ell^2$ (assuming that the top is a reasonably shaped object without any strange tails), and if $\cos \theta$ is of order 1, then this condition can be written as $\omega_3 \gg \sqrt{g/\ell}$, which is the frequency of a pendulum of length ℓ .

REMARKS:

1. The Ω_+ solution is a fairly surprising result. Two strange features of Ω_+ are that it grows with ω_3 , and that it is independent of g . To see what is going on with this precession, note that Ω_+ is the value of Ω that makes the $L_{\perp}$ in Eq. (9.79) essentially equal to zero. So $\mathbf{L}$ points nearly along the vertical axis. The rate of change of $\mathbf{L}$ is the product of a very small horizontal radius (of the tiny circle the tip traces out) and a very large Ω (assuming that ω_3 is large). The product of these equals the “medium sized” torque, $Mg\ell \sin \theta$.
2. In the limit of large ω_3 , the fast precession should look basically like the motion of a free top, because the $\mathbf{L}$ here points essentially in a fixed direction, just as it does for a free top. And indeed, Ω_+ is independent of g . More

precisely, we can see that the value of Ω_+ given in Eq. (9.148) agrees with the free-top precession frequency, by the following reasoning. $I_3\omega_3$ is the component of $\mathbf{L}$ along the symmetry axis, which makes an angle θ with the vertical (which is essentially the direction of $\mathbf{L}$). Therefore, $\mathbf{L}$ has magnitude $L \approx I_3\omega_3/\cos\theta$, which means we can write $\Omega_+ \approx L/I$. This is the precession frequency of a free top, given in Eq. (9.55), as viewed from a fixed frame.

3. We can plot the $\Omega_{\pm}$ in Eq. (9.82) as functions of ω_3 . With the definitions of $\tilde{\omega}_3$ and $\tilde{\Omega}_0$ in Eqs. (9.145) and (9.146), we can rewrite Eq. (9.82) as

$$\Omega_{\pm} = \frac{\omega_3 \tilde{\Omega}_0}{\tilde{\omega}_3} \left(1 \pm \sqrt{1 - \frac{\tilde{\omega}_3^2}{\omega_3^2}} \right). \quad (9.150)$$

Since it is easier to work with dimensionless quantities, let's rewrite this as

$$y_{\pm} = x \pm \sqrt{x^2 - 1}, \quad \text{where } y_{\pm} \equiv \frac{\Omega_{\pm}}{\tilde{\Omega}_0}, \quad \text{and } x \equiv \frac{\omega_3}{\tilde{\omega}_3}. \quad (9.151)$$

A rough plot of $y_{\pm}$ vs. x is shown in Fig. 9.93. The behavior of this graph for large x is found from Eq. (9.151) to be $y_+ \approx 2x$, and $y_- \approx 1/(2x)$. You can show that these are equivalent to the results in Eq. (9.148). ♣

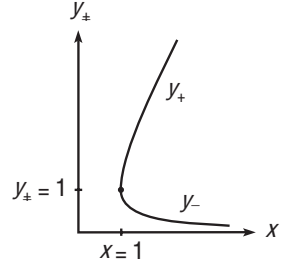


Fig. 9.93

9.18. Many tops

The system is made up of N rigid bodies, each consisting of a disk and a massless stick glued to it on its left (see Fig. 9.94). Label these subsystems as S_i , with S_1 being the one closest to the pole. Let each disk have mass m and moment of inertia I , and let each stick have length ℓ . Let the angular speeds be ω_i . The relevant angular momentum of S_i is then $L_i = I\omega_i$, and it points horizontally.³¹ Let the desired precession frequency be Ω . Then $d\mathbf{L}_i/dt$ has magnitude $L_i\Omega = (I\omega_i)\Omega$ and points into the page at the instant shown in Fig. 9.49.

Consider the torque $\boldsymbol{\tau}_i$ on S_i , around its CM. Let's first look at S_1 . The pole provides an upward force of Nmg (this force is what keeps all the tops up), so it provides a torque of $Nmg\ell$ (into the page) around the CM of S_1 . The downward force from the stick on the right provides no torque around the CM, because it acts at the CM. Likewise, the gravitational force on S_1 provides no torque around the CM. Therefore, $\boldsymbol{\tau}_1 = d\mathbf{L}_1/dt$ gives $Nmg\ell = (I\omega_1)\Omega$, and so

$$\omega_1 = \frac{Nmg\ell}{I\Omega}. \quad (9.152)$$

Now look at S_2 . S_1 provides an upward force of $(N-1)mg$ (this force is what keeps S_2 through S_N up), so it provides a torque of $(N-1)mg\ell$ around the CM of S_2 . As with S_1 , this is the only torque on S_2 . Therefore, $\boldsymbol{\tau}_2 = d\mathbf{L}_2/dt$ gives $(N-1)mg\ell = (I\omega_2)\Omega$, and so

$$\omega_2 = \frac{(N-1)mg\ell}{I\Omega}. \quad (9.153)$$

Similar reasoning applies to the other S_i , so we arrive at

$$\omega_i = \frac{(N+1-i)mg\ell}{I\Omega}. \quad (9.154)$$

The ω_i are therefore in the ratio

$$\omega_1 : \omega_2 : \cdots : \omega_{N-1} : \omega_N = N : (N-1) : \cdots : 2 : 1. \quad (9.155)$$

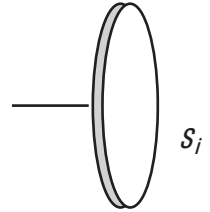


Fig. 9.94

³¹ We're ignoring the angular momentum arising from the precession. This part of $\mathbf{L}$ points vertically (because the tops all point horizontally) and therefore doesn't change. Hence, it doesn't enter into $\boldsymbol{\tau} = d\mathbf{L}/dt$.

Note that we needed to apply $\boldsymbol{\tau} = d\mathbf{L}/dt$ many times, using each CM as the origin. Using only the pivot point on the pole as the origin would have given us only one piece of information, whereas we needed N pieces.

REMARKS: As a double-check, we can verify that the above ω 's make $\boldsymbol{\tau} = d\mathbf{L}/dt$ true, where $\boldsymbol{\tau}$ and $\mathbf{L}$ are the total torque and angular momentum relative to the pivot on the pole. The CM of the entire system is $(N+1)\ell/2$ from the pole, so the torque due to gravity is

$$\tau = Nmg \frac{(N+1)\ell}{2}. \quad (9.156)$$

The total angular momentum is, using Eq. (9.154),

$$\begin{aligned} L &= I(\omega_1 + \omega_2 + \cdots + \omega_N) \\ &= \frac{mg\ell}{\Omega} (N + (N-1) + (N-2) + \cdots + 2 + 1) \\ &= \frac{mg\ell}{\Omega} \left(\frac{N(N+1)}{2} \right). \end{aligned} \quad (9.157)$$

Comparing this with Eq. (9.156), we see that $\tau = L\Omega$, that is, $\tau = |d\mathbf{L}/dt|$.

We can also pose this problem for the setup where all the ω_i are equal (call them ω), and the goal is to find the lengths of the sticks that allow circular precession with the sticks always forming a straight horizontal line. We can use the same reasoning as above, and Eq. (9.154) takes the modified form

$$\omega = \frac{(N+1-i)mg\ell_i}{I\Omega}, \quad (9.158)$$

where ℓ_i is the length of the i th stick. Therefore, the ℓ_i are in the ratio

$$\ell_1 : \ell_2 : \cdots : \ell_{N-1} : \ell_N = \frac{1}{N} : \frac{1}{N-1} : \cdots : \frac{1}{2} : 1. \quad (9.159)$$

Again, we can verify that these ℓ 's make $\boldsymbol{\tau} = d\mathbf{L}/dt$ true, where $\boldsymbol{\tau}$ and $\mathbf{L}$ are the total torque and angular momentum relative to the pivot on the pole. As an exercise, you can show that the CM happens to be a distance ℓ_N from the pole. So the torque due to gravity is, using Eq. (9.158) to obtain ℓ_N ,

$$\tau = Nmg\ell_N = Nmg(\omega I\Omega/mg) = NI\omega\Omega. \quad (9.160)$$

The total angular momentum is simply $L = NI\omega$. So indeed, $\tau = L\Omega = |d\mathbf{L}/dt|$. Note that since the sum $\sum 1/n$ diverges, it is possible to make the setup extend arbitrarily far from the pole. ♣

9.19. Heavy top on a slippery table

In Section 9.7.3, we looked at $\boldsymbol{\tau}$ and $\mathbf{L}$ relative to the pivot point. Such quantities are of no use now, because the pivot is accelerating, so it isn't a legal choice of origin for applying $\boldsymbol{\tau} = d\mathbf{L}/dt$. We will therefore look at $\boldsymbol{\tau}$ and $\mathbf{L}$ relative to the CM, which is always a legal origin for applying $\boldsymbol{\tau} = d\mathbf{L}/dt$.

With the CM as our origin, there are two modifications we need to make to the derivation in Section 9.7.3. First, the moments of inertia are now measured with respect to the CM, instead of the pivot point. I_3 is unchanged, but the parallel-axis theorem gives the new I as

$$I' \equiv I - M\ell^2. \quad (9.161)$$

Second, the torque is modified. The only force from the frictionless floor is the normal force N . But N is not necessarily equal to Mg , because the CM may be accelerating in the vertical direction. The vertical $F = ma$ equation is $N - Mg = M\ddot{y}$, where $y = \ell \cos \theta$. Taking the second derivative of y , we obtain $N = Mg - M\ell(\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta)$. The torque relative to the CM therefore has magnitude

$$\tau = N\ell \sin \theta = Mg\ell \sin \theta - M\ell^2 \sin \theta (\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta), \quad (9.162)$$

and it points in the same direction as in Section 9.7.3. Putting everything together, we see that Eq. (9.69) is unchanged, Eq. (9.70) has I replaced with $I' \equiv I - M\ell^2$, and the second of Eqs. (9.70) also has an extra term and becomes

$$(Mg\ell - M\ell^2(\ddot{\theta} \sin \theta + \dot{\theta}^2 \cos \theta) + I'\dot{\phi}^2 \cos \theta - I_3\omega_3\dot{\phi}) \sin \theta = I\ddot{\theta}. \quad (9.163)$$

Note that if $\dot{\theta} \equiv 0$, the only modification needed is the change in I .

If you prefer to use the Lagrangian method in Section 9.7.4, then I needs to be changed to I' as above, and the other modification comes in the kinetic energy in the Lagrangian. We must add on the kinetic energy of the whole object treated like a point mass at the CM, because so far we have included only the kinetic energy relative to our new origin (the CM). Since the table is frictionless, the CM can move only vertically, so its velocity is $\dot{y} = -\ell\dot{\theta} \sin \theta$. Therefore, we have to add the term $M(\ell\dot{\theta} \sin \theta)^2/2$ to the Lagrangian. You can show that this leads to the extra term in Eq. (9.163).

9.20. Fixed highest point

For the desired motion with P always the highest point, the important thing to note is that every point in the top moves in a fixed circle around the $\hat{z}$ axis. Therefore, $\boldsymbol{\omega}$ points vertically at all times. Hence, if Ω is the frequency of precession, we have $\boldsymbol{\omega} = \Omega\hat{z}$.

Another way to see that $\boldsymbol{\omega}$ points vertically is to view things in the frame that rotates with angular velocity $\Omega\hat{z}$. In this frame, the top has no motion at all. It isn't even spinning, because the point P is always the highest point. In the language of Fig. 9.30, we therefore have $\boldsymbol{\omega}' = 0$, so $\boldsymbol{\omega} = \Omega\hat{z} + \boldsymbol{\omega}'\hat{x}_3 = \Omega\hat{z}$.

The principal moments are (with the pivot as the origin; see Fig. 9.95)

$$I_3 = \frac{MR^2}{2}, \quad \text{and} \quad I \equiv I_1 = I_2 = M\ell^2 + \frac{MR^2}{4}, \quad (9.164)$$

where we have used the parallel-axis theorem in finding I . The components of $\boldsymbol{\omega}$ along the principal axes are $\omega_3 = \Omega \cos \theta$ and $\omega_2 = \Omega \sin \theta$. Therefore (keeping things in terms of the general moments, I_3 and I , for now),

$$\mathbf{L} = I_3\Omega \cos \theta \hat{x}_3 + I\Omega \sin \theta \hat{x}_2. \quad (9.165)$$

The horizontal component of $\mathbf{L}$ is then $L_\perp = (I_3\Omega \cos \theta) \sin \theta - (I\Omega \sin \theta) \cos \theta$, so $d\mathbf{L}/dt$ has magnitude

$$\left| \frac{d\mathbf{L}}{dt} \right| = L_\perp \Omega = \Omega^2 \sin \theta \cos \theta (I_3 - I), \quad (9.166)$$

and it is directed into the page (or out of the page, if this quantity is negative). This must equal the torque, which has magnitude $|\boldsymbol{\tau}| = Mg\ell \sin \theta$ and is directed into the page. Therefore,

$$\Omega = \sqrt{\frac{Mg\ell}{(I_3 - I) \cos \theta}}. \quad (9.167)$$

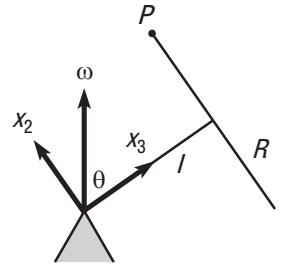


Fig. 9.95

We see that for a general symmetric top, the desired precessional motion (where the same “side” always points up) is possible only if the product $(I_3 - I) \cos \theta$ is greater than zero. That is,

$$\begin{aligned}\theta < \pi/2 &\implies I_3 > I, \\ \theta > \pi/2 &\implies I_3 < I.\end{aligned}\quad (9.168)$$

For the problem at hand, I_3 and I are given in Eq. (9.164), so we find

$$\Omega = \sqrt{\frac{4g\ell}{(R^2 - 4\ell^2) \cos \theta}}. \quad (9.169)$$

The necessary condition for such motion to exist is therefore $R > 2\ell$ if $\theta < \pi/2$, or $R < 2\ell$ if $\theta > \pi/2$.

REMARKS:

1. It is intuitively clear that Ω should become very large as $\theta \rightarrow \pi/2$, although it is by no means intuitively clear that such motion should exist at all for angles near $\pi/2$.
2. If $\theta > \pi/2$ and $R = 0$, we simply have the circular pendulum discussed in Problem 9.12. And indeed, when $R = 0$, the result in Eq. (9.169) reduces to $\Omega = \sqrt{g/\ell \cos \alpha}$, where $\alpha \equiv \pi - \theta$, which agrees with the result in Eq. (9.126).
3. Ω approaches a nonzero constant as $\theta \rightarrow 0$ or $\theta \rightarrow \pi$ (depending on the sign of $I_3 - I$), which isn't entirely obvious.
4. If both R and ℓ are scaled up by the same factor, Eq. (9.169) shows that Ω decreases. This also follows from dimensional analysis.
5. Assuming that $\theta < \pi/2$ (the $\theta > \pi/2$ case can be handled in a similar manner), the condition $I_3 > I$ can be understood in the following way. If $I_3 = I$, then $\mathbf{L} \propto \boldsymbol{\omega}$, so $\mathbf{L}$ points vertically along $\boldsymbol{\omega}$. If $I_3 > I$, then $\mathbf{L}$ points somewhere to the right of the z axis (at the instant shown in Fig. 9.95). This means that the tip of $\mathbf{L}$ is moving into the page, along with the top. This is what we need, because $\boldsymbol{\tau}$ points into the page. If, however, $I_3 < I$, then $\mathbf{L}$ points somewhere to the left of the z axis, so $d\mathbf{L}/dt$ points out of the page, and hence cannot be equal to $\boldsymbol{\tau}$. ♣

9.21. Basketball on a rim

Assume that the precessional motion is counterclockwise when viewed from above, as indicated in Fig. 9.52. Let's look at things in the frame that has the center of the rim as its origin and that rotates with angular velocity $\Omega \hat{\mathbf{z}}$. In this frame, the center of the ball is at rest, and the rim spins clockwise with angular speed Ω . If the contact points are to form a great circle on the ball, the ball must be spinning around the (negative) $\hat{\mathbf{x}}_3$ axis shown in Fig. 9.96. Let the frequency of this spinning be ω' (in the language of Fig. 9.30). Then the nonslipping condition says that $\omega' r = \Omega R$, and so $\omega' = \Omega R/r$. Therefore, the total angular velocity of the ball with respect to the lab frame is

$$\boldsymbol{\omega} = \Omega \hat{\mathbf{z}} - \omega' \hat{\mathbf{x}}_3 = \Omega \hat{\mathbf{z}} - (R/r) \Omega \hat{\mathbf{x}}_3. \quad (9.170)$$

Let us choose the center of the ball as the origin around which we calculate $\boldsymbol{\tau}$ and $\mathbf{L}$. Then every axis in the ball is a principal axis with moment of inertia $I = (2/3)mr^2$. The angular momentum is therefore

$$\mathbf{L} = I\boldsymbol{\omega} = I\Omega \hat{\mathbf{z}} - I(R/r)\Omega \hat{\mathbf{x}}_3. \quad (9.171)$$

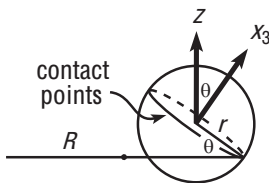


Fig. 9.96

Only the $\hat{x}_3$ piece has a horizontal component which contributes to $d\mathbf{L}/dt$. This component has length $L_{\perp} = I(R/r)\Omega \sin \theta$. Therefore, $d\mathbf{L}/dt$ has magnitude

$$\left| \frac{d\mathbf{L}}{dt} \right| = \Omega L_{\perp} = \frac{2}{3} \Omega^2 m r R \sin \theta, \quad (9.172)$$

and it points out of the page.

The torque (relative to the center of the ball) comes from the force at the contact point. There are two components to this force. The vertical component is mg , and the horizontal component is $m(R - r \cos \theta)\Omega^2$ (pointing to the left), because the center of the ball moves in a circle of radius $(R - r \cos \theta)$. The torque therefore has magnitude

$$|\boldsymbol{\tau}| = mg(r \cos \theta) - m(R - r \cos \theta)\Omega^2(r \sin \theta), \quad (9.173)$$

with out of the page taken to be positive. Equating this $|\boldsymbol{\tau}|$ with the $|d\mathbf{L}/dt|$ in Eq. (9.172) gives

$$\Omega^2 = \frac{g}{\frac{5}{3}R \tan \theta - r \sin \theta}. \quad (9.174)$$

REMARKS:

1. $\Omega \rightarrow \infty$ as $\theta \rightarrow 0$, which makes sense. And $\Omega \rightarrow 0$ as $\theta \rightarrow \pi/2$, which also makes sense.
2. $\Omega \rightarrow \infty$ when $R = (3/5)r \cos \theta$. But this isn't physical, because we must have $R > r \cos \theta$ in order for the other side of the rim to be outside the ball.
3. You can also work out the problem for the case where the contact points trace out a circle other than a great circle (say, one that is inclined at an angle β below the great circle). The expression for the torque in Eq. (9.173) is unchanged, but the value of ω' and the angle of the $\hat{x}_3$ axis both change, so Eq. (9.172) is modified. If you want to play around with this, one thing you can show (with a bit of work) is that if $R \gg r$ and if you want the ball to travel around the rim infinitely fast, then β must equal $\tan^{-1}((5/2) \tan \theta)$. This is larger than θ , so the contact-point circle actually lies below the horizontal. ♣

9.22. Rolling lollipop

We must first find the angular velocity vector $\boldsymbol{\omega}$. Assume that the precessional motion is clockwise when viewed from above, as indicated in Fig. 9.53. We claim that $\boldsymbol{\omega}$ points horizontally to the right (at the instant shown in Fig. 9.97), with magnitude $(R/r)\Omega$. This can be seen in two ways.

The first way is to recognize that we essentially have the same scenario as in the "Rolling cone" setup of Problem 9.3 (imagine the sphere to be a ball of ice cream in a cone whose tip is located at the left end of the stick). The sphere's contact point with the ground is instantaneously at rest (the nonslipping condition), so $\boldsymbol{\omega}$ must pass through this point. But $\boldsymbol{\omega}$ must also pass through the left end of the stick, because that point is also at rest. Therefore, $\boldsymbol{\omega}$ must be horizontal. To find the magnitude, note that the center of the sphere moves with speed ΩR . But since the center may also be considered to be instantaneously moving with frequency ω in a circle of radius r around the horizontal axis, we have $\omega r = \Omega R$. Therefore, $\omega = (R/r)\Omega$.

The second way is to write $\boldsymbol{\omega}$ as $\boldsymbol{\omega} = -\Omega \hat{\mathbf{z}} + \omega' \hat{\mathbf{x}}_3$ (in the language of Fig. 9.30), where ω' is the frequency of the spinning as viewed by someone rotating around the (negative) $\hat{\mathbf{z}}$ axis with frequency Ω . The contact points form a circle of radius R on the ground. But they also form a circle of radius $r \cos \theta$ on the sphere, where θ is the angle between the stick and the ground (this circle is the circle of points where the

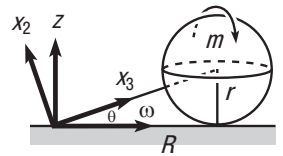


Fig. 9.97

above-mentioned ice cream cone touches the sphere). The nonslipping condition then implies that $\Omega R = \omega'(r \cos \theta)$. Therefore, $\omega' = \Omega R/(r \cos \theta)$, and so

$$\boldsymbol{\omega} = -\Omega \hat{\mathbf{z}} + \omega' \hat{\mathbf{x}}_3 = -\Omega \hat{\mathbf{z}} + \left(\frac{\Omega R}{r \cos \theta} \right) (\cos \theta \hat{\mathbf{x}} + \sin \theta \hat{\mathbf{z}}) = (R/r)\Omega \hat{\mathbf{x}}, \quad (9.175)$$

where we have used $\tan \theta = r/R$.

Now let's calculate the normal force. Choose the pivot as the origin. The principal axes are then $\hat{\mathbf{x}}_3$ along the stick, along with any two directions orthogonal to the stick. Choose $\hat{\mathbf{x}}_2$ to be in the plane of the paper (see Fig. 9.97). Then the components of $\boldsymbol{\omega}$ along the principal axes are

$$\omega_3 = (R/r)\Omega \cos \theta, \quad \text{and} \quad \omega_2 = -(R/r)\Omega \sin \theta. \quad (9.176)$$

The principal moments are

$$I_3 = (2/5)mr^2, \quad \text{and} \quad I_2 = (2/5)mr^2 + m(r^2 + R^2), \quad (9.177)$$

where we have used the parallel-axis theorem. The angular momentum is $\mathbf{L} = I_3 \omega_3 \hat{\mathbf{x}}_3 + I_2 \omega_2 \hat{\mathbf{x}}_2$, so its horizontal component has length $L_\perp = I_3 \omega_3 \cos \theta - I_2 \omega_2 \sin \theta$. Therefore, the magnitude of $d\mathbf{L}/dt$ is

$$\begin{aligned} \left| \frac{d\mathbf{L}}{dt} \right| &= \Omega L_\perp = \Omega (I_3 \omega_3 \cos \theta - I_2 \omega_2 \sin \theta) \\ &= \Omega \left[\left(\frac{2}{5}mr^2 \right) \left(\frac{R}{r} \Omega \cos \theta \right) \cos \theta \right. \\ &\quad \left. - \left(\frac{2}{5}mr^2 + m(r^2 + R^2) \right) \left(-\frac{R}{r} \Omega \sin \theta \right) \sin \theta \right] \\ &= \frac{\Omega^2 m R}{r} \left(\frac{2}{5}r^2 + (r^2 + R^2) \sin^2 \theta \right) \\ &= \frac{7}{5} m r R \Omega^2, \end{aligned} \quad (9.178)$$

where we have used $\sin \theta = r/\sqrt{r^2 + R^2}$. The direction of $d\mathbf{L}/dt$ is out of the page.

The torque (relative to the pivot) is due to the gravitational force acting at the CM, along with the normal force N acting at the contact point. (Any horizontal friction that happens to exist at the contact point produces zero torque relative to the pivot.) Therefore, $\boldsymbol{\tau}$ points out of the page with magnitude $|\boldsymbol{\tau}| = (N - mg)R$. Equating this with the $|d\mathbf{L}/dt|$ in Eq. (9.178) gives

$$N = mg + \frac{7}{5} m r \Omega^2. \quad (9.179)$$

This has the interesting property of being independent of R , and hence also θ . This independence arises because both $|\boldsymbol{\tau}|$ and $|d\mathbf{L}/dt|$ are proportional to R , a fact which is easier to see via the reasoning in the first remark below.

REMARKS:

1. There is actually a quicker way to calculate the $d\mathbf{L}/dt$ in Eq. (9.178). At a given instant, the sphere is rotating around the horizontal x axis with frequency $\omega = (R/r)\Omega$. The moment of inertia around this axis is $I_x = (7/5)mr^2$, from the parallel-axis theorem. Therefore, the horizontal component of $\mathbf{L}$ has magnitude $L_x = I_x \omega = (7/5)mrR\Omega$. Multiplying this by the frequency (namely Ω) at which $\mathbf{L}$ swings around the z axis gives the result for $|d\mathbf{L}/dt|$ in Eq. (9.178). There is also a vertical component of $\mathbf{L}$ relative to the pivot, but this component doesn't change, so it doesn't come into $d\mathbf{L}/dt$. (The vertical component happens to equal $-mR^2\Omega$. This can be obtained by realizing that the sphere behaves like a point

mass for the present purpose, or by using the inertia tensor relative to the pivot, or by calculating $L_z = I_3\omega_3 \sin \theta + I_2\omega_2 \cos \theta$.)

2. The pivot must provide a downward force of $N - mg = (7/5)mr\Omega^2$, to make the net vertical force on the lollipop equal to zero. This result is slightly larger than the $mr\Omega^2$ result for the “sliding” setup in Exercise 9.53.
3. The sum of the horizontal forces at the pivot and the contact point must equal the required centripetal force of $mR\Omega^2$. But it is impossible to say how this force is divided up, without being given more information. ♣

9.23. Rolling coin

Choose the CM as the origin. The principal axes are then $\hat{x}_2$ and $\hat{x}_3$ (as shown in Fig. 9.98), along with $\hat{x}_1$ pointing into the paper. Assume that the precessional motion is counterclockwise when viewed from above, as indicated in Fig. 9.54. Let’s look at things in the frame that has the center of the contact-point circle on the ground as its origin, and that rotates with angular velocity $\Omega\hat{z}$. In this frame, the CM remains fixed, and the coin rotates with frequency ω' (in the language of Fig. 9.30) around the negative $\hat{x}_3$ axis. The nonslipping condition then says that $\omega'r = \Omega R$, and so $\omega' = \Omega R/r$. Therefore, the total angular velocity of the coin with respect to the lab frame is

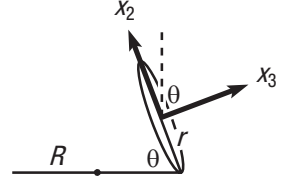


Fig. 9.98

$$\boldsymbol{\omega} = \Omega\hat{z} - \omega'\hat{x}_3 = \Omega\hat{z} - (R/r)\Omega\hat{x}_3. \quad (9.180)$$

But $\hat{z} = \sin \theta \hat{x}_2 + \cos \theta \hat{x}_3$, so we can write $\boldsymbol{\omega}$ in terms of the principal axes as

$$\boldsymbol{\omega} = \Omega \sin \theta \hat{x}_2 - \Omega \left(\frac{R}{r} - \cos \theta \right) \hat{x}_3. \quad (9.181)$$

The principal moments are

$$I_3 = (1/2)mr^2, \quad \text{and} \quad I_2 = (1/4)mr^2. \quad (9.182)$$

The angular momentum is $\mathbf{L} = I_2\omega_2\hat{x}_2 + I_3\omega_3\hat{x}_3$, so its horizontal component has length $L_\perp = I_2\omega_2 \cos \theta - I_3\omega_3 \sin \theta$, with leftward taken to be positive. Therefore, the magnitude of $d\mathbf{L}/dt$ is

$$\begin{aligned} \left| \frac{d\mathbf{L}}{dt} \right| &= \Omega L_\perp \\ &= \Omega (I_2\omega_2 \cos \theta - I_3\omega_3 \sin \theta) \\ &= \Omega \left[\left(\frac{1}{4}mr^2 \right) (\Omega \sin \theta) \cos \theta - \left(\frac{1}{2}mr^2 \right) \left(-\Omega(R/r - \cos \theta) \right) \sin \theta \right] \\ &= \frac{1}{4}mr\Omega^2 \sin \theta (2R - r \cos \theta), \end{aligned} \quad (9.183)$$

with a positive quantity corresponding to $d\mathbf{L}/dt$ pointing out of the page (at the instant shown).

The torque (relative to the CM) comes from the force at the contact point. There are two components to this force. The vertical component is mg , and the horizontal component is $m(R - r \cos \theta)\Omega^2$ leftward, because the CM moves in a circle of radius $(R - r \cos \theta)$. The torque therefore has magnitude

$$|\boldsymbol{\tau}| = mg(r \cos \theta) - m(R - r \cos \theta)\Omega^2(r \sin \theta), \quad (9.184)$$

with out of the page taken to be positive. Equating this $|\boldsymbol{\tau}|$ with the $|d\mathbf{L}/dt|$ in Eq. (9.183) gives

$$\Omega^2 = \frac{g}{\frac{3}{2}R \tan \theta - \frac{5}{4}r \sin \theta}. \quad (9.185)$$

The right-hand side must be positive if a solution for Ω is to exist. Therefore, the condition for the desired motion to be possible is

$$R > \frac{5}{6} r \cos \theta. \quad (9.186)$$

REMARKS:

1. For $\theta \rightarrow \pi/2$, Eq. (9.185) gives $\Omega \rightarrow 0$, as it should. And for $\theta \rightarrow 0$, we obtain $\Omega \rightarrow \infty$, which also makes sense.
2. For $r \cos \theta > R > (5/6)r \cos \theta$, the CM of the coin lies to the *left* of the center of the contact-point circle (at the instant shown). The centripetal force, $m(R - r \cos \theta)\Omega^2$, is therefore negative (which means that it is directed radially outward, to the right), but the motion is still possible. As R gets close to $(5/6)r \cos \theta$, the frequency Ω goes to infinity, which means that the radially outward force also goes to infinity. The coefficient of friction between the coin and the ground must therefore be correspondingly large.
3. We can consider a more general coin, whose density depends only on the distance from the center, and whose I_3 equals βmr^2 . For example, a uniform coin has $\beta = 1/2$, and a coin with all its mass on the edge has $\beta = 1$. By the perpendicular-axis theorem, $I_1 = I_2 = (1/2)\beta mr^2$, and you can show that the above methods yield

$$\Omega^2 = \frac{g}{(1 + \beta)R \tan \theta - (1 + \beta/2)r \sin \theta} \implies R > \left(\frac{1 + \beta/2}{1 + \beta} \right) r \cos \theta. \quad (9.187)$$

The larger β is, the smaller the lower bound on R is. But even if $\beta \rightarrow \infty$ (imagine long radial spokes attached to the coin, and eliminate the ground except for the ring of contact points, so it doesn't get in the way), R still can't be any smaller than $(r/2) \cos \theta$. ♣

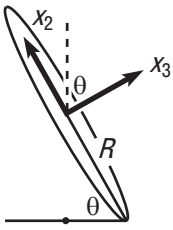


Fig. 9.99

9.24. Wobbling coin

- (a) Choose the CM as the origin. The principal axes are then $\hat{\mathbf{x}}_2$ and $\hat{\mathbf{x}}_3$ (as shown in Fig. 9.99), along with $\hat{\mathbf{x}}_1$ pointing into the paper. Assume that the precessional motion is counterclockwise when viewed from above, as indicated in Fig. 9.55. Consider the setup in the frame rotating with angular velocity $\Omega \hat{\mathbf{z}}$. In this frame, the location of the contact point remains fixed, and the coin rotates with frequency ω' (in the language of Fig. 9.30) around the negative $\hat{\mathbf{x}}_3$ axis. The radius of the circle of contact points on the table is $R \cos \theta$. Therefore, the nonslipping condition says that $\omega' R = \Omega (R \cos \theta)$, and so $\omega' = \Omega \cos \theta$. Hence, the total angular velocity of the coin with respect to the lab frame is

$$\boldsymbol{\omega} = \Omega \hat{\mathbf{z}} - \omega' \hat{\mathbf{x}}_3 = \Omega (\sin \theta \hat{\mathbf{x}}_2 + \cos \theta \hat{\mathbf{x}}_3) - (\Omega \cos \theta) \hat{\mathbf{x}}_3 = \Omega \sin \theta \hat{\mathbf{x}}_2. \quad (9.188)$$

In retrospect, it makes sense that $\boldsymbol{\omega}$ must point in the $\hat{\mathbf{x}}_2$ direction. Both the CM and the instantaneous contact point on the coin are at rest, so $\boldsymbol{\omega}$ must lie along the line containing these two points, that is, along the $\hat{\mathbf{x}}_2$ axis.

- (b) The principal moment around the $\hat{\mathbf{x}}_2$ axis is $I = mR^2/4$. The angular momentum is $\mathbf{L} = I\omega_2 \hat{\mathbf{x}}_2$, so its horizontal component has length $L_\perp = L \cos \theta = (I\omega_2) \cos \theta$. Therefore, $d\mathbf{L}/dt$ has magnitude

$$\left| \frac{d\mathbf{L}}{dt} \right| = \Omega L_\perp = \Omega \left(\frac{mR^2}{4} \right) (\Omega \sin \theta) \cos \theta, \quad (9.189)$$

and it points out of the page.

The torque (relative to the CM) is due to the normal force at the contact point. This normal force is essentially equal to mg , because the CM is assumed to be falling very slowly. Note that there is no sideways friction force at the contact point, because the CM is essentially motionless. The torque therefore has magnitude

$$|\tau| = mgR \cos \theta, \quad (9.190)$$

and it points out of the page. Equating this $|\tau|$ with the $|d\mathbf{L}/dt|$ in Eq. (9.189) gives

$$\Omega = 2\sqrt{\frac{g}{R \sin \theta}}. \quad (9.191)$$

REMARKS:

1. $\Omega \rightarrow \infty$ as $\theta \rightarrow 0$. This is evident if you do the experiment; the contact point travels very quickly around the circle.
 2. $\Omega \rightarrow 2\sqrt{g/R}$ as $\theta \rightarrow \pi/2$, which isn't so obvious. In this case, $\mathbf{L}$ points nearly vertically, and it traces out a tiny cone due to a tiny torque. In this $\theta \rightarrow \pi/2$ limit, Ω is also the frequency at which the plane of the coin spins around the vertical axis. If you spin a coin very fast about a vertical diameter, it initially undergoes a purely spinning motion with only one contact point. But then it gradually loses energy due to friction, until the spinning frequency slows down to $2\sqrt{g/R}$, at which point it begins to wobble (we're assuming that the coin is very thin, so that it can't balance on its edge). In the case where the coin is a US quarter (with $R \approx 0.012$ m), this critical frequency of $2\sqrt{g/R}$ turns out to be $\Omega \approx 57$ rad/s, which corresponds to about 9 Hz.
 3. The result in Eq. (9.191) is a special case of the result in Eq. (9.185) of Problem 9.23. The CM of the coin in Problem 9.23 is motionless if $R = r \cos \theta$. Plugging this into Eq. (9.185) gives $\Omega^2 = 4g/(r \sin \theta)$, which agrees with Eq. (9.191), because r was the coin's radius in Problem 9.23. ♣
- (c) Consider one revolution of the contact point around the z axis. Since the radius of the circle on the table is $R \cos \theta$, the contact point moves a distance $2\pi R \cos \theta$ around the coin during this time. Hence, the new contact point on the coin is a distance $2\pi R - 2\pi R \cos \theta$ away from the original contact point. The coin therefore appears to have rotated by a fraction $(1 - \cos \theta)$ of a full turn during this time. The frequency with which you see it turn is therefore

$$(1 - \cos \theta)\Omega = 2(1 - \cos \theta)\sqrt{\frac{g}{R \sin \theta}}. \quad (9.192)$$

REMARKS:

1. If $\theta \approx \pi/2$, then the frequency of Abe's rotation is essentially equal to Ω . This makes sense, because the top of Abe's head will be, say, always near the top of the coin, and this point will trace out a small circle around the z axis, with nearly the same frequency as the contact point.
2. As $\theta \rightarrow 0$, Abe appears to rotate with frequency $\theta^{3/2}\sqrt{g/R}$ (using $\sin \theta \approx \theta$ and $\cos \theta \approx 1 - \theta^2/2$). Therefore, although the contact point moves infinitely quickly in this limit, we nevertheless see Abe rotating very slowly, as you can experimentally verify.
3. All of the results for frequencies in this problem must be some multiple of $\sqrt{g/R}$, by dimensional analysis. But whether the multiplication factor is zero, infinite, or something in between, is not at all obvious.
4. An incorrect answer for the frequency of Abe's turning (when viewed from above) is that it equals the vertical component of $\boldsymbol{\omega}$, which is

$\omega_z = \omega \sin \theta = (\Omega \sin \theta) \sin \theta = 2(\sin \theta)^{3/2} \sqrt{g/R}$. This does not equal the result in Eq. (9.192). (It agrees at $\theta = \pi/2$ but is off by a factor of 2 for $\theta \rightarrow 0$.) This answer is incorrect because there is simply no reason why the vertical component of ω should equal the frequency of revolution of, say, Abe's nose, around the vertical axis. For example, at moments when ω passes through the nose, then the nose isn't moving at all, so it certainly cannot be described as moving around the vertical axis with frequency $\omega_z = \omega \sin \theta$. The result in Eq. (9.192) is a sort of average measure of the frequency of rotation. Even though any given point on the coin is not undergoing uniform circular motion, your eye sees the coin essentially rotating uniformly as a whole. ♣

9.25. Nutation cusps

- (a) Because both $\dot{\phi}$ and $\dot{\theta}$ are continuous functions of time, we must have $\dot{\phi} = \dot{\theta} = 0$ at a kink. Otherwise, either $d\theta/d\phi = \dot{\theta}/\dot{\phi}$ or $d\phi/d\theta = \dot{\phi}/\dot{\theta}$ would be well defined at the kink. Let the kink occur at $t = t_0$. Then the second of Eqs. (9.92) gives $\sin(\omega_n t_0) = 0$. Therefore, $\cos(\omega_n t_0) = \pm 1$, and the first of Eqs. (9.92) gives

$$\Delta\Omega = \mp\Omega_s, \quad (9.193)$$

as desired. Note that if $\cos(\omega_n t_0) = 1$, then $\Delta\Omega = -\Omega_s$, so Eq. (9.91) says that the kink occurs at the smallest value of θ , that is, at the highest point of the top's motion. And if $\cos(\omega_n t_0) = -1$, then $\Delta\Omega = \Omega_s$, so Eq. (9.91) again says that the kink occurs at the highest point of the top's motion. These results agree with the $\Delta\Omega = \pm\Omega_s$ plots in Fig. 9.38.

- (b) To show that these kinks are cusps, we will show that the slope of the θ vs. ϕ plot is infinite on either side of the kink. That is, we will show that $d\theta/d\phi = \dot{\theta}/\dot{\phi} = \pm\infty$. Consider the case where $\cos(\omega_n t_0) = 1$ and $\Delta\Omega = -\Omega_s$ (the $\cos(\omega_n t_0) = -1$ case proceeds similarly). With $\Delta\Omega = -\Omega_s$, Eq. (9.92) gives

$$\frac{\dot{\theta}}{\dot{\phi}} = \frac{\sin \theta_0 \sin \omega_n t}{1 - \cos \omega_n t}. \quad (9.194)$$

Let $t = t_0 + \epsilon$. Using $\cos(\omega_n t_0) = 1$ and $\sin(\omega_n t_0) = 0$, and expanding Eq. (9.194) to lowest order in ϵ , we find

$$\frac{\dot{\theta}}{\dot{\phi}} = \frac{\sin \theta_0 (\omega_n \epsilon)}{\omega_n^2 \epsilon^2 / 2} = \frac{2 \sin \theta_0}{\omega_n \epsilon}. \quad (9.195)$$

For infinitesimal ϵ , this switches from $-\infty$ to $+\infty$ as ϵ passes through zero. The point here is that although both $\dot{\theta}$ and $\dot{\phi}$ go to zero, $\dot{\phi}$ goes quadratically, whereas $\dot{\theta}$ goes only linearly.

9.26. Nutation circles

- (a) In the limit $\omega_3 \gg \Omega_s$, the original $\mathbf{L}$ points essentially along the x_3 direction, with magnitude $I_3 \omega_3$. So it makes an angle of essentially θ_0 with the vertical z axis. Now consider the quick kick. If $\dot{\phi}$ (which is the angular speed around the vertical z axis) suddenly increases by $\Delta\Omega$, then this corresponds to a sudden increase in angular speed around the x_2 axis of $\sin \theta_0 \Delta\Omega$ (see Fig. 9.100). The kick therefore produces an angular momentum component (relative to the pivot) of $I \sin \theta_0 \Delta\Omega$ in the x_2 direction. So from Fig. 9.100, the angle that the new $\mathbf{L}$ makes with the z axis is (using $\Delta\Omega \ll \omega_3$)

$$\theta'_0 \approx \theta_0 - \frac{I \sin \theta_0 \Delta\Omega}{I_3 \omega_3} = \theta_0 - \frac{\sin \theta_0 \Delta\Omega}{\omega_n}, \quad (9.196)$$

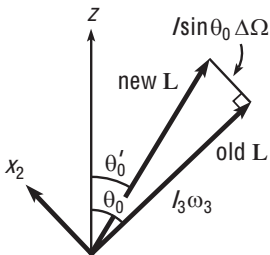


Fig. 9.100

where we have used the definition of ω_n from Eq. (9.86). We see that the effect of the kick is to make $\mathbf{L}$ quickly change its θ value (by only a small amount, because we are assuming $\omega_3 \gg \Delta\Omega$). The ϕ value doesn't immediately change, because immediately after the kick the x_3 axis is still in exactly the same place, because it hasn't had time to move.

- (b) After the kick is finished, $\mathbf{L}$ will trace out a cone around the z axis at approximately the same rate as before, namely Ω_s . (None of the relevant quantities in $\boldsymbol{\tau} = d\mathbf{L}/dt$ have changed much from the original circular-precession case, so the precession frequency is basically the same.) So the new $\mathbf{L}$ has its (ϕ, θ) coordinates given by

$$(\phi(t), \theta(t))_{\mathbf{L}} = \left(\Omega_s t, \theta_0 - \frac{\sin \theta_0 \Delta\Omega}{\omega_n} \right). \quad (9.197)$$

We're assuming that Ω_s is very small, so if you want, you can ignore the Ω_s term and just consider $\mathbf{L}$ to be fixed, at least on the time scale of the nutations (see the remark below). But this term will cancel anyway in producing the following equation. Looking at Eq. (9.91), we see that the coordinates of the CM relative to $\mathbf{L}$ are

$$(\phi(t), \theta(t))_{\text{CM-L}} = \left(\left(\frac{\Delta\Omega}{\omega_n} \right) \sin \omega_n t, \left(\frac{\Delta\Omega}{\omega_n} \sin \theta_0 \right) \cos \omega_n t \right). \quad (9.198)$$

The $\sin \theta_0$ factor in $\theta(t)$ is exactly what is needed for the CM to trace out a circle around $\mathbf{L}$, because a change in ϕ corresponds to a CM spatial change of $\ell \Delta\phi \sin \theta_0$, whereas a change in θ corresponds to a CM spatial change of $\ell \Delta\theta$.

Now let's see how this relates to a free top. For $\omega_n \gg \Omega_s$, the time scale on which $\mathbf{L}$ changes due to the gravitational torque (namely $1/\Omega_s$) is very long compared with the time scale of the nutation (namely $1/\omega_n$). Therefore, since $\mathbf{L}$ is essentially motionless on the time scale of $1/\omega_n$, the effects of gravity are negligible on this time scale. Hence, the system should behave like a free top. So let's verify that the results here do indeed agree with the results from Section 9.6.2.

Equation (9.55) in Section 9.6.2 says that the frequency of the precession of $\hat{\mathbf{x}}_3$ around $\mathbf{L}$ for a free top is L/I . But the frequency of the precession of $\hat{\mathbf{x}}_3$ around $\mathbf{L}$ in the present problem is ω_n , so this had better be equal to L/I . And indeed, L is essentially equal to $I_3\omega_3$, so $\omega_n \equiv I_3\omega_3/I = L/I$. Therefore, for short enough time scales (short enough so that $\mathbf{L}$ doesn't move much), a nutating top with $\omega_3 \gg \Delta\Omega \gg \Omega_s$ looks very much like a free top.

REMARK: We need the $\Delta\Omega \gg \Omega_s$ condition so that the nutation motion looks roughly like circles (that is, like the top plot in Fig. 9.38, and not the others). This requirement can be seen by the following reasoning. The time for one period of the nutation motion is $2\pi/\omega_n$. From Eq. (9.91), $\phi(t)$ increases by $\Delta\phi = 2\pi\Omega_s/\omega_n$ during this time. And also from Eq. (9.91), the diameter d of the "circle" along the ϕ axis is roughly $d = 2\Delta\Omega/\omega_n$. The motion looks approximately like a circle if $d \gg \Delta\phi$, that is, if $\Delta\Omega \gg \Omega_s$. ♣

9.27. Rolling without slipping

Yes, there are indeed other ways. It isn't necessary for the contact points to form a vertical great circle. They can form a smaller tilted circle, as shown in Fig. 9.101. In this case the angular velocity vector points up to the left. There is indeed no slipping here, and the ball rolls straight. You can convince yourself of this in the following way. Imagine the ball hovering and spinning in place with its center at rest. At all times, the point at the bottom of the ball is moving directly out of the page; the collection of these points is the tilted circle in Fig. 9.101. Let the speed of the instantaneous bottom point be v , and now imagine taking a horizontal sheet of paper and sliding it out of the page with constant speed v , just beneath the ball, barely touching it. Since this is the

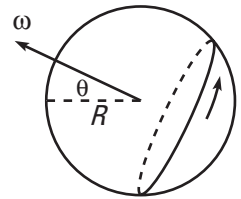


Fig. 9.101

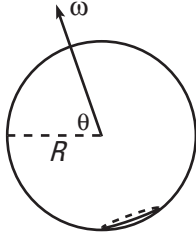


Fig. 9.102

same speed as the bottom point on the ball (which is the contact point, because it's at the bottom), there is no slipping between the ball and the paper. Finally, if we go to the reference frame of the paper, we have the desired motion of a ball rolling without slipping on a flat surface.

If θ is the angle that ω makes with the horizontal, then the radius of the contact circle is $R \cos \theta$. The linear speed of the ball is therefore $\omega(R \cos \theta)$. If you want, you can think of this as $(\omega \cos \theta)R$. That is, only the horizontal component of ω leads to translational motion of the ball. The vertical component simply yields a spinning motion around the vertical diameter. If θ approaches 90° , then the contact circle is very small (see Fig. 9.102), so the speed of the ball is very small (for a given ω).

In the real world, the contact area between the ball and the surface isn't an idealized single point, so if the contact circle isn't a vertical great circle, there is inevitably some slipping at the points near the bottom of the ball. The smaller the contact circle, the more the slipping, because the slipping arises from the twisting motion around the vertical diameter, which comes from the vertical component of ω . If you take an actual ball and roll it with a tilted ω , you will find that the rolling motion is noisier the more that ω is tilted. What you hear is the slipping due to the extended contact region. But in an ideal world (or perhaps in a bowling alley, which is a reasonable approximation thereof), a tilted ω wouldn't be any noisier.

9.28. Rolling straight?

Intuitively, it is fairly clear that the sphere cannot change direction, but it is a little tricky to prove. Qualitatively, we can reason as follows. Assume that there is a nonzero friction force at the contact point. (The normal force is irrelevant here, because it doesn't provide a torque about the center, because the contact point is always directly below the center. This is what is special about a sphere.) Then the sphere accelerates in the direction of this force. However, you can show with the right-hand rule that this force produces a torque that causes the angular momentum to change in a way that corresponds to the sphere accelerating in the direction *opposite* to the friction force (assuming that the sphere is rolling without slipping). There is thus a contradiction, unless the friction force is equal to zero.

Let's now be rigorous. Let the angular velocity of the sphere be ω , which may point diagonally if the contact points don't form a vertical great circle on the sphere. The nonslipping condition says that the sphere's CM velocity equals

$$\mathbf{v} = \omega \times (R\hat{\mathbf{z}}). \quad (9.199)$$

The sphere's angular momentum is

$$\mathbf{L} = I\omega. \quad (9.200)$$

The friction force (if it exists) from the ground at the contact point changes both the momentum and the angular momentum. $\mathbf{F} = d\mathbf{p}/dt$ gives

$$\mathbf{F} = m \frac{d\mathbf{v}}{dt}. \quad (9.201)$$

And $\boldsymbol{\tau} = d\mathbf{L}/dt$ (relative to the center) gives

$$(-R\hat{\mathbf{z}}) \times \mathbf{F} = \frac{d\mathbf{L}}{dt}, \quad (9.202)$$

because the force is applied at position $-R\hat{\mathbf{z}}$ relative to the center. We will now show that the preceding four equations imply that $\dot{\omega} = \mathbf{0}$. From Eq. (9.199), this then implies that $\dot{\mathbf{v}} = \mathbf{0}$, as desired.

Equations (9.199) and (9.201) give $\mathbf{F} = m\dot{\omega} \times (R\hat{\mathbf{z}})$. Plugging this $\mathbf{F}$, along with the $\mathbf{L}$ from Eq. (9.200), into Eq. (9.202) then gives

$$(-R\hat{\mathbf{z}}) \times (m\dot{\omega} \times (R\hat{\mathbf{z}})) = I\dot{\omega}. \quad (9.203)$$

The vector $\dot{\boldsymbol{\omega}}$ must lie in the horizontal plane, because it equals the cross product of two vectors, one of which is the vertical $\hat{\mathbf{z}}$. This implies that (as you can verify) $\hat{\mathbf{z}} \times (\dot{\boldsymbol{\omega}} \times \hat{\mathbf{z}}) = \dot{\boldsymbol{\omega}}$. Therefore, Eq. (9.203) gives

$$-mR^2\dot{\boldsymbol{\omega}} = \frac{2}{5}mR^2\dot{\boldsymbol{\omega}}, \quad (9.204)$$

and so $\dot{\boldsymbol{\omega}} = \mathbf{0}$, as we wanted to show.

9.29. Ball on paper

Our strategy will be to produce, and then equate, two different expressions for the total change in the angular momentum of the ball, relative to its center. The first comes from the effects of the friction force on the ball. The second comes from looking at the general form of the initial and final motion.

To produce our first expression for $\Delta\mathbf{L}$, note that the normal force provides no torque, so we can ignore it. The friction force, $\mathbf{F}$, from the paper changes both $\mathbf{p}$ and $\mathbf{L}$ according to

$$\begin{aligned} \Delta\mathbf{p} &= \int \mathbf{F} dt, \\ \Delta\mathbf{L} &= \int \boldsymbol{\tau} dt = \int (-R\hat{\mathbf{z}}) \times \mathbf{F} dt = (-R\hat{\mathbf{z}}) \times \int \mathbf{F} dt. \end{aligned} \quad (9.205)$$

Both of these integrals run over the entire slipping time, which may include time on the table after the ball leaves the paper. In the second line above, we have used the fact that the friction force always acts at the same location, namely $(-R\hat{\mathbf{z}})$, relative to the center of the ball. The two above equations yield

$$\Delta\mathbf{L} = (-R\hat{\mathbf{z}}) \times \Delta\mathbf{p}. \quad (9.206)$$

To produce our second equation for $\Delta\mathbf{L}$, let's examine how $\mathbf{L}_\perp$ (the horizontal component of $\mathbf{L}$)³² is related to $\mathbf{p}$ when the ball is rolling without slipping, which is the case at both the start and finish. When the ball is not slipping, we have the situation shown in Fig. 9.103 (assuming that the ball is rolling to the right). The magnitudes of $\mathbf{p}$ and $\mathbf{L}_\perp$ are given by

$$\begin{aligned} p &= mv, \\ L_\perp &= I\omega_\perp = \frac{2}{5}mR^2\omega_\perp = \frac{2}{5}Rm(R\omega_\perp) = \frac{2}{5}Rmv = \frac{2}{5}Rp, \end{aligned} \quad (9.207)$$

where we have used the nonslipping condition, $v = R\omega_\perp$. (The actual $I = (2/5)mR^2$ value for a solid sphere won't be important for the final result.) We can now combine the directions of $\mathbf{L}_\perp$ and $\mathbf{p}$ in Fig. 9.103 with the above $L_\perp = (2/5)Rp$ scalar relation to write

$$\mathbf{L}_\perp = \frac{2}{5}R\hat{\mathbf{z}} \times \mathbf{p}, \quad (9.208)$$

where $\hat{\mathbf{z}}$ points out of the page. Since this relation is true at both the start and the finish, it must also be true for the differences in $\mathbf{L}_\perp$ and $\mathbf{p}$. That is,

$$\Delta\mathbf{L}_\perp = \frac{2}{5}R\hat{\mathbf{z}} \times \Delta\mathbf{p}. \quad (9.209)$$

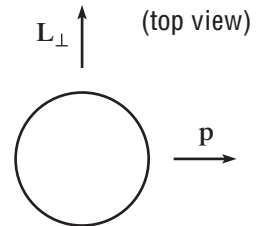


Fig. 9.103

³² The vertical component of $\mathbf{L}$ (which is nonzero only if $\boldsymbol{\omega}$ has a vertical component, which is the case if the contact points don't form a vertical great circle on the ball) is constant, because the torque from friction provides only a horizontal torque. We may therefore ignore it, because we are concerned only with $\Delta\mathbf{L}$.

But $\Delta \mathbf{L}_\perp = \Delta \mathbf{L}$ because the vertical component of $\mathbf{L}$ doesn't change, so Eqs. (9.206) and (9.209) give

$$(-R\hat{\mathbf{z}}) \times \Delta \mathbf{p} = \frac{2}{5}R\hat{\mathbf{z}} \times \Delta \mathbf{p} \implies \mathbf{0} = \hat{\mathbf{z}} \times \Delta \mathbf{p}. \quad (9.210)$$

There are three ways this cross product can be zero:

- $\Delta \mathbf{p}$ is parallel to $\hat{\mathbf{z}}$. But it isn't, because $\Delta \mathbf{p}$ lies in the horizontal plane.
- $\hat{\mathbf{z}} = \mathbf{0}$. Not true.
- $\Delta \mathbf{p} = \mathbf{0}$. This is the only possibility, so it must be true. Therefore, $\Delta \mathbf{v} = \mathbf{0}$, as we wanted to show.

Note that the line of the final motion may very well be shifted sideways from the line of the initial motion. But what we've shown is that the final line is parallel to the initial line, and the speed is the same.

REMARKS:

1. As stated in the problem, it's fine if you move the paper in a jerky motion, so that the ball slips around on it. We assumed nothing about the nature of the friction force in the above reasoning. And we used the nonslipping condition only at the initial and final times. The intermediate motion is arbitrary.
2. As a special case, if you start a ball at rest on a piece of paper, then no matter how you choose to (horizontally) slide the paper out from underneath the ball, the ball will be at rest in the end. The final position will most likely be different from the initial position, but the ball will be at rest wherever it ends up.
3. You are encouraged to experimentally verify that these crazy claims are true. Make sure that the paper doesn't wrinkle, because a wrinkle would allow a force to be applied at a point other than the contact point. And balls that don't squish are much better, of course, because the contact region better resembles a single point. ♣

9.30. Ball on a turntable

Let the angular velocity of the turntable be $\Omega\hat{\mathbf{z}}$, and let the angular velocity of the ball (with respect to the lab frame) be $\boldsymbol{\omega}$, which may point diagonally if the contact points don't form a vertical great circle on the ball. If the ball is at position $\mathbf{r}$ (with respect to the lab frame), then its CM velocity (with respect to the lab frame) may be broken up into the velocity of the turntable (at position $\mathbf{r}$) plus the ball's velocity with respect to the turntable. The nonslipping condition says that the latter is $\boldsymbol{\omega} \times (a\hat{\mathbf{z}})$, where a is the radius of the ball.³³ The ball's velocity with respect to the lab frame is thus

$$\mathbf{v} = (\Omega\hat{\mathbf{z}}) \times \mathbf{r} + \boldsymbol{\omega} \times (a\hat{\mathbf{z}}). \quad (9.211)$$

The important point to realize in this problem is that the friction force from the turntable is responsible for changing both the ball's linear momentum and its angular momentum. In particular, $\mathbf{F} = d\mathbf{p}/dt$ gives

$$\mathbf{F} = m \frac{d\mathbf{v}}{dt}. \quad (9.212)$$

And the angular momentum of the ball is $\mathbf{L} = I\boldsymbol{\omega}$, so $\boldsymbol{\tau} = d\mathbf{L}/dt$ (relative to the center of the ball) gives

$$(-a\hat{\mathbf{z}}) \times \mathbf{F} = I \frac{d\boldsymbol{\omega}}{dt}, \quad (9.213)$$

because the force is applied at position $-a\hat{\mathbf{z}}$ relative to the center.

³³ The velocity with respect to the turntable is actually $\boldsymbol{\omega}_t \times (a\hat{\mathbf{z}})$, where $\boldsymbol{\omega}_t$ is the angular velocity in the turntable frame. But $\boldsymbol{\omega}_t$ differs from $\boldsymbol{\omega}$ simply by the angular velocity of the turntable, which is $\Omega\hat{\mathbf{z}}$. This implies that $\boldsymbol{\omega}_t \times (a\hat{\mathbf{z}}) = \boldsymbol{\omega} \times (a\hat{\mathbf{z}})$.

We will now use the previous three equations to demonstrate that the ball undergoes circular motion. Our goal will be to produce an equation of the form,

$$\frac{d\mathbf{v}}{dt} = \Omega' \hat{\mathbf{z}} \times \mathbf{v}, \quad (9.214)$$

because this describes circular motion with frequency Ω' (to be determined).³⁴ We will eliminate $\mathbf{F}$ first, and then $\boldsymbol{\omega}$. Plugging the expression for $\mathbf{F}$ from Eq. (9.212) into Eq. (9.213) gives

$$(-a\hat{\mathbf{z}}) \times \left(m \frac{d\mathbf{v}}{dt}\right) = I \frac{d\boldsymbol{\omega}}{dt} \implies \frac{d\boldsymbol{\omega}}{dt} = -\left(\frac{am}{I}\right) \hat{\mathbf{z}} \times \frac{d\mathbf{v}}{dt}. \quad (9.215)$$

Taking the derivative of Eq. (9.211) gives

$$\begin{aligned} \frac{d\mathbf{v}}{dt} &= \Omega \hat{\mathbf{z}} \times \frac{d\mathbf{r}}{dt} + \frac{d\boldsymbol{\omega}}{dt} \times (a\hat{\mathbf{z}}) \\ &= \Omega \hat{\mathbf{z}} \times \mathbf{v} - \left(\left(\frac{am}{I}\right) \hat{\mathbf{z}} \times \frac{d\mathbf{v}}{dt}\right) \times (a\hat{\mathbf{z}}). \end{aligned} \quad (9.216)$$

Since we know that the vector $d\mathbf{v}/dt$ lies in the horizontal plane, it is easy to work out the cross product in the second term here (or just use the identity $(\mathbf{A} \times \mathbf{B}) \times \mathbf{C} = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{B} \cdot \mathbf{C})\mathbf{A}$) to obtain

$$\frac{d\mathbf{v}}{dt} = \Omega \hat{\mathbf{z}} \times \mathbf{v} - \left(\frac{ma^2}{I}\right) \frac{d\mathbf{v}}{dt} \implies \frac{d\mathbf{v}}{dt} = \left(\frac{\Omega}{1 + (ma^2/I)}\right) \hat{\mathbf{z}} \times \mathbf{v}. \quad (9.217)$$

For a uniform sphere, $I = (2/5)ma^2$, so we obtain

$$\frac{d\mathbf{v}}{dt} = \left(\frac{2}{7}\Omega\right) \hat{\mathbf{z}} \times \mathbf{v}. \quad (9.218)$$

Therefore, in view of Eq. (9.214), we see that the ball undergoes circular motion, with a frequency equal to $2/7$ times the frequency of the turntable. Note that this result for the frequency doesn't depend on initial conditions. For an extension to this problem, see Weltner (1987) and references therein.

REMARKS:

1. Integrating Eq. (9.218) from the initial time to some later time gives

$$\mathbf{v} - \mathbf{v}_0 = \left(\frac{2}{7}\Omega\right) \hat{\mathbf{z}} \times (\mathbf{r} - \mathbf{r}_0). \quad (9.219)$$

This may be written (as you can verify) in the more suggestive (if not more frightening) form,

$$\mathbf{v} = \left(\frac{2}{7}\Omega\right) \hat{\mathbf{z}} \times \left(\mathbf{r} - \left(\mathbf{r}_0 + \frac{7}{2\Omega}(\hat{\mathbf{z}} \times \mathbf{v}_0)\right)\right). \quad (9.220)$$

This equation describes circular motion, with the center located at the point,

$$\mathbf{r}_c = \mathbf{r}_0 + \frac{7}{2\Omega}(\hat{\mathbf{z}} \times \mathbf{v}_0), \quad (9.221)$$

³⁴ Equation (9.214) describes circular motion because it implies that $\mathbf{v}$ has constant magnitude (since the change in $\mathbf{v}$ is perpendicular to $\mathbf{v}$), and so the direction of $\mathbf{v}$ changes at the constant angular rate Ω' . The only curve with these properties is a circle. If you want to be more mathematical, you can integrate Eq. (9.214) to obtain $\mathbf{v} = \Omega' \hat{\mathbf{z}} \times \mathbf{r} + \mathbf{C}$, which can be written as $d\mathbf{r}/dt = \Omega' \hat{\mathbf{z}} \times (\mathbf{r} - \mathbf{r}_c)$, which implies that $d(\mathbf{r} - \mathbf{r}_c)/dt = \Omega' \hat{\mathbf{z}} \times (\mathbf{r} - \mathbf{r}_c)$, which means that the length of $\mathbf{r} - \mathbf{r}_c$ doesn't change (because its change is perpendicular to itself). And since the changes are also constrained to be perpendicular to $\hat{\mathbf{z}}$, we must have a horizontal circle.

and with radius,

$$R = |\mathbf{r}_0 - \mathbf{r}_c| = \frac{7}{2\Omega} |\hat{\mathbf{z}} \times \mathbf{v}_0| = \frac{7v_0}{2\Omega}. \quad (9.222)$$

2. There are a few special cases to consider:

- If $v_0 = 0$ (that is, if the spinning motion of the ball exactly cancels the rotational motion of the turntable, so that the CM of the ball is at rest in the lab frame), then $R = 0$ and the ball remains in the same place, as it should.
- If the ball is initially not spinning, and just moving along with the turntable, then $v_0 = \Omega r_0$. The radius of the circle is therefore $R = (7/2)r_0$, and its center is located at, from Eq. (9.221),

$$\mathbf{r}_c = \mathbf{r}_0 + \frac{7}{2\Omega} (-\Omega \mathbf{r}_0) = -\frac{5\mathbf{r}_0}{2}. \quad (9.223)$$

The point on the circle diametrically opposite to the initial point is therefore a distance $r_c + R = (5/2)r_0 + (7/2)r_0 = 6r_0$ from the center of the turntable.

- If we want the center of the circle to be the center of the turntable, then Eq. (9.221) says that we need $(7/2\Omega)\hat{\mathbf{z}} \times \mathbf{v}_0 = -\mathbf{r}_0$. This implies that $\mathbf{v}_0$ has magnitude $v_0 = (2/7)\Omega r_0$ and points tangentially in the same direction as the turntable moves. That is, the ball moves at $2/7$ times the velocity of the point on the turntable right beneath it.
3. The fact that the frequency $(2/7)\Omega$ is a rational multiple of Ω means that the ball will eventually return to the same point on the turntable. In the lab frame, the ball traces out two circles in the time it takes the turntable to undergo seven revolutions. From the point of view of someone on the turntable, the ball “spirals” around five times before returning to the original position.
4. If we look at a ball with moment of inertia $I = \beta m a^2$ (so a uniform sphere has $\beta = 2/5$), then Eq. (9.217) shows that the “ $2/7$ ” in Eq. (9.218) gets replaced by “ $\beta/(1 + \beta)$.” If a ball has most of its mass concentrated at its center (so that $\beta \rightarrow 0$), then the frequency of the circular motion goes to 0, and the radius goes to ∞ (as long as $v_0 \neq 0$). ♣